Fundamental Theorem of Matrix Product States

A Formalization Blueprint

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Contents

Chapter 1

Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices $A = \{A^i\}_{i=0}^{d-1}$, which assigns to each chain length N the vector $|V^{(N)}(A)\rangle$ with coefficients indexed by $\sigma = (i_1, \dots, i_N)$,

$$V^{(N)}(A)_\sigma = \operatorname{tr}(A^{i_1} \cdots A^{i_N}).$$

The main non-periodic route follows the canonical-form and basis-of-normal-tensors argument of [CPGSV16, CPGSV21], with the single-block injective case anchored in the earlier representation theorem of [PGVWC07].

Notation

$\mathbb{C}$, $\mathbb{N}$, $\mathbb{Z}$	complex numbers, natural numbers, integers
$M_D(\mathbb{C})$, $\mathrm{GL}_D(\mathbb{C})$	complex matrices and invertible complex matrices
$\operatorname{tr}(X)$, $\operatorname{span}_{\mathbb{C}}(S)$	trace and complex linear span
$A = \{A^i\}_{i=0}^{d-1}$	a translation-invariant MPS tensor
$w = (i_1, \dots, i_L)$, A^w	a word and the product $A^{i_1} \cdots A^{i_L}$
$\lvert V^{(N)}(A)\rangle$	the length-N matrix product vector
$V^{(N)}(A)_\sigma$	the coefficient of $\lvert V^{(N)}(A)\rangle$ at σ
$O_{AB}(N)$	the bilinear overlap $\overline{\langle V^{(N)}(A) \lvert V^{(N)}(B)\rangle}$
$\mathcal{E}_A(X)$	the transfer map $\sum_i A^i X (A^i)^\dagger$
$A^{[L]}$	the length-L blocked tensor

In canonical form one writes

$$A^i = \bigoplus_{j=1}^{g} \mu_j A^i_j, \qquad B^i = \bigoplus_{k=1}^{h} \nu_k B^i_k,$$

where the displayed blocks are normal tensors. If $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every positive chain length N, then the bases of normal tensors have the same size. After relabelling the blocks of B, each matched pair satisfies

$$B^i_j = \zeta_j X_j A^i_j X_j^{-1},$$

for some invertible matrix X_j and nonzero scalar ζ_j, and the equal case compares the coefficient power sums attached to the relabelled bases. The block gauges and the weight comparison then combine into a single gauge conjugacy of the two weighted block-diagonal tensors. For a single injective block this reduces to the familiar conclusion $B^i = XA^iX^{-1}$ for all physical indices i.

Chapters 2–13 contain the aperiodic proof route. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and site-periodic tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4, 5, 6, and 7 develop the positive-map, multiplicative-domain, Perron–Frobenius, and overlap-decay tools used to control normal blocks. Chapters 8 and 9 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapters 11 and 12 isolate block permutation, block separation, bases of normal tensors, and the power-sum comparison. The after-blocking reductions of Chapter 10 give the common primitive block decompositions used immediately in Chapter 13, which records the conditional non-periodic theorem statements. The same-index direct-sum lemmas used after a block correspondence has already been fixed are collected later in Chapter 16, because they are facts about block-diagonal gauges, not part of the canonical-form reduction or of the BNT block-matching proof. The remaining BNT matching and fixed-length span inputs are kept as explicit hypotheses until the corresponding source steps are supplied.

The chapters after Chapter 13 collect applications, alternative formulations, and later tensor-network material. Chapter 14 uses the Fundamental Theorem to obtain virtual symmetry gauges, projective bond representations, and string-order criteria; its Kraus-mixing step appeals only to the later channel-representation chapter, not to any additional input for the Fundamental Theorem. Chapter 15 develops the direct periodic theorem in irreducible form, together with the compatibility theorems for physical-index rotation (Definition 14.1 in Chapter 14). Chapter 16 gives the fixed-length algebraic Fundamental Theorem of [MGRPG+18], and Chapter 17 records the PEPS analogs. Parent Hamiltonians, correlations, and examples are treated after the PEPS material. Channel representations and normal forms are then collected immediately before quantum dynamical semigroups and the operator-convexity material; entropy and matrix product density operators follow. These later topics use the aperiodic theorem, or develop later finite-dimensional channel theory, rather than enter its proof.

Chapter 2

Matrix Product Vectors

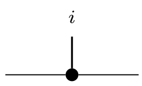
Fix a physical dimension d and a bond dimension D. This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). Lean ✓ Stmt A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \qquad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

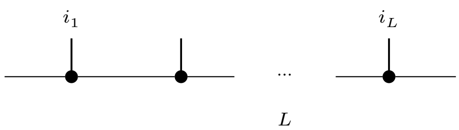


The black node denotes the tensor A, the horizontal legs are virtual, and the upper leg is the physical index i.

Definition 2.2 (Word evaluation). Lean ✓ Stmt Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \cdots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^{\emptyset} = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A, the virtual legs remain open, and the physical legs are labelled by the word $(i_1, \dots, i_L)$.

Lemma 2.3 (Multiplicativity of word evaluation). Lean ✓ Stmt *For any words* w_1, w_2, $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.

Proof. ✓ Proof Immediate from the definition of A^w. $\square$

Definition 2.4 (Matrix product vector). Lean ✓ Stmt The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1,\dots,i_N=0}^{d-1} \mathrm{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

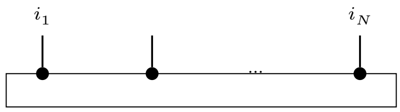
Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \mathrm{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w, we also write

$$c_w(A) := \mathrm{tr}(A^w).$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \ge 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

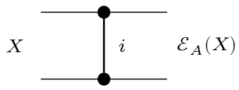


of N copies of the local tensor A, with the outer virtual legs closed by the trace.

Definition 2.5 (Transfer map). Lean ✓ Stmt The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A, the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.6. ✓ Stmt The transfer map is automatically positive: if $X \ge 0$ then $\mathcal{E}_A(X) \ge 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

2.2 Site-periodic tensor families

Definition 2.7 (Site-periodic tensor family). Lean ✓ Stmt For a fixed period m, a *site-periodic tensor family* is a family of local tensors indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D.

Definition 2.8 (Periodic same-state and gauge-equivalence relations). Lean Lean ✓ Stmt For site-periodic tensors A, B at fixed period m:

- equality of the induced site-periodic states;
- cyclic gauge equivalence.

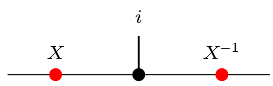
Lemma 2.9 (Equivalence structures for periodic relations). *Lean* *Lean* *✓ Stmt* *Both periodic relations are equivalence relations (reflexive, symmetric, transitive).*

2.3 Gauge equivalence and same MPV

Definition 2.10 (Gauge equivalence). Lean ✓ Stmt Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that, for every $i \in \{0, \dots, d-1\}$,

$$B^i = XA^iX^{-1}.$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index i, and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.11 (Same MPV). Lean ✓ Stmt Two tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N.

Definition 2.12 (Same MPV - different bond dimensions). Lean ✓ Stmt For tensors A and B of possibly different bond dimensions D_1 and D_2, we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every N.

Definition 2.13 (Positive-length same MPV - different bond dimensions). Lean ✓ Stmt For tensors A and B of possibly different bond dimensions, we write $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N > 0$.

Lemma 2.14 (Length-zero MPV coefficient). *Lean* *✓ Stmt* *For any MPS tensor A with bond dimension D, the length-zero MPV coefficient is D, the trace of the identity on the bond space.*

Lemma 2.15 (Positive-length equality with equal bond dimensions). *Lean* *✓ Stmt* *If two tensors have the same positive-length MPV family and the same bond dimension, then they have the same MPV family at every length.*

Remark 2.16. ✓ Stmt Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.17 (Proportional MPV). Lean ✓ Stmt Two tensors A and B (of possibly different bond dimensions) generate weakly proportional MPV families if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$.

Definition 2.18 (Nonzero proportional MPV). Lean ✓ Stmt Two tensors A and B (of possibly different bond dimensions) generate nonzero proportional MPV families if for every N there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. This is the projective reading of the proportionality hypothesis in [CPGSV16, Theorem II.1].

Definition 2.19 (Eventually nonzero proportional MPV). Lean ✓ Stmt Two tensors A and B are eventually nonzero proportional if, for all sufficiently large N, there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. This auxiliary form is used with the eventual linear-independence statement Lemma 12.3.

Lemma 2.20 (Elementary nonzero proportionality rules). *Lean* *Lean* *Lean* *Lean* *✓ Stmt* *Nonzero proportionality implies weak proportionality and eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar* 1.

Proof. ✓ Proof Forgetting the nonvanishing condition gives weak proportionality. The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar c_N at each length, and equality is the case $c_N = 1$. □

Lemma 2.21 (Scaling of word evaluation). *Lean* *✓ Stmt* *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w.

Proof. ✓ Proof Induction on w: the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ. □

Lemma 2.22 (Scaling of MPV components). *Lean* *✓ Stmt* *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for every N and every configuration $\sigma = (i_1, \dots, i_N)$,*

$$V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof Apply Lemma 2.21 to the word $w = (i_1, \dots, i_N)$ of length N, then take the trace; the scalar ζ^N pulls out by linearity. □

Definition 2.23 (Gauge-phase equivalence). Lean Lean ✓ Stmt Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that, for every physical index i,

$$B^i = \zeta X A^i X^{-1}.$$

Proof. ✓ Proof Gauge equivalence is the special case $\zeta = 1$. □

Remark 2.24. ✓ Stmt After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.25 (Gauge covariance of word evaluation). *Lean* *✓ Stmt* *If $B^i = XA^iX^{-1}$ for all i, then for every word w,*

$$B^w = XA^wX^{-1}.$$

Proof. ✓ Proof Induction on the word length, using $XX^{-1} = \mathbb{1}$. □

Theorem 2.26 (Gauge invariance of MPVs). Lean ✓ Stmt *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. ✓ Proof Let $B^i = XA^iX^{-1}$. By Lemma 2.25, $B^w = XA^wX^{-1}$ for every word w. Then $\mathrm{tr}(B^w) = \mathrm{tr}(XA^wX^{-1}) = \mathrm{tr}(A^w)$ by cyclicity of the trace. □

Theorem 2.27 (Gauge-phase scaling of MPVs). Lean ✓ Stmt *If, for every physical index i,*

$$B^i = \zeta X A^i X^{-1},$$

then for every system size N and configuration σ one has

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof Combine gauge covariance of word evaluation with scalar scaling, then take traces. □

2.4 Injectivity and normality

Definition 2.28 (Injectivity). Lean ✓ Stmt A tensor A is *injective* if its matrices span the full matrix algebra: $\mathrm{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.29 (L-block injectivity). Lean ✓ Stmt A tensor A is *L-block injective* if $\mathrm{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Lemma 2.30 (One-site injectivity gives one-block injectivity). Lean ✓ Stmt *If the one-site matrices $\{A^i\}_{i=0}^{d-1}$ span $M_D(\mathbb{C})$, then the length-one word products span $M_D(\mathbb{C})$, so A is 1-block injective.*

Proof. ✓ Proof The length-one words are exactly the letters $i \in \{0, \dots, d-1\}$, and evaluating a length-one word gives the matrix A^i. Thus the length-one word span is the same span as in one-site injectivity. □

Definition 2.31 (Normal tensor). Lean ✓ Stmt A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0-block injective.

Remark 2.32. ✓ Stmt The algebraic characterization of normality in Definition 2.31 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Theorem 8.68, Theorem 8.85, and Theorem 9.22. In particular, every subsequent use of “normal” here refers to this single notion; Definition 9.24 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.33 (Block-injective canonical form). Lean ✓ Stmt Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^{r} \mu_k A_k^i.$$

This block-injective formulation of the canonical form is used in [PGVWC07]; see also [CPGSV21] for the normal canonical form.

Remark 2.34. ✓ Stmt Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.28. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the spectral sense. Chapter 9 treats the normal canonical form needed for the blocking-based multi-block theory, while Chapter 15 develops the separate periodic irreducible-form extension. This section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.35 (Block projection). Lean ✓ Stmt For a finite direct sum $\bigoplus_k \mathbb{C}^{D_k}$, the block projection P_j is the matrix that is the identity on the j-th summand and zero on all other summands.

Definition 2.36 (Block-diagonal matrix). Lean ✓ Stmt A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if it is a direct sum of square matrices, one on each summand.

Lemma 2.37 (Off-block characterization). *Lean* *✓ Stmt* *A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if and only if every entry from one summand to a distinct summand is zero.*

Proof. ✓ Proof The forward direction is immediate from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix. □

Lemma 2.38 (Commutation with block projections). *Lean* *✓ Stmt* *If a matrix on $\bigoplus_k \mathbb{C}^{D_k}$ commutes with every block projection P_j, then it is block diagonal.*

Proof. ✓ Proof Compare the (i, j) block of the identity $XP_j = P_jX$ for $i \neq j$. The right multiplication by P_j keeps that block, while the left multiplication by P_j kills it. Hence every off-block entry is zero, and Lemma 2.37 applies. □

Definition 2.39 (Block-diagonal tensor from blocks). Lean ✓ Stmt Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^{r} \mu_k A_k^i.$$

Lemma 2.40 (Word evaluation of a block-diagonal tensor). Lean ✓ Stmt *For a word w, the word evaluation of a block-diagonal tensor remains block diagonal:*

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w.$$

Proof. ✓ Proof Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor μ_k on the k-th block. □

Theorem 2.41 (MPV decomposition). Lean ✓ Stmt *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle = \sum_{k=1}^{r} \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^{r} \mu_k^N V^{(N)}(A_k)_\sigma.$$

Proof. ✓ Proof By Lemma 2.40, the full word evaluation is block diagonal with blocks $\mu_k^N A_k^w$ for a word w of length N. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\operatorname{tr}(A^w) = \sum_k \mu_k^N \operatorname{tr}(A_k^w)$. □

2.6 Blocking

Definition 2.42 (Blocked tensor). Lean ✓ Stmt Given a tensor A and a blocking length L, the L-*blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D, whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1,\dots,i_L)} = A^{i_1} A^{i_2} \cdots A^{i_L}.$$

Lemma 2.43 (Blocked word evaluation). Lean ✓ Stmt *If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L-tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\widetilde{w}$ of length mL in the original alphabet and one has*

$$(A^{[L]})^w = A^{\widetilde{w}}.$$

Proof. ✓ Proof Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. □

Lemma 2.44 (Blocking preserves MPV coefficients). Lean ✓ Stmt *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L-tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.43). □

Theorem 2.45 (Blocking preserves same MPV). Lean ✓ Stmt *If A and B generate the same MPV family, then for any blocking length L, the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. ✓ Proof By Lemma 2.44, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. □

2.7 MPV overlap

Definition 2.46 (MPV as Hilbert-space vector). Lean ✓ Stmt We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \ldots, d-1\}^N$, with σ-component $V^{(N)}(A)_\sigma$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.47 (MPV inner product). Lean ✓ Stmt The *MPV inner product* between two tensors A and B at system size N is

$$\langle V^{(N)}(A)|V^{(N)}(B)\rangle := \sum_{\sigma\in\{0,\ldots,d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.48 (MPV overlap). Lean ✓ Stmt The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma\in\{0,\ldots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.49, it is the complex conjugate of the Hilbert-space inner product from Definition 2.47.

Lemma 2.49 (Overlap equals conjugate inner product). Lean ✓ Stmt $O_{AB}(N) = \overline{\langle V^{(N)}(A)|V^{(N)}(B)\rangle}$.

Proof. ✓ Proof The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. □

Lemma 2.50 (Overlap determined by positive-length MPV). Lean ✓ Stmt *If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ, then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N.*

Proof. ✓ Proof Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. □

Chapter 3

The Single-Block Fundamental Theorem

This chapter proves the single-block injective Fundamental Theorem of MPS [CPGSV21, Section III.B]: if an injective MPS tensor A generates the same MPV family as another tensor B of the same bond dimension, then there exists an invertible matrix X such that $B^i = XA^iX^{-1}$ for all physical indices i. The proof is purely algebraic and does not require the later overlap or normal-form reductions.

3.1 Trace pairing

Theorem 3.1 (Nondegeneracy of the trace pairing). Lean ✓ Stmt *For $M \in M_D(\mathbb{C})$, $\operatorname{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. ✓ Proof Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\operatorname{tr}(E_{ji}M) = M_{ij}$. Hence if $\operatorname{tr}(MN) = 0$ for all N, then every entry of M vanishes. □

Lemma 3.2 (Same MPV implies trace agreement). Lean ✓ Stmt *If A and B generate the same MPV family, then $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$ for every word w.*

Proof. ✓ Proof Viewing the word w as a configuration of length $|w|$, Definition 2.4 gives the corresponding MPV coefficient as the trace coefficient $c_w(A) = \operatorname{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. □

Definition 3.3 (Trace pairing map). Lean ✓ Stmt For a tensor A, attach to the chosen family $\{A^i\}$ the auxiliary linear map $\Phi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ defined by

$$\Phi_A(M)_i = \operatorname{tr}(M \cdot A^i).$$

Theorem 3.4 (Injectivity of the trace pairing map). Lean ✓ Stmt *If A is injective, then* $\ker \Phi_A = \{0\}$.

Proof. ✓ Proof If $\Phi_A(M) = 0$, then $\operatorname{tr}(M \cdot A^i) = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\operatorname{tr}(MN) = 0$ for all N, so $M = 0$ by Theorem 3.1. □

Lemma 3.5 (Nonvanishing trace on injective tensors). Lean ✓ Stmt *If $D \geq 1$, A is injective, and A and B generate the same MPV family, then B^i cannot be identically zero: $\neg(\forall i, B^i = 0)$.*

Proof. ✓ Proof If $B^i = 0$ for all i, then Lemma 3.2 gives $\mathrm{tr}(A^w) = 0$ for every word w, in particular for all words of length 1. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\mathrm{tr}(\mathbb{1}_D) = D \neq 0$. □

3.2 Linear extension

Theorem 3.6 (Existence and uniqueness of the linear extension). Lean ✓ Stmt *If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i.*

Proof. ✓ Proof Write $\ell_A : \mathbb{C}^d \to M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). Applying Lemma 3.2 to all words of length 2 shows $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\mathrm{range}(\Phi_A) \subseteq \mathrm{range}(\Phi_B)$. By Theorem 3.4, $\ker \Phi_A = \{0\}$, so rank–nullity gives $\dim(\mathrm{range}(\Phi_A)) = \dim(M_D(\mathbb{C}))$. Hence $\dim(\mathrm{range}(\Phi_B)) \geq \dim(M_D(\mathbb{C}))$, and rank–nullity again gives $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. □

Theorem 3.7 (Multiplicativity of the linear extension). Lean ✓ Stmt *If A is injective and A, B generate the same MPV family, and $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is the unique linear map satisfying $T(A^i) = B^i$ for all i, then $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. ✓ Proof **Step 1** ($\Phi_B \circ T = \Phi_A$). Applying Lemma 3.2 to words of length 2 gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^iA^j) = B^iB^j$). Lemma 3.2 applied to the word (i, j, k) gives $\mathrm{tr}(A^iA^jA^k) = \mathrm{tr}(B^iB^jB^k)$ for all i, j, k. Thus, for each k,

$$\begin{aligned} \Phi_B(T(A^iA^j))_k &= \Phi_A(A^iA^j)_k \\ &= \mathrm{tr}(A^iA^jA^k) \\ &= \mathrm{tr}(B^iB^jB^k) \\ &= \Phi_B(B^iB^j)_k. \end{aligned}$$

The same rank–nullity argument used in the proof of Theorem 3.6 gives $\ker \Phi_B = \{0\}$. Hence $T(A^iA^j) = B^iB^j$.

Step 3 (bilinear extension). For fixed j, the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. □

3.3 Simplicity and Skolem–Noether

Theorem 3.8 (Simplicity of $M_D(\mathbb{C})$). Lean ✓ Stmt *A nonzero multiplicative $\mathbb{C}$-linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. □

Theorem 3.9 (Matrix-algebra isomorphisms preserve size). Lean ✓ Stmt *If the full matrix algebras $M_D(\mathbb{C})$ and $M_E(\mathbb{C})$ are isomorphic as $\mathbb{C}$-algebras, then $D = E$.*

Proof. ✓ Proof The isomorphism preserves complex vector-space dimension. Since $\dim_{\mathbb{C}} M_D(\mathbb{C}) = D^2$ and $\dim_{\mathbb{C}} M_E(\mathbb{C}) = E^2$, it follows that $D = E$. $\square$

Theorem 3.10 (Skolem–Noether). Lean ✓ Stmt *Every $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \mathrm{Aut}_{\mathbb{C}\text{-}alg}(M_D(\mathbb{C}))$, there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M.*

Proof. ✓ Proof Via the canonical identification $M_D(\mathbb{C}) \cong \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\mathrm{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X. $\square$

Theorem 3.11 (Matrix-algebra isomorphisms are inner after identifying size). Lean ✓ Stmt *If the full matrix algebras $M_D(\mathbb{C})$ and $M_E(\mathbb{C})$ are isomorphic as $\mathbb{C}$-algebras, then $D = E$. If $h : D = E$ is this equality, then after the canonical index identification along h the isomorphism acts by $\varphi(M) = X\,(\mathrm{reindex}_h\, M)\,X^{-1}$ for some invertible matrix X.*

Proof. ✓ Proof The vector-space dimension comparison gives $D = E$. Transporting along this equality turns the isomorphism into an automorphism of a single full matrix algebra, so Skolem–Noether gives the required conjugating matrix. $\square$

Lemma 3.12 (Unit preservation of multiplicative surjective maps). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$-linear map, then $T(\mathbb{1}) = \mathbb{1}$.*

Proof. ✓ Proof By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$-algebra homomorphism. $\square$

3.4 Proof of the single-block theorem

Theorem 3.13 (Single-block Fundamental Theorem). Lean ✓ Stmt *Let A be an injective MPS tensor and let B be an MPS tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i.*

Proof. ✓ Proof By Theorem 3.6, there is a unique linear T with $T(A^i) = B^i$. By Theorem 3.7, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i, contradicting Lemma 3.5. By Theorem 3.8, T is bijective. By Lemma 3.12, T is a $\mathbb{C}$-algebra automorphism. Theorem 3.10 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. $\square$

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive maps and channels

Definition 4.1 (Positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 4.2. ✓ Stmt We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 4.3 (Trace-preserving map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *trace-preserving* if $\mathrm{tr}(E(X)) = \mathrm{tr}(X)$ for all X.

Definition 4.4 (Completely positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger.$$

We use the Kraus characterisation because it matches the later Kadison–Schwarz arguments. By Choi's theorem, this is equivalent to $E \otimes \mathrm{id}_n$ being positive for all n.

Theorem 4.5 (CP maps are positive). *Lean* *✓ Stmt* *Every completely positive map is positive.*

Proof. ✓ Proof If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. □

Theorem 4.6 (Channels are positive). *Lean* *✓ Stmt* *Every quantum channel is positive.*

Proof. ✓ Proof A quantum channel is completely positive by definition, so Theorem 4.5 applies. □

Theorem 4.7 (Positive maps preserve Hermiticity). *Lean* *✓ Stmt* *If E is positive and $X = X^\dagger$, then $E(X) = E(X)^\dagger$.*

Proof. ✓ Proof Decompose $X = X_+ - X_-$ into its positive and negative parts. Positivity makes $E(X_+)$ and $E(X_-)$ positive semidefinite, hence Hermitian, so their difference $E(X)$ is Hermitian. □

Definition 4.8 (Quantum channel). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 4.9 (Density matrices). Lean Lean ✓ Stmt The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \operatorname{tr}(\rho) = 1\}.$$

Theorem 4.10 (Density matrices are compact). Lean ✓ Stmt *$\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).*

Proof. ✓ Proof The PSD cone is closed (preimage of the closed non-negative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\operatorname{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are non-negative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

Theorem 4.11 (Density matrices are convex). Lean ✓ Stmt *$\mathcal{D}_D$ is convex.*

Proof. ✓ Proof Convex combination of PSD matrices is PSD, and trace is linear. □

Theorem 4.12 (Density matrices are nonempty). Lean ✓ Stmt *For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.*

Proof. ✓ Proof $\frac{1}{D}\mathbb{1}_D$ is a density matrix. □

Theorem 4.13 (Channels preserve density matrices). Lean ✓ Stmt *If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.*

Proof. ✓ Proof Complete positivity implies positivity (Theorem 4.5), so $E(\rho) \geq 0$; trace preservation gives $\operatorname{tr}(E(\rho)) = \operatorname{tr}(\rho) = 1$. □

4.2 Irreducibility

Definition 4.14 (Orthogonal projection). Lean ✓ Stmt A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Definition 4.15 (Irreducible map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Theorem 6.2, condition (1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 4.16 (Injectivity implies irreducibility). Lean ✓ Stmt *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. ✓ Proof If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M, so $P = 0$ or $P = \mathbb{1}$. □

Remark 4.17 (Transfer map is CP). Lean ✓ Stmt The transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is completely positive: it is already written in Kraus form, with Kraus operators $\{A^i\}$.

Theorem 4.18 (Transfer map is a channel). Lean ✓ Stmt *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).*

Proof. ✓ Proof Complete positivity is the observation of Remark 4.17. Trace preservation: $\mathrm{tr}(\mathcal{E}_A(X)) = \sum_i \mathrm{tr}(A^i X (A^i)^\dagger) = \mathrm{tr}((\sum_i (A^i)^\dagger A^i) X) = \mathrm{tr}(X)$ by cyclicity and the normalization hypothesis. □

4.3 Cesàro fixed-point theorem

Definition 4.19 (Cesàro mean). Lean ✓ Stmt The *Cesàro mean* of a linear map E at order N, applied to a starting matrix ρ_0, is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 4.20 (Cesàro telescope). Lean ✓ Stmt $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. ✓ Proof Telescoping: $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^{N} E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. □

Theorem 4.21 (Hermitian fixed-point decomposition). Lean ✓ Stmt *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\mathrm{tr}(P_+ P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Proposition 6.8].*

Proof. ✓ Proof Let Q be the support projection of P_+. Trace preservation and positivity give $\mathrm{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. □

Theorem 4.22 (Cesàro fixed-point theorem). Lean ✓ Stmt *Every quantum channel $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. ✓ Proof Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 4.12). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 4.10), extract a convergent subsequence $\sigma_{N_k} \to \rho$. The telescope identity (Theorem 4.20) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N}\|E^N(\rho_0) - \rho_0\| \to 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. □

4.4 Fixed-point projection

Definition 4.23 (Fixed-point projection). Lean ✓ Stmt Given a linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\mathrm{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)} \rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 4.24 (Power decomposition). Lean ✓ Stmt *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. ✓ Proof P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. □

4.5 Peripheral spectrum and primitivity

Definition 4.25 (Peripheral spectrum). Lean ✓ Stmt The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T.

Definition 4.26 (Peripheral eigenvalues). Lean ✓ Stmt The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1; for a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Theorem 4.27 (1 is a peripheral eigenvalue). Lean ✓ Stmt *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E.*

Proof. ✓ Proof ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Remark 4.28 (Peripheral eigenvalues lie on the unit circle). Lean ✓ Stmt Every peripheral eigenvalue λ satisfies $|\lambda| = 1$. This is simply the defining condition in Definition 4.26.

Theorem 4.29 (Finiteness of peripheral eigenvalues). Lean ✓ Stmt *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. ✓ Proof Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 4.30 (Power-stable peripheral eigenvalues are roots of unity). Lean ✓ Stmt *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. ✓ Proof Since μ^n is an eigenvalue for every n, the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 4.31 (Peripheral powers imply root of unity). Lean ✓ Stmt *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is* closed under powers *($\mu \in \text{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \text{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. ✓ Proof The closure hypothesis provides $\mu^n \in \text{peripheral}(E)$ for all n, which in particular means μ^n is an eigenvalue. Theorem 4.30 then gives $\mu^p = 1$ for some $p > 0$. □

4.5.1 Periodicity removal by powering

Remark 4.32. ✓ Stmt The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p-blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 4.33 (Common exponent for a finite set of roots of unity). Lean ✓ Stmt *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. ✓ Proof Take p to be the least common multiple of the exponents p_μ. □

Lemma 4.34 (Powering collapses peripheral eigenvalues). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then* $\mathrm{peripheral}(E^p) = \{1\}$.

Proof. ✓ Proof The fixed point ρ gives $1 \in \mathrm{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \mathrm{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \mathrm{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 4.35 (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the "bi-canonical" case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 4.41) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.31. This is the simplest argument, but it requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

4.5.2 Kadison–Schwarz input for peripheral and fixed-point arguments

Definition 4.36 (Kraus maps, adjoints, and normalizations). Lean Lean Lean Lean ✓ Stmt For operators $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$, the associated *Kraus map* and *adjoint Kraus map* are

$$E(X) = \sum_i K_i X K_i^\dagger, \qquad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when $\sum_i K_i K_i^\dagger = \mathbb{1}$ and *trace-preserving* when $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Theorem 4.37 (Kadison–Schwarz inequality). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map. Then for every $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. ✓ Proof Apply the unital completely positive map entrywise to the positive 2×2 block matrix $\begin{pmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{pmatrix}$ and take the Schur complement of the lower-right block. □

Theorem 4.38 (Hilbert–Schmidt contraction). Lean ✓ Stmt *If a Kraus map is both unital and trace-preserving, then*

$$\operatorname{tr}(E(X)^\dagger E(X)) \le \operatorname{tr}(X^\dagger X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation then identifies the traces of its two terms. ☐

Theorem 4.39 (KS gap decomposition). Lean ✓ Stmt *For a unital Kraus map,*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \qquad R_i := XK_i^\dagger - K_i^\dagger E(X).$$

Proof. ✓ Proof Expand the right-hand side, distribute the products, and use $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. ☐

Theorem 4.40 (KS equality implies Kraus commutation). Lean ✓ Stmt *Let* E *be a unital Kraus map. If* $E(X^\dagger X) = E(X)^\dagger E(X)$, *then, for every* i,

$$XK_i^\dagger = K_i^\dagger E(X).$$

Proof. ✓ Proof The Kadison–Schwarz gap is a sum of positive semidefinite terms $\sum_i R_i^\dagger R_i$. If the gap vanishes, each R_i must vanish. ☐

Theorem 4.41 (KS equality for peripheral eigenvectors). Lean ✓ Stmt *Let* E *be a Kraus map that is both unital and trace-preserving. If* $E(X) = \mu X$ *with* $|\mu| = 1$, *then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

Proof. ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation and the relation $E(X) = \mu X$ with $|\mu| = 1$ show that its trace is zero, hence the gap vanishes. ☐

Theorem 4.42 (Left multiplicative identity from KS equality). Lean ✓ Stmt *Let* E *be a unital Kraus map. If* $E(X^\dagger X) = E(X)^\dagger E(X)$, *then for every* $Y \in M_D(\mathbb{C})$,

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

Proof. ✓ Proof The commutation relation of Theorem 4.40 lets one move $X^\dagger$ past each Kraus operator inside the sum. ☐

Definition 4.43 (Multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of a Kraus map E is the set of matrices that are simultaneously left- and right-multiplicative for E.

Theorem 4.44 (Multiplicative domain is a $*$-subalgebra). Lean ✓ Stmt *For a unital Kraus map, the multiplicative domain is a* $*$*-subalgebra of* $M_D(\mathbb{C})$.

Proof. ✓ Proof The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix. ☐

4.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV17a, Appendix A].

Lemma 4.45 (Peripheral closure via adjoint fixed point). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:*

1. *the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and*
2. *E is irreducible.*

Then $\mathrm{peripheral}(E)$ *is closed under powers: if $\mu \in \mathrm{peripheral}(E)$ then $\mu^n \in \mathrm{peripheral}(E)$ for every $n \in \mathbb{N}$.*

Proof. ✓ Proof Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\mathrm{tr}(\rho G) = \mathrm{tr}(\rho E(X^\dagger X)) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(E^*(\rho) X^\dagger X) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(\rho X^\dagger X) - |\mu|^2 \,\mathrm{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 6.4). So X is invertible. By Theorem 4.40 (KS equality implies Kraus commutation), X commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 4.42) gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. ☐

Theorem 4.46 (Roots of unity via adjoint fixed point). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.*

Proof. ✓ Proof Combine the closure-under-powers result (Lemma 4.45) with Theorem 4.31. ☐

Remark 4.47. ✓ Stmt Lemma 4.45 and Theorem 4.46 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV17a, Appendix A], which uses a faithful invariant state rather than bi-canonicality.

Definition 4.48 (Channel period). Lean ✓ Stmt The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\mathrm{peripheral}(E)|$.

Definition 4.49 (Primitive map). Lean ✓ Stmt A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\mathrm{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 4.50 (Primitivity equals period one). Lean ✓ Stmt *A linear map with a nonzero fixed point is primitive if and only if its period is* 1.

Proof. ✓ Proof If $\mathrm{peripheral}(E) = \{1\}$, then $|\mathrm{peripheral}(E)| = 1$. Conversely, if $|\mathrm{peripheral}(E)| = 1$, then since $1 \in \mathrm{peripheral}(E)$ (Theorem 4.27), the only element is 1. ☐

Theorem 4.51 (Complementary transfer-map gap for primitive maps). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\operatorname{tr}(\rho) \neq 0$. Let*

$$P(X) = \frac{\operatorname{tr}(X)}{\operatorname{tr}(\rho)}\rho$$

be the fixed-point projection associated to ρ. Assume that:

1. *E is primitive;*
2. *every eigenvalue μ of E satisfies $|\mu| \leq 1$;*
3. *whenever $E(X) = X$ and $\operatorname{tr}(X) = 0$, one has $X = 0$.*

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. ✓ Proof Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\operatorname{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\operatorname{tr}(X) \neq 0$, trace preservation and $\operatorname{tr}(P(X)) = \operatorname{tr}(X)$ give

$$\operatorname{tr}(X) = \operatorname{tr}(E(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(P(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(X),$$

so $\nu \operatorname{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

4.6 Later representation theory

The Choi–Jamiołkowski, Kraus, Stinespring, Naimark, SVD-normal-form, and Lorentz-normal-form material is collected in Chapter 21. The fixed-point algebra remains here because Chapter 7 uses it before that later representation chapter.

4.7 Fixed-point algebra

Definition 4.52 (Kraus fixed points). Lean ✓ Stmt The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\operatorname{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 4.53 (Fixed points lie in the multiplicative domain). Lean ✓ Stmt *Let E be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point $\rho > 0$. Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.*

Proof. ✓ Proof The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. □

Theorem 4.54 (Fixed points form a $*$-subalgebra). Lean ✓ Stmt *Let E be a unital Kraus map whose adjoint map E^* has a positive definite fixed point $\rho > 0$. Then $\operatorname{Fix}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Theorem 6.12].*

Proof. ✓ Proof Closure under addition and scalar multiplication is immediate from linearity. Closure under †: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because fixed points are stable under adjoint. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the weighted Kadison–Schwarz argument gives $E(X^\dagger X) = E(X)^\dagger E(X)$ and, applied to $X^\dagger$, also $E(XX^\dagger) = E(X)E(X)^\dagger$. Thus X lies in the multiplicative domain, and Theorem 4.42 yields $E(XY) = E(X)E(Y) = XY$. □

Remark 4.55. ✓ Stmt Theorem 4.53 is the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

Theorem 4.56 (Projected elements lie in the multiplicative domain). Lean ✓ Stmt *Let E be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose $E^2 = E$. Then $E(X)$ lies in the multiplicative domain of E for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Idempotence gives $E(E(X)) = E(X)$, so $E(X)$ is a fixed point. By the argument in Theorem 4.54, the Kadison–Schwarz gap for the fixed point $E(X)$ vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under †, so the same argument applied to $E(X)^\dagger$ gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

Therefore Theorem 5.28 places $E(X)$ in the multiplicative domain. □

Theorem 4.57 (Choi–Effros left absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(E(X)E(Y)).$$

Proof. ✓ Proof The previous theorem places $E(X)$ in the multiplicative domain, so $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.58 (Choi–Effros right absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(XE(Y)) = E(E(X)E(Y)).$$

Proof. ✓ Proof The previous theorem places $E(Y)$ in the multiplicative domain, so $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.59 (Choi–Effros symmetric absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(XE(Y)).$$

Proof. ✓ Proof Both sides equal $E(E(X)E(Y))$ by the two preceding theorems. □

Theorem 4.60 (Choi–Effros projected product). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)E(Y)) = E(X)E(Y).$$

Proof. ✓ Proof The previous theorem places $E(X)$ and $E(Y)$ in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since E is idempotent under the standing hypotheses, $E(E(X)) = E(X)$ and $E(E(Y)) = E(Y)$, so

$$E(E(X)E(Y)) = E(X)E(Y).$$

□

Definition 4.61 (Adjoint fixed points). Lean ✓ Stmt The *adjoint fixed-point set* $\mathrm{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 4.62 (Fixed points form a $*$-subalgebra in the Heisenberg picture). Lean Lean ✓ Stmt *If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\mathrm{Fix}(E^*)$ is a $*$-subalgebra of $M_D(\mathbb{C})$.*

Proof. ✓ Proof Apply Theorem 4.54 to the adjoint family $\{K_i^\dagger\}$. □

Definition 4.63 (Kraus commutant). Lean Lean ✓ Stmt The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \forall i\}$. It is a $*$-subalgebra of $M_D(\mathbb{C})$.

Theorem 4.64 (Adjoint fixed points equal the Kraus commutant). Lean ✓ Stmt *Under the hypotheses of Theorem 4.62, the adjoint fixed-point $*$-subalgebra equals the Kraus commutant:*

$$\mathrm{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Theorem 6.13].

Proof. ✓ Proof ($\supseteq$): if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. ($\subseteq$): any $*$-subalgebra inside $\mathrm{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \mathrm{Fix}(E^*)$ with $X^\dagger X \in \mathrm{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_i X$ and $XK_i^\dagger = K_i^\dagger X$. □

Theorem 4.65 (Kraus commutant inside the adjoint fixed points). Lean ✓ Stmt *Under the hypotheses above, the Kraus commutant is the largest $*$-subalgebra contained in the adjoint fixed-point set.*

Proof. ✓ Proof By Theorem 4.64, every $*$-subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a $*$-subalgebra. □

4.8 Quantum dynamical semigroups

The one-parameter semigroup theory of Wolf Chapter 7 is treated in Chapter 22. That later chapter develops the exponential representation of norm-continuous semigroups, perturbation bounds, and the GKSL/Lindblad description of generators. We do not duplicate that material here.

Chapter 5

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory following [Wol12, Chapter 5]. We begin with the Kadison–Schwarz inequality for Kraus maps, then describe the multiplicative domain and its algebraic structure. These two parts are used in the non-periodic Fundamental Theorem through peripheral-eigenvector rigidity and normal-block comparison. We also establish the extension to normal operators for positive subunital maps, the resulting order and spectral contractivity statements, and the standard example of a Schwarz map that is not completely positive. The operator-convexity and Jensen material from [Wol12, Chapter 5] is treated separately in Chapter 23.

5.1 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 4.4). The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map is a Kraus map by construction (Remark 4.17), so these results apply directly to transfer maps.

Definitions 5.1–5.4 below restate the Kraus-map notions from Chapter 4 for the reader's convenience.

Definition 5.1 (Kraus map). Lean ✓ Stmt Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 4.4) if and only if it can be written in this form.

Definition 5.2 (Adjoint Kraus map). Lean ✓ Stmt The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A, this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 5.3 (Unital Kraus map). Lean ✓ Stmt A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$. In the later gauge language this is the right-canonical condition.

Definition 5.4 (Trace-preserving Kraus map). Lean ✓ Stmt A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 5.5 (Kadison–Schwarz inequality). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a* unital *Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. ✓ Proof The 2×2 block matrix $\left[\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right]$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$-block maps to $\mathbb{1}$). The Schur complement of the $(2,2)$-block $\mathbb{1}$ gives the result. □

Theorem 5.6 (Kadison–Schwarz for the adjoint channel). Lean ✓ Stmt *If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.*

Proof. ✓ Proof The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 5.5 to $\{K_i^\dagger\}$. □

5.1.1 Two-positive maps and the generalized Schwarz inequality

Definition 5.7 (k-positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is k-positive if $E \otimes \mathrm{id}_k$ is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 5.8 (Two-positive map). Lean ✓ Stmt A linear map is *two-positive* if it is 2-positive.

Theorem 5.9 (Completely positive maps are k-positive). Lean ✓ Stmt *Every completely positive map is k-positive for all $k \geq 1$.*

Theorem 5.10 ((k+1)-positive implies k-positive). Lean ✓ Stmt *Positivity under amplification is monotone in k.*

Theorem 5.11 (Completely positive implies two-positive). Lean ✓ Stmt *Every completely positive map is two-positive.*

Theorem 5.12 (Two-positive implies positive). Lean ✓ Stmt *Every two-positive map is positive.*

Definition 5.13 (Unital linear map). Lean ✓ Stmt A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 5.14 (Kadison–Schwarz for unital two-positive maps). Lean ✓ Stmt *If E is unital and two-positive, then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

Proof. Form the 2×2 block matrix $\left(\begin{smallmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{smallmatrix}\right) \geq 0$. Apply $E \otimes \mathrm{id}_2$ (positive by 2-positivity) and read off the $(2,2)$-block Schur complement: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. □

Theorem 5.15 (Kraus-form Kadison–Schwarz as a special case). Lean ✓ Stmt *The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.*

5.1.2 Douglas-type factorization lemmas

Theorem 5.16 (Range inclusion implies right factorization). Lean ✓ Stmt *If* $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$, *then there exists* C *such that* $A = BC$.

Proof. In finite dimensions, $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. □

Theorem 5.17 (Vectorwise range criterion implies factorization). Lean ✓ Stmt *If* $Av \in \operatorname{ran}(B)$ *for every vector* v, *then there exists* C *with* $A = BC$.

Theorem 5.18 (Hilbert–Schmidt contraction). Lean ✓ Stmt *If a Kraus map is both unital and trace-preserving, then* $\operatorname{tr}(E(X)^\dagger E(X)) \leq \operatorname{tr}(X^\dagger X)$ *for all* $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof By Theorem 5.5, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\operatorname{tr}(E(X^\dagger X)) \geq \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. □

Lemma 5.19 (Trace pairing for Kraus map and adjoint map). Lean ✓ Stmt *For all* $X, Y \in M_D(\mathbb{C})$,

$$\operatorname{tr}(X\, E(Y)) = \operatorname{tr}(E^*(X)\, Y).$$

Lemma 5.20 (Positive-definite trace test). Lean ✓ Stmt *If* $A > 0$, $X \geq 0$, *and* $\operatorname{tr}(AX) = 0$, *then* $X = 0$.

Theorem 5.21 (Fixed-point peripheral eigenvectors satisfy KS equality). Lean ✓ Stmt *Let* E *be unital and let* E^* *be trace-preserving with a positive definite fixed point* $\rho > 0$. *If* $E(X) = \mu X$ *with* $|\mu| = 1$, *then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

5.2 Multiplicative domain

Theorem 5.22 (KS gap decomposition). Lean ✓ Stmt *For a unital Kraus map* E, *the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.

Proof. ✓ Proof Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. □

Theorem 5.23 (KS equality implies Kraus commutation). Lean ✓ Stmt *Let* E *be a unital Kraus map. If* $E(X^\dagger X) = E(X)^\dagger E(X)$ *for some* X, *then* $XK_i^\dagger = K_i^\dagger E(X)$ *for all* i.

Proof. ✓ Proof By Theorem 5.22, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. □

Theorem 5.24 (KS equality for peripheral eigenvectors). Lean ✓ Stmt *Let* E *be a Kraus map that is both unital and trace-preserving. If* $E(X) = \mu X$ *with* $|\mu| = 1$, *then the KS gap vanishes:* $E(X^\dagger X) = E(X)^\dagger E(X)$.

Proof. ✓ Proof The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 5.5. Its trace satisfies $\mathrm{tr}(G) = \mathrm{tr}(E(X^\dagger X)) - \mathrm{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\mathrm{tr}(E(X^\dagger X)) = \mathrm{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\mathrm{tr}(E(X)^\dagger E(X)) = |\mu|^2 \mathrm{tr}(X^\dagger X) = \mathrm{tr}(X^\dagger X)$. So $\mathrm{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. □

Theorem 5.25 (Left multiplicative identity from KS equality)**.** Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. ✓ Proof From Theorem 5.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i. Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

□

5.2.1 Full multiplicative-domain structure

Definition 5.26 (Right and left multiplicative domains)**.** Lean Lean ✓ Stmt For a Kraus map E, define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}),\ E(XY) = E(X)E(Y)\},$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}),\ E(YX) = E(Y)E(X)\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Eqs. (5.11)–(5.12)].

Definition 5.27 (Full multiplicative domain)**.** Lean ✓ Stmt The *multiplicative domain* of E is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

Theorem 5.28 (Multiplicative-domain characterization)**.** Lean Lean ✓ Stmt *Let E be a unital Kraus map. Then*

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger,$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Theorem 5.7].

Proof. ✓ Proof If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 5.25, and the right-domain implication follows by applying Theorem 5.25 to $X^\dagger$. □

Theorem 5.29 (One-sided multiplicative domains are subalgebras)**.** Lean Lean ✓ Stmt *For a unital Kraus map E, both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.*

Proof. ✓ Proof Closure under addition follows from linearity of E. Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z, using Theorem 5.28 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E. □

Theorem 5.30 (Multiplicative domain is a $*$-subalgebra). Lean ✓ Stmt *For a unital Kraus map E, the full multiplicative domain $\mathcal{A}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Theorem 5.7].*

Proof. ✓ Proof If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 5.28 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 5.29, this gives a $*$-subalgebra. □

Theorem 5.31 (Restriction to multiplicative domain is a $*$-homomorphism). Lean ✓ Stmt *If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \to M_D(\mathbb{C})$ is a $*$-algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].*

Proof. ✓ Proof Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 5.23 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. □

5.2.2 Schwarz inequality for normal operators

Theorem 5.32 (CP Schwarz inequality for normal operators). Lean Lean ✓ Stmt *Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap*

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A).$$

This is the completely positive case of [Wol12, Proposition 5.1].

Proof. ✓ Proof The positivity statement is exactly Theorem 5.6. The second formulation is just the corresponding Loewner-order reformulation. □

Theorem 5.33 (Schwarz inequality for normal operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Proposition 5.1].

Proof. ✓ Proof Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} \bar{d}_i d_j P_i P_j$ where $P_i = Ue_ie_i^\dagger U^\dagger$. Each P_i is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown non-negative by a simultaneous-diagonalization argument: positivity of T ensures the scalar PSD matrix $(\langle v_k, T(P_iP_j)v_k\rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i)v_k\rangle\langle v_k, T(P_j)v_k\rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. □

Remark 5.34. The proof follows the positivity-on-abelian argument of [Wol12, Proposition 1.6]: a positive map restricted to the commutative $*$-subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

5.2.3 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 5.35 (Schwarz inequality for subnormal operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0.$$

This is [Wol12, Theorem 5.5].

Proof. ✓ Proof Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunital map $\tilde{T}$ on the larger space (acting as T on the northwest block, zero elsewhere). Apply Theorem 5.33 to $\tilde{T}$ and N, then compress the resulting inequality back to $\mathbb{C}^D$. □

Theorem 5.36 (Schwarz inequality for commuting-dominant operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite* dominant *$B \ge 0$ with $A^\dagger A \le B$ and B commuting with A (in the sense $BA = AB$, $BA^\dagger = A^\dagger B$), then*

$$T(A^\dagger)T(A) \le T(B).$$

This is [Wol12, Theorem 5.6].

Proof. ✓ Proof Let $B^{1/2}$ be the positive square root of B. Since B commutes with A and $A^\dagger$, the contraction $C = B^{-1/2}A$ (defined on the range of $B^{1/2}$) is subnormal. An ε-regularisation $B_\varepsilon = B + \varepsilon\mathbb{1}$ makes $B_\varepsilon > 0$ and the contraction C_ε globally defined. Apply Theorem 5.35 to C_ε, multiply both sides by $T(B_\varepsilon)$, use positivity and monotonicity, and take $\varepsilon \to 0$. □

Theorem 5.37 (CP variant: Kadison–Schwarz for commuting-dominant inputs). Lean ✓ Stmt *Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A, $A^\dagger$, and satisfying $A^\dagger A \le B$, then $E^*(A^\dagger)E^*(A) \le E^*(B)$.*

Proof. ✓ Proof Apply Theorem 5.36 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). □

5.2.4 Positive maps preserve order and spectral intervals

Theorem 5.38 (Positive maps are monotone). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive and $A \le B$, then $T(A) \le T(B)$.*

Proof. ✓ Proof Since $B - A \ge 0$, positivity gives $T(B - A) \ge 0$. By linearity, this is $T(B) - T(A) \ge 0$. □

Theorem 5.39 (Positive maps preserve adjoints). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Write $A = B + iC$ with B and C Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints. □

Theorem 5.40 (Order intervals are preserved). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$. If $a \leq 0 \leq b$ and $a\mathbb{1} \leq A \leq b\mathbb{1}$, then*

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}.$$

This is the matrix form of [Wol12, Equation (5.21)].

Proof. ✓ Proof By monotonicity, $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \leq 0$ and $T(\mathbb{1}) \leq \mathbb{1}$, we have $aT(\mathbb{1}) \geq a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$ since $b \geq 0$. Therefore $a\mathbb{1} \leq T(A) \leq b\mathbb{1}$. □

Theorem 5.41 (Spectral interval contractivity). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\operatorname{spec}(A) \subseteq [a, b]$ with $a \leq 0 \leq b$, then*

$$\operatorname{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Equation (5.21)].

Proof. ✓ Proof For Hermitian matrices, the inclusion $\operatorname{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \leq A \leq b\mathbb{1}$. Apply Theorem 5.40 and translate the resulting inequalities back into spectral bounds. □

5.2.5 A positive Schwarz map that is not completely positive

Example 5.42 (Transpose-trace map on $M_2(\mathbb{C})$). Lean ✓ Stmt Consider the linear map $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\operatorname{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Example 5.3].

Theorem 5.43 (The transpose-trace map is positive). Lean ✓ Stmt *The map from Example 5.42 is positive.*

Proof. ✓ Proof It is a non-negative linear combination of the transpose map and the map $A \mapsto \operatorname{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. □

Theorem 5.44 (The transpose-trace map satisfies the Schwarz inequality). Lean ✓ Stmt *For every $A \in M_2(\mathbb{C})$,*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0,$$

where T is the map from Example 5.42.

Proof. ✓ Proof Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ, so it is enough to treat the case $\operatorname{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. □

Theorem 5.45 (The transpose-trace map is not completely positive). Lean ✓ Stmt *The map from Example 5.42 is not completely positive.*

Proof. ✓ Proof Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. □

Chapter 6

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory used in subsequent chapters: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf's spectral characterization of irreducibility. The conceptual source is Wolf's treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

Theorem 6.1 (Brouwer fixed point on density matrices)**.** Lean Lean Lean Lean Lean ✓ Stmt *Let $D > 0$. Every continuous map from the compact convex set of density matrices in $M_D(\mathbb{C})$ to itself has a fixed point.*

Proof. ✓ Proof The proof starts from Brouwer's theorem for products of simplices, transfers it to closed cubes, and then to compact retracts of finite-dimensional real normed spaces. The density-matrix set is such a compact retract, using the explicit density retraction. □

6.1 Positive definiteness

Remark 6.2. ✓ Stmt Wolf's general theory treats irreducible positive maps first ([Wol12, Theorem 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Theorem 6.3]). The injective-tensor version (Theorem 6.3) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 6.4) follows via the support-projection argument of [Wol12, Theorem 6.2].

Theorem 6.3 (PSD fixed point is positive definite)**.** Lean ✓ Stmt *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. ✓ Proof Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is non-negative; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i: the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M. Given any vector w, choose M with $Mv = w$. Then $\rho w = 0$ for every w, so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Theorem 6.3, item 2], the same conclusion is reached via the growth condition $(\mathbb{1} + T)^{d-1}(X) > 0$ of [Wol12, Theorem 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. □

Theorem 6.4 (PSD fixed point is positive definite — irreducible version). Lean ✓ Stmt *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. ✓ Proof Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q\mathcal{E}_A(QXQ)Q = \mathcal{E}_A(QXQ)$ for all X. By irreducibility of $\mathcal{E}_A$ (Definition 4.15), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite, contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Theorem 6.2, condition (1)]. □

Theorem 6.5 (Orthogonal trace condition). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\mathrm{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D-1$ such that*

$$\mathrm{tr}(B \cdot E^t(A)) > 0.$$

Proof. ✓ Proof The growth theorem for irreducible CP maps ([Wol12, Theorem 6.2, item 2]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\mathrm{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\mathrm{tr}\left(B \cdot \sum_{k=0}^{D-1} \tbinom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\mathrm{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\mathrm{tr}(BA) = 0$). Hence at least one $k \in \{1, \ldots, D-1\}$ contributes a strictly positive term. This is [Wol12, Theorem 6.2, item 4]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. □

6.2 Uniqueness

Theorem 6.6 (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof Both ρ and σ are positive definite by Theorem 6.3. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H. Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 6.3, τ must be zero, giving $\sigma = c_0\rho$. **Relation to Wolf:** In [Wol12, Theorem 6.3, item 2], uniqueness is proved via the growth condition of [Wol12, Theorem 6.2]. The same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 6.3) in place of the growth condition. □

Remark 6.7. ✓ Stmt The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 6.8 (Irreducible fixed-point uniqueness). Lean ✓ Stmt *Let A be an MPS tensor whose transfer map is irreducible. If $\rho, \sigma \geq 0$ are fixed points of $\mathcal{E}_A$, and if ρ is nonzero, then $\sigma = c\rho$ for some scalar c.*

Proof. ✓ Proof The irreducible positive-definiteness theorem (Theorem 6.4) upgrades every nonzero positive semidefinite fixed point to a positive definite one. The same minimal eigenvalue argument used in Theorem 6.6 then applies. □

Theorem 6.9 (Uniqueness of positive eigenvalue for irreducible CP maps). Lean ✓ Stmt *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. ✓ Proof Extract Kraus operators K for E. Irreducibility of E passes to the adjoint transfer map $E^\dagger(\,\cdot\,) = \sum_i K_i^\dagger\,(\,\cdot\,)\,K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 6.20) together with the irreducible positive-definiteness upgrade (Theorem 6.4) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\,\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \operatorname{tr}(\tau X) = \operatorname{tr}(\tau \cdot E(X)) = \operatorname{tr}(E^\dagger(\tau) \cdot X) = t \cdot \operatorname{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\operatorname{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. **Relation to Wolf:** This is [Wol12, Theorem 6.3, item 3], establishing non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps. The proof follows Wolf's dual-map trace argument. □

6.3 Existence and the Perron–Frobenius theorem

Theorem 6.10 (Existence of PSD fixed point). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. ✓ Proof Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 4.18). The Cesàro fixed-point theorem (Theorem 4.22) then provides the fixed point; injectivity is not needed for this step. □

Theorem 6.11 (Quantum Perron–Frobenius). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. ✓ Proof Combine Theorems 6.10, 6.3, and 6.6. □

Theorem 6.12 (Quantum Perron–Frobenius for irreducible transfer maps). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor such that its transfer map is irreducible and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then $\mathcal{E}_A$ has a unique positive definite fixed point, up to scalar multiple.*

Proof. ✓ Proof Existence is the channel fixed-point theorem (Theorem 6.10). Irreducibility upgrades the fixed point to positive definite, and Theorem 6.8 gives uniqueness up to scalar multiple. □

6.4 Right- and left-canonical gauges

Theorem 6.13 (Right-canonical (unital) gauge). Lean ✓ Stmt *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization used in subsequent chapters.*

Proof. ✓ Proof Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_{\rho}(A^i)^\dagger(S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i\rho(A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. □

Theorem 6.14 (Left-canonical (trace-preserving) gauge)**.** Lean ✓ Stmt *Let σ be a fixed point of the adjoint transfer map $\sum_i(A^i)^\dagger\sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i(A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization used in subsequent chapters.*

Proof. ✓ Proof Expand each summand: $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^iS^{-1}$. Summing over i and using $\sum_i(A^i)^\dagger\sigma A^i = \sigma$ gives $(S^\dagger)^{-1}\sigma S^{-1} = (S^\dagger)^{-1}S^\dagger SS^{-1} = \mathbb{1}$. □

Remark 6.15. ✓ Stmt The right-canonical gauge of Theorem 6.13 and the left-canonical gauge of Theorem 6.14 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

6.5 Similarity preserves irreducibility

Lemma 6.16 (Similarity transform preserves irreducibility)**.** Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) = c \cdot C^{-1}\, E(C\,X\,C^\dagger)\,(C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. ✓ Proof Suppose Q is an invariant projection for E': $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1}-Q)E'(QXQ)Q = 0$ for all X. Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E. Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Proposition 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. □

Lemma 6.17 (Similarity is an involution)**.** Lean ✓ Stmt *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transforms by C and C^{-1} compose to the identity:*

$$X \mapsto C(C^{-1}\,E(C\,X\,C^\dagger)\,(C^\dagger)^{-1})\,C^\dagger = E(X).$$

Proof. ✓ Proof Direct computation: the inner conjugation $C \cdot (C^{-1}X(C^\dagger)^{-1}) \cdot C^\dagger$ collapses to X using $CC^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1}C^\dagger = \mathbb{1}$, and the outer conjugation by $C(\cdot)C^\dagger$ then cancels the inner $C^{-1}(\cdot)(C^\dagger)^{-1}$ on $E(X)$ for the same reason. □

Lemma 6.18 (Irreducibility is invariant under similarity)**.** Lean ✓ Stmt *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1}\,E(C\,X\,C^\dagger)\,(C^\dagger)^{-1}$ is irreducible if and only if E is.*

Proof. ✓ Proof The forward direction follows from Lemma 6.16 applied with the inverse C^{-1} (whose determinant is also nonzero): if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1}, which equals E by Lemma 6.17. The reverse direction is Lemma 6.16 itself (with $c = 1$). □

Theorem 6.19 (Scaling preserves irreducibility). Lean ✓ Stmt *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Proposition 6.6].*

Proof. ✓ Proof An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). □

6.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV17a, Appendix A].

Theorem 6.20 (PSD eigenvector for positive maps). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ. Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Theorem 6.5]; see also [EHK78]). The proof applies Brouwer's fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\operatorname{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 6.21 (Adjoint transfer map is nonzero). Lean ✓ Stmt *If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.*

Proof. ✓ Proof If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. □

Theorem 6.22 (Positive-definite adjoint eigenvector for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the general PF existence ([Wol12, Theorem 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3, item 2]); the application to the adjoint transfer map follows [CPGSV17a, Appendix A].

Proof. ✓ Proof By Theorem 6.21, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map. More precisely: if P is invariant for $\mathcal{E}_A$ (i.e. $(\mathbb{1} - P)K_i P = 0$ for all Kraus operators K_i), then taking adjoints gives $PK_i^\dagger(\mathbb{1} - P) = 0$, which means $\mathbb{1} - P$ is invariant for $\mathcal{E}_A^\dagger$. Hence $\mathcal{E}_A$ is irreducible if and only if $\mathcal{E}_A^\dagger$ is irreducible, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ. Hence Theorem 6.20 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by Theorem 6.19, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 6.4) gives σ_0 positive definite. □

Theorem 6.23 (TP-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix σ, and a gauge-equivalent tensor B such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the "spectral rescaling + TP gauge" step of [CPGSV17a, Appendix A]. The similarity is generally non-unitary.

Proof. ✓ Proof By Theorem 6.22, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the TP gauge construction (Theorem 6.14), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. □

Theorem 6.24 (Unital-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix ρ, and a gauge-equivalent tensor B such that*

$$B^i = r^{-1/2} \rho^{-1/2} A^i \rho^{1/2}$$

is unital: $\sum_i B^i (B^i)^\dagger = \mathbb{1}$.

This is the Perron–Frobenius unital-gauge orientation used in [PGVWC07, Theorem Th:TIcanonical, lines 765–770], stated for one irreducible nonzero block.

Proof. ✓ Proof Apply Theorem 6.22 to the conjugate-transposed Kraus family. Irreducibility transfers to that family, and the nonzero Kraus hypothesis is preserved by taking adjoints. Translating the resulting adjoint eigenvector equation back gives $\mathcal{E}_A(\rho) = r\rho$ with ρ positive definite and $r > 0$. The unital gauge theorem then gives the displayed unital representative, and the same square-root similarity gives gauge equivalence to $r^{-1/2}A$. □

6.7 Exponential positivity for irreducible CP maps

Theorem 6.25 (Exponential truncation is positive definite). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is the finite-sum core of [Wol12, Theorem 6.2, item (3)].

Proof. ✓ Proof If a nonzero vector v annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D-1$ would annihilate v. The growth theorem for irreducible CP maps (that is, $(\mathbb{1} + E)^{D-1}(A) > 0$ for irreducible E and nonzero PSD A, as established in the proof of Theorem 6.5) then forces $(\mathbb{1}+E)^{D-1}(A)$ to annihilate v, contradicting its positive definiteness. □

Theorem 6.26 (Exponential positivity for irreducible CP maps). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then*

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Theorem 6.2, item (3)].

Proof. ✓ Proof By Theorem 6.25, the first D terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. □

6.8 Ergodicity of irreducible channels

Theorem 6.27 (Unique density-matrix fixed point for an irreducible channel). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. Moreover σ is positive definite. This is the qualitative form of [Wol12, Corollary 6.3].*

Proof. ✓ Proof Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. □

Theorem 6.28 (Cesàro convergence for irreducible channels). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means*

$$\frac{1}{N}\sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of E. This is [Wol12, Corollary 6.3].

Proof. ✓ Proof Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 6.27, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. □

6.9 Spectral radius at the Perron eigenvalue

Theorem 6.29 (Spectral radius is the Perron eigenvalue). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r. This is [Wol12, Theorem 6.3, item (4)].*

Proof. ✓ Proof Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by r^{-1}. The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map F satisfies $\rho(F) \leq 1$, since $\mathrm{tr}(F^n(X)) = \mathrm{tr}(X)$ for all n bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of E. □

Theorem 6.30 (Real spectral-radius identity). Lean ✓ Stmt *Under the hypotheses of Theorem 6.29, the real-valued spectral radius of E is also equal to r.*

Proof. ✓ Proof This is the real-valued reformulation of Theorem 6.29. □

6.10 Spectral characterization of irreducibility

Definition 6.31 (Spectral properties). Lean ✓ Stmt A completely positive map E on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Theorem 6.4] if there exist a positive real number r, a positive definite right eigenvector ρ, and a positive definite left eigenvector σ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ, and the spectral radius of E is equal to r.

Theorem 6.32 (Irreducibility gives the spectral properties). Lean ✓ Stmt *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.31.*

Proof. ✓ Proof Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 6.29. □

Theorem 6.33 (Spectral properties imply irreducibility). Lean ✓ Stmt *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.31, then it is irreducible.*

Proof. ✓ Proof Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 6.31. □

Theorem 6.34 (Irreducibility iff the spectral properties hold). Lean ✓ Stmt *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the spectral properties of Definition 6.31. This is [Wol12, Theorem 6.4].*

Proof. ✓ Proof Combine Theorems 6.32 and 6.33. □

Chapter 7

Transfer-Operator Gaps and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary transfer-map gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

This chapter concerns gaps in the spectrum of transfer operators: $\rho(F_{AB}) < 1$ for mixed transfer operators, and $\rho(\mathcal{E}_A - P) < 1$ after removing the fixed-point projection. The unqualified phrase *spectral gap* is reserved for the energy gap of a many-body Hamiltonian in Chapter 18.

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

7.1 Mixed transfer operator

Definition 7.1 (Mixed transfer operator). Lean ✓ Stmt The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 7.2 (Rectangular mixed transfer operator). Lean ✓ Stmt For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\mathrm{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \to M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\mathrm{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Remark 7.3 (Self-transfer). Lean ✓ Stmt Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.

Theorem 7.4 (Iterated mixed transfer). Lean ✓ Stmt *For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F_{AB}^N(X) = \sum_{\sigma \in \{0,\dots,d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \cdots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. ✓ Proof Induction on N: the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0,\dots,d-1\}^{N+1} \cong \{0,\dots,d-1\} \times \{0,\dots,d-1\}^N$. □

Theorem 7.5 (Matrix trace of iterated mixed transfer at the identity). Lean ✓ Stmt *For $N \geq 0$,*

$$\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \mathrm{tr}(A^\sigma (B^\sigma)^\dagger).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not *be confused with the operator trace $\mathrm{Tr}(F_{AB}^N)$; its relation to the MPV overlap is proved in the next section.*

Proof. ✓ Proof Apply the matrix trace to each summand in Theorem 7.4 with $X = \mathbb{1}$: $\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \mathrm{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_\sigma \mathrm{tr}(A^\sigma (B^\sigma)^\dagger)$. □

7.2 MPV overlap as transfer trace

Recall from Definition 2.48 that

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 7.6 (Trace expansion over matrix units). Lean ✓ Stmt *For any linear endomorphism T on $M_{D_1 \times D_2}(\mathbb{C})$, the operator trace can be expanded as*

$$\mathrm{Tr}(T) = \sum_{p=0}^{D_1-1} \sum_{q=0}^{D_2-1} (T(E_{pq}))_{pq},$$

where $E_{pq} \in M_{D_1 \times D_2}(\mathbb{C})$ is the matrix with 1 in position (p,q) and 0 elsewhere.

Proof. ✓ Proof The operator trace $\mathrm{Tr}(T)$ equals the sum of $\mathrm{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p,q), and $\mathrm{tr}(M \cdot E_{qp}) = M_{pq}$. □

Theorem 7.7 (Overlap equals transfer trace). Lean ✓ Stmt *For tensors A, B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}(F_{AB}^N).$$

Proof. ✓ Proof Expand the operator trace using Lemma 7.6 as a sum over matrix units E_{pq}. Apply Theorem 7.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p,q)-entry, and reassemble. The resulting double sum over (p,q) and σ equals $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. □

Theorem 7.8 (Rectangular overlap as transfer trace). Lean ✓ Stmt *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N).$$

Proof. ✓ Proof Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Theorem 7.7. □

Lemma 7.9 (Word trace as entrywise inner product). Lean ✓ Stmt *For a word* $\sigma \in \{0, \dots, d-1\}^N$,

$$\operatorname{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \overline{(B^\sigma)_{jk}}.$$

Proof. ✓ Proof Expand the matrix trace and the (j, j)-entry of $A^\sigma (B^\sigma)^\dagger$. □

7.3 Frobenius norm estimates

The transfer-operator gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\mathrm{HS}}$ on finite-dimensional spaces. We use the squared version and an isometric embedding into Euclidean space.

Definition 7.10 (Frobenius norm squared). Lean ✓ Stmt For a (possibly rectangular) matrix $X \in M_{m\times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

Lemma 7.11 (Trace formula for Frobenius norm). Lean ✓ Stmt $\|X\|_F^2 = \operatorname{Re} \operatorname{tr}(X^\dagger X)$.

Proof. ✓ Proof Expand the matrix product and trace entry by entry. □

Definition 7.12 (Euclidean-space embedding). Lean ✓ Stmt The map $\mathrm{vec} : M_{m\times n}(\mathbb{C}) \to \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Lemma 7.13 (Norm of embedded matrix). Lean ✓ Stmt $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Proof. ✓ Proof Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. □

7.4 Eigenvalue bound and transfer-operator gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization (cf. [Wol12, Proposition 6.1] for the general operator-norm version via Russo–Dye). For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument ([Wol12, Theorem 5.3]) to a block embedding of a mixed-transfer eigenvector.

Theorem 7.14 (Eigenvalue bound). Lean ✓ Stmt *Let* A, B *be normalized MPS tensors. Every eigenvalue* μ *of* F_{AB} *satisfies* $|\mu| \leq 1$.

Proof. ✓ Proof Expand $F^n_{AB}(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F^n_{AB}(X)\|_{\mathrm{HS}} \leq C_D \|X\|_{\mathrm{HS}}$ for all n, where C_D depends only on D. If $F_{AB}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n \|X\|_{\mathrm{HS}} = \|F^n_{AB}(X)\|_{\mathrm{HS}} \leq C_D \|X\|_{\mathrm{HS}}$ for every n. Hence $|\mu| \leq 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. □

Theorem 7.15 (Spectral radius bound). Lean ✓ Stmt *$\rho(F_{AB}) \leq 1$ for normalized tensors.*

Proof. ✓ Proof The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 7.14. □

Theorem 7.16 (Spectral radius ≥ 1 implies gauge-phase equivalence). Lean ✓ Stmt *Let A, B be injective normalized MPS tensors. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Since $\rho(F_{AB}) \leq 1$ (Theorem 7.15) and $\rho(F_{AB}) \geq 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \neq 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A, ρ_B of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 6.11) and gauge each tensor separately to a unital Kraus family (Theorem 6.13): set $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B. Let $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

Step 2 (Block Kraus family and off-diagonal embedding). Form the block Kraus operators $C^i = \left(\begin{smallmatrix} A'^i & 0 \\ 0 & B'^i \end{smallmatrix}\right)$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \left(\begin{smallmatrix} 0 & X' \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $E_C(Y) = \mu Y$.

Step 3 (KS equality and Kraus commutation). Because $|\mu| = 1$ and E_C is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 5.24) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 5.23), Y commutes with every block Kraus operator: $C^i Y = Y C^i$ for all i.

Step 4 (Intertwining identity and invertibility). Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every i. Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of A), X' is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 5.25) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence X' is invertible.

Step 5 (Skolem–Noether conclusion). The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ sending B'^i to A'^i. By Skolem–Noether (Theorem 3.10) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for all i, establishing gauge-phase equivalence (Definition 2.23).

The point is that one gauges the individual transfer maps to unital form; one does not require the mixed transfer map itself to be simultaneously unital and trace-preserving. □

Theorem 7.17 (Gauged peripheral eigenvector yields Kraus intertwining). Lean ✓ Stmt *Let A and B be normalized tensors, and let $X \neq 0$ satisfy $F_{AB}(X) = \mu X$ with $|\mu| = 1$. Choose positive-definite fixed points ρ_A, ρ_B of the transfer maps and gauge to unital families $A'^i = S_A^{-1} A^i S_A$ and $B'^i = S_B^{-1} B^i S_B$ with $S_A S_A^\dagger = \rho_A$ and $S_B S_B^\dagger = \rho_B$. Then the transported matrix $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and for every i one has*

$$X'(B'^i)^\dagger = \mu (A'^i)^\dagger X', \qquad A'^i X' = \mu X' B'^i.$$

Proof. ✓ Proof Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure. □

Theorem 7.18 (Intertwining yields gauge-phase equivalence). Lean ✓ Stmt *Let A and B be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families A' and B', an invertible matrix X', and a scalar μ with $|\mu| = 1$ such that, for every physical index i,*

$$A'^i X' = \mu X' B'^i.$$

Then A and B are gauge-phase equivalent.

Proof. ✓ Proof Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating A to a scalar multiple of B. □

Remark 7.19 (Two gauge conventions). ✓ Stmt The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a *unital* Kraus family ($\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$). A different convention, $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$, makes the family *trace-preserving* ($\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the transfer-operator gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 9).

Theorem 7.20 (Strict transfer-operator gap). Lean ✓ Stmt *Let A, B be injective normalized MPS tensors that are* not *gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. ✓ Proof $\rho(F_{AB}) \le 1$ by Theorem 7.15. If $\rho(F_{AB}) = 1$, Theorem 7.16 gives gauge-phase equivalence, contradicting the hypothesis. □

7.5 Transfer-operator gap — rectangular case

Theorem 7.21 (Rectangular intertwining forces equal bond dimension). Lean ✓ Stmt *Let A and B be injective tensors of bond dimensions D_1 and D_2. If there exist $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that, for every physical index i,*

$$X (B^i)^\dagger = \mu (A^i)^\dagger X, \qquad A^i X = \mu X B^i,$$

then $D_1 = D_2$.

Proof. ✓ Proof The intertwining relation shows that $\ker(X)$ is invariant under all generators of B. Since B is injective, its generators span the full matrix algebra, so $\ker(X)$ is trivial and $D_2 \le D_1$. Applying the same argument to $X^\dagger$ gives $D_1 \le D_2$. □

Theorem 7.22 (Rectangular transfer-operator gap). Lean ✓ Stmt *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2, with $D_1 \ne D_2$. Then $\rho(F^{\mathrm{rect}}_{AB}) < 1$.*

Proof. ✓ Proof The same word-expansion argument as in Theorem 7.14 gives $|\mu| \le 1$ for every eigenvalue of F^{rect}_{AB}. If $\rho(F^{\mathrm{rect}}_{AB}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A and B separately to unital Kraus families and apply the block-embedding Kadison–Schwarz argument as in the square case. The resulting intertwining relations satisfy the hypotheses of Theorem 7.21, so $D_1 = D_2$, contradicting the hypothesis. □

Theorem 7.23 (Rectangular overlap decay). Lean ✓ Stmt *If $D_1 \ne D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof The transfer-operator gap $\rho(F^{\mathrm{rect}}_{AB}) < 1$ (Theorem 7.22) implies $(F^{\mathrm{rect}}_{AB})^N \to 0$, so the operator trace converges to 0. By Theorem 7.8, $O_{AB}(N) = \mathrm{Tr}((F^{\mathrm{rect}}_{AB})^N) \to 0$. □

7.6 MPV overlap decay

Theorem 7.24 (Transfer powers converge to zero)**.** Lean ✓ Stmt *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof The transfer-operator gap $\rho(F_{AB}) < 1$ (Theorem 7.20) implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$ by the Gelfand formula, so $F_{AB}^n(X) \to 0$ for every X. □

Theorem 7.25 (Overlap decay)**.** Lean ✓ Stmt *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof By Theorem 7.7, $O_{AB}(N) = \mathrm{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 7.6): $\mathrm{Tr}(F_{AB}^N) = \sum_{p,q}(F_{AB}^N(E_{pq}))_{pq}$. By Theorem 7.24, each $F_{AB}^N(E_{pq}) \to 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \to 0$. □

7.7 Block separation

These asymptotic facts are the ingredients used later in the canonical-form separation arguments.

Remark 7.26 (Cross-correlation decay)*.* Lean ✓ Stmt If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\mathrm{tr}(F_{AB}^N(X)) \to 0$ as $N \to \infty$. This is the scalar trace consequence of the operator convergence $F_{AB}^N(X) \to 0$ from Theorem 7.24.

Remark 7.27 (Self-correlation persists)*.* Lean ✓ Stmt If ρ is a fixed point of $\mathcal{E}_A$, then $\mathrm{tr}(\mathcal{E}_A^N(\rho)) = \mathrm{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 6.11.

7.8 Transfer-operator gap under irreducible-TP hypotheses

The preceding transfer-operator gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with trace-preserving normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), following Cirac et al. [CPGSV17b] and the irreducibility theory of [Wol12, Section 6.2]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 7.28 (Gauge-equivalence from irreducible-TP spectral radius)**.** Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized MPS tensors of bond dimension D. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Choose a modulus-one eigenvector as in Theorem 7.16. The proof follows the five-step structure of Theorem 7.16, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 6.11) and the right-canonical gauge (Theorem 6.13). The change is in Step 4: the Kadison–Schwarz equality gives $X'^\dagger X'$ as a nonzero positive semidefinite fixed point of $\mathcal{E}_{B'}$. Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 6.4 upgrades it to positive definite, giving invertibility of X'. Formally, Theorem 7.17 combines the separately gauged eigenvector into the required intertwining relations. Once X' is invertible, Theorem 7.18 upgrades the intertwining to gauge-phase equivalence. □

Theorem 7.29 (Strict transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. ✓ Proof Contrapositive of Theorem 7.28, combined with $\rho(F_{AB}) \le 1$ (Theorem 7.15). ☐

Theorem 7.30 (Transfer power decay (irreducible-TP)). Lean ✓ Stmt *Under the hypotheses of Theorem 7.29, $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof By the Gelfand formula, $\rho(F_{AB}) < 1$ implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$. ☐

Theorem 7.31 (Overlap decay (irreducible-TP)). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof Combine the transfer-operator gap (Theorem 7.29) with the overlap–trace identity (Theorem 7.7). ☐

Theorem 7.32 (Rectangular transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let A (bond dimension D_1) and B (bond dimension D_2) be irreducible trace-preserving normalized tensors with $D_1 \ne D_2$. Then $\rho(F_{AB}^{\mathrm{rect}}) < 1$.*

Proof. ✓ Proof Assume for contradiction that $\rho = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. The Cauchy–Schwarz rigidity argument produces an intertwiner X' with $A'^i X' = X' B'^i$ for all i. Since $\ker(X')$ is invariant under all B'^i, irreducibility forces $\ker(X') = \{0\}$; similarly for $X'^\dagger$. This gives $D_1 = D_2$, contradicting the hypothesis. ☐

Theorem 7.33 (Rectangular overlap decay (irreducible-TP)). Lean ✓ Stmt *If $D_1 \ne D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof Combine Theorem 7.32 with Theorem 7.8. ☐

7.9 Conditional expectation from a faithful fixed point

This section follows [Wol12, Section 6.4]; the conditional expectation is built from the fixed-point algebra of [Wol12, Theorem 6.14] via [Wol12, (1.40)]. The scalar conditional expectation E_σ is the special case where the fixed-point $*$-subalgebra is the scalar algebra $\mathbb{C} \cdot \mathbb{1}$ (the primitive-channel case). The general irreducible case with nontrivial period requires the Wedderburn block decomposition of [Wol12, Theorem 6.14].

Definition 7.34 (Conditional expectation). Lean ✓ Stmt Given a $*$-subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a conditional expectation onto S if it is idempotent, unital, has range contained in S, and satisfies $P(X) = X$ for all $X \in S$.

Definition 7.35 (Scalar conditional expectation). Lean ✓ Stmt For $\sigma \ge 0$ with $\mathrm{tr}(\sigma) \ne 0$, define

$$E_\sigma(X) := \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}.$$

Lemma 7.36 (Evaluation formula). Lean ✓ Stmt $E_\sigma(X) = \dfrac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}$.

Theorem 7.37 (Unitality). Lean ✓ Stmt $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Theorem 7.38 (Idempotence). Lean ✓ Stmt *$E_\sigma(E_\sigma(X)) = E_\sigma(X)$ for all X.*

Lemma 7.39 (Range is scalar). Lean ✓ Stmt *For every X, there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c\,\mathbb{1}$.*

Lemma 7.40 (Fixes scalar operators). Lean ✓ Stmt *For $c \in \mathbb{C}$, $E_\sigma(c\,\mathbb{1}) = c\,\mathbb{1}$.*

Theorem 7.41 (Absorbs the adjoint map). Lean ✓ Stmt *If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.*

Theorem 7.42 (Adjoint map absorbs scalar projection). Lean ✓ Stmt *If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.*

Lemma 7.43 (Image lies in adjoint fixed points). Lean ✓ Stmt *For all X, $E_\sigma(X)$ is fixed by T^*.*

Theorem 7.44 (Scalar map is a conditional expectation). Lean ✓ Stmt *Let K be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

Theorem 7.45 (Complete positivity of the scalar conditional expectation). Lean ✓ Stmt *For any positive semidefinite σ, the scalar conditional expectation E_σ is a completely positive map.*

Proof. ✓ Proof Write $S = \sqrt{\sigma}$ and define the Kraus family $K_{j,i} = |j\rangle\langle i|\,S$ indexed by $(j,i) \in \{1,\dots,D\}^2$. Using $|j\rangle\langle i|\,M\,|i\rangle\langle j| = M_{ii}|j\rangle\langle j|$ and summing first over i and then over j,

$$\sum_{j,i} K_{j,i}\,X\,K_{j,i}^\dagger = \sum_j |j\rangle\langle j|\,\mathrm{tr}(SXS) = \mathrm{tr}(\sigma X)\mathbb{1},$$

where we used cyclicity of the trace and $S^2 = \sigma$. Hence $E_\sigma = (\mathrm{tr}(\sigma))^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$ is the Kraus map scaled by the non-negative real number $(\mathrm{tr}(\sigma))^{-1}$, and therefore completely positive. ☐

Theorem 7.46 (σ-trace preservation of the scalar conditional expectation). Lean ✓ Stmt *If $\mathrm{tr}(\sigma) \neq 0$, then for every X*

$$\mathrm{tr}(\sigma\,E_\sigma(X)) = \mathrm{tr}(\sigma X).$$

Proof. ✓ Proof Direct computation: $\mathrm{tr}(\sigma \cdot \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)}\mathbb{1}) = \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)}\,\mathrm{tr}(\sigma) = \mathrm{tr}(\sigma X)$. ☐

7.10 Stationary support

This section develops the support-projection theory behind stationary states, following [Wol12, Section 6.4, Propositions 6.10–6.11].

Lemma 7.47 (Lower-triangular Kraus condition implies corner invariance). Lean ✓ Stmt *Let A be an MPS tensor and let P be an orthogonal projection. If, for every physical index i,*

$$(\mathbb{1} - P)A^i P = 0$$

then

$$P\mathcal{E}_A(PXP)P = \mathcal{E}_A(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof Expand the transfer map and use $(\mathbb{1} - P)A^i P = 0$ to reduce each Kraus term to the P-corner. ☐

Theorem 7.48 (Support projection invariance). Lean ✓ Stmt *Let E be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let P denote the support projection of ρ. Then*

$$PE(PXP)P = E(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all i. Apply Lemma 7.47. □

Definition 7.49 (Stationary state). Lean ✓ Stmt For an irreducible channel E on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\mathrm{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 6.11).

Definition 7.50 (Stationary support). Lean ✓ Stmt The *stationary support* of an irreducible channel E is the support projection of the stationary state (Definition 7.49).

Theorem 7.51 (Stationary support is full). Lean ✓ Stmt *For an irreducible channel E, the stationary support equals the identity:* $\mathrm{supp}(\rho_\infty) = \mathbb{1}$.

Proof. ✓ Proof The support projection P of ρ_∞ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all X (Theorem 7.48). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. □

Remark 7.52 (Stationary-support formulations). The non-trivial formulations needed here are the following results from Wolf:

- the converse direction of [Wol12, Proposition 6.10]: if the support projection of every positive semidefinite fixed point is full, then the channel is irreducible;
- [Wol12, Proposition 6.11]: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are not used elsewhere in this chapter.

7.10.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra, $*$-structure, and compression of a PSD fixed point onto its support sector needed for [Wol12, Corollary 6.6]: the corner carries a $\mathbb{C}$-algebra and $*$-structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

Lemma 7.53 (Support projection annihilates the kernel). Lean ✓ Stmt *Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \mathrm{supp}(\rho)$. For every $v \in \mathbb{C}^D$, if $\rho\, v = 0$, then $P\, v = 0$.*

Proof. ✓ Proof Diagonalize $\rho = U\ \mathrm{diag}(\lambda)\, U^\dagger$ with $\lambda_j \geq 0$. Writing $w := U^\dagger v$, the hypothesis $\rho v = 0$ gives $\lambda_j\, w_j = 0$ for every j; hence $w_j = 0$ whenever $\lambda_j > 0$. The support projection acts as $\mathrm{diag}(\chi_{\lambda_j > 0})$ in the eigenbasis, so $P\, v = U\ \mathrm{diag}(\chi_{\lambda_j > 0})\, w = 0$. □

Definition 7.54 (Corner submodule / idempotent-corner identification). Lean ✓ Stmt For an idempotent $P \in M_D(\mathbb{C})$, the subspace $\{X \mid PXP = X\}$ and the corner $\{P\, X\, P \mid X \in M_D(\mathbb{C})\}$ are identified as $\mathbb{C}$-linear spaces by the identity on underlying matrices.

Remark 7.55 ($\mathbb{C}$-algebra and $*$-structure on the corner). Lean Lean Lean Lean ✓ Stmt The corner algebra associated to an idempotent already carries a ring structure with unit P. In the matrix case it also has $\mathbb{C}$-module and $\mathbb{C}$-algebra structures and, under the additional assumption $P^\dagger = P$, star, star ring, and star module structures over $\mathbb{C}$.

Theorem 7.56 (Faithful compression onto the support). Lean ✓ Stmt *Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \operatorname{supp}(\rho)$, and let $V : \mathbb{C}^D \to \mathbb{C}^k$ be a compression isometry satisfying $V\,V^\dagger = \mathbb{1}_k$ and $V^\dagger\,V = P$. Then the compression*

$$V\,\rho\,V^\dagger \in M_k(\mathbb{C})$$

is positive definite.

Proof. ✓ Proof Hermiticity follows from Hermiticity of ρ and the product-star identity. For strict positivity, let $w \neq 0$, set $u := V^\dagger w$, and note that $VV^\dagger = \mathbb{1}_k$ forces $u \neq 0$, while $V^\dagger V = P$ forces $Pu = u$. The quadratic form computes as $w^\dagger(V\rho V^\dagger)w = u^\dagger\rho u$, which is non-negative by positive semidefiniteness. If it vanishes, then $\rho u = 0$, and by Lemma 7.53 also $Pu = 0$, contradicting $Pu = u \neq 0$. ☐

7.11 Wedderburn decomposition of the fixed-point algebra

This section proves the structure theorem for the fixed-point algebra of a trace-preserving Kraus map, following [Wol12, Theorems 6.12–6.14]. The proof uses the Jacobson radical characterization of semisimplicity together with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.

Theorem 7.57 (Fixed-point algebra is semisimple). Lean ✓ Stmt *The adjoint-fixed-point $*$-subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.*

Proof. ✓ Proof The subalgebra is finite-dimensional (as a subalgebra of $M_D(\mathbb{C})$), hence Artinian. By [Wol12, Theorem 6.14], it suffices to show the Jacobson radical J vanishes. If $x \in J$, then $x^*x \in J$ (left-ideal closure). The radical of an Artinian ring is nilpotent, so x^*x is nilpotent. A self-adjoint nilpotent matrix satisfies $\|x^*x\|^{2^k} = \|(x^*x)^{2^k}\| = 0$, hence $x^*x = 0$, and the C*-identity gives $x = 0$. ☐

Theorem 7.58 (Abstract Wedderburn–Artin decomposition). Lean ✓ Stmt *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $*$-subalgebra is $\mathbb{C}$-algebra isomorphic to*

$$\prod_{k=1}^{n} M_{d_k}(\mathbb{C}).$$

Proof. ✓ Proof Combine semisimplicity (Theorem 7.57) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field. ☐

Theorem 7.59 (Fixed-point algebra decomposition with multiplicities). Lean ✓ Stmt *There exist integers n, block sizes $d_1, \dots, d_n \geq 1$, multiplicities $m_1, \dots, m_n \geq 1$, and a $\mathbb{C}$-algebra isomorphism from the adjoint-fixed-point $*$-subalgebra to*

$$\prod_{k=1}^{n} M_{d_k}(\mathbb{C})$$

such that

$$\sum_{k=1}^{n} d_k m_k \leq D.$$

Proof. ✓ Proof Apply the abstract Wedderburn decomposition (Theorem 7.58). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra $M_D(\mathbb{C})$, this realization also yields multiplicities m_k and the bound $\sum_k d_k m_k \le D$. □

Remark 7.60 (Block form of the fixed-point algebra). Wolf's block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with

$$0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form $M_{d_k}(\mathbb{C}) \otimes \rho_k$. The theorem above gives the product decomposition, multiplicities, and dimension bound needed from this block form.

Theorem 7.61 (Weighted fixed points form a $*$-subalgebra)**.** Lean ✓ Stmt *Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, and let $\rho > 0$ satisfy $T(\rho) = \rho$. If*

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

then $\rho^{-1/2} F_T \rho^{-1/2}$ is a $*$-subalgebra of $M_D(\mathbb{C})$.

Proof. ✓ Proof Set $B_i = \rho^{-1/2} K_i \rho^{1/2}$. The equation $T(\rho) = \rho$ implies that the family $\{B_i\}$ is unital, and trace preservation of T implies that ρ is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 4.54 to B. The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with $\rho^{-1/2} F_T \rho^{-1/2}$. □

7.12 Further irreducibility and primitivity equivalences

This section collects several criteria and equivalences from [Wol12, Theorems 6.2, 6.7, 6.8] that are used later.

Theorem 7.62 (Exponential positivity condition)**.** Lean Lean ✓ Stmt *Let E be an irreducible completely positive map, $A \ge 0$ with $A \ne 0$, and $t > 0$. Then $\exp(tE)(A)$ is positive definite. Conversely, if $\exp(tE)(A)$ is positive definite for every such t, A, then E is irreducible (full equivalence* Lean *).*

Theorem 7.63 (Kraus-span equivalence)**.** Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.*

Theorem 7.64 (Combined primitivity equivalence)**.** Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

Remark 7.65 (Pairwise primitivity equivalences). Lean Lean Lean Lean ✓ Stmt These are the pairwise equivalences in [Wol12, Theorem 6.8].

Theorem 7.66 (Scalar fixed-point algebra case)**.** Lean ✓ Stmt *In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\mathrm{tr}(\rho X)}{\mathrm{tr}(\rho)} \mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

7.13 Multi-cycle block-permutation structure

This section develops the block-permutation structure underlying the cycle decomposition of [Wol12, Theorem 6.16]. The multi-cycle decomposition refines the single-cycle case to handle arbitrary peripheral periods.

Definition 7.67 (Block-permutation structure). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$. A *block-permutation structure* for T consists of a finite index set ι, a family of orthogonal projections $P_k \in M_D(\mathbb{C})$ for $k \in \iota$, and a permutation $\sigma \in \mathrm{Sym}(\iota)$ (not necessarily a single cycle—in general σ may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k)\,T(X), \quad T(XP_k) = T(X)\,T(P_k)$$

for every $k \in \iota$ and $X \in M_D(\mathbb{C})$.

Definition 7.68 (Multi-cycle block-permutation structure). Lean ✓ Stmt A *multi-cycle decomposition* of $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ consists of a finite cycle index ι, a per-cycle period $m : \iota \to \mathbb{N}_{>0}$, a projection family $P_{c,k} \in M_D(\mathbb{C})$ for $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$ consisting of orthogonal projections. For every $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$, the per-cycle cyclic action is

$$T(P_{c,k+1}) = P_{c,k}.$$

The multiplicative-domain factorisations are $T(P_{c,k}\,X) = T(P_{c,k})\,T(X)$ and $T(X\,P_{c,k}) = T(X)\,T(P_{c,k})$.

Theorem 7.69 (Per-cycle corner preservation). Lean ✓ Stmt *For every cycle $c \in \iota$ and index $k \in \{0, \dots, m(c)-1\}$, the iterate $T^{m(c)}$ preserves the corner $P_{c,k} M_D(\mathbb{C})\,P_{c,k}$.*

Proof. ✓ Proof By induction on n, the n-th iterate T^n sends $P_{c,k+n}\,X\,P_{c,k+n}$ to $P_{c,k}\,T^n(X)\,P_{c,k}$, using the multiplicative-domain factorisations and the cyclic action $T(P_{c,j+1}) = P_{c,j}$ at each induction step. Specialising to $n = m(c)$ and using that the indices $k+m(c) = k$ in $\{0, \dots, m(c)-1\}$ yields corner preservation of $T^{m(c)}$. $\square$

Theorem 7.70 (Common-period corner preservation). Lean ✓ Stmt *If $N \in \mathbb{N}$ is divisible by every per-cycle period $m(c)$, then T^N preserves every corner $P_{c,k} M_D(\mathbb{C})\,P_{c,k}$ of the decomposition.*

Proof. ✓ Proof Write $N = m(c)\,q$. Corner preservation is stable under taking powers, so $T^N = (T^{m(c)})^q$ preserves each corner by Theorem 7.69. $\square$

Definition 7.71 (Flattening to a single-index block-permutation structure). Lean ✓ Stmt A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index $\Sigma_{c\in\iota}\{0, \dots, m(c)-1\}$: the permutation is the sigma-product of per-cycle cyclic shifts $k \mapsto k+1$ (whose cycle decomposition has one cycle per $c \in \iota$), the projections are $(c, k) \mapsto P_{c,k}$, and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.

7.14 Peripheral eigenvalue group structure

This section proves the peripheral spectrum structure of [Wol12, Theorem 6.6] for irreducible Schwarz maps: the peripheral eigenvalues form a finite cyclic group. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 7.75.

Lemma 7.72 (Peripheral eigenvectors are invertible). Lean ✓ Stmt *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.*

Proof. ✓ Proof The KS equality gives $E(X^\dagger X) = X^\dagger X$, so $X^\dagger X$ is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so $\det(X) \neq 0$. □

Lemma 7.73 (Peripheral eigenvalues: product closure). *Lean* *✓ Stmt* *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E, then so is $\mu\nu$.*

Proof. ✓ Proof Take eigenvectors X, Y for μ, ν. The KS equality at X puts X in the multiplicative domain, so $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$. Since X, Y are units (Lemma 7.72), $YX \neq 0$ and $|\mu\nu| = 1$. □

Theorem 7.74 (Peripheral eigenvalues: cyclic structure). *Lean* *✓ Stmt* *Let E be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is* not *required). Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. Any finite subgroup of $\mathbb{C}^\times$ is cyclic. The generator γ has order $m = |\text{peripheral spectrum}|$. □

Theorem 7.75 (Peripheral eigenvalues form a cyclic group). *Lean* *✓ Stmt* *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof Apply Theorem 7.74 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \mathrm{tr}(P_0) = D$ with $\mathrm{tr}(P_0) \in \mathbb{N}$. □

Theorem 7.76 (Period divides bond dimension). *Lean* *✓ Stmt* *Let E be as in Theorem 7.75, and let γ be a primitive m-th root of unity generating the peripheral eigenvalues. Then $m \mid D$.*

Proof. ✓ Proof Follows from the cyclic projection decomposition and trace preservation, as in the final step of Theorem 7.75. □

7.14.1 Cyclic decomposition of irreducible Schwarz maps

Definition 7.77 (Corner preservation). Lean ✓ Stmt For an orthogonal projection P and linear map E, the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for all X.

Definition 7.78 (Corner subspace). Lean ✓ Stmt The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 7.79 (Restricted corner map). Lean ✓ Stmt The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 7.80 (Irreducible restriction on a corner). Lean ✓ Stmt Irreducibility of the corner-restricted map.

Definition 7.81 (Rank of an orthogonal projection). Lean ✓ Stmt The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, i.e. the dimension of the corner algebra $PM_D(\mathbb{C})P$.

Theorem 7.82 (Compression isometry for an orthogonal projection). Lean ✓ Stmt *Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n. The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P.*

Theorem 7.83 (Rank equals trace for an orthogonal projection). Lean ✓ Stmt *The rank of an orthogonal projection P equals its trace: $n = \operatorname{tr} P$.*

Theorem 7.84 (Peripheral unitary eigenvector). Lean ✓ Stmt *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Theorem 7.85 (Powers on a peripheral-unitary orbit). Lean ✓ Stmt *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for all n.*

Theorem 7.86 (Normalized peripheral unitary). Lean ✓ Stmt *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Theorem 7.87 (Cyclic projections from a peripheral unitary). Lean ✓ Stmt *A primitive m-th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Theorem 7.88 (Cyclic decomposition for irreducible Schwarz maps). Lean ✓ Stmt *Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.*

Theorem 7.89 (Power maps preserve each cyclic sector). Lean ✓ Stmt *Appropriate powers of the channel preserve each sector corner.*

Theorem 7.90 (Irreducibility of restricted sector dynamics). Lean ✓ Stmt *Each sector restriction is irreducible.*

Theorem 7.91 (Primitivity of restricted sector dynamics). Lean ✓ Stmt *Each sector restriction is primitive.*

Theorem 7.92 (Corner invariance at the period). Lean ✓ Stmt *The channel raised to the period preserves every cyclic corner.*

7.14.2 Group structure of the peripheral spectrum

Theorem 7.93 (Peripheral product closure). Lean ✓ Stmt *The product of two peripheral eigenvalues is peripheral.*

Theorem 7.94 (Peripheral inverse closure). Lean ✓ Stmt *The inverse of a peripheral eigenvalue is peripheral.*

Theorem 7.95 (Peripheral cyclic group with period divisibility). Lean ✓ Stmt *The peripheral eigenvalues form a finite cyclic group whose order divides the dimension.*

Theorem 7.96 (Peripheral eigenvalue multiplicity one). Lean ✓ Stmt *Each peripheral eigenvalue has algebraic multiplicity one.*

Proof. ✓ Proof Given a peripheral unitary eigenvector U for γ, the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ-eigenvector X. By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c \cdot \mathbb{1}$, hence $X = c \cdot U$ and the eigenspace is one-dimensional. □

7.15 Primitive overlap convergence (complementary transfer-map gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument, corresponding to the primitive-case specialization of [Wol12, Theorem 6.7] where T_ϕ is a rank-one projection. For primitive normalized transfer maps, Chapter 4 supplies this complementary transfer-map input under explicit hypotheses.

Theorem 7.97 (Complementary transfer-map gap implies trace convergence to 1). Lean ✓ Stmt *Let E be trace-preserving with fixed point ρ ($\operatorname{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 4.23). If $\rho_{\text{spec}}(E - P) < 1$, then $\operatorname{Tr}(E^n) \to 1$ as $n \to \infty$.*

Proof. ✓ Proof By the power decomposition (Theorem 4.24), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\text{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \to 0$, so $\operatorname{Tr}(E^n) \to \operatorname{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\operatorname{tr}(X)}{\operatorname{tr}(\rho)}\rho$, we have $\operatorname{Tr}(P) = \frac{\operatorname{tr}(\rho)}{\operatorname{tr}(\rho)} = 1$. ☐

Theorem 7.98 (Self-overlap convergence from a complementary transfer-map gap). Lean ✓ Stmt *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \to 1$ as $N \to \infty$.*

Proof. ✓ Proof By Theorem 7.7 (with $A = B$), $O_{AA}(N) = \operatorname{Tr}(\mathcal{E}_A^N)$. By Theorem 7.97, $\operatorname{Tr}(\mathcal{E}_A^N) \to 1$. ☐

Lemma 7.99 (Norm of self-overlap convergence). Lean ✓ Stmt *If A is an MPS tensor and $O_{AA}(N) \to 1$ in $\mathbb{C}$ as $N \to \infty$, then $\|O_{AA}(N)\| \to 1$ as $N \to \infty$.*

Proof. ✓ Proof Follows from the continuity of the norm: $O_{AA}(N) \to 1$ implies $\|O_{AA}(N)\| \to \|1\| = 1$. ☐

Chapter 8

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D, the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

8.1 Cumulative span

Definition 8.1 (Word span). Lean ✓ Stmt The *word span* at length n is the subspace

$$S_n(A) = \mathrm{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Section II].

Definition 8.2 (Cumulative span). Lean ✓ Stmt The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 8.3 (Monotonicity of cumulative span). Lean ✓ Stmt *$T_n(A) \leq T_{n+1}(A)$ for all n.*

Proof. ✓ Proof T_{n+1} includes all words of length $\leq n+1$, which contains all words of length $\leq n$. □

Lemma 8.4 (Dimension bound). Lean ✓ Stmt *$\dim T_n(A) \leq D^2$ for all n.*

Proof. ✓ Proof $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. □

Theorem 8.5 (Cumulative span stabilisation). Lean ✓ Stmt *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.*

Proof. ✓ Proof If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2}, so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. □

Theorem 8.6 (Dimension growth). Lean ✓ Stmt *If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

Proof. ✓ Proof $T_n \leq T_{n+1}$ (Lemma 8.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. □

Lemma 8.7 (Word span characterises block injectivity). Lean ✓ Stmt *$S_L(A) = M_D(\mathbb{C})$ if and only if A is L-block injective.*

Proof. ✓ Proof Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. □

Lemma 8.8 (Zero-length word span). Lean ✓ Stmt *If $D \geq 2$, then $S_0(A) \neq M_D(\mathbb{C})$.*

Proof. ✓ Proof The zero-length products consist only of the identity, so $S_0(A) = \mathrm{span}_{\mathbb{C}}\{\mathbb{1}\}$ has dimension 1. Since $\dim M_D(\mathbb{C}) = D^2 \geq 4$, this span cannot be all of $M_D(\mathbb{C})$. □

Corollary 8.9 (0-block injectivity). Lean ✓ Stmt *If $D \geq 1$ and A is 0-block injective, then $D = 1$.*

Proof. ✓ Proof By Lemma 8.7, 0-block injectivity gives $S_0(A) = M_D(\mathbb{C})$. Lemma 8.8 excludes $D \geq 2$, hence the nonzero bond dimension must be 1. □

Lemma 8.10 (Two-sided span from one nonzero matrix). Lean ✓ Stmt *If $C \in M_D(\mathbb{C})$ is nonzero, then*

$$\mathrm{span}_{\mathbb{C}}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}).$$

This is the matrix-factorization input used in [PGVWC07, Lemma `lem1`].

Proof. ✓ Proof Choose a nonzero entry C_{pq}. Then every matrix unit E_{ij} is a scalar multiple of $E_{ip}CE_{qj}$, so the displayed span contains the standard matrix basis. □

Lemma 8.11 (Exact words around a nonzero middle matrix). Lean ✓ Stmt *If A is L-block injective and $X \neq 0$, then*

$$\mathrm{span}_{\mathbb{C}}\{A^u X A^v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

Proof. ✓ Proof By L-block injectivity, the length-L word products span $M_D(\mathbb{C})$. Bilinearity of $(R, S) \mapsto RXS$ reduces the preceding lemma to products with R and S chosen among those word products. □

8.2 Fitting decomposition

Definition 8.12 (Fitting decomposition). Lean ✓ Stmt A *Fitting decomposition* of a linear endomorphism $f : V \to V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ-eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V,
4. the generalized eigenspaces are linearly independent.

Theorem 8.13 (Fitting decomposition exists). Lean ✓ Stmt *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. ✓ Proof Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. □

Theorem 8.14 (Nilpotent power bound). Lean ✓ Stmt *If f is nilpotent on a space of dimension n, then $f^n = 0$.*

Proof. ✓ Proof A nilpotent endomorphism on an n-dimensional space has nilpotency index $\leq n$. □

8.3 Nonzero trace product

Theorem 8.15 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). Lean ✓ Stmt *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 8.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 8.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0, hence $T_{L_0} = M_D(\mathbb{C})$). □

Theorem 8.16 (Cumulative span reaches $M_D(\mathbb{C})$). Lean ✓ Stmt *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Take n from Theorem 8.15; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. □

Theorem 8.17 (Nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2$ such that $\operatorname{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.*

Proof. ✓ Proof By Theorem 8.16, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\operatorname{tr}(\mathbb{1}) = D \neq 0$). □

Theorem 8.18 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). Lean ✓ Stmt *If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \operatorname{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.*

Proof. ✓ Proof The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. □

Theorem 8.19 (Sharp nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2 - \operatorname{kr}(A) + 1$ such that $\operatorname{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.*

Proof. ✓ Proof By Theorem 8.18, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. □

Theorem 8.20 (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If A is normal and $D \geq 2$, then there exists a word w of* positive *length $1 \leq |w| \leq D^2 - \operatorname{kr}(A) + 1$ such that $\operatorname{tr}(A^w) \neq 0$. This strengthens Theorem 8.19 by requiring $|w| \geq 1$, a feature needed for the blocking argument in the general Wielandt bound below.*

Proof. ✓ Proof Define the *positive-level* cumulative span $V_n = \mathrm{span}\{A^w : 1 \le |w| \le n\}$. For $D \ge 2$, the span $S_0(A) = \mathrm{span}\{\mathbb{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\ge d' \ge 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left-multiplication by each A^i, so $S_N(A) \subseteq V_n \ne M_D(\mathbb{C})$ for all $N \ge 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbb{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\mathrm{tr}(\mathbb{1}) = D \ge 2 \ne 0$). ☐

8.4 Eigenvector extraction

Theorem 8.21 (Nonzero trace implies eigenvector). Lean ✓ Stmt *If $M \in M_D(\mathbb{C})$ has $\mathrm{tr}(M) \ne 0$, then M has a nonzero eigenvalue $\mu \ne 0$ and a corresponding eigenvector $\varphi \ne 0$ with $M\varphi = \mu\varphi$.*

Proof. ✓ Proof A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 8.13) ensures that the corresponding generalized eigenspace is nontrivial. On that generalized eigenspace, $M - \mu\mathbb{1}$ is nilpotent, so its kernel is nontrivial. Any nonzero vector in $\ker(M - \mu\mathbb{1})$ is therefore a genuine eigenvector with eigenvalue μ. ☐

Theorem 8.22 (Eigenvalue extraction). Lean ✓ Stmt *If A is normal, then there exist a word w_0 with $|w_0| \le D^2$, a nonzero scalar $\mu \ne 0$, and a nonzero vector $\varphi \ne 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. ✓ Proof By Theorem 8.17, there is a word w_0 with $\mathrm{tr}(A^{w_0}) \ne 0$ and $|w_0| \le D^2$. By Theorem 8.21, A^{w_0} has a nonzero eigenvector. ☐

8.5 Eigenvector spreading

Definition 8.23 (Cumulative vector span). Lean ✓ Stmt Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w\varphi : |w| \le n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 8.24 (Vector span stabilisation). Lean ✓ Stmt *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \ge n$.*

Proof. ✓ Proof Same argument as Theorem 8.5: left-multiplication by A^i cannot escape the stabilised span. ☐

Lemma 8.25 (Vector span from matrix span). Lean ✓ Stmt *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \ne 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. ✓ Proof Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N. Applying such matrices to φ recovers all of $\mathbb{C}^D$. ☐

Theorem 8.26 (Eigenvector spreading). Lean ✓ Stmt *Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \ne 0$ and $\varphi \ne 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. ✓ Proof The vector span K_n is monotone and $\dim K_n \le D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 8.24). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 8.16, so Lemma 8.25 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. ☐

8.6 Proof of the cumulative Wielandt bound

Theorem 8.27 (Wielandt chain). Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension $D \geq 1$. Then the following hold simultaneously:*

1. *$T_{D^2}(A) = M_D(\mathbb{C})$ (cumulative span stabilises);*
2. *there exists a word w_0 with $|w_0| \leq D^2$ and $\operatorname{tr}(A^{w_0}) \neq 0$;*
3. *there exist $w_0, \mu \neq 0, \varphi \neq 0$ with $|w_0| \leq D^2$ and $A^{w_0}\varphi = \mu\,\varphi$;*
4. *for every $\varphi \neq 0$, $K_{D^2}(A, \varphi) = \mathbb{C}^D$.*

Proof. ✓ Proof Items (1)–(3) follow from Theorems 8.16 and 8.22. Item (4) combines these with the eigenvector spreading bound (Theorem 8.26). □

Theorem 8.28 (Wielandt bound). Lean ✓ Stmt *If A is a normal MPS tensor with bond dimension D, then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2. This is the cumulative-span part of [SPGWC10, Theorem 1]. The cited theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, stated below as a separate statement.*

Proof. ✓ Proof Immediate from Theorem 8.16. □

Remark 8.29 (Explicit blocking-length estimates). ✓ Stmt Theorem 8.28 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2-d+1)\cdot D^2$ ([SPGWC10, Theorem 1, case (1)]) is proved in Theorem 8.42 below. The sharp blocking-length estimates for injectivity (D^4) and bi-canonical form ($3D^5$) from [SPGWC10] are not proved here.

Definition 8.30 (One-step augmentation). Lean ✓ Stmt Let A be an MPS tensor with Kraus family $\{A^i\}_{i=0}^{d-1}$, and let $X \in M_D(\mathbb{C})$. The *one-step augmentation* of A by X is the tensor $\widetilde{A}$ with physical dimension $d+1$ whose first Kraus operator is X and whose remaining Kraus operators are those of A:

$$\widetilde{A}^0 = X, \qquad \widetilde{A}^{i+1} = A^i.$$

Lemma 8.31 (Kraus operators lie in the one-step span). Lean ✓ Stmt *Every Kraus operator A^i belongs to the one-step span $S_1(A)$.*

Proof. ✓ Proof This is the length-one word with letter i. □

Lemma 8.32 (Redundant one-step augmentation). Lean Lean Lean ✓ Stmt *Let $X \in S_1(A)$, and let $\widetilde{A}$ be the one-step augmentation of A by X. Then*

$$S_n(\widetilde{A}) = S_n(A)$$

for every $n \geq 0$.

Proof. ✓ Proof The new generator X is already in $S_1(A)$, and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length. □

Lemma 8.33 (Normality and Kraus rank under one-step augmentation). Lean Lean ✓ Stmt *Let $X \in S_1(A)$, and let $\widetilde{A}$ be the one-step augmentation of A by X. If A is normal, then $\widetilde{A}$ is normal, and*

$$\operatorname{kr}(\widetilde{A}) = \operatorname{kr}(A).$$

Proof. ✓ Proof Since all exact word spans are unchanged by Lemma 8.32, eventual full word span for A gives eventual full word span for $\widetilde{A}$. The equality of Kraus ranks is the equality of the dimensions of the common one-step span. □

Theorem 8.34 (Wielandt case (2), invertible generator)**.** Lean Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension D, and suppose that some Kraus operator A^{i_0} is invertible. Then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}), \qquad \iota(A) \leq D^2 - \mathrm{kr}(A) + 1.$$

Proof. ✓ Proof The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument applied at all later lengths would contradict normality. Since $\dim S_1(A) = \mathrm{kr}(A)$ and $\dim M_D(\mathbb{C}) = D^2$, the full algebra is reached by length $D^2 - \mathrm{kr}(A) + 1$. The index bound follows from the definition of $\iota(A)$. □

Theorem 8.35 (Wielandt case (2), one-step invertible element)**.** Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If $S_1(A)$ contains an invertible matrix X, then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}).$$

In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].

Proof. ✓ Proof Adjoin X as an extra first Kraus operator, obtaining $\widetilde{A}$. Primitivity with a positive-definite fixed point and normalization give normality of A, hence normality of $\widetilde{A}$; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of $\widetilde{A}$ is invertible, so Theorem 8.34 gives $S_{D^2-\mathrm{kr}(\widetilde{A})+1}(\widetilde{A}) = M_D(\mathbb{C})$. Transfer this equality back to A. □

Corollary 8.36 (Wielandt case (2), index bound)**.** Lean ✓ Stmt *Under the hypotheses of Theorem 8.35,*

$$\iota(A) \leq D^2 - \mathrm{kr}(A) + 1,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$.

Proof. ✓ Proof The displayed word-span equality gives an admissible value in the defining minimum for $\iota(A)$. □

Corollary 8.37 (Wielandt case (2), invertible Kraus operator)**.** Lean Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is invertible, then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}), \qquad \iota(A) \leq D^2 - \mathrm{kr}(A) + 1.$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.31. □

Lemma 8.38 (Rank-one extraction from a non-invertible eigenvector)**.** Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension $D \geq 1$. Suppose that a Kraus operator A^{i_0} is non-invertible and that there exists a vector $\varphi \in \mathbb{C}^D$ and a nonzero scalar μ such that $A^{i_0}\varphi = \mu\varphi$. Then, for every $\psi \in \mathbb{C}^D$,*

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

Proof. ✓ Proof Decompose A^{i_0} into its nilpotent and nonzero generalized eigenspaces. On the nonzero part, multiplication by A^{i_0} is injective, so the rectangular spans grow strictly until they fill the whole range. The nilpotent index contributes the remaining length. Since the eigenvalue on φ is nonzero, eigenvector padding moves the resulting cumulative membership to the exact length $D^2 - D + 1$. □

Theorem 8.39 (Wielandt case (3), one-step non-invertible element). Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If $S_1(A)$ contains a non-invertible matrix X with a nonzero eigenvalue, then*

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

More explicitly, it is enough to have $X\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].

Proof. ✓ Proof Adjoin X as an extra first Kraus operator. The augmented tensor is normal and has the same exact word spans as A. The eigenvector relation for X is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length $D - 1$, and Lemma 8.38 gives the rank-one operators $|\varphi\rangle\langle\psi|$ in length $D^2 - D + 1$. Multiplying these two fixed-length spans gives S_{D^2} for the augmented tensor, hence for A. □

Corollary 8.40 (Wielandt case (3), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.39,*

$$\iota(A) \leq D^2.$$

Proof. ✓ Proof The equality $S_{D^2}(A) = M_D(\mathbb{C})$ gives an admissible value in the defining minimum for $\iota(A)$. □

Corollary 8.41 (Wielandt case (3), non-invertible Kraus operator). Lean Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is non-invertible and has an eigenvector $\varphi \neq 0$ with nonzero eigenvalue μ, then*

$$S_{D^2}(A) = M_D(\mathbb{C}), \qquad \iota(A) \leq D^2.$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.31. □

Theorem 8.42 (Wielandt's inequality — general bound). Lean ✓ Stmt *Let A be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive with a positive-definite fixed point. Then*

$$\iota(A) \leq (D^2 - \mathrm{kr}(A) + 1) \cdot D^2,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof. ✓ Proof

1. $D = 1$: In this case $\iota(A) = 0$, so the bound holds trivially.

2. $D \geq 2$: By Theorem 8.20, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\mathrm{tr}(A^w) \neq 0$, where $d' = \mathrm{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ (Theorem 8.21). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal (Theorem 8.53), and the encoding of w as a single block index gives $B^{i_0} = A^w$.

   (a) If A^w is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   (b) If A^w is non-invertible (non-invertible case), the eigenvector φ spans a proper subspace, and the rank-one extraction argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \mathrm{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate qualitative fixed-length conclusion stated below does not supply this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$.

□

Remark 8.43 (Quantitative and qualitative Wielandt bounds). ✓ Stmt The quantitative bound $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$ is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint's own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 8.59: there exists N with $S_N(A) = M_D(\mathbb{C})$. The quantitative bound is not on the fundamental-theorem critical path.

8.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 8.44 (Fixed-length vector span). Lean ✓ Stmt The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 8.45 (Word span products). *Lean* *✓ Stmt* *$S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$: the product of word spans at lengths m and n is contained in the word span at length $m + n$.*

Proof. ✓ Proof Every product $A^{w_1} A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1 w_2}$ with $|w_1 w_2| = m + n$. □

Lemma 8.46 (Eigenvector padding). *Lean* *✓ Stmt* *Suppose $A^{i_0} \varphi = \mu \varphi$ with $\mu \neq 0$. Then for all n,*

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0. The eigenvector relation gives $A^{w \cdot i_0^k} \varphi = \mu^k \cdot A^w \varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n. The reverse inclusion $H_n \leq K_n$ is immediate. □

Theorem 8.47 (Fixed-length vector spanning implies matrix spanning). Lean ✓ Stmt *Assume:*

1. $H_n(A, \varphi) = \mathbb{C}^D$ *(length-n word products applied to φ span all of $\mathbb{C}^D$), and*
2. *for each standard basis vector e_j, the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.*

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof Fix a matrix unit E_{ij}. By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 8.45. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. ☐

Lemma 8.48 (Blocking transfer). Lean ✓ Stmt *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Each length-n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length-nL word in the original alphabet $\{0, \dots, d-1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. ☐

Theorem 8.49 (Bounded injective blocking for left-canonical normal tensors). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then there is a positive blocking length $L \le D^4$ such that the blocked tensor $A^{[L]}$ is one-site injective.*

This is the blocked-injectivity form of the quantum Wielandt input used in the non-periodic Fundamental Theorem chain. It corresponds to the statement in [CPGSV16, Section II, line 332] that every normal tensor becomes injective after at most D^4 blockings, using the index estimate of [SPGWC10, Theorem 1]. The Lean statement includes the left-canonical/trace-preserving hypothesis because that is the canonical-form context in which the blocked-injectivity input is consumed.

Proof. ✓ Proof The proof uses the full-Kraus-rank index bound from the quantum Wielandt inequality. Since A is normal, the index is attained; the one-step Kraus rank is nonzero in the left-canonical case, so the bound $(D^2 - \operatorname{rank} S_1(A) + 1)D^2$ is at most D^4. The passage from exact word-span fullness to injectivity of the blocked tensor is Lemma 8.48. ☐

Corollary 8.50 (Uniform injective blocking for normal BNT families). Lean Lean ✓ Stmt *Let $(A_x)_{x\in X}$ be a finite normal-canonical BNT family whose blocks have positive virtual bond dimension. Then there is a positive integer L such that every blocked tensor $A_x^{[L]}$ is one-site injective. The same conclusion holds simultaneously for two finite normal-canonical BNT families whose blocks have positive virtual bond dimension.*

This is a common-blocking theorem for the scope-restricted one-copy-per-sector surface of Definition 12.7. It assumes that the normal-CF-BNT representative families have already been selected and strictly ordered; it does not reconstruct the full CPSV16 BNT sector decomposition with repeated copies inside one sector. The restriction is recorded in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`.

Proof. ✓ Proof Apply Theorem 8.49 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking. ☐

Remark 8.51 (Relation with the rank-one extraction lemma). ✓ Stmt In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 8.47 is the second step. Theorem 8.58 supplies the rank-one hypothesis in blocked form, and Theorem 8.59 then gives a blocked, fixed-length word-span saturation result.

Remark 8.52 (Use of the quantum Wielandt theorem in the MPS route). The source papers use the quantum Wielandt theorem as an input converting normality into injectivity after blocking. In the notation of this chapter, the input has two distinct conclusions. First, Theorem 8.28 gives cumulative saturation $T_{D^2}(A) = M_D(\mathbb{C})$. This alone is not the injectivity statement used in canonical form, because injectivity requires products of one fixed length. Second, Lemma 2(b) of [SPGWC10], formalized here through Theorem 8.59, supplies a length N for which $S_N(A) = M_D(\mathbb{C})$. This is the statement that matches the injectivity of Γ_N and can be transported through blocking by Lemma 8.48. Thus any later use of Wielandt in the Fundamental Theorem must specify whether it is using cumulative span, exact-length span, or the blocked version of the exact-length conclusion.

8.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor* $A^{[L]}$ and develops the "rectangular span" used to produce rank-one elements.

Theorem 8.53 (Blocking preserves normality). Lean ✓ Stmt *If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

Proof. ✓ Proof Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A)$$

by Lemma 8.45. Iterating this inclusion gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 8.54, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. □

Lemma 8.54 (Chunking: word span embeds into blocked word span). Lean ✓ Stmt $S_{nL}(A) \subseteq S_n(A^{[L]})$ *for all n, L.*

Proof. ✓ Proof Every length-nL word in the original alphabet can be split into n consecutive chunks of length L, each of which is a single blocked Kraus operator. By induction on n, using the word span product lemma (Lemma 8.45). □

Theorem 8.55 (Word eigenvector gives blocked eigenvector). Lean ✓ Stmt *If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L, then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0.*

Proof. ✓ Proof By definition, $(A^{[L]})^{j_0} = A^{w_0}$. □

Theorem 8.56 (Blocked fixed-length matrix spanning). Lean ✓ Stmt *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$),*

2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$),*

3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m.*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.
Remark. *The proof gives the explicit bound $N = ((D-1)+m+(D-1)) \cdot L$; the statement above gives only the existential conclusion.*

Proof. ✓ Proof Apply Theorem 8.47 to the blocked tensor $A^{[L]}$, which is normal by Theorem 8.53. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 8.55. Transfer back to the original tensor via Lemma 8.48. □

Theorem 8.57 (Blocked fixed-length matrix spanning from word eigenvectors)**.** Lean ✓ Stmt *Suppose A is normal and $L > 0$. If a length-L word has a nonzero eigenvector and a length-L word has a nonzero transpose eigenvector, then there exists N such that $S_N(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof The rank-one extraction theorem supplies the blocked rank-one element required by Theorem 8.56; apply the blocked assembly theorem. □

Theorem 8.58 (Rank-one extraction for blocked tensor)**.** Lean ✓ Stmt *Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0, nonzero vectors φ, ψ, nonzero scalars μ, ν, and $m \in \mathbb{N}$ such that:*

1. *$A^{\sigma_0}\varphi = \mu\varphi$,*

2. *$(A^{\tau_0})^T\psi = \nu\psi$,*

3. *$\varphi\psi^T \in S_m(A^{[N_0]})$.*

Proof. ✓ Proof Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose j_0 with $\operatorname{tr}(B^{j_0}) \neq 0$. By Theorem 8.21, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0, a nonzero vector ψ, and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem 8.53. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). □

Theorem 8.59 (Fixed-length word-span saturation via blocking)**.** Lean ✓ Stmt *If A is a normal MPS tensor with bond dimension D, then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

Proof. ✓ Proof By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 8.58 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 8.56 then gives $S_{((D-1)+m+(D-1))\cdot N_0}(A) = M_D(\mathbb{C})$. □

Remark 8.60 (Blocking argument for fixed-length spanning). ✓ Stmt Theorem 8.59 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 8.58) and then applying the fixed-length matrix-spanning theorem (Theorem 8.56). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

8.9 Primitive MPS tensors

Definition 8.61 (Primitive MPS tensor). Lean ✓ Stmt Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \qquad \rho \geq 0, \qquad \rho \neq 0, \qquad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ, then $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This is weaker than the primitive condition of [SPGWC10], which requires $\rho > 0$; see Remark 8.67. The stronger positive-definite hypothesis is imposed separately in Theorems 8.68 and 8.78.

Theorem 8.62 (Primitive overlap convergence). *Lean* *✓ Stmt* *If A is a primitive MPS tensor, then $O_{AA}(N) \to 1$.*

Proof. ✓ Proof Immediate from the self-overlap convergence under a complementary transfer-map gap (Theorem 7.98). □

8.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 8.61: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1.

Theorem 8.63 (Primitive MPS tensors have nonzero word products). *Lean* *✓ Stmt* *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length* exactly *n with $A^w \neq 0$.*

Proof. ✓ Proof The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. □

Theorem 8.64 (Iterated transfer does not vanish). *Lean* *✓ Stmt* *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. ✓ Proof By Theorem 8.63, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. □

8.10.1 From primitivity to normality

The primitivity condition of Definition 8.61 combines TP normalization, a nonzero PSD fixed point, and a complementary transfer-map gap. It is connected to normality through the following results.

Theorem 8.65 (Fixed-point uniqueness from a complementary transfer-map gap). *Lean* *✓ Stmt* *If A is a primitive MPS tensor with PSD fixed point ρ, then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ: $\sigma = \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$.*

Proof. ✓ Proof Set $\sigma' = \sigma - \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$. Then $\operatorname{tr}(\sigma') = 0$ and $(\mathcal{E}_A - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $\mathcal{E}_A - P_\rho$, contradicting the complementary transfer-map gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. □

Theorem 8.66 (Complement powers tend to zero). *Lean* *✓ Stmt* *If A is a primitive MPS tensor with PSD fixed point ρ, then $(\mathcal{E}_A - P_\rho)^n \to 0$ in operator norm.*

Proof. ✓ Proof The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the complementary transfer-map gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. $\square$

Remark 8.67 (Primitivity: transfer-map gap vs. positive definiteness). ✓ Stmt The primitivity condition used in Section 8.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ, and the complementary transfer-map gap $\rho_{\mathrm{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of "primitive" in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the transfer-map gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \mathrm{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Theorem 8.68 gives irreducibility. Together with the Burnside argument below (Section 8.10.3) and Theorem 8.84, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \mathrm{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 8.79. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies that primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

8.10.2 Primitive implies irreducible

Theorem 8.68 (Primitive tensors with positive-definite fixed point are irreducible)**.** Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV17a, Section 2.3]).*

Proof. ✓ Proof By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1} - P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1}-P)A^i P = 0$ implies $(\mathbb{1}-P)\cdot\mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).
2. *Convergence:* By Theorem 8.66, $\mathcal{E}_A^n(\sigma) \to \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\mathrm{tr}(P\rho P) > 0$, so $\frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.
4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

$\square$

8.10.3 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 9.15) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 8.69 (Algebra span)**.** Lean ✓ Stmt For an MPS tensor A, let $\mathrm{alg}(A)$ be the unital $\mathbb{C}$-subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}_{i=0}^{d-1}$:

$$\mathrm{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1\rangle \subseteq M_D(\mathbb{C}).$$

Definition 8.70 (Invariant submodule). Lean ✓ Stmt A subspace $W \le \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i.

Definition 8.71 (Irreducible action). Lean ✓ Stmt The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A-invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 8.72 (Irreducible tensor implies irreducible action). *Lean* ✓ *Stmt* *If A is irreducible as a tensor (Definition 9.15), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 8.71).*

Proof. ✓ Proof Let $W \le \mathbb{C}^D$ be an A-invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i. Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. ☐

Theorem 8.73 (Burnside's theorem for Kraus algebras). *Lean* ✓ *Stmt* *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:*

$$\mathrm{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].

Proof. ✓ Proof Transport $\mathrm{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. ☐

Remark 8.74. ✓ Stmt This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

Theorem 8.75 (Irreducible action implies eventual cumulative spanning). *Lean* ✓ *Stmt* *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 8.73, $\mathrm{alg}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some n. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \le T_1(A) \le \cdots$ stabilizes, so one can choose a uniform N with $T_N(A) = M_D(\mathbb{C})$. ☐

8.10.4 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect primitivity in Definition 8.61, formulated via a complementary transfer-map gap, directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Theorem 8.76 (Primitive tensors with positive-definite fixed point have irreducible transfer map). *Lean* ✓ *Stmt* *If A is a primitive MPS tensor with positive-definite fixed point ρ, then the transfer map $\mathcal{E}_A$ is irreducible.*

The proof uses fixed-point uniqueness (Theorem 8.65): every PSD fixed point is proportional to ρ, so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

Proof. ✓ Proof By Theorem 8.65, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. □

Theorem 8.77 (Primitive tensors with positive-definite fixed point are strongly irreducible)**.** Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.*

Proof. ✓ Proof Combine Theorem 8.76 (irreducibility) with the peripheral-spectrum primitivity from the transfer-map gap hypothesis (Theorem 8.62). □

Theorem 8.78 (Primitive MPS with positive-definite fixed point is normal)**.** Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.*

This is the key connection between primitivity in Definition 8.61, formulated via a complementary transfer-map gap, and normality; it removes the need for the aperiodicity-based argument.

Proof. ✓ Proof By Theorem 8.77, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 7.64 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank). □

8.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). An aperiodicity hypothesis upgrades cumulative spanning to a single-length word span.

Remark 8.79 (Aperiodicity condition). ✓ Stmt We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}_{i=0}^{d-1}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the primitive condition with positive-definite fixed point (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 7.64) without passing through aperiodicity.

Remark 8.80 (Counterexample: cumulative spanning does not imply normality). ✓ Stmt In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\operatorname{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n. Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n, and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \operatorname{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 8.81 (Word span monotonicity from aperiodicity)**.** Lean ✓ Stmt *If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all n.*

Proof. ✓ Proof For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 8.45. □

Theorem 8.82 (Cumulative span equals word span under aperiodicity)**.** Lean ✓ Stmt *If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all n.*

Proof. ✓ Proof By Theorem 8.81, the word spans are monotone: $S_0 \leq S_1 \leq \cdots$. The supremum $T_n = S_0 + \cdots + S_n$ therefore equals the largest term S_n. □

Theorem 8.83 (Cumulative spanning plus aperiodicity implies normality). Lean ✓ Stmt *If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. ✓ Proof By Theorem 8.82, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. □

Theorem 8.84 (Algebra span plus aperiodicity implies normality). Lean ✓ Stmt *If* $\mathrm{alg}(A) = M_D(\mathbb{C})$ *and* $\mathbb{1} \in S_1(A)$, *then A is normal.*

Proof. ✓ Proof Since $\mathrm{alg}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 8.83. □

Theorem 8.85 (Irreducible tensor plus aperiodicity implies normality). Lean ✓ Stmt *If A is an irreducible tensor, $D > 0$, and $\mathbb{1} \in S_1(A)$, then A is normal. This assembles the full chain:*

$$\textit{irreducible tensor} \xrightarrow{8.72} \textit{irreducible action} \xrightarrow{\textit{Burnside}} \mathrm{alg}(A) = M_D(\mathbb{C}) \xrightarrow{8.84} \textit{normal.}$$

This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. The Burnside step is supplied by Theorems 8.72 and 8.73.

Proof. ✓ Proof By Theorem 8.72, A acts irreducibly on $\mathbb{C}^D$. By Theorem 8.73, $\mathrm{alg}(A) = M_D(\mathbb{C})$. By Theorem 8.84 with the aperiodicity hypothesis, A is normal. □

Remark 8.86 (Normal canonical form avoids the aperiodicity step). The aperiodicity results in Section 8.11 are needed only for the block-injective argument through Definition 2.31, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 9–12 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV17a]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

8.12 Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9)

Theorem 8.87 (TP-primitive irreducible tensor is injective after blocking). Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$, irreducible tensor, and primitive transfer map. Then some positive blocking $A^{[L]}$ is one-site injective.*

Proof. ✓ Proof If $D = 1$, trace preservation gives a nonzero one-site Kraus matrix, and every nonzero scalar matrix spans $M_1(\mathbb{C})$; hence A is one-site injective, so Lemma 2.30 gives the positive witness $L = 1$. Suppose now that $D \geq 2$. The normality theorem (Theorem 9.39) gives a block-injectivity witness L. If $L = 0$, Corollary 8.9 would force $D = 1$, a contradiction. Thus $L > 0$, and the blocked-chain equivalence identifies this L-block injectivity with one-site injectivity of the blocked tensor $A^{[L]}$. □

Remark 8.88. See Theorem 8.53 for the formalized statement that blocking preserves normality.

Chapter 9

Canonical Form Reduction

This chapter and Chapter 10 carry out a single reduction of an arbitrary translation-invariant MPS tensor to a weighted sum of primitive normal blocks, following [PGVWC07], [CPGSV17a], [CPGSV21], and [Wol12, Chapter 6]. The reduction chain is

invariant-subspace split → irreducible block decomposition → zero-block separation → TP gauge → period removal →

The all-zero summand contributes only at length $N = 0$; it is recorded not as a tensor but as a bond-dimension count D_0 in the identity $D = D_0 + \sum_k D_k$, together with a positive-length equality of MPV families.

The present chapter develops the per-tensor steps. Chapter 10 continues the same chain past a common blocking length and produces the primitive-block decomposition used in the Fundamental Theorem in Chapter 13. The grouping of equal-modulus and repeated-copy sectors into bases of normal tensors is carried out in Chapter 12.

Two normalizations appear. The unital orientation $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ is used in [PGVWC07, Theorem Th:TIcanonical]; the trace-preserving (left-canonical) orientation $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ is used in the after-blocking reductions of Chapter 10. They exchange under the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, whose transfer map is the Frobenius adjoint of $\mathcal{E}_A$.

The source PGVWC07 theorem is recorded as a single headline statement (Theorem 9.1). The reduction that feeds the Fundamental Theorem does not pass through it: it goes through Theorem 9.35, Theorem 9.62, and the after-blocking reductions of Chapter 10. The principal outputs of this chapter are Theorem 9.1 (the PGVWC07 translation-invariant canonical form), Theorems 9.30 and 9.38 (TP-gauge and primitive-irreducible reductions), and Theorem 9.33 (the normal canonical form from primitive blocks).

Theorem 9.1 (Translation-invariant canonical form)**.** Lean ✓ Stmt *Let A be a translation-invariant MPS representation of bond dimension D. There is a positive scalar s and a finite family of blocks A_k, possibly empty, such that the globally rescaled tensor $s^{-1}A$ has the same MPV coefficients on every positive-length ring as*

$$\bigoplus_k \mu_k A_k^i.$$

The weights are positive and satisfy $1 \geq |\mu_k|$; if the family is nonempty, then some weight has norm 1. For each block,

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}, \qquad \sum_i (A_k^i)^\dagger \Lambda_k A_k^i = \Lambda_k,$$

where Λ_k is diagonal, positive, and full-rank. Each block has positive bond dimension. Moreover the only fixed points of $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$ are the scalar multiples of $\mathbb{1}$: for every X,

$$\mathcal{E}_{A_k}(X) = X \Longrightarrow \exists\, c \in \mathbb{C}, X = c\,\mathbb{1}.$$

The total bond dimension of the decomposition is at most D.

This is the positive-length form of [PGVWC07, Theorem Th:TIcanonical], with the global spectral-radius normalization step made explicit.

Remark 9.2 (Source proof order). The proof follows [PGVWC07, Theorem Th:TIcanonical] in order: normalize the spectral radius and use a full-rank positive fixed point to obtain the unital gauge; derive the invariant support projection from a singular positive fixed point by the positivity contradiction; split the finite-ring trace into the supported and orthogonal summands; iterate, using a non-scalar fixed point to split further until each block has only scalar fixed points; and finally repeat the fixed-point argument for the dual map and diagonalize the resulting full-rank positive fixed point by a unitary gauge. The primitive-block results below are used only after their hypotheses have been obtained from this argument.

Remark 9.3 (CPSV blocked canonical form (source statement)). The blocked canonical-form reduction of [CPGSV17a, Section II.C] and [CPGSV21, Section IV.A] states that an arbitrary translation-invariant MPS tensor becomes, after blocking by some period $p \geq 1$, the direct sum of an all-zero summand and a weighted family of normal tensors $\bigoplus_k \mu_k B_k$ with $|\mu_k| \leq 1$ and at least one $|\mu_k| = 1$. The closest proved results here are Theorem 10.12 and Theorem 9.51; the source upper normalization $|\mu_k| \leq 1$ with a unit-modulus witness is the one remaining input, recorded in Remark 9.53.

9.1 Normal tensors, canonical forms, and bases of normal tensors

This section records the CPSV definitions of *normal tensor*, *canonical form*, and *basis of normal tensors* from [CPGSV17a]. Write

$$\sigma_\partial(\mathcal{E}) = \{\lambda \in \sigma(\mathcal{E}) : |\lambda| = r(\mathcal{E})\}$$

for the peripheral spectrum of a transfer map $\mathcal{E}$, where $r(\mathcal{E})$ is its spectral radius. The stronger predicates used later (Definition 9.24 and the canonical-form BNT predicate of Chapter 12) add left-canonical normalization, strict modulus ordering, and one copy per sector; these are subcases of the CPSV definitions, treated by the bases of normal tensors of Chapter 12.

Definition 9.4 (CPSV normal tensor (NT))**.** Lean ✓ Stmt A tensor A of physical dimension d and bond dimension D is a *normal tensor* if, after the spectral-radius normalization of [CPGSV17a], (i) A admits no nontrivial invariant orthogonal projection and (ii) the associated CPM $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ has peripheral spectrum $\sigma_\partial(\mathcal{E}_A) = \{1\}$.

Remark 9.5. Condition (i) cannot be dropped. A reducible tensor may have peripheral spectrum $\{1\}$ while still admitting a nontrivial invariant projection, so condition (ii) alone does not characterize normality.

The CPSV *canonical form* (CF) of a tensor A is the normal-block decomposition

$$A^i \cong \bigoplus_{k=1}^{r} \mu_k A_k^i$$

with weights normalized by $|\mu_k| \leq 1$ and with at least one $|\mu_k| = 1$; see [CPGSV17a, Section II.C]. In the statements below the structural part is supplied by the stronger normal canonical form of Definition 9.24, and the global weight normalization is invoked as a separate source hypothesis where needed.

Definition 9.6 (CPSV basis of normal tensors (BNT)). Lean ✓ Stmt A finite family $\{A_j\}_{j=1}^g$ of normal tensors, with bond dimension allowed to depend on j, is a *basis of normal tensors* for A if:

1. each A_j is normal in the sense of Definition 9.4;
2. for every system length N there exist coefficients $c_j^{(N)} \in \mathbb{C}$ with
$$|V^{(N)}(A)\rangle = \sum_{j=1}^{g} c_j^{(N)} |V^{(N)}(A_j)\rangle;$$
3. there exists N_0 such that for all $N > N_0$ the family $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ is linearly independent.

Remark 9.7 (Relation to the strong predicates). A tensor A which is both irreducible (Definition 9.15) and has primitive transfer map is normal in the sense of Definition 9.4. The reverse implications from the stronger predicates of Definition 9.24 and of Chapter 12 require the fundamental-theorem step or the algebraic-to-spectral normality equivalence, established later.

9.2 Transfer map normalization

Remark 9.8 (Scalar normalization conventions). The reductions use four scalar facts: scaling preserves injectivity; the transfer map under a scalar ζ rescales by $|\zeta|^2$; unit-modulus phase rescaling preserves $\sum_i (A^i)^\dagger A^i = \mathbb{1}$; and the MPV of a weighted block-diagonal tensor factorizes as $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$ with $\eta_k := \mu_k/|\mu_k|$.

Left-canonical normalization is required before block-by-block comparison. Without it the implication $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k,\ \mathcal{V}(A_k) = \mathcal{V}(B_k)$ already fails for two scalar blocks: for $\mu = (1, 2)$, $A = (1, 3/2)$, $B = (3, 1/2)$ the two direct sums generate identical MPV families but the first blocks differ at $N = 1$. The strengthened canonical-form hypotheses remove this ambiguity by combining left-canonical normalization with ordered nonzero weight moduli.

9.3 Invariant subspace decomposition

The block decomposition rests on one principle: a positive semidefinite fixed point of $\mathcal{E}_A$ that is not full rank has a proper invariant support, which block-diagonalizes the Kraus operators without changing the MPV family.

Theorem 9.9 (Unitary conjugation preserves the MPV family). Lean ✓ Stmt *If U is a unitary matrix, then A and $U^\dagger AU$ generate the same MPV family.*

Proof. ✓ Proof By cyclicity of the trace, $\operatorname{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \operatorname{tr}(A^{i_1} \cdots A^{i_N})$. □

Definition 9.10 (Support projection). Lean ✓ Stmt For a positive semidefinite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ. Via the spectral decomposition $\rho = U \operatorname{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$,
$$P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger,$$

where $\mathbf{1}_{\lambda_j>0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 9.11 (Invariant projection predicate). Lean ✓ Stmt An MPS tensor A *has an invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1}-P)A^iP = 0$ for all i. A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 9.15).

Theorem 9.12 (PSD fixed point gives invariant projection). Lean ✓ Stmt *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be its support projection. Then $(\mathbb{1} - P)A^iP = 0$ for all i: the range of P is invariant under every A^i.*

Proof. ✓ Proof From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i\rho(A^i)^\dagger$, one gets $(\mathbb{1} - P)A^i\rho(A^i)^\dagger(\mathbb{1}-P) = 0$ for each i. Since ρ is supported on the range of P, this forces $(\mathbb{1}-P)A^iP = 0$.

This is the finite-dimensional case of [Wol12, Proposition 6.10]: the support projection of any stationary state is sub-harmonic. The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Theorem 6.2]; see also [CPGSV21, Section IV.A.1]. □

Theorem 9.13 (Two-block decomposition). Lean ✓ Stmt *If $\mathcal{E}_A$ has a nonzero PSD fixed point ρ that is not positive definite, then there exist MPS tensors A_1, A_2 of smaller bond dimensions with $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.*

Proof. ✓ Proof The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not positive definite). Theorem 9.12 gives $(\mathbb{1} - P)A^iP = 0$, so a unitary change of basis block-diagonalizing P puts A^i in upper-triangular block form with diagonal blocks B^i_{11}, B^i_{22}. Theorem 9.9 preserves the MPV under the conjugation, and the trace of an upper-triangular product equals the sum of the diagonal-block traces, so deleting the off-diagonal blocks preserves all MPV coefficients. □

Theorem 9.14 (Non-scalar fixed point gives a further split). Lean Lean ✓ Stmt *Suppose A is unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If X is a Hermitian fixed point of $\mathcal{E}_A$ which is not a scalar multiple of $\mathbb{1}$, then there is a nonzero PSD fixed point ρ which is not positive definite, and consequently A admits a two-block MPV-equivalent decomposition with both block dimensions strictly smaller than D. This is the fixed-point shift and further decomposition step in [PGVWC07, Theorem Th:TIcanonical].*

Proof. ✓ Proof Let $\lambda_{\max}$ be the largest eigenvalue of X. Then $\rho := \lambda_{\max}\mathbb{1} - X$ is positive semidefinite, singular, and nonzero (since X is not scalar), and unitality with the fixed-point equation for X gives $\mathcal{E}_A(\rho) = \rho$. Apply Theorem 9.13. The non-scalar hypothesis is needed because scalar multiples of $\mathbb{1}$ are fixed by every unital map and yield no nontrivial support projection. □

9.4 Iterated reduction

Definition 9.15 (Irreducible tensor). Lean ✓ Stmt An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 9.16 (Irreducible block decomposition). Lean ✓ Stmt *Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ with each A_k irreducible and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$.*

Proof. ✓ Proof By strong induction on D. If A is irreducible, take $r = 1$. Otherwise a nontrivial invariant projection P gives the upper-triangular splitting of Theorem 9.13; deleting the off-diagonal part preserves the MPV family and yields two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block.

Both [PGVWC07, Theorem 4] and [CPGSV21, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting. □

Theorem 9.17 (Scalar fixed points in an irreducible unital block). Lean ✓ Stmt *If A is irreducible and unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and $\mathcal{E}_A(X) = X$, then X is a scalar multiple of $\mathbb{1}$. This is the final fixed-point assertion in the proof of [PGVWC07, Theorem Th:TIcanonical].*

Proof. ✓ Proof Tensor irreducibility gives irreducibility of the completely positive transfer map. By Wolf's Theorem 6.6 for irreducible unital Kraus maps, every fixed point is scalar. □

Theorem 9.18 (Unital gauge from a full-rank Perron–Frobenius eigenvector). Lean Lean Lean ✓ Stmt *Let ρ be positive definite with $\mathcal{E}_A(\rho) = r\rho$ for some $r > 0$. For $B^i := r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$,*

$$\sum_i B^i(B^i)^\dagger = \mathbb{1}.$$

If $r = 1$, then the unscaled gauge $B^i := \rho^{-1/2}A^i\rho^{1/2}$ is unital and generates the same MPV family as A. This is the spectral-radius normalization and full-rank fixed-point gauge of [PGVWC07, Theorem Th:TIcanonical].

Proof. ✓ Proof The positive square root of ρ is invertible. Substituting $B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$ into $\sum_i B^i(B^i)^\dagger$ and using $\mathcal{E}_A(\rho) = r\rho$ gives the identity. For $r = 1$ the change from A to B is a similarity, so cyclicity of the trace gives equality of all MPV coefficients. □

Theorem 9.19 (Unitary diagonal positive fixed point in TP gauge). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1}, \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. ✓ Proof The TP hypothesis makes $\mathcal{E}_A$ a channel with a nonzero PSD fixed point. Irreducibility makes $\mathcal{E}_A$ an irreducible CP map, and Theorem 6.4 upgrades the fixed point to positive definite. The spectral theorem diagonalizes it by a unitary conjugation, which preserves the TP normalization. □

The passage from irreducible tensors to full algebra spanning uses Burnside's theorem: an irreducible subalgebra of $M_D(\mathbb{C})$ is all of $M_D(\mathbb{C})$. It is developed alongside the Wielandt material in Chapter 8; see Theorem 8.75.

9.5 Canonical form from primitivity

The self-overlap convergence $O_{A_k A_k}(N) \to 1$ for each block is a derived consequence of block normality, obtained from the transfer-map gap criterion of Chapter 7; it is not part of the CPSV canonical-form definition. The result of this section is Theorem 9.22: blocking an irreducible TP block by the least common multiple of its peripheral periods makes the transfer map primitive, removing periodicity.

Remark 9.20 (Two ways to obtain the canonical-form predicate from normal blocks). Given injective, left-canonical, strictly-ordered nonzero-weight blocks (μ_k, A_k), the CPSV canonical-form conditions of [CPGSV17a, Section II.C] follow from either a positive-definite primitive fixed point or the peripheral-spectrum condition $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$; in both cases the overlap clause $O_{A_k A_k}(N) \to 1$ comes from the complementary transfer-map gap of Chapter 7. These statements assume the blocks are already in normal form; they are not the arbitrary-input existence theorem of [PGVWC07, Theorem Th:TIcanonical].

Remark 9.21. Consequently the overlap-convergence hypothesis is redundant once peripheral-spectrum primitivity is known: the argument factors through the complementary transfer-map gap formulation of Chapter 7. In [CPGSV21] primitivity is obtained from irreducibility plus periodicity removal by blocking to period one; Theorem 9.22 is exactly that periodicity-removal step under irreducibility and TP normalization.

Theorem 9.22 (Blocking yields a peripheral-spectrum primitive transfer map)**.** Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$.*

Proof. ✓ Proof Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, which is unital with irreducible Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint of K, so Theorem 4.46 shows every peripheral eigenvalue is a root of unity. Since $\sigma_\partial(\mathcal{E}_A)$ is finite (Theorem 4.29), Lemma 4.33 gives a common exponent p with $\mu^p = 1$ for all $\mu \in \sigma_\partial(\mathcal{E}_A)$, and Lemma 4.34 gives $\sigma_\partial(\mathcal{E}_A^p) = \{1\}$. Blocking by p takes the pth power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. □

Remark 9.23. Theorem 9.22 is the periodicity-removal-by-blocking step of [CPGSV17a, Appendix A] for an irreducible block already in TP gauge.

9.6 Normal canonical form

Definition 9.24 (Normal canonical form conditions)**.** Lean ✓ Stmt Scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfy the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each $\mathcal{E}_{A_k}$ is primitive (equivalently, $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$),
4. $|\mu_1| \geq |\mu_2| \geq \cdots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here "normal" is the spectral sense of [CPGSV17a], which by [SPGWC10, Proposition 3] is equivalent to the eventual block-injectivity formulation of Definition 2.31.

Remark 9.25. The comparison theorems built on Definition 9.24 additionally assume pairwise distinct moduli and one copy per sector. This is the distinct-modulus, one-copy subcase: equal-modulus sectors are first grouped by the common value of $|\mu_k|$, and repeated copies are then treated by the phase-class BNT sector construction of Chapter 12 (Definition 12.29) and the collapsed sector theorem of Chapter 13 (Theorem 13.30). This caveat is stated once here and referenced where the subcase recurs.

Theorem 9.26 (Self-overlap is derived in normal canonical form)**.** Lean ✓ Stmt *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \to 1$ for every block k.*

Proof. ✓ Proof Peripheral-spectrum primitivity, together with irreducibility and TP normalization, yields the complementary transfer-map gap. Hence, for every block k,

$$\lim_{N\to\infty} O_{A_k A_k}(N) = 1.$$

□

Remark 9.27. See Theorem 7.28 for the modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

Lemma 9.28 (Mixed-transfer spectral radius from unit-modulus overlap). Lean ✓ Stmt *Let A and B have the same bond dimension. If $\|\langle V^{(N)}(A)|V^{(N)}(B)\rangle\| \to 1$, then the mixed transfer map has spectral radius at least 1.*

Proof. ✓ Proof If the mixed transfer spectral radius were < 1, the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace; via $\mathrm{tr}(F_{AB}^N) = \langle V^{(N)}(A)|V^{(N)}(B)\rangle$ the overlap would tend to 0, contradicting the hypothesis. □

Theorem 9.29 (Gauge-phase equivalence from unit-modulus overlap). Lean ✓ Stmt *Let A and B be irreducible MPS tensors of the same bond dimension with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If $\|\langle V^{(N)}(A)|V^{(N)}(B)\rangle\| \to 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof By Lemma 9.28 the mixed transfer spectral radius is at least 1, and Theorem 7.28 then gives gauge-phase equivalence. □

9.7 Steps toward canonical form existence

Combining Theorem 9.30 with the Perron–Frobenius gauge of Chapter 6 and the irreducible-block scalar-fixed-point statement Theorem 9.17 yields the PGVWC07 unital orientation and a diagonal positive-definite dual fixed point on each irreducible block; the full PGVWC07 statement is Theorem 9.1.

Theorem 9.30 (Left-canonical normalization for irreducible TP blocks). Lean ✓ Stmt *Let A be an irreducible MPS tensor in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that:*

1. *$B^i := U^\dagger A^i U$ is TP,*
2. *$\mathcal{E}_B(\Lambda) = \Lambda$,*
3. *Λ is diagonal and positive definite,*
4. *A and B generate the same MPV family.*

This is the left-canonical normalization step (Canonical Form II) of [CPGSV17a, Appendix A]. The unitary conjugation is a second step inside TP gauge, performed only after TP normalization.

Proof. ✓ Proof Theorem 9.19 provides the unitary and the diagonal PD fixed point, and unitary conjugation preserves trace-preservation and the MPV family (Theorem 9.9). □

Theorem 9.31 (Dual diagonalization for irreducible unital blocks). Lean ✓ Stmt *Let A be an irreducible MPS tensor in unital gauge $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}, \qquad \sum_i (B^i)^\dagger \Lambda B^i = \Lambda,$$

and A, B generate the same MPV family. This is the dual-map fixed-point and unitary-diagonalization step of [PGVWC07, Theorem Th:TIcanonical].

Proof. ✓ Proof Apply Theorem 9.30 to the conjugate-transposed family $i \mapsto (A^i)^\dagger$: unitality of A is trace preservation for this family, whose transfer map is the dual of $\mathcal{E}_A$. Translating back gives the displayed equations. □

Lemma 9.32 (Common scalar rescaling of block weights). Lean ✓ Stmt *Multiplying every block weight in a block-diagonal MPS tensor by a scalar c multiplies every length-N MPV coefficient by c^N.*

Proof. ✓ Proof Expand the block-diagonal MPV coefficient as a finite sum over blocks and factor c^N out of the sum. □

Theorem 9.33 (Normal canonical form from a primitive block decomposition). Lean ✓ Stmt *Suppose A generates the same MPV family as $\bigoplus_k \mu_k A_k$, and that each A_k is irreducible, satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, and has primitive transfer map, the moduli $|\mu_k|$ are pairwise distinct, all $\mu_k \neq 0$, and all bond dimensions are positive. Then there exist a blocking length $p > 0$, weights (ν_k), and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions. The pairwise-distinct-modulus hypothesis is the distinct-modulus, one-copy subcase discussed in the remark after Definition 9.24.*

Proof. ✓ Proof The given blocks are already primitive, so take $p = 1$. Distinctness of the moduli gives a reordering making them strictly decreasing; set $\nu_k := \mu_k$ and $B_k := A_k$ in that order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are preserved, so the reordered family satisfies Definition 9.24 and generates the same MPV family as A. □

Remark 9.34 (Scope of Theorem 9.33)*.* Theorem 9.33 assumes all six block hypotheses in advance and produces the normal canonical form by reordering. It is not the arbitrary-input existence theorem of [CPGSV17a, Section II.C]; that primitive-block reduction, including the zero-tail count, is Theorem 10.12 of Chapter 10.

9.8 TP-gauge reduction for arbitrary tensors

Theorem 9.35 (TP-gauge reduction for arbitrary tensors with zero-block separation). Lean ✓ Stmt *For any MPS tensor A, there exist a zero-block bond dimension D_0 and nontrivial blocks $(B_k)_{k=1}^r$ with nonzero weights $(\mu_k)_{k=1}^r$, each block irreducible, left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), and of positive bond dimension, such that for every $N \geq 1$ and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r \mu_k B_k)_\sigma,$$

and the zero-block contribution is recorded by the length-zero identity

$$D = D_0 + \sum_{k=1}^{r} D_k.$$

This is the furthest unconditional reduction available before periodicity removal.

Proof. ✓ Proof Theorem 10.2 supplies D_0 and the nonzero irreducible blocks $(C_k)_{k=1}^r$. For every $N \geq 1$ and every σ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r C_k)_\sigma.$$

The blockwise TP-gauge theorem then replaces each C_k by a left-canonical block B_k with weight μ_k. Again for every $N \geq 1$ and every σ,

$$V^{(N)}(\bigoplus_{k=1}^{r} C_k)_\sigma = V^{(N)}(\bigoplus_{k=1}^{r} \mu_k B_k)_\sigma.$$

The length-zero identity $D = D_0 + \sum_{k=1}^{r} D_k$ is preserved. □

9.9 Blocking and primitivity reduction

Blocking by an integer p carries MPV equalities, weights, transfer maps, and identifications of the blocked physical alphabet. The consequence needed later is the existence of a common blocking period under which every transfer map in a finite family becomes primitive.

Theorem 9.36 (Common blocking period for a block family). Lean ✓ Stmt *Let $(B_k)_{k=1}^{r}$ be a finite family of positive-bond-dimension MPS tensors, each admitting a blocking period p_k making its transfer map primitive. Then $p = \mathrm{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k.*

Proof. ✓ Proof Each p_k divides p, so primitivity of $\mathcal{E}_{B_k^{[p_k]}}$ carries over to $\mathcal{E}_{B_k^{[p]}}$ via the divisibility identity for transfer-map powers. □

9.10 Period removal (cyclic-sector decomposition)

In the non-periodic proof of [CPGSV17a] the required step is periodicity removal by blocking: after blocking each irreducible TP block by the least common multiple of its peripheral periods, the result has no nontrivial p-periodic vectors [CPGSV17a, Section 2.3]. The cyclic-sector decomposition comes from the peripheral spectrum of the adjoint transfer map [Wol12, Theorem 6.6].

Theorem 9.37 (Period removal by blocking). Lean *Let A be a left-canonical irreducible tensor of period m. Then $A^{[m]}$ admits:*

1. *left-canonical sector tensors $(C_k)_{k=1}^{m}$,*
2. *orthogonal projections $(P_k)_{k=1}^{m}$ summing to $\mathbb{1}$ with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,*
3. *compression linear equivalences $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,*
4. *a direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^{m} C_k$.*

This is the period-removal step of [CPGSV17a, Section 2.3]: after blocking by m, the result admits a unit-weight decomposition with no nontrivial p-periodic vectors, the cyclic projections P_k coming from the peripheral spectrum of the adjoint transfer map ([Wol12, Theorem 6.6]).

Proof. The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating the cyclic relation m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and the blocked-transfer identity $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$ identifies these with the adjoint transfer map of $A^{[m]}$. Apply the block decomposition from adjoint-fixed projections. □

Theorem 9.38 (Period removal with primitive irreducible sectors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period $m \geq 1$ and a unit-weight decomposition of $A^{[m]}$ into sector blocks $(C_k)_{k=1}^m$, each of which is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension. Equivalently, the cyclic-sector decomposition of $A^{[m]}$ already satisfies the trace-preservation, primitivity, and irreducibility hypotheses needed for the common-blocking step (Lemma 9.63).*

Proof. ✓ Proof Left-canonicality makes $K_i := (A^i)^\dagger$ unital, irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point; the cyclic peripheral-spectrum theorem and Theorem 9.37 then produce the cyclic-sector decomposition. Keep the projections and compression equivalences, and apply Theorem 10.11 to the sector blocks. □

9.11 Reduction to primitive blocks

The after-blocking statements that take an arbitrary tensor or a TP-primitive-irreducible block to a primitive block decomposition appear in Chapter 10, and the block-injectivity statement in Chapter 8. The single statement recorded here is the normality of a TP-primitive irreducible block.

Theorem 9.39 (TP-primitive irreducible tensor is normal). Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then A is normal.*

Proof. ✓ Proof Peripheral primitivity and irreducibility yield the complementary transfer-map gap. The PSD fixed point is positive definite by Theorem 6.4, so Theorem 8.78 applies. □

9.11.1 Basis of Normal Tensors

This subsection records the single-copy case and the elementary regrouping by common weight modulus. It is not the general equal-modulus comparison: that is the SectorBNT material of Chapter 12 and its use in the equal-MPV theorem of Chapter 13, as discussed in the remark after Definition 9.24. Equal-modulus and repeated-copy sectors do occur: for $A = C \oplus (-C)$ the length-N MPV coefficients are $(1 + (-1)^N)\, V^{(N)}(C)_\sigma$, which is not a single scalar power.

Definition 9.40 (Single-copy sector decomposition). Lean ✓ Stmt Given nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(B_k)_{k=1}^r$, the *single-copy sector decomposition* has r sectors, one basis block B_k per sector, multiplicity 1, and sector weight μ_k. As a finite sum,

$$\sum_{k=1}^{r} \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k B_k)_\sigma.$$

Lemma 9.41 (Single-copy sector decomposition preserves the represented family). Lean ✓ Stmt *The single-copy sector decomposition of $(\mu_k, B_k)_k$ represents the same MPS family as $\bigoplus_k \mu_k B_k$.*

Proof. ✓ Proof Expanding the sector decomposition gives $\sum_k \mu_k^N V^{(N)}(B_k)_\sigma$, the finite block-sum expression for $\bigoplus_k \mu_k B_k$. □

The sorted distinct-modulus sector statement and the regrouping of blocks by common weight modulus are the subcase discussed in the remark after Definition 9.24; they are subsumed by the phase-class BNT sector construction of Chapter 13 and the development of Chapter 12, and are not recorded separately here.

9.12 Scope and BNT input reductions

Remark 9.42 (Canonical-form reduction before the equal-case comparison). The reduction separates the all-zero summand, which contributes only to the length-zero trace, from the nonzero part; decomposes the nonzero part into irreducible blocks; places each block in TP gauge; removes the period of each irreducible TP block by blocking; and restricts to the cyclic spectral sectors [CPGSV17a, Section 2.3 and Appendix A]. The invariant-subspace decomposition, TP gauges, and one-block period-removal steps are recorded above; Chapter 10 contains the common-blocking construction, the zero-tail identities, and the common-sector decompositions over one blocked alphabet. The equal-case comparison is the subcase discussed in the remark after Definition 9.24.

Remark 9.43 (Scope relative to [PGVWC07], Section III). The reduction follows [CPGSV17a] and [CPGSV21], using site-independent tensors, transfer-map primitivity, and CPM fixed-point normalization. The period-removal step of [PGVWC07, Theorem Th:periodic] is Theorem 9.37; the explicit decomposition of the state vector into p-periodic summands lies outside this chapter.

9.12.1 Building a BNT from normal blocks with normalized weights

The BNT predicate of [CPGSV17a] (Definition 9.6) is strengthened in Chapter 12 by multiplicity, span, and weight-norm conditions. The results below separate the after-blocking preparation of suitable blocks from the normalized-weight BNT construction.

Lemma 9.44 (Common injective blocking for primitive irreducible families). Lean ✓ Stmt *Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension MPS tensors, each trace-preserving, primitive, and tensor-irreducible. Then there is a single $L \geq 1$ such that each $B_k^{[L]}$ is one-site injective, and this blocking preserves trace preservation, primitivity, and irreducibility for every block.*

Proof. ✓ Proof For each block, Theorem 8.87 gives a positive length at which it becomes one-site injective; the product of these lengths is a common length. Injectivity persists under positive further blocking, and Theorem 10.5 preserves the other hypotheses. □

Theorem 9.45 (Prepared BNT blocks after blocking). Lean ✓ Stmt *Let A be an MPS tensor. Then there is a blocking length $p \geq 1$, nonzero weights μ_k, and blocks B_k over the p-blocked alphabet such that:*

1. *each block has positive bond dimension;*
2. *each block is trace-preserving, primitive, tensor-irreducible, and one-site injective;*
3. *$A^{[p]}$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family at every positive length.*

No weight-modulus normalization is asserted at this stage.

Proof. ✓ Proof Apply the common primitive irreducible block decomposition to (A, A), giving a positive blocking length and a nonzero weighted family of trace-preserving primitive irreducible blocks with the positive-length MPV identity. Lemma 9.44 supplies one common further blocking making every block one-site injective while preserving the other hypotheses; finally identify the iterated blocked alphabet with the direct one. □

Theorem 9.46 (BNT from normal blocks with normalized weights). Lean ✓ Stmt *Let $r \in \mathbb{N}$, and for each $k = 1, \dots, r$ let $D_k \in \mathbb{N}$, $\mu_k \in \mathbb{C}$, and let B_k be an MPS tensor of physical dimension d and bond dimension D_k. Assume:*

*1. **positive bond dimensions:** $D_k > 0$ for all k,*

*2. **TP normalization:** $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$,*

*3. **primitivity:** $\mathcal{E}_{B_k}$ is primitive for each k,*

*4. **irreducibility:** each B_k is irreducible,*

*5. **injectivity:** each B_k is injective,*

*6. **nonzero weights:** $\mu_k \neq 0$ for all k,*

*7. **bounded weights:** $|\mu_k| \leq 1$ for all k,*

*8. **global unit witness:** $|\mu_k| = 1$ for some k.*

Then there exists a sector decomposition P such that P and $\bigoplus_{k=1}^r \mu_k B_k$ generate the same MPV family at every length, and P satisfies the BNT conditions of Definition 9.6 together with the multiplicity, span, and weight-bound conditions of Chapter 12.

Proof. ✓ Proof The phase-class quotient groups gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars. The TP, primitive, and irreducible hypotheses are those under which the phase-class construction applies, while injectivity gives the eventual linear-independence condition. The resulting sector decomposition inherits the span and multiplicity conditions, and the bounded-weight and global-unit hypotheses give the weight-norm conditions. □

Lemma 9.47 (Total dimension of a dimension-preserving phase-class quotient)**.** Lean ✓ Stmt *In the phase-class quotient of the prepared-block BNT supplier, suppose every original copy weight is nonzero and every block in a phase class has the bond dimension of its representative. Then the total bond dimension of the collapsed sector decomposition equals the sum of the bond dimensions of the original prepared blocks.*

Proof. ✓ Proof Sum the representative bond dimensions over the copies in each phase class. The same-dimension hypothesis replaces each representative dimension by the enumerated original block dimension, and the phase-class regrouping identity recovers the original direct-sum dimension. □

Lemma 9.48 (Prepared phase-equivalent blocks have the same bond dimension)**.** Lean ✓ Stmt *Let X and Y be trace-preserving, primitive, irreducible MPS blocks of positive bond dimension. If their MPV families differ by a nonzero scalar power, then their bond dimensions are equal.*

Proof. ✓ Proof Self-overlap normalization forces the scalar relating the two MPV families to have unit modulus, so the rectangular overlap has norm tending to 1. If the bond dimensions differed, the rectangular overlap-decay theorem would force that overlap to tend to 0. □

Lemma 9.49 (Prepared phase classes preserve bond dimension)**.** Lean ✓ Stmt *In a prepared family of positive-bond-dimension trace-preserving primitive irreducible blocks, every block in an MPV phase class has the bond dimension of its representative.*

Proof. ✓ Proof Apply Lemma 9.48 to each representative and each member of its phase class. □

Lemma 9.50 (Prepared collapsed BNT total dimension)**.** Lean ✓ Stmt *For prepared trace-preserving primitive irreducible blocks of positive bond dimension with all copy weights nonzero, the phase-class quotient preserves the total bond dimension: the collapsed sector decomposition has total bond dimension equal to the sum of the original block dimensions.*

Proof. ✓ Proof Lemma 9.49 supplies the same-dimension hypothesis in Lemma 9.47. ☐

Theorem 9.51 (Conditional BNT form after blocking for an arbitrary tensor)**.** Lean ✓ Stmt *Let A be an MPS tensor. Then there exists a blocking length $p \geq 1$, nonzero weights μ_k, and blocks B_k over the blocked alphabet, such that:*

1. *every block has positive bond dimension;*
2. *every block is TP-normalized, primitive, irreducible, and injective;*
3. *$A^{[p]}$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family at every positive length.*

Moreover, under $|\mu_k| \leq 1$ for every k and $|\mu_k| = 1$ for at least one k, there exists a sector decomposition P over the blocked alphabet in the strengthened BNT canonical form of Chapter 12 such that $A^{[p]}$ and P generate the same MPV family at every positive length.

Proof. ✓ Proof The after-blocking reduction supplies the TP-normalized primitive irreducible injective blocks and the positive-length MPV identity. The normalized-weight hypotheses are the choice that [CPGSV17a, Section II.C] makes; Theorem 9.46 then produces the sector decomposition, following the BNT selection of [CPGSV17a, Appendix A]. The missing rescaling step is recorded in Remark 9.53. ☐

Theorem 9.52 (Conditional BNT form after blocking, completed at length zero)**.** Lean ✓ Stmt *In the setting of Theorem 9.51, the same prepared blocks may be chosen so that the positive-length MPV identity upgrades to all lengths whenever the total bond dimension of P equals the bond dimension of the blocked input tensor.*

Proof. ✓ Proof The positive-length statement is Theorem 9.51. At length zero the MPV coefficient is the bond dimension, so equality of the two total bond dimensions supplies the missing length-zero coefficient, and the positive-length identity gives all remaining lengths. ☐

Remark 9.53 (Arbitrary-input reduction)*.* Theorem 9.51 gives the arbitrary-input reduction, conditional on the two weight-normalization hypotheses displayed there. The preparatory part, Theorem 9.45, produces the TP-normalized primitive irreducible injective blocks and the positive-length MPV identity but not the weight normalization. Since the all-zero summand contributes only at length zero, the nonzero primitive blocks identify the MPV family for $N \geq 1$. The proof that the prepared weights can always be rescaled to satisfy $|\mu_k| \leq 1$ with a unit-modulus witness is the one remaining step, recorded in `docs/paper-gaps/cpsv16_cf_normalization_and_proportional_comparison.`
the length-zero trace identity and the fully unblocked canonical form remain part of the reduction in Chapter 10.

9.12.2 Auxiliary lemmas

The lemmas below collect the structural facts about transfer maps, blocking, and cyclic-sector reblocking used in the after-blocking constructions of Chapter 10 and Chapter 13 and in the periodic extension of Chapter 15.

Lemma 9.54 (Canonical form from peripheral primitivity)**.** *Let $(\mu_k, A_k)_{k=1}^r$ be nonzero weights and injective left-canonical blocks with positive bond dimensions, strictly decreasing moduli $|\mu_1| > \cdots > |\mu_r| > 0$, and $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$. Then $(\mu_k, A_k)_{k=1}^r$ satisfies the canonical-form conditions of Definition 11.12, and $O_{A_k A_k}(N) \to 1$ for every k. The strict ordering is the distinct-modulus subcase discussed in the remark after Definition 9.24.*

Lemma 9.55 (Adjoint primitivity equivalence). Lean ✓ Stmt *Let $\mathcal{E}$ be a finite-dimensional completely positive map. Then $\mathcal{E}$ is primitive if and only if its Frobenius adjoint $\mathcal{E}^\dagger$ is primitive; equivalently, $\sigma_\partial(\mathcal{E}) = \{1\}$ iff $\sigma_\partial(\mathcal{E}^\dagger) = \{1\}$, since the peripheral spectrum of $\mathcal{E}^\dagger$ is the complex conjugate of that of $\mathcal{E}$.*

Lemma 9.56 (Primitivity under blocking by a multiple). Lean ✓ Stmt *Let A have positive bond dimension and let $p_0, p \geq 1$ with $p_0 \mid p$. If $\mathcal{E}_{A^{[p_0]}}$ is primitive, then $\mathcal{E}_{A^{[p]}}$ is primitive.*

Lemma 9.57 (Blocking the weighted block sum). Lean ✓ Stmt *For nonzero weights $(\mu_k)_{k=1}^r$, blocks $(B_k)_{k=1}^r$, and any $p \geq 1$,*

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p \, B_k^{[p]},$$

where $\sim$ denotes generation of the same MPV family at every length.

Lemma 9.58 (Positive-length equality survives blocking). Lean ✓ Stmt *If two tensors have the same MPV coefficients at every positive length, then so do their blocked tensors after any positive blocking length.*

Lemma 9.59 (Positive-length blocking of a weighted block sum). Lean ✓ Stmt *If A and $\bigoplus_k \mu_k B_k$ have the same MPV coefficients at every positive length, then after any positive blocking length p, $A^{[p]}$ and $\bigoplus_k \mu_k^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma 9.60 (Positive-length equality of powered block sums). Lean ✓ Stmt *If $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ have the same MPV coefficients at every positive length, then after any positive common blocking length p, $\bigoplus_j (\mu_j^A)^p A_j^{[p]}$ and $\bigoplus_k (\mu_k^B)^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma 9.61 (Nonzero weights survive blocking). Lean ✓ Stmt *If $\mu \in \mathbb{C}$ is nonzero and $p \geq 1$, then $\mu^p \neq 0$; blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.*

Lemma 9.62 (Primitive cyclic sectors of irreducible TP tensors). Lean ✓ Stmt *Let A be a left-canonical irreducible MPS tensor with $D > 0$. Then there exist a period $m \geq 1$ and left-canonical blocks $(C_k)_{k=1}^m$ with primitive transfer maps, tensor-irreducible structure, and positive bond dimensions such that $A^{[m]}$ generates the same MPV family as $\bigoplus_{k=1}^m C_k$ (unit weights).*

Lemma 9.63 (Common reblocking of cyclic-sector families). Lean ✓ Stmt *Given a finite family $(B_k)_{k=1}^r$ of left-canonical irreducible TP tensors with positive bond dimensions and per-block period-removal lengths $(m_k)_{k=1}^r$, there is a common blocking length p such that each $B_k^{[p]}$ admits a unit-weight direct-sum decomposition over a single common blocked alphabet $d^{[p]}$ into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.*

Lemma 9.64 (Nonzero unit weight of the common reblocked family). Lean ✓ Stmt *The unit weight on each sector of the common reblocked cyclic-sector family of Lemma 9.63 is nonzero.*

Lemma 9.65 (Sector irreducibility for the adjoint blocked map). Lean ✓ Stmt *Let A be a left-canonical irreducible MPS tensor with $D > 0$, let m be the period of its cyclic-sector decomposition, and let $(P_u)_{u=1}^m$ be the cyclic projections of the adjoint transfer map with $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ and $\sum_u P_u = \mathbb{1}$. Then the restriction of $(\mathcal{E}_{A^\dagger})^m$ to each corner $P_u M_D(\mathbb{C}) P_u$ is irreducible.*

Lemma 9.66 (Iterated blocking at the transfer-map level). Lean ✓ Stmt *For any MPS tensor A and $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

Definition 9.67 (Primitive irreducible cyclic-sector decomposition). Lean ✓ Stmt *A has primitive irreducible cyclic sectors* of period $m \geq 1$ if there is a family $(C_k)_{k=1}^{m}$ with $A^{[m]} \sim \bigoplus_{k=1}^{m} C_k$ (unit weights), each C_k left-canonical, with primitive transfer map, tensor-irreducible, and of positive bond dimension.

Lemma 9.68 (Common reblocking at a prescribed multiple). Lean ✓ Stmt *With hypotheses as in Lemma 9.63, and given any common multiple p of the per-block period-removal lengths (m_k), the common reblocking can be performed at the prescribed length p.*

Lemma 9.69 (Structural properties of the common reblocked sectors). Lean ✓ Stmt *Each sector tensor in the common reblocked cyclic-sector family of Lemma 9.63 is trace-preserving ($\sum_i (B^i)^\dagger B^i = \mathbb{1}$), has primitive transfer map ($\sigma_\partial(\mathcal{E}_B) = \{1\}$), is tensor-irreducible, and has positive bond dimension.*

Lemma 9.70 (Common-alphabet flattening). Lean ✓ Stmt *With $p = m_k e_k$ for each k, the iterated blocked tensor $(B_k^{[m_k]})^{[e_k]}$ agrees with $B_k^{[p]}$ in MPV family, after the canonical identification of the p-blocked alphabet $d^{[p]}$ with the e_k-fold tensor power of the m_k-blocked alphabet $d^{[m_k]}$.*

Lemma 9.71 (Reindexed nonzero-part identity). Lean ✓ Stmt *The powered nonzero-part decomposition $\bigoplus_k \mu_k^p B_k^{[p]}$ carries over under the common-alphabet identification of Lemma 9.70: the common-alphabet sectors have the same nonzero-part decomposition as the direct p-blocked tensors.*

Lemma 9.72 (Direct blocking in the common alphabet). Lean ✓ Stmt *For every original block B_k, direct blocking at the common length $p = m_k e_k$ agrees, as an MPV family, with the same block transported to the common blocked alphabet. Write $\widetilde{B}_k$ for this common-alphabet image. Then*

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}(\widetilde{B}_k)_\sigma.$$

Chapter 10

Canonical-form reductions after blocking

10.1 Scope

This chapter continues the canonical-form reduction begun in Chapter 9. There the per-tensor steps were established: invariant-subspace splitting, irreducible block decomposition, separation of the all-zero summand, blockwise trace-preserving gauge, and one-block period removal. Here the same chain is carried past a common blocking length, producing the primitive-block decomposition that closes the reduction.

The chapter opens with the formal zero-block separation (Theorem 10.2), which is the statement used by the arbitrary-tensor TP gauge of Chapter 9 (Theorem 9.35). It then assembles the after-blocking decompositions and reaches two final statements: the primitive-block decomposition of a single tensor (Theorem 10.12) and the joint decomposition of two same-MPV tensors at a common blocking period (Theorem 10.9). Both feed the proof of the Fundamental Theorem in Chapter 13.

10.2 Zero-block separation

In an irreducible block decomposition some blocks may have all-zero Kraus operators. Such blocks contribute only at length $N = 0$, where the contribution equals their bond dimension. There is no zero tensor in the output: the contribution is isolated as a single bond-dimension count D_0 in the length-zero identity $D = D_0 + \sum_k D_k$, while the positive-length content is an equality of MPV families.

Lemma 10.1 (All-zero irreducible blocks have bond dimension at most one). Lean ✓ Stmt *If A is irreducible with every Kraus operator $A^i = 0$, then $D \leq 1$.*

Proof. ✓ Proof If $D \geq 2$, irreducibility forces some Kraus operator to be nonzero; otherwise the bond space splits as a direct sum of any two complementary subspaces. Hence $A^i = 0$ for every i implies $D \leq 1$. □

Theorem 10.2 (Zero-block separation). Lean ✓ Stmt *Every MPS tensor A admits a block decomposition into a zero-block bond dimension D_0 and a finite family of nonzero blocks $(B_k)_{k=1}^{r}$, each irreducible, with at least one nonzero Kraus operator, and of positive bond dimension,* *recorded by two facts. The positive-length equality of MPV families: for every $N \geq 1$ and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma.$$

The length-zero dimension identity

$$D = D_0 + \sum_{k=1}^{r} D_k,$$

where D_0 is the total bond dimension of the all-zero blocks.

Proof. ✓ Proof Apply the irreducible block decomposition (Theorem 9.16) and partition the blocks by whether each has a nonzero Kraus operator; D_0 accumulates the bond dimensions of the all-zero blocks. For every all-zero irreducible block C_j and every $N \geq 1$, $V^{(N)}(C_j)_\sigma = 0$, so

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma.$$

At $N = 0$ the MPV coefficient is the bond dimension, giving $D = D_0 + \sum_{k=1}^r D_k$. □

10.3 After-blocking decompositions and zero-tail identities

Theorem 10.3 (Blockwise TP-gauge normalization of irreducible blocks). Lean ✓ Stmt *Suppose A generates the same MPV family as $\bigoplus_{k=1}^r B_k$, where each B_k is irreducible with at least one nonzero Kraus operator. Then there are nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(C_k)_{k=1}^r$ such that, for every N and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k C_k)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(C_k)_\sigma,$$

each C_k irreducible and left-canonical, $\sum_i (C_k^i)^\dagger C_k^i = \mathbb{1}$, with positive bond dimension. This is the blockwise form of the trace-preserving gauge; the arbitrary-tensor statement is Theorem 9.35.

Proof. ✓ Proof The Perron–Frobenius TP-gauge construction is applied to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the weights μ_k. □

For a primitive nonzero block satisfying the same hypotheses, normality is exactly Theorem 9.39.

Theorem 10.4 (Blocking preserves irreducibility for primitive irreducible blocks). Lean ✓ Stmt *Let A be left-canonical, irreducible, primitive, of positive bond dimension. For every $P > 0$, the blocked tensor $A^{[P]}$ is irreducible: there is no orthogonal projection Q with $Q \neq 0$, $Q \neq \mathbb{1}$ such that, for every P-blocked index α,*

$$(\mathbb{1} - Q)(A^{[P]})^\alpha Q = 0.$$

Proof. ✓ Proof A positive-definite fixed point of $\mathcal{E}_A$ remains fixed by the blocked transfer map; primitivity gives uniqueness of PSD fixed points for the blocked map, hence its transfer map is irreducible, which is the irreducibility criterion for $A^{[P]}$. □

Theorem 10.5 (Extra blocking of primitive irreducible sector blocks). Lean ✓ Stmt *Let A be trace-preserving, primitive, and tensor-irreducible. For every $k > 0$,*

$$\sum_\alpha ((A^{[k]})^\alpha)^\dagger (A^{[k]})^\alpha = \mathbb{1}, \qquad \sigma_\partial(\mathcal{E}_{A^{[k]}}) = \{1\},$$

and $A^{[k]}$ is tensor-irreducible. This k is a later common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.

Proof. ✓ Proof Trace preservation follows from blocking a trace-preserving tensor, primitivity from a positive power of a primitive transfer map, and irreducibility from Theorem 10.4. □

Theorem 10.6 (Common blocking period for two primitive normal tensors)**.** Lean ✓ Stmt *Let A and B be left-canonical normal tensors of positive bond dimensions, with $A^{[p_A]}$, $B^{[p_B]}$ having primitive transfer maps for some $p_A, p_B > 0$. Then there is a common $p > 0$ such that both blocked tensors are left-canonical,*

$$\sum_\alpha ((A^{[p]})^\alpha)^\dagger (A^{[p]})^\alpha = \mathbb{1}, \qquad \sum_\beta ((B^{[p]})^\beta)^\dagger (B^{[p]})^\beta = \mathbb{1},$$

both have primitive transfer maps, $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$ and $\sigma_\partial(\mathcal{E}_{B^{[p]}}) = \{1\}$, and both are normal.

Proof. ✓ Proof Take $p = \mathrm{lcm}(p_A, p_B)$. Primitivity follows from divisibility, and left-canonical normalization and normality are preserved under blocking. □

Remark 10.7 (Two-tensor primitive block decomposition). Applying Theorem 10.12 to each of two same-MPV tensors A and B produces, after a positive blocking length, a left-canonical primitive nonzero block decomposition for each side, with the separate zero-block count. This is a preliminary step toward Theorem 10.9, where the two sides are matched.

Theorem 10.8 (Cyclic sectors after the zero-block split)**.** Lean ✓ Stmt *If A and B generate the same MPV family, then after the zero-block split and the trace-preserving gauge there are weighted nonzero-block tensors $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ with, for every $N \geq 1$ and σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma, \qquad V^{(N)}(B)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k^B B_k)_\sigma,$$

and the two nonzero parts agree:

$$V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k^B B_k)_\sigma.$$

Each nonzero block has primitive irreducible cyclic sectors: for each j, there is $m_j \geq 1$ with a unit-weight sector decomposition $A_j^{[m_j]} \sim \bigoplus_{u=1}^{m_j} C_{j,u}^A$, and similarly for B_k.

Proof. ✓ Proof Apply Theorem 9.35 separately to A and B. Its positive-length conclusion gives, for $N \geq 1$,

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma, \qquad V^{(N)}(B)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k^B B_k)_\sigma.$$

Chaining these through $V(A) = V(B)$ identifies the two nonzero parts. Finally apply Theorem 9.62 to every trace-preserving irreducible nonzero-weight block. □

Theorem 10.9 (Common blocking length for cyclic sectors)**.** Lean ✓ Stmt *If two tensors generate the same MPV family, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length p. For every $N \geq 1$ and every blocked σ, each blocked tensor equals its powered nonzero part,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma, \qquad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_k (\mu_k^B)^p (B_k)^{[p]})_\sigma,$$

and the two powered nonzero parts agree,

$$V^{(N)}\big(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\big)_\sigma = V^{(N)}\big(\textstyle\bigoplus_k (\mu_k^B)^p (B_k)^{[p]}\big)_\sigma.$$

Proof. ✓ Proof Start from Theorem 10.8. With p_A, p_B the least common multiples of the period-removal lengths on the two sides, use $p = \mathrm{lcm}(p_A, p_B)$. The prescribed-common-multiple lemma constructs both one-sided sector families at this length, and the positive-length blocking lemmas carry the one-sided tensor/nonzero-part equalities over to the common alphabet; for every $N \geq 1$,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\big(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\big)_\sigma.$$

The same applies on the B side, and the two-sided blocking lemma gives equality of the two powered nonzero parts. □

Theorem 10.10 (Unconditional common primitive block decompositions)**.** Lean ✓ Stmt *Let A and B have the same MPV family. Then there is a positive blocking length at which both blocked tensors have the same positive-length MPV family as weighted direct sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two weighted nonzero-sector families have the same positive-length MPV family.*

Proof. ✓ Proof Apply Theorem 10.9 to choose the common blocking length and the two common cyclic-sector families. For either one of these families, write G_k for its cyclic-sector blocks, and write (ν_x, C_x) for the weights and blocks of the flattened common-sector family from Lemma 9.71. The common-alphabet nonzero-part identity gives, for every length N and blocked word σ,

$$V^{(N)}\big(\textstyle\bigoplus_k \mu_k^p G_k^{[p]}\big)_\sigma = V^{(N)}\big(\textstyle\bigoplus_x \nu_x C_x\big)_\sigma,$$

Applying this on the two sides of the comparison yields a positive length p and weighted primitive block families such that, for every $N \geq 1$ and blocked word σ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_x \mu_x^A A_x)_\sigma, \qquad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_y \mu_y^B B_y)_\sigma,$$

with the two weighted nonzero parts equal at every positive length. The structural properties and the nonvanishing of the weights come from the common-sector families. □

10.4 Primitive-block decomposition of an arbitrary tensor

This section gives the primitive sector theorem used in the period-removal step and the resulting primitive-block decomposition of an arbitrary tensor after blocking.

Theorem 10.11 (Unconditional primitive blocked sectors)**.** Lean ✓ Stmt *Let A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive mth root of unity γ, with blocked cyclic-sector decomposition $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$, and $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$. Then every sector tensor satisfies $\sigma_\partial(\mathcal{E}_{C_u}) = \{1\}$ and is tensor-irreducible.*

Proof. ✓ Proof Theorem 9.65 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u. The compression equivalences φ_u intertwine the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$, and the cyclic peripheral-spectrum argument shows each corner restriction is primitive. Primitivity and irreducibility are preserved across φ_u; then use Theorem 9.55 to pass from the adjoint blocked map to the primal transfer map of C_u. □

Theorem 10.12 (Arbitrary tensor to primitive block decomposition)**.** Lean ✓ Stmt *For any MPS tensor A, there exist $p \geq 1$, a zero-tail bond dimension D_0, nonzero weights $(\mu_k)_{k=1}^r$, and blocks $(B_k)_{k=1}^r$ such that, for every blocked σ of positive length N,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

The zero-block contribution is recorded separately by the length-zero identity $D_0 + \sum_{k=1}^{r} D_k = D$. *Each block is left-canonical and primitive,*

$$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}, \qquad \sigma_\partial(\mathcal{E}_{B_k}) = \{1\},$$

and each bond dimension is positive.

Proof. ✓ Proof Theorem 9.35 supplies the zero-block dimension, the left-canonical nontrivial blocks, the positive-length equality, and the identity $D = D_0 + \sum_k D_k$. Theorem 9.36 provides a common blocking period p making all transfer maps primitive, Lemma 9.61 keeps the blocked weights nonzero, and Lemma 9.59 carries the positive-length equality through blocking. □

Chapter 11

Block Permutation and Separation

This chapter records two reduction outputs: the algebra-automorphism decomposition Theorem 11.7; and the strict-weight block separation Lemmas 11.19 and 11.20, together with the canonical-form gauge comparison Lemma 11.21. The per-block linear-extension construction is stated once in Chapter 16, Section 16.9.

11.1 Block permutation algebra

Definition 11.1 (Block ideal). Lean ✓ Stmt Let $\{R_k\}_{k=1}^r$ be simple rings. The k-th *block ideal* in $\prod_{j=1}^r R_j$ is

$$I_k := \{x \in \prod_{j=1}^r R_j : x_j = 0 \text{ for } j \neq k\}.$$

Theorem 11.2 (Each block ideal is an atom). Lean ✓ Stmt *Each block ideal I_k is an atom in the lattice of two-sided ideals of $\prod_{j=1}^r R_j$.*

Proof. ✓ Proof Under the order isomorphism between two-sided ideals of $\prod_{j=1}^r R_j$ and products of two-sided ideal lattices, I_k corresponds to the tuple with $\top$ in the k-th coordinate and $\bot$ elsewhere. This tuple is an atom, hence so is I_k. □

Theorem 11.3 (Ring isomorphisms permute block ideals). Lean ✓ Stmt *Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \to \prod_k R_k$ permutes the block ideals: there exists a permutation π such that T maps the k-th block ideal to the $\pi(k)$-th block ideal.*

Proof. ✓ Proof By Theorem 11.2, each block ideal is an atom in the lattice of two-sided ideals of $\prod_k R_k$. A ring isomorphism induces an order automorphism of this lattice, so it permutes the atoms. Since the atoms are exactly the block ideals, this gives the required permutation π. □

Remark 11.4 (Ring level versus algebra level). Theorem 11.3 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorem 11.6 stays at this ring level: once the block ideals are permuted, one can extract the corresponding single-block maps without using scalar multiplication. Theorems 11.5 and 11.7 require an upgrade to $\mathbb{C}$-algebra automorphisms. The reason is threefold: dimension preservation compares the blocks as $\mathbb{C}$-vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$-algebras; and the final per-block maps must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

Theorem 11.5 (Dimension preservation). Lean ✓ Stmt *Assume $D_k \geq 1$ for all k. If T is an algebra automorphism of $\prod_{k=1}^{r} M_{D_k}(\mathbb{C})$, and π is the induced permutation, then $D_{\pi(k)} = D_k$ for all k.*

Proof. ✓ Proof The restriction of T to the k-th block ideal induces a $\mathbb{C}$-algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$, which forces $D_k = D_{\pi(k)}$ by comparing dimensions as $\mathbb{C}$-vector spaces. □

Theorem 11.6 (Per-block component maps are bijections). Lean ✓ Stmt *Assume $D_k \geq 1$ for all k. Let T be a ring automorphism of $\prod_{k=1}^{r} M_{D_k}(\mathbb{C})$, and let π be the induced permutation from Theorem 11.3. For each block index k, the map*

$$M \mapsto (T(0, \dots, 0, M, 0, \dots, 0))_{\pi(k)}$$

from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ is bijective.

Proof. ✓ Proof Theorem 11.3 forces a single-support element in the k-th block ideal to remain supported on the $\pi(k)$-th block after applying T. Injectivity follows from injectivity of T. For surjectivity, apply the same argument to T^{-1} and the inverse permutation. □

Theorem 11.7 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$). Lean ✓ Stmt *Assume $D_k \geq 1$ for all k. Every $\mathbb{C}$-algebra automorphism T of $\prod_{k=1}^{r} M_{D_k}(\mathbb{C})$ decomposes as a block permutation π followed by per-block inner automorphisms: there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that, for every tuple $M = (M_1, \dots, M_r)$, $T(M)_k = X_k M_{\pi(k)} X_k^{-1}$, where $T(M)_k$ denotes the k-th block component of $T(M)$.*

Proof. ✓ Proof Theorem 11.3 gives the permutation π. Theorem 11.5 identifies the matrix sizes $D_{\pi(k)} = D_k$. For each k, Theorem 11.6 gives a bijective multiplicative map from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ extracted from the elements supported on one factor of T; after reindexing by the dimension equality, this is a $\mathbb{C}$-algebra automorphism of $M_{D_k}(\mathbb{C})$. Skolem–Noether (Theorem 3.10) makes each such automorphism inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, Section IV.A] for the MPS context. □

11.2 Burnside and invariant projections

Lemma 11.8 (The algebra span reaches a finite cumulative span). Lean ✓ Stmt *If the generated unital algebra satisfies $\mathrm{alg}(A) = M_D(\mathbb{C})$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Every element of $\mathrm{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n\geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \cdots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. □

Lemma 11.9 (Irreducible action gives an irreducible tensor). Lean ✓ Stmt *If the letters of A act irreducibly on $\mathbb{C}^D$, then A has no nontrivial invariant orthogonal projection.*

Proof. ✓ Proof If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A. Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. □

Theorem 11.10 (Invariant projection gives a two-block decomposition). Lean ✓ Stmt *Suppose P is an orthogonal projection satisfying, for every physical index i,*

$$(\mathbb{1} - P)A^i P = 0.$$

Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. ✓ Proof Diagonalize P by a unitary U. Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem 9.9. In this basis the relation $(\mathbb{1} - P)A^i P = 0$ makes each letter upper block triangular. Discarding the off-diagonal blocks does not change the trace of any word product, so the resulting block-diagonal tensor has the same MPV coefficients. Reindex the two diagonal blocks by $\{0, \dots, n-1\}$ and $\{0, \dots, m-1\}$. □

Theorem 11.11 (Nontrivial invariant projection gives strict two-block reduction). Lean ✓ Stmt *Under the hypotheses of Theorem 11.10, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2, such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.*

Proof. ✓ Proof Apply Theorem 11.10. In the diagonal form of P, the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. □

11.3 Block separation

Recall from Definition 2.48 and Theorem 7.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \mathrm{Tr}(F_{AB}^N).$$

We use this notation throughout the block-separation argument.

Definition 11.12 (Canonical form conditions). Lean ✓ Stmt A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form conditions* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \cdots \geq |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \to 1$ for each k.

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Theorem 6.7, item 3]), $T^n(\rho) \to \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Theorem 6.8] for the completely-positive characterisation and Chapter 7 for the transfer-operator gap proof.

Remark 11.13 (Canonical-form theorem versus local predicates). The translation-invariant canonical-form theorem of [PGVWC07, Theorem Th:TIcanonical, lines 742–763] starts from an arbitrary TI-MPS representation and constructs a block canonical form with positive weights, unital block normalization, full-rank diagonal invariant densities, and uniqueness of the identity fixed point. Definition 11.12 is a local predicate used after such structure has already been supplied or separately proved. In particular, it assumes injectivity and the self-overlap limit, and the block-separation lemmas below additionally require strict separation of the weight moduli. These results are therefore conditional consequences, not the source canonical-form existence or uniqueness theorems.

Remark 11.14 (Normalization convention). The convention here differs from [PGVWC07, Theorem 4], which uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$. This chapter instead works in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides a gauge transformation between these two normalizations (Theorem 6.14).

Remark 11.15 (Overlap hypothesis). In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum $= \{1\}$). Theorem 9.54 gives this implication: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form conditions retain condition (5) explicitly because it can be verified independently of primitivity.

Remark 11.16 (Normal-canonical companion). Definition 11.12 gives the block-injective conditions. The normal-canonical analog instead uses Definition 9.24: injectivity and the explicit self-overlap hypothesis are replaced by irreducibility, TP normalization, and primitive transfer maps. Theorem 9.26 then supplies the self-overlap limit automatically.

Lemma 11.17 (Repeated-word evaluation). Lean ✓ Stmt *For any word w and any $L \geq 0$, the evaluation of A on the L-fold concatenation w^L is $(A^w)^L$.*

Proof. ✓ Proof By induction on L, using the multiplicativity of word evaluation under concatenation. □

Lemma 11.18 (Block separation core). Lean ✓ Stmt *Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. Assume moreover that the weight moduli are strictly ordered:*

$$|\mu_1| > |\mu_2| > \cdots > |\mu_r|.$$

If $\sum_{k=1}^r \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all N and σ, then each pair (A_k, B_k) generates the same MPV family [CPGSV16, Appendix `AppendixMPV`*, lines 1054–1192].*

Proof. ✓ Proof By induction on the number of blocks r. In the inductive step, take bilinear overlaps with the leading block A_1: from $\sum_{k=1}^r \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ one obtains

$$\sum_{k=1}^r \overline{\mu_k}^N (O_{A_1 A_k}(N) - O_{A_1 B_k}(N)) = 0.$$

After division by $\overline{\mu_1}^N$, the contributions from $k \geq 2$ decay because $|\mu_k/\mu_1| < 1$ (strict weight ordering) and the bilinear overlaps $O_{A_1 A_k}(N)$ and $O_{A_1 B_k}(N)$ are bounded (Theorems 7.23 and 7.25 supply the relevant bounds), leaving $O_{A_1 A_1}(N) - O_{A_1 B_1}(N) \to 0$. Condition (5) gives $O_{A_1 A_1}(N) \to 1$, hence $O_{A_1 B_1}(N) \to 1$. Since $O_{A_1 B_1}(N) \to 1 \neq 0$, the bond dimensions must match: $D_{A_1} = D_{B_1}$; otherwise Theorem 7.23 would give $O_{A_1 B_1}(N) \to 0$, a contradiction. With equal bond dimensions established, the contrapositive of Theorem 7.25 forces A_1 and B_1 to be gauge-phase equivalent; the limit $O_{A_1 B_1}(N) \to 1$ then makes the phase trivial, so A_1 and B_1 generate the same MPV family. Peeling off the leading block and dividing by μ_1^N, the induction hypothesis applies. $\square$

Lemma 11.19 (Block separation from canonical form)**.** Lean ✓ Stmt *Let $((\mu_k),(A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. Assume moreover that the weight moduli are strictly ordered:*

$$|\mu_1| > |\mu_2| > \cdots > |\mu_r|.$$

If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family [CPGSV16, Appendix `AppendixMPV`, lines 1054–1192].

Proof. ✓ Proof Apply Lemma 11.18 to the identity obtained from the equality of the two block-diagonal MPV families. The induction peels off the dominant block and treats the remainder as a decaying tail. Concretely, one isolates the leading component $\mu_1^N V^{(N)}(A_1)_\sigma$ and bounds the remaining contribution by $O(|\mu_2/\mu_1|^N)$ after dividing by μ_1^N. The canonical form conditions supply injectivity, TP normalization, strict nonzero weights, and self-overlap convergence for (A_k), while the hypotheses on (B_k) supply the comparison injectivity and TP normalization. $\square$

Lemma 11.20 (Block separation from normal canonical form)**.** Lean ✓ Stmt *Let $((\mu_k),(A_k))$ satisfy the normal canonical form conditions, and let (B_k) be irreducible tensors of the same bond dimensions, satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. Assume moreover that the weight moduli are strictly ordered:*

$$|\mu_1| > |\mu_2| > \cdots > |\mu_r|.$$

If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family [CPGSV16, Appendix `AppendixMPV`, lines 1054–1192].

Proof. ✓ Proof The proof follows the same peeling argument as Lemma 11.19, with Theorem 7.28 (modulus-one eigenvalue rigidity for irreducible tensors) replacing the injective overlap-decay argument at the identification step, since the blocks are irreducible rather than injective. Theorem 9.26 replaces the explicit self-overlap hypothesis on the leading block. Since both block families already have the same bond dimensions, the conclusion is direct per-block same MPV. $\square$

Lemma 11.21 (Canonical-form gauge comparison, same-structure case)**.** Lean ✓ Stmt *Under the hypotheses of Lemma 11.19, including strict ordering of the weight moduli, each pair (A_k, B_k) is gauge equivalent. Hence the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent. This is a same-structure local comparison theorem, not the full uniqueness theorem of [PGVWC07, Theorem thm-uniq, lines 1002–1015].*

Proof. ✓ Proof Lemma 11.19 gives per-block same MPV. Lemma 16.51 then converts this to per-block gauge equivalence and then to global gauge equivalence of the block-diagonal tensors. $\square$

Remark 11.22 (Relation to the literature). The block-permutation part follows the decomposition of automorphisms of semisimple matrix algebras used in [CPGSV21, Section IV.A]. The separation part is the local strict-weight peeling argument used in [CPGSV16, Appendix `AppendixMPV`, lines 1054–1192]; the proof here formulates it as a same-structure block-separation lemma using overlap bounds and mixed-transfer decay.

Chapter 12

Basis of Normal Tensors

This chapter introduces the *basis of normal tensors* (BNT) notion, the canonical-form variants with BNT separation, the cross-overlap decay theorem, and the permutation-rigidity arguments used in the proof of the Fundamental Theorem. It is the reference for the repeated-copy and equal-modulus coefficient comparison used later: sector coefficients are compared here, and the Newton–Girard identities recover the copy weights from those coefficients. The main references are [PGVWC07], [CPGSV16], and [CPGSV21].

12.1 Basis of normal tensors

Definition 12.1 (Basis of Normal Tensors (BNT))**.** Lean ✓ Stmt A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each A_j is normal (Definition 2.31),
2. at each system size N, the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$,
3. for sufficiently large N, the block MPV vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent.

This is [CPGSV21, Definition 4.2]. In the canonical-form characterization of [CPGSV16, Section 2.3 and Appendix A], a BNT is a minimal family of representatives for the relation

$$\widetilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$$

among the normal blocks in the canonical form.

Lemma 12.2 (Finite-index overlap-orthonormality implies eventual linear independence)**.** Lean ✓ Stmt *Let I be a finite index set and let $(A_i)_{i\in I}$ be a finite family of tensors such that $O_{A_iA_i}(N) \to 1$ for each i, and $O_{A_iA_j}(N) \to 0$ whenever $i \neq j$. Then the MPV states $\{|V^{(N)}(A_i)\rangle\}_{i\in I}$ are linearly independent for all sufficiently large N.*

Proof. ✓ Proof The Gram matrix of the family $\{|V^{(N)}(A_i)\rangle\}_{i\in I}$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. ☐

Lemma 12.3 (Overlap-orthonormality implies eventual linear independence). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g}$ be a finite family of tensors such that $O_{A_jA_j}(N) \to 1$ for each j and $O_{A_iA_j}(N) \to 0$ for $i \neq j$. Then the MPV states $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent for all sufficiently large N.*

Proof. ✓ Proof Apply Lemma 12.2 with $I = \{1, \ldots, g\}$. □

Lemma 12.4 (Eventual coefficient extraction from linear independence). Lean ✓ Stmt *If a finite family of vectors is linearly independent for all sufficiently large N, and two finite linear combinations of that family are equal for all sufficiently large N, then the corresponding coefficients are equal for all sufficiently large N.*

Proof. ✓ Proof Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length. □

Lemma 12.5 (Overlap-orthonormality for two families). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap $O_{A_jB_k}(N)$ tends to 0. Then the combined MPV family*

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large N.

Proof. ✓ Proof Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks. □

Definition 12.6 (No gauge-phase-equivalent distinct blocks). Lean ✓ Stmt A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

Definition 12.7 (Normal canonical form with BNT separation). Lean ✓ Stmt A family is in *normal canonical form with BNT separation* if it satisfies Definition 9.24 and, in addition:

1. the weight moduli $|\mu_1| > |\mu_2| > \cdots > |\mu_r|$ are strictly decreasing (a one-representative restriction),
2. distinct blocks of the same bond dimension are not gauge-phase equivalent,
3. the dominant block has unit modulus, $|\mu_1| = 1$ ([CPGSV16, paragraph after `eq:II_CF1`]).

This is a scope-restricted one-copy-per-sector surface: it keeps one representative for each strict weight-modulus class and does not encode the full CPSV16 BNT multiplicity data. The missing general multiplicity route is recorded in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`. The strict ordering describes an already selected or grouped representative family. Equal-modulus copies are not separated by this surface; they appear as the individual weights $\mu_{j,q}$ in the coefficient sum $c_N^{(j)} = \sum_q \mu_{j,q}^N$ used in the sector-BNT surface below. This is not the source BNT expansion

$$A^i = \bigoplus_{j=1}^{g} \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1},$$

where repeated copies of the same basis tensor remain visible through their coefficients, multiplicities, and gauges.

Theorem 12.8 (Cross-overlap decay from explicit BNT hypotheses). Lean ✓ Stmt *Let $(A_j)_{j=1}^r$ be injective and trace-preserving normalized. Assume that distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \to 0$ for every pair $j \neq k$.*

Proof. ✓ Proof If the bond dimensions differ, Theorem 7.23 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 7.25 gives the decay. □

Lemma 12.9 (BNT from explicit blockwise hypotheses). Lean ✓ Stmt *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume that the blocks are injective and trace-preserving normalized, that $O_{A_k A_k}(N) \to 1$ for each k, and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. ✓ Proof Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 12.8 supplies the off-diagonal decay. Therefore Lemma 12.3 gives the eventual linear independence required in Definition 12.1. □

Remark 12.10 (Comparing the two normality conditions). Definition 12.7 references the normal canonical form conditions (Definition 9.24), which encode normality via the spectral characterization (primitive transfer map). Definition 12.1 requires normality via the algebraic characterization (eventual block injectivity, Definition 2.31). These are equivalent by Remark 2.20, but the equivalence must be invoked explicitly: the Wielandt theory of Chapter 8 provides the implication from primitivity to eventual block injectivity needed here.

Lemma 12.11 (Restricted normal-CF-BNT hypotheses give a BNT). Lean ✓ Stmt *If a family is in normal canonical form with BNT separation and each block is also known to be normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor. This is the one-copy-per-sector variant documented in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`, not the full CPSV16 BNT multiplicity statement.*

Proof. ✓ Proof The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 12.1. □

12.2 Permutation rigidity for bases of normal tensors

Lemma 12.12 (Invertible change of basis). Lean ✓ Stmt *Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \mathrm{GL}_g(\mathbb{C})$ such that*

$$w_j = \sum_{i=1}^{g} U_{ij} v_i$$

for every j.

Proof. ✓ Proof Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^g$; these coefficients form the columns of U. If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. □

Lemma 12.13 (Eventual change of basis from equal spans). Lean ✓ Stmt *If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size N, then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.*

Proof. ✓ Proof By Lemma 12.3, both families are linearly independent for all sufficiently large N. At such an N, the span equality lets us apply Lemma 12.12. □

Theorem 12.14 (Permutation rigidity for equal block counts). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g}$ and $(B_j)_{j=1}^{g}$ be injective families of positive bond dimension satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size N, then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.*

Proof. ✓ Proof For large N, Lemma 12.13 gives an invertible matrix U_N relating the two MPV families. Fix j and write $|V^{(N)}(B_j)\rangle = \sum_i u_{ij}^{(N)} |V^{(N)}(A_i)\rangle$. With $G_A(N) = (O_{A_i A_\ell}(N))_{i,\ell}$ and $b_j(N) = (O_{A_i B_j}(N))_i$, this gives

$$G_A(N)\overline{u_j(N)} = b_j(N), \qquad O_{B_j B_j}(N) = u_j(N)^\top G_A(N)\overline{u_j(N)}.$$

If all mixed overlaps $O_{A_i B_j}(N)$ tended to 0, then $b_j(N) \to 0$; since $G_A(N) \to I$, one obtains $\overline{u_j(N)} \to 0$ and hence $u_j(N) \to 0$. Thus $O_{B_j B_j}(N) \to 0$, contradicting $O_{B_j B_j}(N) \to 1$. Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 7.23 and 7.25 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B-family makes f injective, hence bijective, and its inverse is the required permutation. □

Lemma 12.15 (Permutation rigidity with asymptotically orthonormal overlaps). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors of positive bond dimension satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size N,*

$$\operatorname{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \le j \le g_A\} = \operatorname{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \le k \le g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j.

Proof. ✓ Proof By Lemma 12.3, both families are linearly independent for all sufficiently large N. At such an N, the common span has the same dimension when computed from the A-family or the B-family, so $g_A = g_B$. After identifying the two index sets, Theorem 12.14 yields the permutation conclusion. □

Theorem 12.16 (Non-equivalent irreducible TP blocks are eventually non-proportional). Lean Lean ✓ Stmt *Let A and B be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound N_0 there is $N \ge N_0$ such that $|V^{(N)}(A)\rangle$ is not proportional to $|V^{(N)}(B)\rangle$. Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.*

Proof. ✓ Proof Suppose that for all sufficiently large N one had $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. Then

$$O_{AA}(N) = |c_N|^2 O_{BB}(N), \qquad O_{AB}(N) = c_N O_{BB}(N).$$

Since $O_{AA}(N) \to 1$ and $O_{BB}(N) \to 1$, the first identity gives $|c_N| \to 1$, and hence $|O_{AB}(N)| \to 1$. The irreducible trace-preserving overlap decay theorem gives $O_{AB}(N) \to 0$ when $A \not\sim_{\mathrm{gp}} B$, a contradiction. □

12.3 Newton–Girard identities

Theorem 12.17 (Newton–Girard trace recursion). Lean Lean ✓ Stmt *If $A, B \in M_n(\mathbb{C})$ satisfy $\mathrm{tr}(A^k) = \mathrm{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial. It is enough to assume this equality for $1 \leq k \leq n$.*

Proof. ✓ Proof Let e_m denote the mth elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^{m} (-1)^{i-1} e_{m-i} \,\mathrm{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\mathrm{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. □

Remark 12.18 (How the power-sum equality is used)*.* The equal-MPV corollary in the source compares scalar power sums only after the BNT representatives have already been matched. If the paired basis tensors satisfy $B_j = e^{i\phi_j} Y_j A_j Y_j^{-1}$, then for every positive length N the coefficient $\sum_q \mu_{j,q}^N$ of $|V^{(N)}(A_j)\rangle$ in the BNT expansion of A equals the coefficient $\sum_q (\nu_{j,q} e^{i\phi_j})^N$ coming from B. This scalar equality is the input to the Newton–Girard step.

Thus the power-sum lemma determines the multiset of weights with multiplicity, but it is not a replacement for the BNT matching argument. The block permutation and the phase gauges must first identify which basis tensor is being compared. Once that matching is available, the power-sum lemma supplies the multiplicity and weight equality needed to identify the weighted block-diagonal gauge.

The finite-range statement below includes the same-cardinality form, the nonzero-entry unequal-cardinality form, and the positive-power conclusion after deleting zeros. The nonzero-entry form pads the shorter family by zeros, applies the same-cardinality result at length $\max(m, n)$, and then uses the nonzero-entry hypothesis to count the padded zeros. The Appendix lemma in the source fixes an ordering convention for the two families; here the conclusion is stated as multiset equality, so no sorting hypothesis is needed. If equality at exponent 0 is also assumed, the nonzero-entry hypothesis can be omitted because $\sum_k \lambda_{a,k}^0 = x_a$ and $\sum_k \lambda_{b,k}^0 = x_b$.

Theorem 12.19 (Finite-range equal power sums imply equal multisets)**.** Lean Lean Lean Lean ✓ Stmt *Let $\alpha, \beta : \{0, \dots, n-1\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for $1 \leq k \leq n$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal. More generally, if $\alpha : \{0, \dots, m-1\} \to \mathbb{C}$ and $\beta : \{0, \dots, n-1\} \to \mathbb{C}$ have no zero entries and their power sums agree for $1 \leq k \leq \max(m, n)$, then $m = n$ and the two multisets are equal. There is also an exponent-zero list form: for lists $(\lambda_{a,k})_{k=1}^{x_a}$ and $(\lambda_{b,k})_{k=1}^{x_b}$, the hypothesis $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$ for $1 \leq N \leq \max(x_a, x_b)$ implies*

$$\{\lambda_{a,k} \mid \lambda_{a,k} \neq 0\}_{k=1}^{x_a} = \{\lambda_{b,k} \mid \lambda_{b,k} \neq 0\}_{k=1}^{x_b}$$

as multisets. If the equality is also assumed at $N = 0$, then the original lists have the same multiset.

Proof. ✓ Proof Apply the finite-range Newton–Girard theorem to $\mathrm{diag}(\alpha)$ and $\mathrm{diag}(\beta)$, then extract the roots of $\prod_i (X - \alpha_i)$ and $\prod_i (X - \beta_i)$. For unequal cardinalities, pad both lists with zeros to length $\max(m, n)$. The equal-root conclusion compares the padded multisets, and the nonzero-entry hypothesis makes the number of padded zeros exactly $\max(m, n) - m$ and $\max(m, n) - n$, forcing $m = n$. Without a nonzero-entry hypothesis, first delete the zero entries; for positive N, this does not change $\sum_k \lambda_k^N$. The nonzero-entry form then compares the remaining multisets. In the exponent-zero list form, the equality at $N = 0$ gives $x_a = x_b$, so the same-cardinality finite-range theorem applies directly. ☐

Theorem 12.20 (Equal power sums imply equal multisets)**.** Lean ✓ Stmt *Let $\alpha, \beta : \{0, \dots, n - 1\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Proof. ✓ Proof Construct diagonal matrices $\mathrm{diag}(\alpha)$ and $\mathrm{diag}(\beta)$. The power-sum hypothesis gives $\mathrm{tr}(\mathrm{diag}(\alpha)^k) = \mathrm{tr}(\mathrm{diag}(\beta)^k)$. By Theorem 12.17, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. ☐

Remark 12.21. Theorems 12.17 and 12.20 remain relevant for the equal-MPV source corollary discussed in Section 13.3, where the paper uses a power-sum step to recover the multiplicities r_j and the weights $\mu_{j,q}$. The BNT multiplicity lemmas handle this recovery directly rather than through a separate explicit-coefficient equal-case theorem.

12.4 The BNT canonical form on a sector decomposition

This section introduces the BNT canonical-form predicate following [CPGSV16, §II, lines 287–301] and [CPGSV21, Definition 4.2, Proposition 4.3, and lines 1864–1884]. The predicate is defined on a sector decomposition with raw sector weights $\mu_{j,q}$ and sector coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$; no equal-modulus factorization is assumed.

12.4.1 Sector decompositions and phase matching

The sector-decomposition data type, the gauge-phase matching predicates, and the MPV phase-class apparatus are collected here because they are used both by the BNT canonical-form predicate below and by the equal-MPV comparison results in Chapter 13.

Definition 12.22 (Sector weights and sector decomposition)**.** Lean Lean ✓ Stmt A sector-weight tuple over g basis blocks consists of multiplicities $r_j > 0$, nonzero weights $\mu_{j,q}$ for $q \in \{1, \dots, r_j\}$, and the sector coefficients

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with these multiplicities and weights.

Definition 12.23 (Sector basis matching)**.** Lean ✓ Stmt A *sector basis matching* between two sector decompositions P and Q consists of

1. a basis permutation $\pi : \{1, \dots, g_P\} \simeq \{1, \dots, g_Q\}$,
2. per-block multiplicity equalities $r_j^P = r_{\pi(j)}^Q$,
3. per-block bond-dimension equalities $D_j^P = D_{\pi(j)}^Q$, and

4. per-block gauge-phase equivalences $P_j \simeq_{\mathrm{gp}} Q_{\pi(j)}$ after identifying the two matrix algebras by the bond-dimension equality.

The hypotheses of Lemma 13.11 are exactly these listed components.

Definition 12.24 (Sector basis pre-matching). Lean ✓ Stmt A *sector basis pre-matching* between sector decompositions P and Q consists of a permutation of their basis blocks, equality of the paired bond dimensions, and gauge-phase equivalence of each corresponding pair. It does not include the copy-count equality; that equality is recovered from total MPV equality and BNT linear independence in Lemma 13.13.

Definition 12.25 (Single-family overlap-orthogonality hypotheses for sector bases). Lean ✓ Stmt For one sector decomposition P, this condition states positive basis dimensions, trace-preserving normalization of all basis blocks, self-overlap limits equal to 1, and off-overlap limits equal to 0.

Definition 12.26 (Primitive overlap-span hypotheses for sector bases). Lean ✓ Stmt For sector decompositions P and Q, this condition states the primitive overlap-rigidity hypotheses for their basis blocks: nonzero bond dimensions, injectivity, trace-preserving normalization, self-overlap limits equal to 1, off-overlap limits equal to 0, and equality of the finite-length spans of the basis MPV states for every system size.

Definition 12.27 (MPV phase equivalence for nonzero-weight blocks). Lean ✓ Stmt Two blocks A and B are MPV-phase equivalent if there is a nonzero scalar ζ such that, for every system size N and every physical word,

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

The two bond dimensions are allowed to differ.

Definition 12.28 (Common MPV phase cover). Lean ✓ Stmt For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

Definition 12.29 (MPV phase classes). Lean Lean Lean ✓ Stmt For a finite family of blocks, put $j \sim k$ when there is a nonzero scalar factor ζ such that

$$|V^{(N)}(B_k)\rangle = \zeta^N |V^{(N)}(B_j)\rangle$$

for every length N. The associated tuple enumerates the finite quotient, chooses one representative per class, records the scalar relating the representative to each member, and proves that distinct representatives are not MPV-phase equivalent.

Definition 12.30 (BNT canonical form). Lean ✓ Stmt Given a sector decomposition P, the BNT canonical-form predicate records the minimal BNT hypotheses of [CPGSV16, CPGSV21] without any equal-modulus or strict-ordering condition on the raw sector weights. It asserts that each basis block has positive bond dimension, is injective, irreducible, and left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent after identifying equal bond dimensions; and the [CPGSV16, §II.C, line 246] normalization holds,

$$\forall (j,q),\ |\mu_{j,q}| \le 1, \qquad \exists\, (j_*, q_*) : |\mu_{j_*,q_*}| = 1.$$

The MPV expansion takes the raw two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^{g} \left(\sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma,$$

matching [CPGSV16, § II.C, lines 271–301] and [CPGSV21, Definition 4.2, Proposition 4.3, and lines 1864–1884]. The line-246 normalization admits every example of the source paper (in particular $C \oplus D$, $C \oplus (-C)$, $C \oplus (1/2)C$, and $C \oplus (-C) \oplus (1/2)C$) and is not derivable from the per-block normality hypotheses alone.

Lemma 12.31 (Raw two-layer sector coefficient identity). Lean ✓ Stmt *For every sector decomposition P, every length N, and every basis index j, the sector coefficient is the raw power sum*

$$c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

This is [CPGSV16, two-layer BNT and coefficient-expansion displays, lines 287–301] and [CPGSV21, lines 1864–1884].

Proof. ✓ Proof Definitional unfolding of the sector coefficient. □

Lemma 12.32 (Sector coefficient bound by multiplicity). Lean ✓ Stmt *Under the optional global modulus normalization [CPGSV16, §II, lines 217–246], every raw weight satisfies $|\mu_{j,q}| \le 1$, and the triangle inequality bounds the norm of the sector coefficient by the multiplicity:*

$$\left|c_N^{(j)}\right| \le r_j.$$

Proof. ✓ Proof Apply the triangle inequality to the raw power sum from Lemma 12.31 and bound each term by 1 using $|\mu_{j,q}| \le 1$. □

Lemma 12.33 (Cross-overlap decay between distinct basis blocks). Lean ✓ Stmt *If P is a BNT canonical form and $j \ne k$, then $O_{P_j P_k}(N) \to 0$ as $N \to \infty$, where P_j and P_k are the basis blocks at indices j and k. The argument follows the normal-tensor overlap dichotomy of [CPGSV16, lines 1080–1091]: differing bond dimensions force decay, and matching bond dimensions are ruled out by the gauge-phase distinctness clause of [CPGSV16, lines 264–279].*

Proof. ✓ Proof Case on whether the bond dimensions match. If they differ, apply the dimension-mismatch overlap-decay theorem. If they agree, the basis-distinctness clause of the BNT canonical form rules out gauge-phase equivalence after identifying the common bond dimension, so the equal-dimension overlap-decay theorem applies. □

Lemma 12.34 (Combined-family eventual linear independence). Lean ✓ Stmt *Let P and Q each be a BNT canonical form, and suppose every mixed overlap $O_{P_j Q_k}(N)$ tends to 0. Then the union family*

$$\{|V^{(N)}(P_j)\rangle\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\rangle\}_{k=1}^{g_Q}$$

is linearly independent for all sufficiently large N. This is the corollary at [CPGSV16, lines 1121–1132], matching the BNT linear-independence input recorded at [CPGSV21, line 1850].

Proof. ✓ Proof Feed the per-family normalized self-overlaps (from the canonical-form hypotheses), the within-family cross-overlap decay (Lemma 12.33), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma 12.5). □

Lemma 12.35 (Sector coefficient bound from the line-246 normalization). Lean ✓ Stmt *Under the BNT canonical form, the [CPGSV16, § II.C, line 246] normalization yields directly*

$$|c_N^{(j)}| \leq r_j$$

for every N and basis index j, without an additional explicit modulus-bound hypothesis.

Proof. ✓ Proof Feed the global modulus normalization hypothesis into Lemma 12.32. □

Lemma 12.36 (Unit-modulus weight witness from the line-246 normalization). Lean ✓ Stmt *Under the BNT canonical form, the [CPGSV16, § II.C, line 246] unit-modulus existential provides indices (j_*, q_*) with $|\mu_{j_*,q_*}| = 1$.*

Proof. ✓ Proof Direct projection onto the unit-modulus existential hypothesis of the canonical-form predicate. □

Example 12.37 ($C \oplus (1/2)C$, unequal-modulus admissibility). Lean ✓ Stmt The single-sector two-copy decomposition with raw weights $(1, 1/2)$ demonstrates that the BNT canonical form admits unequal-modulus copies: the modulus bound holds because $|1| \leq 1$ and $|1/2| = 1/2 \leq 1$, and the unit-modulus existential is witnessed by the first copy with weight 1. This is the [CPGSV16, § II.C, line 246] normalization in positive form, applied to the $C \oplus (1/2)C$ example.

12.5 Cesàro non-decay and unit-block coefficient

This section formalizes the analytic input that supplies, for any sector j_0 carrying a unit-modulus weight, the "sector-j_0 coefficient does not asymptotically vanish" ingredient used in the matching step of [CPGSV16, Appendix MPV proof, line 1182]. The reduction is to a pure-analytic statement: a finite sum of N-th powers of complex numbers in the closed unit disk, with at least one unit-modulus summand, cannot tend to zero.

The proof is the elementary Cesàro-mean identity $M_T := T^{-1} \sum_{N=0}^{T-1} |c(N)|^2 = \sum_{p,q} T^{-1} \sum_{N=0}^{T-1} (\mu_p \overline{\mu_q})^N$, which splits into: T^{-1} times the partial geometric sum for each pair (p, q); each pair with $\mu_p \overline{\mu_q} = 1$ contributes the limit 1, while each pair with $\mu_p \overline{\mu_q} \neq 1$ contributes the limit 0 (since $|(z^T - 1)/(z - 1)| \leq 2/|z - 1|$ divided by $T \to 0$). The diagonal pair (q^*, q^*) at a unit-modulus index q^* contributes, so $M_T \to \#\{\text{matching pairs}\} \geq 1 > 0$, contradicting $c(N) \to 0 \Rightarrow |c(N)|^2 \to 0 \Rightarrow M_T \to 0$.

Lemma 12.38 (Cesàro non-decay of a finite power sum). Lean ✓ Stmt *Let $\mu : \{0, \ldots, r-1\} \to \mathbb{C}$ satisfy $|\mu_q| \leq 1$ for every q and let some q^* satisfy $|\mu_{q^*}| = 1$. Then the sequence $N \mapsto \sum_{q=0}^{r-1} (\mu_q)^N$ does not tend to 0 as $N \to \infty$. This is the analytic content of the unit-block non-decay step at [CPGSV16, § II.C, line 246] and [CPGSV16, Appendix MPV proof, line 1182].*

Proof. ✓ Proof Cesàro-mean argument as described above. □

Lemma 12.39 (Unit-block coefficient non-decay). Lean ✓ Stmt *Let P be a BNT canonical form and let j_0 be a sector index such that some copy of sector j_0 carries a unit-modulus weight, that is, there exists q with $|\mu_{j_0,q}| = 1$. Then the sector-j_0 coefficient $c_N^{(j_0)} = \sum_q \mu_{j_0,q}^N$ does not tend to 0 as $N \to \infty$.*

This reads the [CPGSV16, § II.C, line 246] normalization sector by sector, together with the convergence use at [CPGSV16, Appendix "Proofs of Section MPS", line 1244]. The unit-modulus witness is supplied externally (as the hypothesis above) because the source statement is global ("at least one of them equals one" over the two-layer index (j, q)) and does not pin the witness to a specific sector.

Proof. ✓ Proof Apply Lemma 12.38 to the family $q \mapsto \mu_{j_0,q}$, using the modulus bound $|\mu_{j_0,q}| \le 1$ from the canonical-form normalization together with the per-sector unit-modulus hypothesis. ☐

Remark 12.40. Used inside the matching theorem of [CPGSV16, Appendix MPV proof, line 1182] this lemma supplies the non-decay step on $c_N^{(j_0)}$. The matching theorem takes the unit-modulus sector and witness as explicit parameters, mirroring the global character of the source normalization at [CPGSV16, § II.C, line 246].

12.5.1 Matched-sector weight multiset equality

Given a matched basis-sector pair $(j_0, \tilde{k}_0)$ and a nonzero scalar $\zeta \in \mathbb{C} \setminus \{0\}$ such that the matched coefficient identity $c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q)$ holds for every N past some threshold N_0, this subsection records the two algebraic conclusions: the *matched-sector weight multiset equality* derived from the coefficient identity by power-sum recovery (Theorem 12.20), and the *matched-sector weight-permutation extractor* upgrading the multiset equality to an explicit per-copy indexing. Together they realise the $\mu_{j,q} = \nu_{j,q} \cdot e^{i\varphi_j}$ recovery of [CPGSV16, Appendix MPV proof, lines 1184–1188]. At the source level, the preceding block-selection argument supplies the matched normal sectors [CPGSV16, Appendix MPV proof, line 1182]; the sector-coefficient identity used here is the equal-MPV coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1184–1188], followed by the global-gauge construction in [CPGSV16, Appendix MPV proof, lines 1189–1192]. The coefficient identity is supplied, in the equal-MPV case, by Lemma 12.48 below. In the proportional case it is kept as the remaining hypothesis of Theorem 12.55.

Lemma 12.41 (Matched-sector weight multiset equality)**.** Lean ✓ Stmt *Let P, Q be sector decompositions with weights $\mu_{j,q}$ and $\nu_{k,q'}$, and fix indices j_0, $\tilde{k}_0$, and a nonzero scalar $\zeta \in \mathbb{C} \setminus \{0\}$. Suppose that for every $N > N_0$,*

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q).$$

Then the per-sector multiplicities agree, $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$, and the rescaled per-copy weight multisets coincide. This is [CPGSV16, Appendix MPV proof, lines 1184–1188] ($\mu_{j,q} = \nu_{j,q} \cdot e^{i\varphi_j}$ and matching multiplicities $r_{a,j} = r_{b,j}$) read on a single matched sector.

Proof. ✓ Proof Unfold both sides of the coefficient identity to per-copy power sums $\sum_q \mu_{j_0,q}^N$ and $\sum_{q'} \nu_{\tilde{k}_0,q'}^N$, and distribute ζ^N through the right-hand sum to obtain, for every $N > N_0$,

$$\sum_q \mu_{j_0,q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0,q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 12.19), using $\zeta \neq 0$ together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality. ☐

Lemma 12.42 (Matched-sector weight-permutation extractor)**.** Lean ✓ Stmt *Under the hypotheses of Lemma 12.41, the per-sector multiplicities agree and there is a permutation τ of the per-copy index set identifying the individual weights up to the gauge phase: for every q,*

$$\nu_{\tilde{k}_0,\tau(q)} = \zeta^{-1} \cdot \mu_{j_0,q}.$$

This is the per-copy realisation of [CPGSV16, Appendix MPV proof, line 1188] ($\mu_{j,q} = \nu_{j,q} \cdot e^{i\varphi_j}$): the multiset equality from Lemma 12.41 is upgraded to a concrete indexing between the matched P- and Q-copies.

Proof. ✓ Proof Invoke Lemma 12.41 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality identification and cancel ζ via $\zeta \neq 0$ to convert the gauge factor on the right of the pointwise identity to ζ^{-1} on the left. □

12.6 Strong existential matching under per-sector unit weights

This section lifts the weak existential matching $\exists j, \exists k$ to the strong existential $\forall k, \exists j$ on the original pair of BNT canonical forms, without recursion, partial sector dropping, or any combined-linear-independence obligation on a partial union. The result is a single $\forall k, \exists j$ statement on the original (P, Q), under the explicit hypothesis that every Q-side sector carries a unit-modulus copy weight.

The per-sector unit-weight hypothesis is the formal version used in the current full-basis theorem: [CPGSV16, § II.C, line 246] records the modulus-one hypothesis as a single *global* existential over the two-layer (j, q) index of the BNT display, whereas the statement below assumes the corresponding witness in each sector. This is why the result is labelled as a full-basis form rather than as the unrestricted source theorem. The bijective-matching theorem below applies the strong existential theorem twice (with (P, Q) swapped) to derive the cardinality identity and the per-pair bijection and gauges on the full basis, assuming the same per-sector unit-weight condition on both sides.

Theorem 12.43 (Strong existential matching, full-basis form). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. Assume that every sector k of Q has a unit-modulus copy weight:*

$$\forall k \in J(Q),\ \exists q,\ |\nu_{k,q}| = 1.$$

Then for every sector k of Q there exist a sector j of P with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.

Proof. ✓ Proof The contrapositive of [CPGSV16, Appendix MPV proof, line 1182], combined with linear independence on the full family $\{P_j\}_j \cup \{Q_k\}_k$, is iterated externally over each sector k. Each iteration is discharged by the full-basis weak existential matching theorem with the roles of P and Q swapped: supply the canonical-form hypotheses in reversed order, the basis non-emptiness witness from the retained global unit-modulus witness, the per-sector unit-modulus witness for the chosen Q-sector, and the same-MPV hypothesis through its pointwise-symmetric flip. The resulting conclusion is re-oriented by symmetry of gauge-phase equivalence under the bond-dimension identification and symmetry of overlap convergence. □

Remark 12.44 (Linear independence on the full family). Theorem 12.43 consumes *only* the weak existential matching theorem, which itself routes through the linear-independence step (Lemma 12.34) on the *full* family $\{P_j\}_j \sqcup \{Q_k\}_k$, the corollary input of [CPGSV16, Appendix MPV corollary, lines 1131–1133], not on any partial union.

12.6.1 Bijective matching by symmetry

This subsection derives the symmetry argument of [CPGSV16, Appendix MPV proof, line 1182]: $g_a \geq g_b$ and $g_b \geq g_a$ together imply $g_a = g_b$.

The strong existential matching theorem (Theorem 12.43) gives one direction of the full-basis matching when every Q-sector has a unit-modulus copy weight. Applying the same theorem once more with the roles of P and Q swapped, using the corresponding per-sector hypothesis on P and the symmetric flip of the same-MPV hypothesis, gives the other direction. The two matchings, combined with the basis-distinctness clause of the canonical-form predicate [CPGSV16, § II.C, lines 264–279], force injectivity in both directions.

Scope. The literal source conclusion "$g_a = g_b$ and a relabelling $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$" ([CPGSV16, Appendix MPV proof, line 1182]) is formalized below under the explicit per-sector unit-weight hypotheses

$$\forall j \in J(P),\ \exists q,\ |\mu_{j,q}| = 1, \qquad \forall k \in J(Q),\ \exists q,\ |\nu_{k,q}| = 1.$$

Under these hypotheses every sector lies in the unit-modulus part, so the symmetric injective matchings give a bijection on the full BNT basis. The coefficient comparison is not folded into this matching step; it is recorded separately in the following subsection.

Theorem 12.45 (Bijective matching by symmetry)**.** Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. Assume that every sector of P and every sector of Q has a unit-modulus copy weight. Then there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

such that for every $k \in J(Q)$ the bond dimensions match, the basis tensors $P_{\beta(k)}$ and Q_k are gauge-phase equivalent after identifying the matched bond dimensions, and the basis cross-overlap does not tend to zero.

Proof. ✓ Proof Apply Theorem 12.43 first with (P, Q) and the original hypothesis to obtain a raw matched-sector function via classical choice, and then again with (Q, P) and the symmetric flipped hypothesis to obtain the symmetric raw function.

Forward injectivity. Suppose $\varphi_0(k_1) = \varphi_0(k_2) = j$. The two matching witnesses give a common centre P_j with two gauge-phase equivalences, after identifying the bond dimensions, to Q_{k_1} and Q_{k_2}. Composing through this centre (transitivity of gauge-phase equivalence under these bond-dimension identifications) yields a gauge-phase equivalence between Q_{k_1} and Q_{k_2} under the dimension equality $\dim_Q(k_1) = \dim_Q(k_2)$. The basis-distinctness clause of the canonical-form predicate on Q ([CPGSV16, § II.C, lines 264–279]) then forces $k_1 = k_2$. Backward injectivity is symmetric.

Bijection. The two injections give equality of the two finite sector counts. Hence the forward injection is a bijection, and the matching witnesses are extracted by re-applying the classical-choice specification at each index in the domain. ☐

12.6.2 Full-basis coefficient identity and global gauge

The matching theorem above gives the sector phases and gauges. The remaining coefficient comparison is the step at [CPGSV16, Appendix MPV proof, lines 1187–1188]. In the equal-MPV setting, the full BNT basis and eventual linear independence give the comparison directly. In the proportional setting below, the same comparison is recorded as the explicit hypothesis needed before the copy weights can be assembled into the global gauge. The hypotheses in this subsection require a unit-modulus copy in every sector on both sides; under this normalization, the matching covers the whole BNT basis.

Lemma 12.46 (Unit phase from matched normalized BNT blocks)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms. If, for every length N and word σ, a sector P_j and a sector Q_k have MPV families related by*

$$V^{(N)}(Q_k)_\sigma = \zeta^N V^{(N)}(P_j)_\sigma$$

then $|\zeta| = 1$.

Proof. ✓ Proof The BNT normalization gives self-overlap limits of modulus 1 for both sectors. The displayed MPV identity scales the Q_k self-overlap by $(\zeta\bar{\zeta})^N$ times the P_j self-overlap. Taking limits forces $|\zeta| = 1$. $\square$

Theorem 12.47 (Proportional sector matching with unit phases)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Assume that every sector of P and every sector of Q has at least one copy weight of modulus 1. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$, scalars $\zeta_k \in \mathbb{C}$, and block gauges X_k such that $|\zeta_k| = 1$ and, for every sector k and physical index i,*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$$

Equivalently, for every length N and word σ,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$$

This records the per-block phase conclusion of [CPGSV16, Appendix MPV theorem statement, lines 1167–1170]; the proof step at [CPGSV16, Appendix MPV proof, line 1182] is the corresponding argument. The following coefficient-comparison step is separate.

Proof. ✓ Proof Apply the proportional sector-matching theorem to obtain the bijection and dimension-compatible gauge-phase equivalences. Choosing the gauges and scalars gives $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, and therefore

$$O_{Q_k Q_k}(N) = |\zeta_k|^{2N} O_{P_{\beta(k)} P_{\beta(k)}}(N).$$

The normalized self-overlap limits give $O_{Q_k Q_k}(N) \to 1$ and $O_{P_{\beta(k)} P_{\beta(k)}}(N) \to 1$, so $|\zeta_k|^{2N} \to 1$ and hence $|\zeta_k| = 1$. The MPV scalar-power identity follows from the same gauge equation: $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. $\square$

Lemma 12.48 (Full-basis coefficient identity from matched phases)**.** Lean ✓ Stmt *Let P be a BNT canonical form, let Q be a sector decomposition, and assume P_{tot} and Q_{tot} have the same MPV family. Suppose that there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

and scalars $\zeta_k \in \mathbb{C}$ such that, for every $k \in J(Q)$, every length N, and every spin configuration σ of length N,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N \, V^{(N)}(P_{\beta(k)})_\sigma.$$

Then for every $k \in J(Q)$ there is N_0 such that, for all $N > N_0$,

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q).$$

This is the coefficient-comparison step of [CPGSV16, Appendix MPV proof, lines 1187–1188] in full-basis form.

Proof. ✓ Proof For every N, expand the equality of total MPVs as

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

For every N, the phase relation gives $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$, hence

$$\sum_{j \in J(P)} c_N^{(j)}(P)|V^{(N)}(P_j)\rangle = \sum_{j \in J(P)} \zeta_{\beta^{-1}(j)}^N c_N^{(\beta^{-1}(j))}(Q)|V^{(N)}(P_j)\rangle.$$

Lemma 12.4 applied to the P-basis gives $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N. □

Lemma 12.49 (Full-basis coefficient identity from global gauges). Lean ✓ Stmt *Let P and Q be BNT canonical forms with the same MPV family. Suppose that there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

and, for every $k \in J(Q)$, a bond-dimension equality and a gauge-phase equivalence between Q_k and $P_{\beta(k)}$. Then for every k there are a unit-modulus scalar ζ_k and a natural number N_0 such that, for all $N > N_0$,

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q).$$

This is the full-basis equal-MPV coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188].

Proof. ✓ Proof For each matched sector, the gauge-phase equivalence gives $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. Lemma 12.46 gives $|\zeta_k| = 1$. Applying Lemma 12.48 to the same phases gives $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N. □

Definition 12.50 (Copy-weight matching for matched BNT sectors). Lean ✓ Stmt Let the sectors of two decompositions P and Q be matched by $\beta : J(Q) \simeq J(P)$, and let ζ_k be the phase attached to sector k. A copy-weight matching consists of copy permutations τ_k and the following identities: for all sectors k and copies q,

$$\nu_{k,q} = \zeta_k^{-1} \, \mu_{\beta(k),\tau_k(q)}.$$

This is the copy-level conclusion supplied by the equal-MPV coefficient comparison in [CPGSV16, Appendix MPV proof, line 1188]. For the proportional theorem it remains the separate coefficient-comparison input.

Lemma 12.51 (Copy-weight matching from matched-sector coefficient identities). Lean ✓ Stmt *Let the sectors of two decompositions P and Q be matched by $\beta : J(Q) \simeq J(P)$, and let $\zeta_k \neq 0$ be the phase attached to sector k. Suppose that for every k there is N_0 such that, for all $N > N_0$,*

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q).$$

Then the phases ζ_k determine a copy-weight matching: after a copy permutation inside each matched sector,

$$\nu_{k,q} = \zeta_k^{-1} \, \mu_{\beta(k),\tau_k(q)}.$$

This is, sector by sector, the content of the coefficient comparison in [CPGSV16, Appendix MPV proof, line 1188].

Proof. ✓ Proof For all sufficiently large N, the coefficient identity in sector k is

$$\sum_r \mu_{\beta(k),r}^N = \zeta_k^N \sum_q \nu_{k,q}^N = \sum_q (\zeta_k \nu_{k,q})^N.$$

Lemma 12.42 applies the finite power-sum comparison sector by sector, producing a copy permutation τ_k with $\mu_{\beta(k),\tau_k(q)} = \zeta_k \nu_{k,q}$. Since $\zeta_k \neq 0$, this is equivalent to $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$. □

Theorem 12.52 (Global gauge from matched sectors and copy weights). Lean ✓ Stmt *Suppose that the sectors of two sector decompositions P and Q are matched by a bijection $\beta : J(Q) \simeq J(P)$, with bond-dimension equalities, copy permutations τ_k, nonzero phases ζ_k, and block gauges X_k. If, for every sector k, copy q, and physical index i,*

$$Q_k^i = \zeta_k\, X_k P_{\beta(k)}^i X_k^{-1}, \qquad \nu_{k,q} = \zeta_k^{-1}\, \mu_{\beta(k),\tau_k(q)},$$

then, in the flattened Q-coordinates,

$$Q_{\text{flat}}^i = X\, P_{\text{matched}}^i\, X^{-1}, \qquad X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is the direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192], separated from the coefficient-comparison step that supplies the weight identities.

Proof. ✓ Proof For each matched copy,

$$\nu_{k,q} Q_k^i = \left(\zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}\right)\left(\zeta_k X_k P_{\beta(k)}^i X_k^{-1}\right) = \mu_{\beta(k),\tau_k(q)} X_k P_{\beta(k)}^i X_k^{-1}.$$

Thus every flattened weighted block is conjugated by its sector gauge X_k. Lemma 16.41 assembles these block conjugacies into the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. □

Theorem 12.53 (Global gauge from a copy-weight matching). Lean ✓ Stmt *Suppose that the sectors of P and Q are matched, with bond-dimension equalities, nonzero phases, and block gauges satisfying*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

If the same phases also give a copy-weight matching, then the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is only a reformulation of Theorem 12.52.

Proof. ✓ Proof The copy-weight matching supplies $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$. Together with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, these are exactly the two displayed hypotheses of Theorem 12.52. □

Theorem 12.54 (Conditional proportional global gauge from a copy-weight matching). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Assume that every sector of P and every sector of Q has at least one copy weight of modulus 1. Then the proportional sector-matching theorem gives β, the bond-dimension equalities, the unit phases ζ_k, and the block gauges X_k. If the remaining proportional coefficient comparison supplies a copy-weight matching for these same phases, then the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge*

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

Proof. ✓ Proof The proportional sector-matching theorem supplies β, X_k, and unit phases ζ_k satisfying $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $|\zeta_k| = 1$, each ζ_k is nonzero. A copy-weight matching gives $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$, so Theorem 12.53 gives the direct-sum gauge. □

Theorem 12.55 (Conditional proportional global gauge from coefficient identities). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Assume that every sector of P and every sector of Q has at least one copy weight of modulus 1. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k, and block gauges X_k such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$$

for every sector k and physical index i. If these same phases satisfy the matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q)$$

for all sufficiently large N, then they determine a copy-weight matching and hence the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

Proof. ✓ Proof The sector-matching theorem gives β, X_k, and $|\zeta_k| = 1$ with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $\zeta_k \neq 0$, the coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ produce $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$ by Lemma 12.51. Theorem 12.54 then applies to this copy-weight matching. □

Lemma 12.56 (Equal-MPV matched copy-weight witnesses). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Assume that every sector of P and every sector of Q has at least one copy weight of modulus 1. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k, and block gauges X_k such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

For the same phases there is a copy-weight matching: after a permutation of the copies inside each matched sector,

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}.$$

Proof. ✓ Proof The full-basis matching theorem gives the bijection and block gauges. The self-overlap comparison

$$\langle V_N(Q_k), V_N(Q_k)\rangle = |\zeta_k|^{2N} \langle V_N(P_{\beta(k)}), V_N(P_{\beta(k)})\rangle$$

together with the BNT normalization limits on both sides forces $|\zeta_k| = 1$. The matched MPV phase identities also give the eventual coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Lemma 12.51 applies the finite power-sum comparison in each sector and gives the copy-weight identities. □

Theorem 12.57 (Full-basis global gauge in matched coordinates). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Assume that every sector of P and every sector of Q has at least one copy weight of modulus 1. Then there exist:*

- *a bijection $\beta : J(Q) \simeq J(P)$,*
- *bond-dimension equalities* $\dim P_{\beta(k)} = \dim Q_k$,

- *copy-number equalities $r_{\beta(k)}(P) = r_k(Q)$ and copy permutations τ_k,*
- *unit-modulus phases ζ_k with $|\zeta_k| = 1$ and block gauges X_k,*

such that, for every sector k, copy q, and physical index i,

$$Q_k^i = \zeta_k\, X_k P^i_{\beta(k)} X_k^{-1}, \qquad \nu_{k,q} = \zeta_k^{-1}\, \mu_{\beta(k),\tau_k(q)}.$$

Moreover, if

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$$

is the flattened block-diagonal gauge, then in the flattened Q-coordinates,

$$Q^i_{\text{flat}} = X\, P^i_{\text{matched}}\, X^{-1}$$

for every physical index i. This is the explicit direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192].

Proof. ✓ Proof Lemma 12.56 gives β, the bond-dimension identifications, the phases ζ_k, the block gauges X_k, and copy permutations τ_k satisfying

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}.$$

Theorem 12.53 assembles the block conjugacies into the flattened direct-sum conjugation by $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. □

Theorem 12.58 (Literal equal-MPV gauge after sector-coordinate permutation). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Assume that every sector of P and every sector of Q has at least one copy weight of modulus 1. Then the total bond dimensions agree; write their common value as D. There is*

$$Y \in \mathrm{GL}_D(\mathbb{C})$$

such that, for every physical index i,

$$Q^i_{\text{tot}} = Y\, P^i_{\text{tot}}\, Y^{-1},$$

where P^i_{tot} and Q^i_{tot} are both regarded as matrices in $M_D(\mathbb{C})$ using the total bond-dimension equality. This is the normalized BNT form of [CPGSV16, Corollary II.2 and Appendix MPV proof, lines 1189–1192], under the same unit-modulus-copy normalization as Theorem 12.57.

Proof. ✓ Proof Apply Theorem 12.57 to obtain the flattened gauge after the sectors of P have been reindexed by the bijection between basis blocks of P and Q. The sector-coordinate permutation ρ, induced by this bijection, the copy permutations, and the bond-dimension equalities, identifies the reindexed P-side direct sum with the total tensor P_{tot}, transported along the total bond-dimension equality. Composing the permutation gauge for ρ with the sector-permuted gauge gives the displayed single matrix Y. □

Chapter 13

Proof of the Fundamental Theorem

The final aperiodic proof of the Fundamental Theorem is separated here into three kinds of statements.

1. Section 13.1 records the source statements from the literature. They remain marked as not ready until the canonical-form reduction, BNT representative comparison, repeated-copy coefficient comparison, and global gauge conclusion have all been proved together from the source hypotheses.

2. Sections 13.2 and 13.3 state the proved comparison results: homogeneous pair-span padding, BNT coefficient comparison, sector-weight multiset recovery, sector-basis matching, and the conditional equal-MPV sector theorem.

3. Section 13.4 records the overlap and BNT preparation used to pass from primitive irreducible blocks to linearly independent BNT representatives.

The equal-MPV comparison used here is the SectorBNT coefficient comparison of Chapter 12. This chapter records where that comparison enters the Fundamental Theorem proof. The comparison is made at fixed sufficiently large system size. If a coefficient belongs to a non-dominant sector, it may decay with N, but BNT linear independence still detects whether it is zero. No projection-limit argument for non-dominant sectors is used.

Normality appears in two forms in this proof. The algebraic form is eventual exact-length spanning, $S_L(A) = M_D(\mathbb{C})$ for some L. Spectral primitivity enters only through separate implications, such as the primitive-to-normality results of Chapter 8. When the statements below assume trace-preserving, primitive, irreducible blocks, the passage to BNT representatives uses those separate implications rather than redefining normality.

The proof follows the aperiodic comparison in [CPGSV17a]: block to remove non-trivial periodicity, choose BNT representatives, compare coefficients using linear independence, recover weights by power sums, and then construct the block-diagonal gauge. The periodic irreducible-form theorem with its additional Z-gauge degree of freedom is treated separately in Chapter 15.

13.1 Source theorems

The next two statements are the literature-level canonical-form theorems. The proved intermediate statements used toward them appear in the later sections of this chapter and in Chapter 12.

Theorem 13.1 (Multi-block fundamental theorem for normal tensors). Not ready *Let A and B be MPS tensors in canonical form. Write their bases of normal tensors as $(A_k)_{k=1}^{g_a}$ and $(B_\ell)_{\ell=1}^{g_b}$. If, for every system size N, the MPVs generated by A and B are proportional, then $g_a = g_b$. After relabelling the basis of normal tensors of A, there are scalars $\zeta_k \in \mathbb{C}$ with $|\zeta_k| = 1$ and invertible matrices Y_k such that, for every k and every physical index i,*

$$B_k^i = \zeta_k \, Y_k \, A_k^i \, Y_k^{-1}.$$

Thus the paired normal tensors have the same bond dimension and the same gauge-phase class.

This is the multi-block theorem of [CPGSV17a, Theorem II.1]; see also [CPGSV21, Theorem 4.5 and Corollary IV.5]. The same-MPV input is reduced to common primitive block decompositions by Theorem 10.9 of Chapter 10. The BNT matching and coefficient comparison are developed in Chapter 12. The per-block gauge-phase witnesses, copy-weight identities, and block-diagonal gauge construction are then combined in Theorem 12.57. The same-index direct-sum corollaries of Chapter 16 are useful algebraic consequences, but they are not the BNT block-matching theorem itself.

Theorem 13.2 (Equal canonical-form representations are globally conjugate). Not ready *Let A and B be canonical-form representations generating the same MPV family for every system size N. Then the matrix dimensions of A^i and B^i coincide, and there is one invertible matrix X such that, for every physical index i,*

$$A^i = X \, B^i \, X^{-1}.$$

Thus the equal case removes the blockwise phase freedom of the proportional theorem and gives a single global conjugacy.

This is the equal-case corollary of [CPGSV17a, Corollary II.2]; see also [CPGSV21, Corollary IV.5]. The coefficient identities for the paired sectors are supplied by the BNT comparison of Chapter 12; the weight recovery uses Newton–Girard. The resulting global conjugacy is recorded in Theorem 12.58.

13.2 Homogeneous Pair-Span Padding

The pair-span padding lemmas below are algebraic statements about fixed-length word spans in the product algebra. They convert a full homogeneous pair-word span at one positive length into the identity pair at all sufficiently large lengths, which is the exact-length form needed in BNT separation.

Lemma 13.3 (Identity pair from full homogeneous pair-word span). Lean Lean Lean Lean ✓ Stmt *Let A and B be MPS tensors of bond dimensions D_1 and D_2.*

1. *If the homogeneous pair-word span is the full product algebra at length n,*

$$\mathrm{span}_{\mathbb{C}}\{(A^w, B^w) : |w| = n\} = M_{D_1}(\mathbb{C}) \times M_{D_2}(\mathbb{C}),$$

then the identity pair $(\mathbb{1}_{D_1}, \mathbb{1}_{D_2})$ belongs to the same homogeneous span.

2. *If the span is full at one positive length $S > 0$, then there is $L \in \mathbb{N}$ such that, for every $n \geq L$,*

$$(\mathbb{1}_{D_1}, \mathbb{1}_{D_2}) \in \mathrm{span}_{\mathbb{C}}\{(A^w, B^w) : |w| = n\}.$$

Proof. ✓ Proof Statement (1) is immediate from equality with the full product algebra. For (2), fullness at a positive length propagates to all larger lengths by the positive-length span-propagation lemma. Applying (1) at each propagated length gives the identity pair in every sufficiently large homogeneous span. ☐

13.3 Equal-MPV Comparisons

The equal-MPV proof is a coefficient comparison. In the one-copy aligned case, write

$$A^i = \bigoplus_{j=1}^{g} \mu_j^A A_j^i, \qquad B^i = \bigoplus_{j=1}^{g} \mu_{\pi(j)}^B B_{\pi(j)}^i,$$

and suppose the paired basis blocks satisfy

$$B_{\pi(j)}^i = e^{\mathrm{i}\phi_j} X_j A_j^i X_j^{-1}.$$

Then equality of the total MPVs gives, for every N,

$$\begin{aligned} 0 &= |V^{(N)}(A)\rangle - |V^{(N)}(B)\rangle \\ &= \sum_{j=1}^{g} \left[(\mu_j^A)^N - (\mu_{\pi(j)}^B e^{\mathrm{i}\phi_j})^N \right] |V^{(N)}(A_j)\rangle. \end{aligned}$$

For all sufficiently large N, BNT linear independence forces each displayed coefficient to vanish. Since the weights are nonzero and the equality holds for all consecutive sufficiently large N, one obtains $\mu_j^A = \mu_{\pi(j)}^B e^{\mathrm{i}\phi_j}$. The phases are then absorbed into the weights, and the block gauges X_j form the block-diagonal global gauge.

The repeated-copy and equal-modulus part of the comparison is not restated as a separate theorem in this chapter. Its reference is Chapter 12: the coefficient identity Lemma 12.48, the copy-weight recovery Lemma 12.51, and the global-gauge Theorem 12.57. No strict ordering of the moduli is used in these SectorBNT statements; strict-modulus or one-copy comparisons are special cases, not substitutes for the literature theorem.

The source equal-MPV corollary says that canonical-form tensors generating the same MPV family for all system sizes are globally conjugate [CPGSV17a, Corollary II.2]. The BNT block-count, bond-dimension, gauge-phase, copy-count, and weight-multiset conclusions are recorded in Chapter 12. The coordinate form of the same result is recorded there under the unit-modulus-copy normalization.

Theorem 13.4 (Bond dimension equality from unit overlap)**.** Lean ✓ Stmt *Let A and B be irreducible trace-preserving blocks, possibly with different bond dimensions. If*

$$\lim_{N\to\infty} \left|\langle V^{(N)}(B)|V^{(N)}(A)\rangle\right| = 1,$$

then the two bond dimensions are equal.

Proof. ✓ Proof The displayed quantity is the modulus of the Hilbert-space inner product; it agrees in modulus with the mixed MPV overlap. The proof is the contrapositive of rectangular overlap decay: if the two bond dimensions differ, then

$$\lim_{N\to\infty} \left|\langle V^{(N)}(B)|V^{(N)}(A)\rangle\right| = 0,$$

contradicting the assumed convergence of this modulus to 1. ☐

Lemma 13.5 (Common-block equal-MPV comparison)**.** *Fix a common block structure: the same number of blocks g, the same bond dimensions $(D_k)_{k=1}^g$, and the same weights $(\mu_k)_{k=1}^g$. Let $(\mu_k, A_k)_{k=1}^g$ and $(\mu_k, B_k)_{k=1}^g$ be two families in canonical form with BNT separation on this common structure. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) is gauge equivalent, and hence the corresponding block-diagonal tensors are gauge equivalent.*

Proof. This is the one-copy, already aligned subcase of the BNT global-gauge theorem. In Chapter 12, specialize Theorem 12.57 to sector decompositions P and Q with the same (g, D_k, μ_k) block structure. The basis permutation is the identity, the per-block matrices X_k give the pairwise gauge equivalences, and the block-diagonal matrix $X = \bigoplus_k X_k$ gives the global conjugation on $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$. □

13.3.1 Sector multiplicities and weight recovery

This subsection records the coefficient comparison for sectors with repeated copies. If the jth basis block occurs with nonzero weights $\mu_{j,1}, \dots, \mu_{j,r_j}$, its contribution at length N is

$$c_{S,j}^{(N)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

After the basis blocks have been paired up to gauge phase, both sides are written in the same BNT basis. The full family of basis MPV states is linearly independent for all sufficiently large N, so equality of total MPVs separates the coefficient arrays $c_{S,j}^{(N)}$. Equality of those power sums for all sufficiently large N determines both the multiplicity r_j and the multiset of weights.

The same scalar power-sum algebra appears in the PEPS application of [CPGSV17a, Theorem IV.13], where structure coefficients are written as

$$c_{\alpha,\beta,\gamma}^{(L)} = \operatorname{tr}\left(\chi_{\alpha,\beta,\gamma}^L\right), \qquad m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta.$$

In the source appendix the vertical canonical forms are $\bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ and $\bigoplus_\beta \nu_\beta \otimes A_\beta$, with $m_\alpha = \operatorname{tr}(\mu_\alpha)$ and $n_\beta = \operatorname{tr}(\nu_\beta)$. There μ_α and ν_β are diagonal positive matrices; only their traces enter as the scalar coefficients compared here.

The base sector-weight and sector-decomposition data types are recorded once, in Chapter 12 (Definition 12.22).

Lemma 13.6 (Sector decomposition formula)**.** Lean ✓ Stmt *For a sector decomposition S with associated tensor A_{tot}, the MPV satisfies*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g} c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Lemma 13.7 (Eventual coefficient comparison from BNT linear independence)**.** Lean ✓ Stmt *If two sector-weight tuples S, T over the same basis, with coefficient arrays $c_{S,j}^{(N)}$ and $c_{T,j}^{(N)}$, produce equal total MPVs and the basis is eventually linearly independent (the BNT property), then there is N_0 such that*

$$\forall N > N_0, \ \forall j, \quad c_{S,j}^{(N)} = c_{T,j}^{(N)}.$$

Lemma 13.8 (BNT recovery of copy counts and sector weights). Lean ✓ Stmt *Let S and T be two sector-weight tuples over the same basis, without assuming their multiplicities agree. If the basis is eventually linearly independent and the total MPVs of S and T agree, then there are copy-count equalities $r_j^S = r_j^T$ for every basis block j, and, after identifying the corresponding copy index sets, the sector weight multisets are equal for every j:*

$$\{\mu_{j,q}^S : 1 \le q \le r_j^S\} = \{\mu_{j,q}^T : 1 \le q \le r_j^T\}.$$

Proof. ✓ Proof BNT linear independence and equality of total MPVs give eventual coefficient equality by Lemma 13.7. Thus, for each basis sector j, the two finite nonzero weight families satisfy

$$\sum_{q=1}^{r_j^S} (\mu_{j,q}^S)^N = \sum_{q=1}^{r_j^T} (\mu_{j,q}^T)^N$$

for all sufficiently large N. The signed-geometric-sequence argument extends this to every positive exponent. Applying the finite-range unequal-cardinality power-sum theorem (Theorem 12.19) gives both $r_j^S = r_j^T$ and equality of the sector weight multisets. □

Lemma 13.9 (Phase matching recovers copy counts and sector weights). Lean ✓ Stmt *Let P and Q be sector decompositions. Assume there is a permutation π pairing the basis blocks and, for each j, a nonzero complex number ζ_j such that*

$$V^{(N)}(Q_{\pi(j)})_\sigma = \zeta_j^N V^{(N)}(P_j)_\sigma$$

for every N and σ. If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then the copy counts satisfy $r_j^P = r_{\pi(j)}^Q$, and after multiplying the weights of $Q_{\pi(j)}$ by ζ_j the sector weight multisets agree.

Proof. ✓ Proof Absorb the factors ζ_j into the sector weights of Q and use the basis blocks of P. The phase-matching hypothesis and the sector decomposition formula show that the resulting decomposition has the same total MPV family as Q, hence also as P. This is the shared-basis case of Lemma 13.8, which recovers the copy-count equalities and the claimed equality of weight multisets. □

Lemma 13.10 (Two-basis sector comparison after phase matching). Lean ✓ Stmt *Let P and Q be sector decompositions. Assume there is a permutation π pairing the basis blocks, the multiplicities satisfy $r_j^P = r_{\pi(j)}^Q$, and for each j there is a nonzero complex number ζ_j such that*

$$V^{(N)}(Q_{\pi(j)})_\sigma = \zeta_j^N V^{(N)}(P_j)_\sigma$$

for every N and σ. If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ by multiplication by ζ_j.

Proof. ✓ Proof Lemma 13.9 gives copy-count equalities and the corresponding weight-multiset equality. The recovered copy-count equalities coincide with the assumed ones, so the weight-multiset equality gives the conclusion. □

Lemma 13.11 (Gauge-phase paired bases imply phase-matched sector comparison). Lean ✓ Stmt *Let P and Q be sector decompositions. Assume there is a permutation π pairing the basis blocks, the multiplicities satisfy $r_j^P = r_{\pi(j)}^Q$, and each corresponding pair of basis blocks is gauge-phase equivalent. If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ after multiplication by the corresponding gauge phase.*

Proof. ✓ Proof Gauge-phase equivalence gives the required phase-scaling identity for the MPVs by Theorem 2.27. The conclusion then follows from Lemma 13.10. □

The sector-basis matching predicates used by the comparison results below are recorded in Chapter 12: Definition 12.23 (sector basis matching), Definition 12.24 (sector basis pre-matching), Definition 12.25 (single-family overlap-orthogonality hypotheses), and Definition 12.26 (primitive overlap-span hypotheses).

Lemma 13.12 (Single-family overlap hypotheses plus span give overlap-span hypotheses). Lean ✓ Stmt *If two sector bases each satisfy the single-family overlap-orthogonality hypotheses, and one also supplies one-site injectivity of all their basis blocks together with equality of the finite-length MPV spans, then the two bases satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof Combine the one-family hypotheses on the two sides with the remaining injectivity and span-comparison hypotheses. These are exactly the assertions in Definition 12.26. □

Lemma 13.13 (A pre-matching gives a sector basis matching). Lean ✓ Stmt *Let M be a sector basis pre-matching between P and Q. If the total MPV families agree and the basis MPVs of P are eventually linearly independent, then the copy-count equalities recovered by the phase-matched BNT coefficient comparison turn M into a full sector basis matching.*

Proof. ✓ Proof Gauge-phase equivalence gives the required phase-scaling identities for the corresponding basis MPVs:

$$V^{(N)}(Q_{\pi(j)})_\sigma = \zeta_j^N V^{(N)}(P_j)_\sigma.$$

Lemma 13.9 then recovers the missing copy-count equalities, which are precisely the extra assertions needed for Definition 12.23. □

Theorem 13.14 (Overlap rigidity gives a sector basis matching). Lean ✓ Stmt *Assume the basis blocks of two sector decompositions have nonzero bond dimensions, are trace-preserving and injective, have asymptotically orthonormal overlaps within each family, and span the same MPV subspace at every system size. If the total MPV families also agree, then there exists a sector basis matching between the two decompositions.*

Proof. ✓ Proof The permutation-rigidity lemma for bases of normal tensors gives a permutation, the paired bond dimensions, and the gauge-phase equivalences of the basis blocks, hence a sector basis pre-matching. Asymptotic orthonormality gives the BNT linear-independence condition, and Lemma 13.13 recovers the copy-count equalities from equality of the total MPV families. □

Lemma 13.15 (Primitive overlap-span hypotheses give a sector basis matching). Lean ✓ Stmt *If two sector decompositions satisfy the primitive overlap-span hypotheses and their total MPV families agree, then there exists a sector basis matching between them.*

Proof. ✓ Proof The collected hypotheses are exactly the nonzero-dimension, injectivity, normalization, overlap, and span assumptions of Theorem 13.14. Applying that theorem gives the matching witness. □

Theorem 13.16 (Span equality for representative sector bases). Lean ✓ Stmt *Let two nonzero-weight block families be trace-preserving, primitive, irreducible, injective, and carry nonzero weights. If their finite-length MPV spans agree for every system size, then there exist sector decompositions P and Q whose total tensors have the same MPVs as the two weighted nonzero-block tensors, whose bases satisfy BNT linear independence, and whose sector bases satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof Choose one MPV phase-class representative on each side. The one-sided positive-dimension, injectivity, normalization, and overlap hypotheses give all one-family assertions in Definition 12.26. The assumed nonzero-block span equality is preserved when each block is replaced by a phase-equivalent representative. This supplies the two-family span equality for the sector bases. □

The MPV phase-equivalence and common-phase-cover predicates are recorded in Chapter 12 (Definition 12.27 and Definition 12.28).

Lemma 13.17 (Common MPV phase cover gives nonzero-block span equality)**.** Lean ✓ Stmt *Let two finite nonzero-weight block families map onto a third finite family of common blocks. Suppose both maps are surjective, and each block has finite-length MPV vectors equal to a nonzero scalar power times the MPV vectors of its common block. Then, for every system size, the two nonzero-weight block families span the same MPV subspace.*

Proof. ✓ Proof Each nonzero-block MPV vector lies in the span of the common family because it is a scalar multiple of its common block's MPV vector. Surjectivity gives the reverse inclusion, so each nonzero-block span equals the common span; transitivity gives the equality of the two nonzero-block spans. □

Lemma 13.18 (Common phase cover gives nonzero-block span equality)**.** Lean ✓ Stmt *A common MPV phase cover for two nonzero-weight block families gives equality of their finite-length MPV spans at every system size.*

Proof. ✓ Proof The common cover specifies the common family, the two surjective class maps, and the MPV-phase equivalences. The span lemma then applies directly. □

Lemma 13.19 (Common phase cover from a BNT block-matching conclusion)**.** Lean ✓ Stmt *If a BNT block-matching comparison has produced a permutation of the two basis families together with paired bond dimensions and gauge-phase equivalences, then the two basis families have a common MPV phase cover.*

Proof. ✓ Proof Take the left basis family as the common family. The comparison permutation and its inverse give the two class maps, while each gauge-phase equivalence gives the required MPV-phase equivalence. □

Lemma 13.20 (Common phase cover from a bijective phase matching)**.** Lean ✓ Stmt *Let $(A_s)_{s\in S}$ and $(B_t)_{t\in T}$ be two finite block families. Suppose there is a bijection $\sigma : S \to T$ and, for each $s \in S$, a scalar $\omega_s \in \mathbb{C}^\times$ such that*

$$|V^{(N)}(B_{\sigma(s)})\rangle = \omega_s^N |V^{(N)}(A_s)\rangle$$

for every length $N \geq 0$, with $|V^{(0)}(C)\rangle$ interpreted as the bond dimension of C. For each corresponding pair, the case $N = 0$ records equality of bond dimensions. Then the two block families admit a common MPV phase cover.

Proof. ✓ Proof Use the first family as the common family. The two covering maps are the identity on S and the inverse of σ. The displayed phase relation provides the required phase factor for the corresponding block of the second family. □

Lemma 13.21 (Span equality from a BNT block-matching comparison)**.** Lean ✓ Stmt *A BNT block-matching comparison gives equality, for every system size, of the finite-length MPV spans of the two nonzero-weight block families.*

Proof. ✓ Proof First form the common MPV phase cover from the block-matching comparison. The common phase-cover span lemma then gives the stated span equality. □

Lemma 13.22 (Span equality from a bijective phase matching). Lean ✓ Stmt *A bijective phase matching gives equality, for every system size, of the finite-length MPV spans of the two block families.*

Proof. ✓ Proof First form the common MPV phase cover from the bijective phase matching. The common phase-cover span lemma then gives the stated span equality. □

Theorem 13.23 (Overlap-span hypotheses from a common phase cover). Lean ✓ Stmt *Let two block families be trace-preserving, primitive, irreducible, injective, and carry nonzero weights. If they have a common MPV phase cover, then there exist BNT sector decompositions P and Q for the two weighted nonzero parts, and the sector bases satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof The common cover gives finite-length span equality for the two nonzero-weight block families. The phase-class representative construction then carries this span equality to the BNT sector bases and supplies the remaining one-family overlap hypotheses. □

Theorem 13.24 (Sector comparison via a sector-basis matching). Lean ✓ Stmt *Given two sector decompositions P and Q, a sector basis matching (Definition 12.23) between them, eventual linear independence of the basis MPV states of P, and equality of the total MPV families of P and Q, one obtains nonzero scalars $(\zeta_j)_j$ such that for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ by multiplication by ζ_j.*

Proof. ✓ Proof The sector-basis matching contains exactly the permutation, copy, and gauge-phase hypotheses of Lemma 13.11, so the conclusion follows directly from that lemma. □

The blocked-word identification and the resulting unconditional common primitive block decomposition are part of the after-blocking reduction; see Theorem 10.10.

Remark 13.25 (Equal-case comparison after canonical reduction). The source proof starts after both tensors have been put in canonical form and after a basis of normal tensors has been chosen. In that notation the BNT expansion has the form

$$|V^{(N)}(A)\rangle = \sum_{j=1}^{g}\left(\sum_{q=1}^{r_j^A}(\mu_{j,q}^A)^N\right)|V^{(N)}(A_j)\rangle,$$

and similarly for B. The proportional theorem of [CPGSV16, CPGSV21] first pairs the BNT representatives: after relabelling,

$$B_j^i = e^{\mathrm{i}\phi_j}Y_jA_j^iY_j^{-1}.$$

The equal case then uses the SectorBNT coefficient comparison of Chapter 12: Lemma 12.49 gives the matched coefficient identities, Lemma 12.51 recovers multiplicities and weight multisets, and Theorem 12.57 assembles the block gauges Y_j into the block-diagonal global conjugating matrix. This is the equal-case corollary [CPGSV17a, Corollary II.2], restated in [CPGSV21, Corollary IV.5].

The comparison theorems above start after the common blocked nonzero sectors and their BNT block comparison have been supplied. The structural reduction gives a common positive blocking length and decompositions

$$A^{[p]} \sim \mathbf{0}_{d^p,z_A} \oplus A_{\mathrm{nz}}, \qquad B^{[p]} \sim \mathbf{0}_{d^p,z_B} \oplus B_{\mathrm{nz}}, \qquad z_A = z_B,$$

with trace-preserving, primitive, irreducible nonzero sectors. The remaining inputs for the unrestricted source theorem are one-site injectivity of those sectors and the BNT block comparison for the representative families.

Common phase covers are only a criterion for the finite-length span equality needed by the sector-comparison theorem. Lemma 13.18 turns a common phase cover into span equality, while Lemma 13.19 constructs such covers from BNT block-matching information.

13.4 Overlap and BNT preparation for the Fundamental Theorem

The following results supply the single-block and basis-of-normal-tensors inputs for the sector comparison. The proportional single-block theorem gives gauge-phase equivalence from proportional MPVs. The overlap-limit criterion then gives eventual linear independence of the chosen BNT representatives. The same-index direct-sum constructions are collected in Chapter 16.

13.4.1 Proportional single-block ingredient

Theorem 13.26 (Proportional MPV implies gauge-phase equivalence). Lean ✓ Stmt *Let A and B be injective MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If A and B generate proportional MPV families ($V^{(N)}(A)_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$) and $O_{AA}(N) \to 1$, $O_{BB}(N) \to 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Suppose A and B are not gauge-phase equivalent. Then the overlap-decay theorem gives $O_{AB}(N) \to 0$. The proportionality hypothesis gives

$$O_{AB}(N) = c_N O_{BB}(N), \qquad O_{AA}(N) = |c_N|^2 O_{BB}(N).$$

Since $O_{BB}(N) \to 1$, the first identity implies $c_N \to 0$. The second identity then gives $O_{AA}(N) \to 0$, contradicting $O_{AA}(N) \to 1$. □

13.4.2 Equal-norm primitive irreducible blocks (BNT input)

Theorem 13.27 (Gauge phases have unit modulus for primitive irreducible blocks). Lean ✓ Stmt *For trace-preserving primitive irreducible blocks, every gauge-phase equivalence has a phase factor of modulus 1.*

Proof. ✓ Proof The gauge-phase equivalence $B^i = \zeta\, X A^i X^{-1}$ scales every MPV coefficient by ζ^N,

$$V^{(N)}(B)_\sigma = \zeta^N\, V^{(N)}(A)_\sigma,$$

so the self-overlaps satisfy

$$O_{BB}(N) = |\zeta|^{2N}\, O_{AA}(N).$$

For primitive irreducible trace-preserving blocks the self-overlaps converge in modulus, $\|O_{AA}(N)\| \to 1$ and $\|O_{BB}(N)\| \to 1$, so $|\zeta|^{2N} \to 1$, which forces $|\zeta| = 1$. □

Theorem 13.28 (Existential BNT independence from overlap limits). Lean ✓ Stmt *If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.*

Proof. ✓ Proof This is Lemma 12.3, restated with an explicit threshold N_0. □

Theorem 13.29 (Granular sector decomposition from eventual BNT independence). Lean ✓ Stmt *If the MPV states of the blocks are linearly independent for all sufficiently large system sizes and all weights are nonzero, then the granular sector decomposition represents the same MPS family and satisfies the BNT linear-independence condition.*

Proof. ✓ Proof The equality of represented families is Lemma 9.41. The assumed eventual linear independence is exactly the BNT condition for the basis blocks of the granular sector decomposition. □

The MPV phase-class data type is recorded in Chapter 12 (Definition 12.29).

Theorem 13.30 (Phase-class BNT sector construction). Lean ✓ Stmt *For trace-preserving primitive irreducible blocks of positive bond dimension with nonzero weights, there is a sector decomposition representing the same weighted block sum and satisfying the BNT linear-independence condition. The construction chooses one representative per MPV phase class and absorbs the nonzero scalar factors into the sector weights.*

Proof. ✓ Proof Finite phase classes partition the original block indices. Expanding the sector decomposition gives the original finite sum after replacing each class member by its representative and moving the factor ζ^N into $(\zeta\mu_k)^N$. If two chosen representatives were gauge-phase equivalent, they would lie in the same MPV phase class, contradicting their being distinct classes. Therefore the representatives satisfy the separated primitive-normal hypothesis, and the overlap-limit criterion (Theorem 13.28) gives the BNT linear-independence condition. □

Chapter 14

Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem [PGVWC07], following the symmetry analysis of [PGWS+08]. For injective tensors, physical on-site symmetries determine virtual gauges, those gauges define projective representations on the bond space, and virtual intertwiners enter the string-order criteria.

14.1 Physical Symmetries Give Virtual Gauges

Definition 14.1 (Physical-index rotation). Lean ✓ Stmt Given a matrix $u \in M_d(\mathbb{C})$ and an MPS tensor A, the *physical-index rotation* is the tensor $(A_u)^i := \sum_j u_{ij} A^j$.

Corollary 14.2 (Periodic symmetry yields gauge equivalence up to a root of unity). *Lean* *✓ Stmt* *Let A be in irreducible form II and let u be a matrix such that $B := A_u$ is also in irreducible form II and $\mathcal{V}(A) = \mathcal{V}(B)$. Then there exists $m > 0$ such that A and B are gauge equivalent up to an m-th root of unity.*

Proof. ✓ Proof Direct application of the periodic equal-case Fundamental Theorem (Theorem 15.24) to A and B. □

14.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 14.3 (Twisted tensor). Lean ✓ Stmt Given a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g-twisted tensor* is defined, for each $i \in \{0, \dots, d-1\}$, by

$$\widetilde{A}^i_g := \sum_{j=0}^{d-1} U(g)_{ij}\, A^j.$$

Definition 14.4 (On-site symmetry). Lean ✓ Stmt A tensor A is *on-site symmetric* under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$ for every $g \in G$.

Lemma 14.5 (Functoriality of twisting). *Lean* *✓ Stmt* *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g,$$

i.e. twisting by gh is the same as first twisting by h and then by g.

Proof. ✓ Proof Expanding both sides using $U(gh) = U(g)\,U(h)$, then swapping the order of summation. ☐

Lemma 14.6 (Identity twist). Lean ✓ Stmt *Twisting by the identity group element is trivial: $\widetilde{A}_1 = A$.*

Proof. ✓ Proof Since $U(1) = \mathbb{1}_d$, each component reduces to $\sum_j \delta_{ij}\, A^j = A^i$. ☐

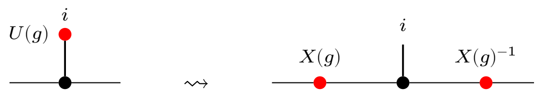
Lemma 14.7 (Symmetric twists are gauge equivalent). Lean ✓ Stmt *If A is injective and on-site symmetric under U, then for every $g \in G$ the twisted tensor $\widetilde{A}_g$ is gauge equivalent to A.*

Proof. ✓ Proof On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$, and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence. ☐

Remark 14.8 (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from string-order rigidity. The on-site symmetry condition first identifies A and $\widetilde{A}_g$ as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 14.7. Gauge uniqueness and the twist composition law then give Theorem 14.15. The string-order results later in this chapter use this virtual representation as an input.

Theorem 14.9 (Virtual symmetry equation). Lean ✓ Stmt *If A is injective and on-site symmetric under U, then for every $g \in G$ there exist an invertible matrix $X(g) \in \mathrm{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\widetilde{A}^i_g = \varphi(g)\, X(g)\, A^i\, X(g)^{-1}$ for all i.*

Proof. ✓ Proof The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 14.7 to obtain $X(g)$, then set $\varphi(g) = 1$. ☐

Lemma 14.10 (Gauge uniqueness up to scalar). Lean ✓ Stmt *Let A be an injective tensor. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\,A^i\,X^{-1} = Y\,A^i\,Y^{-1}$ for all i, then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. ✓ Proof The matrix $Z = Y^{-1}X$ commutes with every A^i. Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 16.12, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. ☐

Theorem 14.11 (Gauge uniqueness from equality of MPV families). Lean ✓ Stmt *Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\,A^i\,X^{-1} = Y\,A^i\,Y^{-1}$ for all i, then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. ✓ Proof This is exactly Lemma 14.10; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced. ☐

Definition 14.12 (Scalar 2-cochain). Lean ✓ Stmt A *scalar 2-cochain* on a group G is a function $\omega : G \times G \to \mathbb{C}^\times$.

Definition 14.13 (Projective representation). Lean ✓ Stmt Given a scalar 2-cochain ω, a *projective representation* of G on $\mathbb{C}^D$ is a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh).$$

Lemma 14.14 (Associativity forces the cocycle condition). Lean ✓ Stmt *If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity:*

$$\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k).$$

Proof. ✓ Proof Expanding $\rho(g)\,(\rho(h)\rho(k))$ and $(\rho(g)\rho(h))\rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. □

Theorem 14.15 (Virtual representation theorem for injective MPS). Lean ✓ Stmt *Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. Then there exist:*

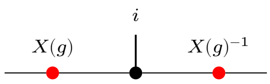
1. *a scalar 2-cochain $\omega : G \times G \to \mathbb{C}^\times$, and*
2. *a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,*

$$\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh).$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has, for every $g \in G$ and $i \in \{0, \dots, d-1\}$,

$$\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}.$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. ✓ Proof Since A is injective and on-site symmetric, each twisted tensor $\widetilde{A}_g$ is gauge equivalent to A (Lemma 14.7). Choose $X(g) \in \mathrm{GL}_D(\mathbb{C})$ with $\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}$ for all i.

By functoriality (Lemma 14.5), the product $X(h)\,X(g)$ also intertwines A with $\widetilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 14.10) gives a nonzero scalar $\omega'(h,g)$ with $X(h)\,X(g) = \omega'(h,g)\,X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh)$ where $\omega(g,h) := \omega'(h^{-1}, g^{-1})$. □

Corollary 14.16 (2-cocycle identity for the factor system). Lean ✓ Stmt *If $D > 0$, the factor system ω from Theorem 14.15 satisfies the 2-cocycle identity for all $g, h, k \in G$:*

$$\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k).$$

Proof. ✓ Proof Theorem 14.15 gives ρ and ω. By Lemma 14.14, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. □

14.3 Cohomology Classes of Virtual Symmetry Cocycles

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathbb{C}^\times)$.

Definition 14.17 (Coboundary). Lean ✓ Stmt A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi \colon G \to \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega(g,h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}.$$

Definition 14.18 (Cohomologous cocycles). Lean ✓ Stmt Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi \colon G \to \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega_1(g,h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g,h).$$

Theorem 14.19 (Cohomology relation is an equivalence relation). *Lean* *✓ Stmt* *The relation "cohomologous to" is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.*

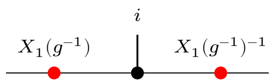
Proof. ✓ Proof Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1}. Transitivity: compose the witnesses φ and ψ pointwise. □

Lemma 14.20 (Coboundary iff cohomologous to the trivial cocycle). *Lean* *✓ Stmt* *A cocycle ω is a coboundary if and only if $\omega \sim \mathbb{1}$, where $\mathbb{1}(g,h) = 1$ for all g, h.*

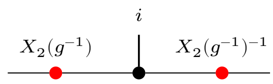
Proof. ✓ Proof Both directions follow by absorbing or introducing the factor $\mathbb{1}(g,h) = 1$ via $x \cdot 1 = x$. □

Theorem 14.21 (Gauge independence of the cocycle class). *Lean* *✓ Stmt* *If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U, then $\omega_1 \sim \omega_2$.*

Proof. ✓ Proof For each g, gauge uniqueness gives a nonzero scalar $f(g)$ with $X_2(g^{-1}) = f(g)\, X_1(g^{-1})$. Equivalently, for each fixed g the two local gauges



and



describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g,h) = f(g)\, f(h)\, f(gh)^{-1}\omega_1(g,h)$. □

Definition 14.22 (Multiplicative 2-cocycle condition). Lean ✓ Stmt A scalar 2-cochain $\omega \colon G \times G \to \mathbb{C}^\times$ is a *2-cocycle* if it satisfies, for all $g, h, k \in G$,

$$\omega(g,h)\omega(gh,k) = \omega(g,hk)\omega(h,k).$$

Definition 14.23 (Second cohomology quotient). Lean ✓ Stmt The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\!\sim,$$

where $\sim$ is the coboundary-equivalence relation.

Definition 14.24 (Projective equivalence). Lean ✓ Stmt Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

Theorem 14.25 (Projective equivalence and cohomologous cocycles). Lean ✓ Stmt *Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2. Then*

$$\rho_1 \sim_{\mathrm{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

Proof. ✓ Proof This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems. □

14.4 Permutation-Based Physical Symmetry

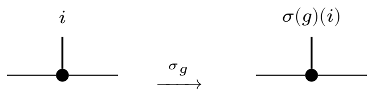
The linear-combination twist $\widetilde{A}^i_g = \sum_j U(g)_{ij} A^j$ is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is useful for symmetries that act by reordering basis states rather than by a general linear map.

Definition 14.26 (Permutation symmetry datum). Lean ✓ Stmt An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u : G \to M_d(\mathbb{C})$ and a physical-index action $\sigma : G \to \mathrm{End}(\{0, \ldots, d-1\})$.

Definition 14.27 (Permutation-twisted tensor). Lean ✓ Stmt Given an on-site symmetry datum (u, σ), the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:



Lemma 14.28 (Identity permutation twist). Lean ✓ Stmt *If $\sigma(1) = \mathrm{id}$, then $(A^{\sigma_1})^i = A^i$ for all i.*

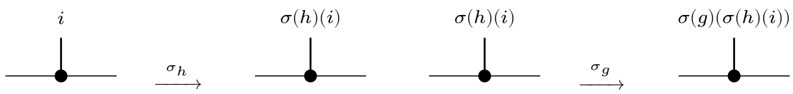
Proof. ✓ Proof Immediate from $\sigma(1)(i) = i$ for all i. □

Composition order. The linear-combination twist (Lemma 14.5) composes as $\widetilde{(\widetilde{A}_h)}_g = \widetilde{A}_{gh}$, applying h first then g. The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma 14.29 (Composition of permutation twists). Lean ✓ Stmt *If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then*

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

Proof. ✓ Proof Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

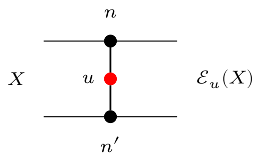
14.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS+08] detects whether an on-site unitary u acts as a local symmetry on the state generated by an injective MPS tensor.

Definition 14.30 (Twisted transfer map)**.** Lean ✓ Stmt Given an MPS tensor A and a physical-index matrix u, the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n'|u|n\rangle A_n \, X \, A_{n'}^\dagger.$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

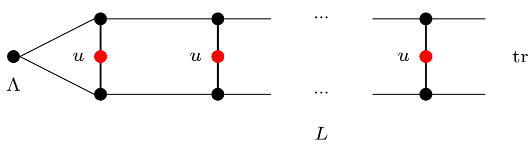


The upper and lower black nodes denote A_n and $A_{n'}^\dagger$, and the red node labelled u couples the physical legs.

Definition 14.31 (String order parameter)**.** Lean ✓ Stmt The *string order parameter* for twist u and stationary state Λ at length L is

$$R_L(u) := \operatorname{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length-L doubled chain with a copy of u inserted on each physical rung:



The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 14.32 (Boundary string order parameter)**.** Lean ✓ Stmt For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *boundary string order parameter* is

$$R_L(u; X, Y) := \operatorname{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The ordinary string order parameter is the special case $X = Y = \mathbb{1}$.

Definition 14.33 (Local symmetry)**.** Lean ✓ Stmt An MPS tensor A has *local symmetry* under a unitary u with respect to a boundary state Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and, for every physical index i,

$$\sum_j u_{ij} \, A^j = \mu \, V \, A^i \, V^\dagger.$$

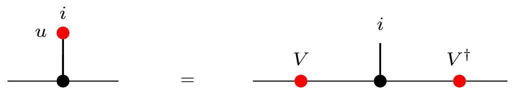
The unitary V intertwines the physical action of u with a virtual conjugation, and the boundary state Λ is invariant under V.

Definition 14.34 (String order). Lean ✓ Stmt String order *exists* for u with respect to a boundary state Λ if there exist boundary matrices X, Y and a positive constant $c > 0$ such that $c \leq \|R_L(u; X, Y)\|$ for all L, where $R_L(u; X, Y)$ is the boundary-refined string-order parameter. Thus some boundary-refined string-order sequence does not decay to zero.

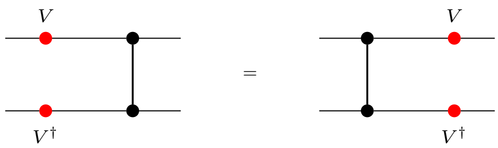
Lemma 14.35 (String order prevents decay to zero). Lean ✓ Stmt *If A has string order for u with respect to a boundary state Λ, then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \to \infty$.*

Proof. ✓ Proof A uniform lower bound $c \leq \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. □

Definition 14.36 (Local physical intertwining). Lean ✓ Stmt The local physical intertwining relation is $\sum_j u_{ij} A^j = V A^i V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



Definition 14.37 (Transfer-map covariance). Lean ✓ Stmt Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation: $\mathcal{E}(VXV^\dagger) = V\,\mathcal{E}(X)\,V^\dagger$. In doubled-layer notation, the virtual operators slide through the local transfer cell:

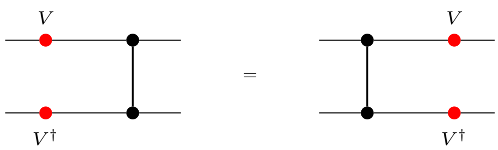


Definition 14.38 (Commutation in the doubled transfer picture). Lean ✓ Stmt In the transfer-map language, the commutation of the doubled transfer matrix $E = \sum_i A_i \otimes \bar{A}_i$ with $V \otimes \bar{V}$ is

$$V\,\mathcal{E}_1(X)\,V^\dagger = \mathcal{E}_1(VXV^\dagger).$$

Theorem 14.39 (Transfer-map covariance and doubled commutation are equivalent). Lean ✓ Stmt *Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V.*

Proof. ✓ Proof Both sides express the same local picture with opposite orientation:



The two formulas are the same identity read from opposite sides. □

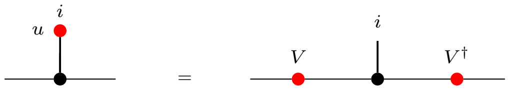
Lemma 14.40 (Unitary Kraus mixing). Lean ✓ Stmt *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel:*

$$\sum_i \left(\sum_j u_{ij} A_j\right) Y \left(\sum_j u_{ij} A_j\right)^\dagger = \sum_i A_i\,Y\,A_i^\dagger.$$

Proof. ✓ Proof This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 21.20. □

Theorem 14.41 (Local physical intertwining implies transfer-map covariance). Lean ✓ Stmt *If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

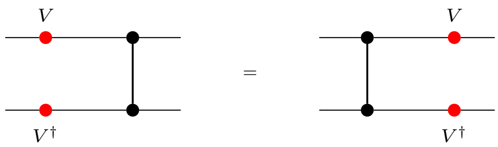
Proof. ✓ Proof Starting from



one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u-boxes disappear by the unitary Kraus-mixing identity. □

Theorem 14.42 (Transfer-map covariance implies local physical intertwining). Lean ✓ Stmt *For an injective MPS, if V is unitary and the transfer map is covariant, then there exists a unitary u satisfying the local physical intertwining relation.*

Proof. ✓ Proof Set $B^i := V\,A^i\,V^\dagger$. The transfer-map covariance relation



implies that the two Kraus families B and A define the same transfer map. Kraus freedom for equal channels converts $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. □

Definition 14.43 (Twisted companion tensor). Lean ✓ Stmt For an MPS tensor A and a physical-index matrix u, define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}}\, A^{n'}.$$

Lemma 14.44 (Twisted transfer as a mixed transfer operator). Lean ✓ Stmt *The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u):*

$$\mathcal{E}_u = F_{A,B_u}.$$

Proof. ✓ Proof Expand both definitions and rearrange the double sum. □

Theorem 14.45 (Twisted-transfer eigenvalues have modulus at most one). Lean ✓ Stmt *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \le 1$.

Proof. ✓ Proof By Lemma 14.44, the twisted transfer map is the mixed transfer operator F_{A,B_u}. Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. Theorem 7.14 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ. □

Theorem 14.46 (Modulus-one twisted eigenvalue implies gauge-phase equivalence)**.** Lean ✓ Stmt *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u.

Proof. ✓ Proof By Lemma 14.44, the twisted transfer map is the mixed transfer operator F_{A,B_u}. Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. In that gauge the eigenvalue equation becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 7.28 yields gauge-phase equivalence. Undoing the common trace-preserving gauge gives the claim for A and B_u. □

Lemma 14.47 (Spectral radius < 1 forces boundary-string decay)**.** Lean ✓ Stmt *If $\rho(\mathcal{E}_u) < 1$, then, as $L \to \infty$, for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has*

$$R_L(u; X, Y) \longrightarrow 0.$$

Proof. ✓ Proof In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \to 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \operatorname{tr}(\Lambda X Z)$ gives the stated scalar convergence. □

Theorem 14.48 (String order gives gauge-phase equivalence with the companion tensor)**.** Lean ✓ Stmt *Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ, then A is gauge-phase equivalent to the companion tensor B_u.*

Proof. ✓ Proof If $\rho(\mathcal{E}_u) < 1$, Lemma 14.47 forces every boundary string-order sequence to decay to zero, contradicting Definition 14.34. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 14.45 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 14.46 then gives the gauge-phase equivalence. □

Theorem 14.49 (Gauge-phase equivalence with the companion tensor yields a virtual unitary)**.** Lean ✓ Stmt *Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u, then there exist a unitary V and a phase μ with $|\mu| = 1$ such that, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger.$$

Proof. ✓ Proof Start from an invertible intertwiner X between A and B_u. The positive matrix $Q := XX^\dagger$ is a positive fixed point of $\mathcal{E}_A$. By uniqueness of positive fixed points for an injective normalized tensor, Q is a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V, and the original intertwining relation becomes the phased covariance equation above, with $|\mu| = 1$. □

Theorem 14.50 (Virtual unitary gives a twisted-transfer eigenmatrix). Lean ✓ Stmt *Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and that, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu\, V A^i V^\dagger$$

with V unitary. Then V is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V.$$

Proof. ✓ Proof Expand $\mathcal{E}_u(V)$, insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. □

Theorem 14.51 (Boundary-state invariance of the virtual unitary). Lean ✓ Stmt *Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\mathrm{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu\, V A^i V^\dagger,$$

then

$$V^\dagger \Lambda V = \Lambda.$$

Proof. ✓ Proof The covariance relation shows that $V^\dagger \Lambda V$ is another positive fixed point of the adjoint transfer map. Since Λ is the normalized positive fixed point, uniqueness forces $V^\dagger \Lambda V = \Lambda$. □

Lemma 14.52 (Local symmetry implies string order). Lean ✓ Stmt *Assume $\mathrm{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If u is a local symmetry of A with respect to Λ, then string order exists for u.*

Proof. ✓ Proof Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the local-symmetry datum. By Theorem 14.50, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L\, \mathrm{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. □

Theorem 14.53 (Local symmetry iff a unitary peripheral eigenmatrix exists). Lean ✓ Stmt *Let A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\mathrm{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then u is a local symmetry if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that*

$$\mathcal{E}_u(V) = \mu V.$$

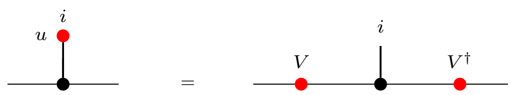
By Theorem 14.45, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. ✓ Proof The forward direction is Theorem 14.50. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 14.46 gives gauge-phase equivalence with the companion tensor; then Theorem 14.49 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 14.51 gives the boundary-state invariance needed for local symmetry. □

Theorem 14.54 (String order ⇔ local symmetry). Lean ✓ Stmt *For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\mathrm{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order for u exists iff u is a local symmetry.*

Proof. ✓ Proof If string order exists, Theorem 14.48 gives a gauge-phase equivalence with the companion tensor. Theorem 14.49 extracts a virtual unitary and phase from that equivalence, and Theorem 14.51 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 14.52 turns local symmetry into a non-decaying boundary string-order sequence. □

Theorem 14.55 (Virtual unitary from string order)**.** Lean ✓ Stmt *For an injective MPS, let* $\Lambda \in M_D(\mathbb{C})$*, and assume* $uu^\dagger = \mathbb{1}$ *and* $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$*. If string order exists for* u *with boundary parameter* Λ*, then there exists a virtual unitary* V *and a unit-modulus scalar* μ *such that* $\sum_j u_{ij} A^j = \mu\, V A^i V^\dagger$ *for all* i*. Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*



Proof. ✓ Proof String order first gives gauge-phase equivalence with the companion tensor by Theorem 14.48. Theorem 14.49 then normalizes the resulting intertwiner to a unitary and produces the phase μ. □

14.6 SPT Phase Classification

Two injective symmetric tensors belong to the same symmetry-protected topological phase when their virtual representation cocycles are cohomologous [CGW11].

Definition 14.56 (Same SPT phase)**.** Lean ✓ Stmt Injective tensors A and B, both symmetric under an on-site representation $U : G \to \mathrm{GL}_d(\mathbb{C})$, are in the *same SPT phase* if there exist virtual cocycles ω_A, ω_B with corresponding projective representations that intertwine the respective twisted tensors and satisfy $\omega_A \sim \omega_B$ (cohomologous).

Theorem 14.57 (String order holds for injective symmetric MPS)**.** Lean ✓ Stmt *Let* A *be an injective tensor, symmetric under a unitary representation* U*. Assume* $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$*,* Λ *is positive definite,* $\mathrm{tr}(\Lambda) = 1$*, and* $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$*. Then string order exists for every group element* g*.*

Proof. ✓ Proof The virtual representation theorem produces $\rho(g^{-1}) \in \mathrm{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij} A^j = \rho(g^{-1})\, A^i\, \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 14.46 gives gauge-phase equivalence with the companion tensor, and Theorem 14.49 extracts a virtual unitary and phase. By Theorem 14.51, this unitary preserves the boundary state, so Lemma 14.52 gives string order. □

Theorem 14.58 (String order is an SPT invariant)**.** Lean ✓ Stmt *If* A *and* B *are injective, both on-site symmetric under a unitary representation* U*, and if* $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$*,* $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$*,* Λ_A *and* Λ_B *are positive definite,* $\mathrm{tr}(\Lambda_A) = 1$*,* $\mathrm{tr}(\Lambda_B) = 1$*,* $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$*, and* $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$*, then for every* $g \in G$*, string order exists for* A *with twist* $U(g)$ *if and only if it exists for* B*. In particular, since Definition 14.56 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase.*

Proof. ✓ Proof Both directions follow from Theorem 14.57, which shows that string order holds universally for injective symmetric tensors with canonical normalisations. □

Chapter 15

Periodic MPS: irreducible form and the periodic Fundamental Theorem

This chapter develops the periodic equal-case Fundamental Theorem of MPS (Theorem 15.24) following [DlCCSPG17], and gives its two applications: the symmetry corollary lifting a physical on-site symmetry of a tensor in irreducible form II to a virtual Z-gauge, and the equivalence between p-refinability of the matrix-product-vector family and p-divisibility of the transfer map.

15.1 Periodic irreducible form and physical-index rotation

Definition 15.1 (Periodic tensor). Lean ✓ Stmt An irreducible MPS tensor A of bond dimension D is *m-periodic* (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the m-th roots of unity $\{e^{2\pi ik/m} : 0 \leq k < m\}$. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 4.49).

Definition 15.2 (Irreducible form). Lean ✓ Stmt A tensor A is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j \, A^i_j,$$

where each block A_j is an m_j-periodic tensor, each weight $\mu_j \neq 0$, and the blocks are pairwise non-repeated (no two distinct j, j' have A_j gauge-equivalent to $A_{j'}$). This is the standard form introduced in [DlCCSPG17, Section II]. It extends the normal canonical form (Definition 9.24), which requires the stronger condition that every block is 1-periodic.

Remark 15.3 (Relation to normal canonical form). Not ready Every tensor in normal canonical form (Definition 9.24) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic.

Physical-index rotation was introduced in Definition 14.1 (Chapter 14). The following compatibility theorems are proved in the present chapter.

Theorem 15.4 (Transfer map preserved by physical rotation). *Lean* *✓ Stmt* *If $uu^\dagger = I$, then the transfer map is invariant under physical-index rotation: $\mathcal{E}_{A_u} = \mathcal{E}_A$.*

Proof. ✓ Proof This is the unitary freedom of Kraus representations from [Wol12, Theorem 2.18]. □

Theorem 15.5 (Left-canonical preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is left-canonical, then A_u is left-canonical.*

Proof. ✓ Proof Expand $\sum_i B_i^\dagger B_i$ where $B_i = \sum_j u_{ij} A_j$, swap sums, apply orthogonality $u^\dagger u = I$, and collapse to $\sum_j A_j^\dagger A_j = I$. □

Theorem 15.6 (Irreducibility preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is irreducible, then A_u is irreducible.*

Proof. ✓ Proof Contrapositive: any nontrivial invariant projection P for the rotated tensor yields $(1 - P)\, A_k\, P = 0$ for all k via the orthogonality of u, contradicting irreducibility of A. □

Theorem 15.7 (Periodicity preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is periodic with period m, then A_u is periodic with the same period.*

Proof. ✓ Proof Assembles transfer-map invariance, left-canonical preservation, and irreducibility preservation. □

Theorem 15.8 (Equality of MPV families preserved by physical rotation). Lean ✓ Stmt *If $\mathcal{V}(A) = \mathcal{V}(B)$, then $\mathcal{V}(A_u) = \mathcal{V}(B_u)$.*

Proof. ✓ Proof Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation. □

Theorem 15.9 (Physical rotation distributes over block-diagonal composition). Lean ✓ Stmt *The rotated block-diagonal tensor satisfies $(\bigoplus_k \mu_k A_k)_u = \bigoplus_k \mu_k (A_k)_u$.*

Proof. ✓ Proof Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing. □

Theorem 15.10 (Irreducible form preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is in irreducible form, then A_u is in irreducible form with the same block structure and rotated blocks.*

Proof. ✓ Proof Each block remains periodic by Theorem 15.7, equality of MPV families transfers via Theorem 15.8, and block-diagonal compatibility follows from Theorem 15.9. □

15.1.1 The Z-gauge degree of freedom

Definition 15.11 (Entrywise periodic Z-gauge ratio). Lean ✓ Stmt Given complex weights μ, ν, define the entrywise Z-gauge ratio by

$$z(\mu, \nu) := \mu/\nu.$$

Lemma 15.12 (Matched powers imply root-of-unity ratio). Lean ✓ Stmt *If $\mu^m = \nu^m$ and $\nu \neq 0$, then*

$$z(\mu, \nu)^m = 1.$$

Lemma 15.13 (Ratio rescales denominator back to numerator). Lean ✓ Stmt *If $\nu \neq 0$, then*

$$z(\mu, \nu)\, \nu = \mu.$$

Definition 15.14 (Diagonal periodic Z-gauge matrix). Lean ✓ Stmt For matched weight families $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix

$$Z := \operatorname{diag}\big(z(\mu_i, \nu_i)\big)_i.$$

Lemma 15.15 (Diagonal Z-gauge has finite order under matched powers). Lean ✓ Stmt *If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all i, then*

$$Z^m = \mathbb{1}.$$

Lemma 15.16 (Diagonal Z-gauge transports diagonal weights). Lean ✓ Stmt *Under $\nu_i \neq 0$ for all i,*

$$Z \operatorname{diag}(\nu) = \operatorname{diag}(\mu).$$

Definition 15.17 (Z-gauge equivalence). Lean ✓ Stmt Let A be an m-periodic irreducible tensor of bond dimension D. A *Z-gauge transformation* of A is a diagonal unitary matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. Z commutes with every Kraus operator: $[A^i, Z] = 0$ for all physical indices i, and
2. $Z^m = \mathbb{1}_D$ (all diagonal entries are m-th roots of unity).

Replacing A by ZA (i.e. $A^i \mapsto ZA^i$) leaves the MPV family invariant for lengths N that are multiples of m, because the factor $\operatorname{tr}(Z^N) = \operatorname{tr}(\mathbb{1}) = D$ for $N \equiv 0 \pmod m$. Two tensors A, B are *Z-gauge equivalent* if there exists an invertible Y and a Z-gauge transformation Z for A such that $ZA^i = YB^iY^{-1}$ for all i.

Remark 15.18 (Z-gauge is trivial for period-1 blocks). Lean ✓ Stmt When $m = 1$ the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so Z-gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The Z-gauge is therefore a genuine new degree of freedom that appears only for $m > 1$.

Theorem 15.19 (Blocking a $\mathbb{Z}_m$-gauge gives an ordinary gauge). Lean ✓ Stmt *If A and B are $\mathbb{Z}_m$-gauge equivalent, then their m-blocked tensors $A^{[m]}$ and $B^{[m]}$ are gauge equivalent in the ordinary sense.*

Proof. ✓ Proof Write the $\mathbb{Z}_m$-gauge relation as $ZA^i = YB^iY^{-1}$ with $Z^m = \mathbb{1}$ and $[A^i, Z] = 0$. For a blocked letter, commute one factor of Z past each physical letter in the associated length-m word. The total prefactor becomes $Z^m = \mathbb{1}$, so the entire blocked word is related by the same ordinary gauge Y. □

Corollary 15.20 (Blocking a $\mathbb{Z}_m$-gauge preserves the MPV family). Lean ✓ Stmt *If A and B are $\mathbb{Z}_m$-gauge equivalent, then the blocked tensors $A^{[m]}$ and $B^{[m]}$ have the same MPV family.*

Proof. ✓ Proof Apply Theorem 15.19 and then the gauge invariance of MPV families. □

15.2 Periodic Fundamental Theorem

The theorems in Chapter 13 all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly $\{1\}$. This is achieved by blocking the original tensor by a common period. [DlCCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

15.2.1 Common-period blocking lemmas

Definition 15.21 (LCM period for a finite block family). Lean ✓ Stmt Given a finite family of blocking periods (p_i) indexed by a finite set, their least common multiple serves as the shared blocking period.

Definition 15.22 (Common-period blocked family). Lean ✓ Stmt Given a finite family of tensors (B_i) with designated periods (p_i), block each B_i to the shared period $\operatorname{lcm}(p_i)$, so every block lives in the same physical space.

Theorem 15.23 (Common-period blocking preserves transfer-map primitivity). *Lean* *✓ Stmt* *If each tensor B_i has a primitive transfer map after blocking by its own positive period p_i, then blocking the entire family to the common LCM period preserves primitivity of each member's transfer map.*

Proof. ✓ Proof Each p_i divides $\operatorname{lcm}(p_i)$. Apply the divisibility monotonicity theorem for primitive blocked transfer maps to each family member separately. □

15.2.2 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 ("equal case") of [DlCCSPG17]. The statements recorded here are conditional components of that theorem: the final comparison still assumes periodic-overlap and power-equality hypotheses, so the literature theorem has not yet been obtained here as a single unconditional result.

Theorem 15.24 (Periodic Fundamental Theorem). *Not ready* *Lean* *Lean* *Lean* *Lean* *Let A and B be MPS tensors in irreducible form, with bases of periodic blocks $\{A_j\}_{j\in J}$ and $\{B_k\}_{k\in K}$ respectively. Suppose that $\mathcal{V}(A) = \mathcal{V}(B)$ (equal MPV families for all lengths N).*

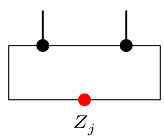
Then $|J| = |K|$ and there is a bijection $\pi : J \to K$ such that A_j and $B_{\pi(j)}$ are m_j-periodic for the same period m_j. Moreover, for each $j \in J$ there exists:

1. *an invertible matrix Y_j of size $D_j \times D_{\pi(j)}$, and*
2. *a diagonal unitary matrix Z_j of size $D_j \times D_j$ satisfying $Z_j^{m_j} = \mathbb{1}$ and $[A_j^i, Z_j] = 0$ for all i,*

such that, for every physical index i,

$$Z_j\, A_j^i = Y_j\, B_{\pi(j)}^i\, Y_j^{-1}$$

For each periodic block, the extra freedom is encoded by



rather than by an ordinary aperiodic gauge alone: the operator Z_j sits on the closing virtual bond and satisfies $Z_j^{m_j} = \mathbb{1}$. Equivalently, setting $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$ and $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$, one has $ZA^i = YB^iY^{-1}$ for all i, with $[A^i, Z] = 0$ and $\mathcal{V}(A) = \mathcal{V}(ZA)$.

Remark. *The proof is conditional on the periodic-overlap dichotomy [DlCCSPG17, Proposition 3.3], which is left here as a separate hypothesis.*

15.2.3 Periodic overlap dichotomy

Definition 15.25 (Cyclic sector decomposition). Lean ✓ Stmt A family of compressed sector tensors $(C_k)_{k=0}^{m-1}$ on corner bond spaces forms a *cyclic sector decomposition* of the blocked tensor $A^{(m)}$ if there exist orthogonal projections $P_0, \dots, P_{m-1}$ on the ambient bond space such that $\sum_k P_k = \mathbb{1}$, the projections satisfy the cyclic-shift relation $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$ with the index $k+1$ taken cyclically modulo m, each P_k commutes with every blocked letter $P_k A_i^{(m)} = A_i^{(m)} P_k$, and $V^{(N)}(C_k)_\sigma = \mathrm{tr}(P_k (A^{(m)})^\sigma)$ for every word σ, and each block carries a $*$-algebra isomorphism $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k \cdot M_D(\mathbb{C}) \cdot P_k$ (multiplicative and adjoint-preserving) intertwining the compressed adjoint transfer map with the sector adjoint transfer map on the corner of P_k. This is the inverse cyclic indexing of the off-diagonal convention in [DlCCSPG17, Appendix A]. There the same adjoint transfer map is written E^*, and the source projectors satisfy $E^*(P_u) = P_{u+1}$ together with $A^i = \sum_u P_u A^i P_{u+1}$. Here P_k corresponds to the source sector $u = -k \pmod m$, which turns the source shift into the displayed adjoint-transfer relation.

Theorem 15.26 (Self-overlap along period multiples). Lean ✓ Stmt *If A is m-periodic, then $\lim_{N\to\infty} O_{AA}(mN) = m$.*

Theorem 15.27 (Different periods imply asymptotic orthogonality). Lean ✓ Stmt *If A and B have periods $m_A \neq m_B$, then $\lim_{N\to\infty} O_{AB}(N) = 0$.*

Proof. ✓ Proof If $D_A \neq D_B$, use Theorem 15.33. Assume $D_A = D_B$ and $A \sim_{\mathrm{gp}} B$, say $B^i = \zeta X A^i X^{-1}$ with $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \neq 0$. The transfer maps satisfy

$$\mathcal{E}_B(Y) = \zeta\bar{\zeta}\, X\, \mathcal{E}_A(X^{-1} Y X^{-*})\, X^*.$$

With $\mathcal{E}_A(\rho_A) = \rho_A$, put $\sigma = X \rho_A X^*$. Then $\mathcal{E}_B(\sigma) = |\zeta|^2 \sigma$, while $\mathcal{E}_B(\rho_B) = \rho_B$ for a nonzero positive fixed point ρ_B. Perron–Frobenius eigenvalue uniqueness [Wol12, Theorem 6.3] gives $|\zeta|^2 = 1$. Hence $\mathcal{E}_B$ is conjugate to $\mathcal{E}_A$, so $\sigma_{\mathrm{per}}(\mathcal{E}_A) = \sigma_{\mathrm{per}}(\mathcal{E}_B)$. Since

$$\sigma_{\mathrm{per}}(\mathcal{E}_A) = \{e^{2\pi i r/m_A} : 0 \le r < m_A\}, \qquad \sigma_{\mathrm{per}}(\mathcal{E}_B) = \{e^{2\pi i r/m_B} : 0 \le r < m_B\},$$

equality of the two finite sets gives $m_A = m_B$, a contradiction. Thus $A \nsim_{\mathrm{gp}} B$, and the irreducible trace-preserving overlap dichotomy gives $\lim_{N\to\infty} O_{AB}(N) = 0$. □

Lemma 15.28 (Sector-block normality for periodic tensors). Lean ✓ Stmt *For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.*

Theorem 15.29 (No sector match implies asymptotic orthogonality). Lean ✓ Stmt *Suppose A and B have the same total bond dimension and the same period m. Let A_u and B_v be period-blocked sector tensors of virtual dimensions d_u and e_v, indexed by $u, v \in \mathbb{Z}_m$, such that*

$$\bar{A} \sim \bigoplus_{u\in\mathbb{Z}_m} A_u, \qquad \bar{B} \sim \bigoplus_{v\in\mathbb{Z}_m} B_v$$

with unit weights, and the two decompositions are cyclic sector decompositions. Assume each sector is left-canonical,

$$\sum_\alpha (A_u^\alpha)^\dagger A_u^\alpha = I_{d_u}, \qquad \sum_\alpha (B_v^\alpha)^\dagger B_v^\alpha = I_{e_v},$$

Assume $\forall u,\ d_u \neq 0$ and $\forall v,\ e_v \neq 0$, and whenever $d_u = e_v$, the corresponding sector tensors are not gauge-phase equivalent after identifying their virtual spaces:

$$d_u = e_v \implies A_u \nsim_{\mathrm{gp}} B_v.$$

Then $\lim_{N\to\infty} O_{AB}(N) = 0$.

Proof. ✓ Proof Write $\bar{A} \sim \bigoplus_{u\in\mathbb{Z}_m} A_u$ and $\bar{B} \sim \bigoplus_{v\in\mathbb{Z}_m} B_v$. For every N,

$$O_{\bar{A}\bar{B}}(N) = \sum_{u\in\mathbb{Z}_m} \sum_{v\in\mathbb{Z}_m} O_{A_uB_v}(N).$$

For each sector pair (u, v),

$$d_u = e_v, \quad A_u \nsim_{\text{gp}} B_v \quad \Longrightarrow \quad \lim_{N\to\infty} O_{A_uB_v}(N) = 0,$$

and

$$d_u \neq e_v \quad \Longrightarrow \quad \lim_{N\to\infty} O_{A_uB_v}(N) = 0.$$

The finite double sum therefore gives

$$\lim_{N\to\infty} O_{\bar{A}\bar{B}}(N) = \sum_{u\in\mathbb{Z}_m} \sum_{v\in\mathbb{Z}_m} \lim_{N\to\infty} O_{A_uB_v}(N) = 0.$$

If $A \sim_{\text{gp}} B$, then $\bar{A} \sim_{\text{gp}} \bar{B}$, so for some $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \neq 0$, $(\bar{B})^\alpha = \zeta X(\bar{A})^\alpha X^{-1}$. Thus $V^{(N)}(\bar{B})_\sigma = \zeta^N V^{(N)}(\bar{A})_\sigma$, and

$$O_{\bar{A}\bar{B}}(N) = \bar{\zeta}^N O_{\bar{A}\bar{A}}(N), \qquad O_{\bar{B}\bar{B}}(N) = |\zeta|^{2N} O_{\bar{A}\bar{A}}(N).$$

Since $\lim_{N\to\infty} O_{\bar{A}\bar{A}}(N) = \lim_{N\to\infty} O_{\bar{B}\bar{B}}(N) = m > 0$,

$$1 = \lim_{N\to\infty} \frac{O_{\bar{B}\bar{B}}(N)}{O_{\bar{A}\bar{A}}(N)} = \lim_{N\to\infty} |\zeta|^{2N},$$

hence $|\zeta| = 1$. Therefore

$$\lim_{N\to\infty} |O_{\bar{A}\bar{B}}(N)| = \lim_{N\to\infty} O_{\bar{A}\bar{A}}(N) = m \neq 0,$$

contradicting $\lim_{N\to\infty} O_{\bar{A}\bar{B}}(N) = 0$. Thus $A \nsim_{\text{gp}} B$, and the irreducible trace-preserving overlap dichotomy gives $\lim_{N\to\infty} O_{AB}(N) = 0$. □

Lemma 15.30 (Sector-match propagation under translation)**.** Lean Not ready *Let A and B be left-canonical tensors with period m, and let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical. Suppose $d_{u_0} = e_{v_0}$, $d_{u_0} \neq 0$, and, after identifying the virtual spaces, $A_{u_0} \sim_{\text{gp}} B_{v_0}$. Then for every $l \in \mathbb{Z}_m$ there is an equality $d_{u_0+l} = e_{v_0+l}$ and, after this identification,*

$$A_{u_0+l} \sim_{\text{gp}} B_{v_0+l}.$$

Lemma 15.31 (Per-site proportionality from blocked sector match)**.** Lean Not ready *Let A and B be left-canonical tensors with period m, and let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical. Assume $d_u \neq 0$ and A_u is normal for every u. Fix $q \in \mathbb{Z}_m$ and assume that for every u there are $d_u = e_{u+q}$, $\zeta_u \neq 0$, and $X_u \in \mathrm{GL}_{d_u}(\mathbb{C})$ such that $B^\alpha_{u+q} = \zeta_u X_u A^\alpha_u X_u^{-1}$ for every blocked physical index α. The source obtains a phase-only product identity after absorbing these sector gauges into the B-sectors: with Q_v the cyclic sector projectors of B, it writes*

$$\widetilde{B}^i_v = U_v Q_v B^i Q_{v+1} U^\dagger_{v+1}.$$

The contraction input is then, for every physical block $(i_1, \dots, i_m)$,

$$A_u^{i_1} A_{u+1}^{i_2} \cdots A_{u+m-1}^{i_m} = e^{i\lambda_{u+q}} \widetilde{B}_{u+q}^{i_1} \widetilde{B}_{u+q+1}^{i_2} \cdots \widetilde{B}_{u+q+m-1}^{i_m}.$$

Then there are a phase ξ and an invertible matrix U such that $A^i = e^{i\xi} U B^i U^{-1}$ for every physical index i.

Theorem 15.32 (Sector match yields repeated-block equivalence). Lean Not ready *Let A and B be periodic tensors with the same period m. Let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical, and assume every sector A_u has nonzero virtual dimension. If there are sectors u_0, v_0 with $d_{u_0} = e_{v_0}$ and $A_{u_0} \sim_{\mathrm{gp}} B_{v_0}$ after identifying their virtual spaces, then A and B are equivalent up to repeated blocks.*

Theorem 15.33 (Different bond dimensions imply asymptotic orthogonality). Lean ✓ Stmt *If two periodic tensors have different bond dimensions, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \to 0$ as $N \to \infty$.*

Theorem 15.34 (Periodic overlap dichotomy). Lean Not ready *Let A and B be periodic tensors. Then at least one of the following alternatives holds:*

1. *$\langle V^{(N)}(A) | V^{(N)}(B) \rangle \to 0$ as $N \to \infty$;*
2. *A and B are equivalent up to repeated blocks.*

Proof. Not ready If $m_A \neq m_B$, then $\lim_{N\to\infty} O_{AB}(N) = 0$ by Theorem 15.27. Assume $m_A = m_B = m$. If $D_A \neq D_B$, then $\lim_{N\to\infty} O_{AB}(N) = 0$ by Theorem 15.33. Assume $D_A = D_B$.

Choose period-blocked sector decompositions

$$\bar{A} \sim \bigoplus_{u\in\mathbb{Z}_m} A_u, \qquad \bar{B} \sim \bigoplus_{v\in\mathbb{Z}_m} B_v.$$

If no pair (u, v) satisfies

$$d_u = e_v, \qquad A_u \sim_{\mathrm{gp}} B_v,$$

then Theorem 15.29 gives $\lim_{N\to\infty} O_{AB}(N) = 0$.

Otherwise there are $u_0, v_0 \in \mathbb{Z}_m$ such that

$$d_{u_0} = e_{v_0}, \qquad A_{u_0} \sim_{\mathrm{gp}} B_{v_0}.$$

Theorem 15.32 then gives repeated-block equivalence between A and B. □

Theorem 15.35 (Eventual linear independence of a periodic basis). Lean Not ready *For a finite family of pairwise non-equivalent periodic tensors whose periods divide a common integer p, there is N_0 such that for every $N \geq N_0$ the vectors $\{|V^{(pN)}(A_j)\rangle\}_j$ are linearly independent.*

Remark 15.36 (Comparison with the blocking approach). Not ready Theorem 15.24 and the equal-MPV source corollary discussed in Section 13.3 both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Section 13.3, following [CPGSV17a]): for a common multiple p of the periods,

   $$A^{[p]} \sim \bigoplus_a C_a, \qquad B^{[p]} \sim \bigoplus_b D_b,$$

   with all C_a, D_b normal. The aperiodic theorem gives, after a permutation of the normal summands,

   $$D_{\pi(a)}^{\alpha} = X_a C_a^{\alpha} X_a^{-1}.$$

   This is a gauge statement for $A^{[p]}$ and $B^{[p]}$; the cyclic phases have been absorbed into the period-p words.

2. *Periodic approach* (Theorem 15.24, following [DlCCSPG17]): keep the cyclic sectors

$$A^i = \sum_{u\in\mathbb{Z}_m} P_u A^i P_{u+1}, \qquad B^i = \sum_{v\in\mathbb{Z}_m} Q_v B^i Q_{v+1}.$$

Sector matching gives phases ϕ_v, a shift q, and unitaries U_v such that

$$P_u A^i P_{u+1} = e^{i\xi} e^{i\phi_{u+q}} U_{u+q} Q_{u+q} B^i Q_{u+q+1} U_{u+q+1}^{-1} e^{-i\phi_{u+q+1}}.$$

In the equal-MPV theorem the proportional phase is absorbed, so one obtains $ZA^i = UB^iU^{-1}$ with $Z^m = \mathbb{1}$; the trace only sees powers of the corresponding m-th roots of unity.

Chapters 9 and 13 follow only the blocking argument: remove periodicity by blocking, then apply the aperiodic theory. The periodic extension treats periodic blocks directly.

Theorem 15.37 (Peripheral proportional case from exact MPV equality)**.** Lean Not ready *Let A and B be periodic tensors with the same matrix product vectors. Then their bond dimensions agree, and after identifying the bond spaces, A and B are equivalent up to repeated blocks.*

Proof. Not ready By Theorem 15.34, either the overlap $\langle V^{(N)}(A)|V^{(N)}(B)\rangle$ tends to 0 or the bond dimensions agree and A and B are equivalent up to repeated blocks. In the decay case, equality of the MPV families gives, for every N,

$$\langle V^{(N)}(A)|V^{(N)}(B)\rangle = \langle V^{(N)}(A)|V^{(N)}(A)\rangle.$$

Along multiples of the period m_a this would imply

$$0 = \lim_{N\to\infty} \langle V^{(m_aN)}(A)|V^{(m_aN)}(B)\rangle = \lim_{N\to\infty} \langle V^{(m_aN)}(A)|V^{(m_aN)}(A)\rangle = m_a,$$

contradicting $m_a > 0$. □

Theorem 15.38 (Peripheral proportional case from phase rescaling)**.** Lean Not ready *Assume that whenever two periodic tensors have proportional MPV families, one can rescale one tensor by a unit-modulus phase so that the MPV families agree exactly. Then proportional periodic MPV families already force the tensors to agree up to repeated blocks.*

Proof. Not ready Apply the phase-rescaling hypothesis to replace A by a periodic tensor A' with the same MPV family as B. Theorem 15.37 then yields repeated-block equivalence between A' and B. Since A' differs from A only by a unit-modulus phase, A and A' are themselves equivalent up to repeated blocks, and transitivity gives the claim. □

Remark 15.39 (Proportional periodic FT). Not ready There is also a proportional periodic Fundamental Theorem (Theorem 3.4, "proportional case", in [DlCCSPG17]). The former formulation of this theorem assumed externally supplied coefficient arrays with nonzero limits as hypotheses. That is not the hypothesis set of the source theorem. A faithful statement must instead derive the nondecaying cross-overlap witnesses from the proportional MPV hypothesis and the periodic BNT structure.

15.3 Physical symmetries on periodic MPS

Theorem 15.40 (Physical symmetry yields a virtual Z-gauge). Lean ✓ Stmt *Let A be an MPS tensor in irreducible form II and let $U : G \to \mathrm{GL}_d(\mathbb{C})$ be a representation of a group G on the physical leg, acting unitarily ($U(g)\,U(g)^\dagger = \mathbb{1}$). Assume moreover the periodic equal-case Fundamental Theorem of MPS (Theorem 15.24) as an explicit hypothesis on the pair (d, D), expressed by the hypothesis* Lean *$d\,D$. If A is on-site symmetric under U — that is, the twisted tensor $\widetilde{A}_g$ has the same MPV family as A for every $g \in G$ — then for each $g \in G$ there exists a positive period m_g and a $\mathbb{Z}_{m_g}$-gauge equivalence between A and $\widetilde{A}_g$.*

Concretely there are matrices Z_g and Y_g, with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$, satisfying, for every physical index i,

$$Z_g\, A^i = Y_g\, \widetilde{A}^i_g\, Y_g^{-1}$$

Proof. ✓ Proof The twisted tensor $\widetilde{A}_g$ coincides with the physical-index rotation $A_{U(g)}$ of Definition 14.1. By Theorem 15.10 the rotated tensor remains in irreducible form II, and the on-site symmetry hypothesis supplies the equality $\mathcal{V}(A) = \mathcal{V}(A_{U(g)})$. Apply the periodic equal-case Fundamental Theorem (Theorem 15.24) to obtain the asserted $\mathbb{Z}_{m_g}$-gauge equivalence. □

Definition 15.41 (Periodic projective-rigidity hypothesis). Lean ✓ Stmt Let A be an MPS tensor with physical dimension d and bond dimension D, and assume it is on-site symmetric under a group action $U : G \to \mathrm{GL}_d(\mathbb{C})$. The tensor A satisfies the *periodic projective-rigidity hypothesis* if one may choose a family of virtual gauges $Y : G \to \mathrm{GL}_D(\mathbb{C})$ and a scalar 2-cochain $u : G \times G \to \mathbb{C}^\times$ such that

1. every Y_g realises the intertwining from [DlCCSPG17, Corollary 4.1] with the twisted tensor $\widetilde{A}_g$ for some accompanying Z_g (with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$); and
2. the family obeys the projective multiplication law on the bond space: $Y_h\, Y_g = u(g, h)\, Y_{gh}$.

The injective case reduces the rigidity to scalar uniqueness of gauges (cf. §14.2); in the genuinely periodic setting the bond-space commutant of A is generated by the Z_g matrices, so coherence of the factor system u is an independent analytic assumption.

Theorem 15.42 (Projective-representation upgrade). Lean ✓ Stmt *Under the hypotheses of Corollary 15.40 together with the periodic projective-rigidity hypothesis (Definition 15.41), the Y_g family defines a projective representation of G on the bond space: there exist a scalar 2-cochain $\omega : G \times G \to \mathbb{C}^\times$ and virtual gauges $X : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying*

$$X(g)\, X(h) \;=\; \omega(g, h)\, X(gh),$$

while for each g the matrix $X(g^{-1})$ still realises the $\mathbb{Z}_{m_g}$-intertwining from [DlCCSPG17, Corollary 4.1] between A and the twisted tensor $\widetilde{A}_g$.

Proof. ✓ Proof Unpack the rigidity hypothesis to extract the family (Y, u) together with the intertwiners from [DlCCSPG17, Corollary 4.1]. Set $X(g) := Y(g^{-1})$ and $\omega(g, h) := u(h^{-1}, g^{-1})$. The projective multiplication law $X(g)\, X(h) = \omega(g, h)\, X(gh)$ is then exactly the (h^{-1}, g^{-1})-instance of the factor-system identity, using $(gh)^{-1} = h^{-1} g^{-1}$. The intertwining relation between A and $\widetilde{A}_g$ transports unchanged. □

Theorem 15.43 (Cocycle identity for the upgraded factor system). Lean ✓ Stmt *If the bond dimension D is positive, the factor system ω produced by Theorem 15.42 satisfies the multiplicative 2-cocycle identity $\omega(g, h)\, \omega(gh, k) = \omega(g, hk)\, \omega(h, k)$.*

Proof. ✓ Proof Associativity of matrix multiplication applied to $X(g)\,X(h)\,X(k)$, combined with the projective multiplication law, gives

$$\omega(g,h)\,\omega(gh,k)\,X((gh)k) = \omega(g,hk)\,\omega(h,k)\,X(g(hk)).$$

Since $(gh)k = g(hk)$, both sides are scalar multiples of the same matrix $X((gh)k)$. The matrix lies in $\mathrm{GL}_D(\mathbb{C})$ and is therefore invertible, but the scalar cancellation leading from an equality of scalar multiples of a matrix to equality of the scalar coefficients additionally requires $D > 0$ so that the underlying index set is inhabited. Cancelling then yields $\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k)$ in $\mathbb{C}^\times$, hence in $\mathrm{Units}(\mathbb{C})$. ☐

Remark 15.44 (SPT classification)*.* The cohomology class of ω in $H^2(G, U(1))$ classifies the SPT phase of the symmetric MPS (cf. Chen–Gu–Wen and Pérez-García et al.). This classification statement is not established here; only the underlying 2-cocycle identity of Theorem 15.43 is available.

Theorem 15.45 (Projective representation and cocycle from symmetry)**.** Lean ✓ Stmt *Theorems 15.42 and 15.43 combine into a single statement: from the periodic projective-rigidity hypothesis (Definition 15.41), one obtains a projective representation of G on the bond space whose factor system is a genuine 2-cocycle whenever $D > 0$.*

15.4 Refinement and divisibility

The next theorem characterises when the matrix-product-vector family of a periodic tensor admits a finer description in terms of a smaller-period tensor whose blocks have been merged. This was the original motivation for the irreducible-form framework in [DlCCSPG17].

Definition 15.46 (p-divisible CP map)**.** Lean ✓ Stmt A linear endomorphism $\mathcal{E}$ of $M_D(\mathbb{C})$ is *p-divisible* if there exists a CPTP map $\mathcal{E}'$ of $M_D(\mathbb{C})$ such that $\mathcal{E} = (\mathcal{E}')^p$.

Definition 15.47 (p-refinable MPS tensor)**.** Lean ✓ Stmt An MPS tensor B with physical dimension d and bond dimension D is *p-refinable* if there exist another such tensor A and an isometry $W : \mathbb{C}^d \to (\mathbb{C}^d)^{\otimes p}$ (encoded as a matrix $W \in M_{d^p\times d}(\mathbb{C})$ with $W^\dagger W = \mathbb{1}$) such that the p-blocked tensor $A^{[p]}$ matches the W-image of B at the level of MPV coefficients:

$$V^{(N)}(A^{[p]})_\tau \;=\; \sum_{\sigma\in\{0,\dots,d-1\}^N} \Big(\prod_k W_{\tau(k),\,\sigma(k)}\Big)\, V^{(N)}(B)_\sigma$$

for every $N \in \mathbb{N}$ and every $\tau \in \{0,\dots,d^p-1\}^N$.

In Dirac notation, this encodes $|V_{pN}(A)\rangle = W^{\otimes N}\,|V_N(B)\rangle$ for all N.

Theorem 15.48 (Pullback of a refinement witness)**.** Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II. If B is p-refinable, then there exist a tensor A, an isometry $W : \mathbb{C}^d \to (\mathbb{C}^d)^{\otimes p}$, and the pullback tensor*

$$C^\alpha := \sum_{i=0}^{d-1} W_{\alpha,i} B^i \qquad (\alpha \in \{0,\dots,d^p-1\})$$

such that C is again in irreducible form II, $\mathcal{E}_C = \mathcal{E}_B$, and C has the same MPV family as $A^{[p]}$.

Proof. ✓ Proof Choose (A, W) from the refinement data. The pullback construction gives $\mathcal{E}_C = \mathcal{E}_B$ and equality of MPV families between C and $A^{[p]}$. To see that C is still in irreducible form II, transport each periodic block B_j of B through the same physical isometry W. The left-canonical normalization is preserved because $W^\dagger W = \mathbb{1}$, while irreducibility and the peripheral spectrum are preserved because the transfer map of each transported block agrees with that of B_j. Reassembling these transported blocks with the same weights gives an irreducible-form decomposition of C. □

Definition 15.49 (Blocked equal-case Z-gauge extraction hypothesis). Lean ✓ Stmt Fix (d, D, p). This hypothesis says that whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, there exists an integer $m \geq 1$ and a $\mathbb{Z}_m$-gauge equivalence between C and $A^{[p]}$.

Definition 15.50 (Blocked-to-root reconstruction hypothesis from Z-gauge). Lean ✓ Stmt Fix (d, D, p). This hypothesis says that if $m \geq 1$ and a blocked tensor C in irreducible form II is $\mathbb{Z}_m$-gauge equivalent to $A^{[p]}$, then there exists a tensor A' with $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$ and $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Definition 15.51 (Blocked equal-case root hypothesis). Lean ✓ Stmt Fix (d, D, p). This states the combined root-existence statement: whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, one can replace A by a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Theorem 15.52 (Blocked equal-case root from Z-gauge extraction). Lean ✓ Stmt *If the blocked equal-case Z-gauge extraction hypothesis and the blocked-to-root reconstruction hypothesis both hold, then the combined blocked equal-case root hypothesis holds.*

Proof. ✓ Proof Apply the Z-gauge extraction hypothesis to obtain a blocked $\mathbb{Z}_m$-gauge witness between C and $A^{[p]}$, and then apply the reconstruction hypothesis to that witness. □

Theorem 15.53 (Reduction to the blocked equal-case root hypothesis). Lean ✓ Stmt *Assume the blocked equal-case root hypothesis. Then every p-refinable tensor B in irreducible form II admits a left-canonical root A' with $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$.*

Proof. ✓ Proof Apply Theorem 15.48 to the refinement witness of B. This produces a blocked tensor C in irreducible form II with $\mathcal{E}_C = \mathcal{E}_B$ and the same MPV family as $A^{[p]}$. The blocked equal-case root hypothesis then gives a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$. Combining the two transfer-map identities yields $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$. □

Theorem 15.54 (Forward direction from the blocked equal-case root hypothesis). Lean ✓ Stmt *Assume the blocked equal-case root hypothesis. Then for every tensor B in irreducible form II, p-refinability of B implies p-divisibility of $\mathcal{E}_B$.*

Proof. ✓ Proof Combine Theorem 15.53 with the witness-level forward implication: once $\mathcal{E}_B$ is identified with $\mathcal{E}_{A'^{[p]}}$ for a left-canonical A', the blocking identity gives $\mathcal{E}_B = (\mathcal{E}_{A'})^p$. □

The following conditional statements isolate the refinability–divisibility equivalence from [DlCCSPG17, Theorem 4.1].

Theorem 15.55 (p-refinability iff transfer-map p-divisibility). Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the two root hypotheses used in the one-way statements: the forward left-canonical root hypothesis for refinability $\Rightarrow$ divisibility and the reverse transfer-map root hypothesis for divisibility $\Rightarrow$ refinability. Then B is p-refinable iff its transfer map $\mathcal{E}_B$ is p-divisible.*

Proof. ✓ Proof Bundle the forward implication (Theorem 15.56) with the reverse implication (Theorem 15.59) into a single biconditional. The forward direction assumes the left-canonical root hypothesis; the reverse direction assumes only the transfer-map root hypothesis. Together they give the full equivalence under those hypotheses. □

Theorem 15.56 (p-refinability implies transfer-map p-divisibility)**.** Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II. Assume a left-canonical root hypothesis: every refinement witness gives a tensor A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. Then p-refinability of B implies p-divisibility of $\mathcal{E}_B$.*

Proof. ✓ Proof The left-canonical root hypothesis supplies a witness A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (left-canonical) and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. The left-canonical property makes $\mathcal{E}_A$ a CPTP channel, and the blocking identity $\mathcal{E}_{A^{[p]}} = (\mathcal{E}_A)^p$ then identifies $\mathcal{E}_B$ with the p-th power of the channel $\mathcal{E}_A$. □

Definition 15.57 (Blocked-to-root channel-root hypothesis from Z-gauge)**.** Lean ✓ Stmt Fix (d, D, p). This sharpened reconstruction hypothesis asks for a CPTP map $\mathcal{E}'$ such that $\mathcal{E}_C = (\mathcal{E}')^p$ and the Choi/Kraus rank of $\mathcal{E}'$ is at most d, whenever C is related to $A^{[p]}$ by a blocked $\mathbb{Z}_m$-gauge witness.

Theorem 15.58 (Channel-root criterion for blocked-to-root reconstruction)**.** Lean ✓ Stmt *If the blocked-to-root channel-root hypothesis holds, then the tensor-level blocked-to-root reconstruction hypothesis holds as well.*

Proof. ✓ Proof Choose a tensor A' realizing the bounded Kraus family of the root channel. Because the channel is trace-preserving, the Kraus operators of A' satisfy $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$. The identity $\mathcal{E}_C = (\mathcal{E}_{A'})^p$ is then equivalent to $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ by the blocking formula. □

Theorem 15.59 (Transfer-map p-divisibility implies p-refinability)**.** Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the reverse transfer-map root hypothesis: there is a tensor A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ whenever $\mathcal{E}_B$ is p-divisible. Then p-divisibility of $\mathcal{E}_B$ implies p-refinability of B.*

Proof. ✓ Proof From the reverse transfer-map root hypothesis, extract A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. This matches two finite Kraus families of the same completely positive map: the blocked family $A^{[p]}$ with d^p operators and the original family B with d operators. The cardinality comparison $d \leq d^p$ holds whenever $p \geq 1$, so [Wol12, Theorem 2.18] supplies an isometry $V \in M_{d^p \times d}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $(A^{[p]})^\alpha = \sum_j V_{\alpha,j} B^j$. Expanding coeff$(A^{[p]})$ (offFn τ) by linearity of matrix multiplication and of the trace, the contributions reorganise into the V-weighted sum over length-N words of B required by Definition 15.47, exhibiting V as the isometry W in the statement of p-refinability. □

Remark 15.60 (Continuum-limit application)*.* Not ready The original motivation for Theorem 15.55 is a continuum-limit construction in the spirit of [DlCCSPG17]: refinability along arbitrarily large p is equivalent to infinite divisibility of the transfer map, which selects out continuum limits of translationally invariant MPS. The further consequences are explored elsewhere.

Chapter 16

The Algebraic Fundamental Theorem

This chapter develops the one-dimensional MPS results of [MGRPG+18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly with $M_D(\mathbb{C})$. Two preparatory lemmas—the virtual bond gauge (Lemma 16.13) and the proportionality lemma (Lemma 16.14)—carry the entire argument, yielding the main theorem and its corollaries for translation-invariant chains, non-translation-invariant descriptions of translation-invariant states, and blocked chains.

Throughout, D is the bond dimension and d is the physical dimension with $d \geq D^2$. The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ($2L + 1$ in place of $3L$) are not pursued in this chapter. The non-translation-invariant chain data, including same chain state, injectivity, and cyclic gauge equivalence, is set up below.

16.1 Alternative fixed-length non-TI setup

The algebraic proof of [MGRPG+18] keeps a fixed periodic chain length and allows the local tensors to vary from site to site. The first step is therefore a separate setup for finite non-translation-invariant chains.

Definition 16.1 (Periodic non-TI chain data). Lean Lean Lean Lean Lean ✓ Stmt Fix a chain length n. A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \qquad k = 0, \dots, n-1,$$

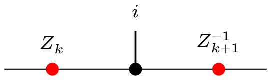
where each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \operatorname{tr}\left(A_0^{\sigma_0} A_1^{\sigma_1} \cdots A_{n-1}^{\sigma_{n-1}}\right).$$

Two such chains generate the *same chain state* if these coefficients agree for every σ. The chain is *injective* if each local tensor A_k is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site k and physical index i, with indices understood modulo n so that $Z_n = Z_0$. Locally, this gauge relation is represented by



where the black node denotes the site tensor named in the surrounding formula and the two red side nodes denote the bond gauges on the neighboring virtual legs.

Lemma 16.2 (Cyclic gauge equivalence is an equivalence relation). Lean Lean Lean ✓ Stmt *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

Proof. ✓ Proof Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. □

16.2 One-sided inverse and decomposition

If A is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \to M_D(\mathbb{C})$ is surjective and admits a right inverse.

Lemma 16.3 (Existence of a right inverse). Lean ✓ Stmt *If A is injective, then there exists a $\mathbb{C}$-linear map $\Psi : M_D(\mathbb{C}) \to \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \mathrm{id}_{M_D(\mathbb{C})}$.*

Proof. ✓ Proof Surjectivity of Φ_A yields a linear section by choice. □

Definition 16.4 (Decomposition map). Lean ✓ Stmt Let A be an injective MPS tensor. The *decomposition map* $\Psi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ is a fixed choice of right inverse of Φ_A.

Lemma 16.5 (Decomposition property). Lean ✓ Stmt *If A is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that*

$$X = \sum_{i=0}^{d-1} c_i A^i.$$

Proof. ✓ Proof Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. □

Remark 16.6 (Non-uniqueness of the decomposition). When $d > D^2$, the map Φ_A has a nontrivial kernel, so the section Ψ_A is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \mathrm{id}$, so the choice does not affect the conclusions.

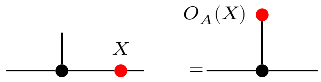
16.3 Physical realisation of virtual insertions

For an injective tensor A, the physical realisation map encodes the effect of inserting a virtual matrix X on the bond as a $d \times d$ operation on the physical indices.

Definition 16.7 (Physical realisation map). Lean ✓ Stmt Let A be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row i of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:



The map $X \mapsto O_A(X)$ is $\mathbb{C}$-linear.

Lemma 16.8 (Defining property of the physical realisation map). Lean ✓ Stmt *For every $X \in M_D(\mathbb{C})$ and every physical index i,*

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} \, A^j.$$

Proof. ✓ Proof By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying Φ_A gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \mathrm{id}$. □

Theorem 16.9 (Physical realisation is multiplicative). Lean ✓ Stmt *Let A be an injective MPS tensor. For all $X, Y \in M_D(\mathbb{C})$,*

$$O_A(XY) = O_A(X) \, O_A(Y).$$

Proof. ✓ Proof Applying Lemma 16.8 to X yields

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} \, A^j.$$

Multiply by Y on the right and apply the linear map Ψ_A:

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij} \, \Psi_A(A^j Y).$$

Taking the kth component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij} \, O_A(Y)_{jk} = (O_A(X) \, O_A(Y))_{ik}.$$

Since this holds for all i and k, the matrices are equal. □

Remark 16.10 (General injective case). When $d > D^2$, the decomposition map Ψ_A need not be unique. Nevertheless, Theorem 16.9 is already the general injective statement: for the fixed choice of Ψ_A in Definition 16.4, the associated physical realisation map O_A is multiplicative. This supersedes the earlier basis-only formulation.

16.4 Intermediate lemmas

Lemma 16.11 (Blocking preserves injectivity). Lean ✓ Stmt *Let $A_0, \ldots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor $(A_0 \otimes \cdots \otimes A_{m-1})^{(i_0, \ldots, i_{m-1})} := A_0^{i_0} A_1^{i_1} \cdots A_{m-1}^{i_{m-1}}$ is again injective.*

Proof. ✓ Proof The span of the blocked matrices contains all products $A_0^{i_0} \cdots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. □

Lemma 16.12 (Centre of $M_D(\mathbb{C})$). Lean ✓ Stmt *If $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}$.*

Proof. ✓ Proof Standard: Z commutes with every matrix unit E_{ij}, which forces all off-diagonal entries to vanish and all diagonal entries to agree. □

16.5 Same state forces a virtual bond gauge

This section presents the 3-site virtual-bond statement used later in the proof of the PEPS fundamental theorem. Here the hypothesis is already the all-length MPV equality of the combined tensors, and the conclusion is the conjugacy of the middle-bond insertion data.

Lemma 16.13 (Virtual bond gauge – 3-site case)**.** Lean ✓ Stmt *Let $A = (A_0, A_1, A_2)$ and $B = (B_0, B_1, B_2)$ be two 3-site injective periodic chains with common bond dimension $D \geq 1$ and $\mathcal{V}(A) = \mathcal{V}(B)$ (same MPV family). Then at the middle bond there exists an invertible matrix $Z \in \mathrm{GL}_D(\mathbb{C})$ such that the virtual-insertion trace coefficients agree after conjugation:*

$$\mathrm{tr}(A_0^{i_0} X\, A_1^{i_1} A_2^{i_2}) = \mathrm{tr}(B_0^{i_0} Z^{-1} X Z\, B_1^{i_1} B_2^{i_2})$$

for all $X \in M_D(\mathbb{C})$ and all physical indices i_0, i_1, i_2.

In [MGRPG+18, Lemma 1], the statement is given for arbitrary chain length $n \geq 3$ and any bond position, with uniqueness of Z up to scalar. The chain-level generalization to arbitrary $n \geq 3$ and any bond position is not pursued here.

Proof. ✓ Proof The $\mathcal{V}(A) = \mathcal{V}(B)$ hypothesis gives, for every $X \in M_D(\mathbb{C})$, equality of the virtual-insertion trace coefficients: $\mathrm{tr}(A_0^{i_0} X\, A_1^{i_1} A_2^{i_2}) = \mathrm{tr}(B_0^{i_0} Y\, B_1^{i_1} B_2^{i_2})$ for some Y depending on X. Since all three site tensors on each side are injective, the physical realisation maps O_A, O_B are algebra homomorphisms by Theorem 16.9. Setting $T(X) = Y$ defines a $\mathbb{C}$-linear multiplicative map, hence a nonzero $\mathbb{C}$-algebra endomorphism of $M_D(\mathbb{C})$, which is bijective by Theorem 3.8. Skolem–Noether (Theorem 3.10) gives $Z \in \mathrm{GL}_D(\mathbb{C})$ with $T(X) = Z^{-1} X Z$. ☐

16.6 Aligned gauges force proportionality

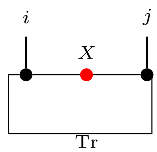
After applying Lemma 16.13 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG+18] shows that this agreement forces the local tensors to be proportional.

Lemma 16.14 (Aligned gauges force proportionality – Lemma 2 of [MGRPG+18])**.** Lean ✓ Stmt *Let A_1, A_2 and B_1, B_2 be injective MPS tensors with the same bond dimension D. Suppose that for all $X \in M_D(\mathbb{C})$ (on the internal bond), all $Y \in M_D(\mathbb{C})$ (on the external bond), and all physical indices i, j,*

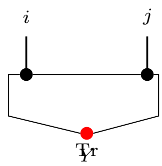
$$\mathrm{tr}(A_1^i\, X\, A_2^j) = \mathrm{tr}(B_1^i\, X\, B_2^j) \tag{16.1}$$

$$\mathrm{tr}(A_2^j\, Y\, A_1^i) = \mathrm{tr}(B_2^j\, Y\, B_1^i) \tag{16.2}$$

Diagrammatically, inserting X on the bond between the two sites (Eq. 16.1):



while the insertion Y on the outer trace bond (Eq. 16.2) is:



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that

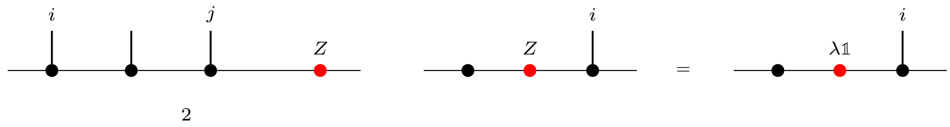
$$A_1^i = \lambda\, B_1^i \qquad \text{and} \qquad A_2^j = \lambda^{-1}\, B_2^j$$

for all physical indices i, j.

Proof. ✓ Proof By cyclicity and non-degeneracy of the trace pairing, the conditions in Eqs. (16.1) and (16.2) give, for all physical indices i, j,

$$\begin{aligned} A_2^j A_1^i &= B_2^j B_1^i, \\ A_1^i A_2^j &= B_1^i B_2^j. \end{aligned}$$

Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map Ψ_{A_2}, and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity gives $A_1^i = Z B_1^i$, and the second gives $A_1^i = B_1^i Z$. Hence Z commutes with every B_1^i; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, Z is central. By Lemma 16.12, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into the first product identity forces $A_2^j = \lambda^{-1} B_2^j$. ☐

16.7 The Fundamental Theorem for injective chains

The chain result below combines the site and physical indices into a single physical index and applies the single-block Fundamental Theorem to the resulting tensor. The fixed-length same-state statement requires a separate same-state reduction hypothesis.

Theorem 16.15 (Fundamental Theorem for injective chains)**.** Lean ✓ Stmt *Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two n-site chains with common bond dimension D, and assume that $\mathbf{A}$ is injective. Suppose that, after combining the site index and physical index into a single MPS tensor, the resulting two tensors generate the same MPV family. Then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.*

Proof. ✓ Proof Choose one site of $\mathbf{A}$; its injectivity implies injectivity of the combined tensor by Lemma 16.11. Apply the single-block Fundamental Theorem (Theorem 3.13) to the two combined tensors. This yields one invertible matrix X such that every combined matrix of $\mathbf{B}$ equals the corresponding combined matrix of $\mathbf{A}$ conjugated by X. Reading this identity back at each site gives $B_k^i = X A_k^i X^{-1}$ for all k and i, so the constant choice $Z_k = X$ is a cyclic gauge. ☐

Lemma 16.16 (Uniform site rescaling rescales the combined tensor)**.** Lean ✓ Stmt *Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an n-site chain and let $\zeta \in \mathbb{C}$. If $\zeta\mathbf{A}$ denotes the chain obtained by replacing each site tensor A_k with ζA_k, and if $\widetilde{\mathbf{A}}$ denotes the combined tensor of $\mathbf{A}$, then*

$$\widetilde{\zeta\mathbf{A}} = \zeta\, \widetilde{\mathbf{A}}.$$

Proof. ✓ Proof Combining the site and physical indices does not change which local matrix is read at the component (k, i), so both sides evaluate to ζA_k^i there. ☐

Theorem 16.17 (Injective-chain Fundamental Theorem up to a global phase). Lean ✓ Stmt *Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two n-site chains with common bond dimension D, and assume that $\mathbf{A}$ is injective. Suppose that the combined tensors $\widetilde{\mathbf{A}}$ and $\widetilde{\mathbf{B}}$ are gauge-phase equivalent. Then there exist invertible matrices $Z_0, \dots, Z_{n-1} \in \mathrm{GL}_D(\mathbb{C})$ and a nonzero scalar $\zeta \in \mathbb{C}^\times$ such that*

$$B_k^i = \zeta\, Z_k A_k^i Z_{k+1}^{-1}$$

for all sites k and physical indices i, where $k+1$ is interpreted cyclically modulo n.

Proof. ✓ Proof If the combined tensor of $\mathbf{B}$ differs from that of $\mathbf{A}$ by a gauge and a nonzero scalar ζ, rescale each site of $\mathbf{B}$ by ζ^{-1}. Lemma 16.16 removes this common scalar from the combined tensor, so Theorem 16.15 applies to the rescaled chain. Multiplying the resulting sitewise relations by ζ gives the displayed conclusion. ☐

16.7.1 From fixed-length chain states to combined-tensor MPV agreement

Definition 16.18 (Same-state reduction hypothesis). Lean ✓ Stmt A *same-state reduction hypothesis* for dimensions (d, D) is the assertion that whenever $\mathbf{A}$ is an injective n-site chain with $n \geq 3$ and $\mathbf{A}, \mathbf{B}$ generate the same chain state, then the combined tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.

Lemma 16.19 (Same state gives combined-tensor MPV agreement). Lean ✓ Stmt *Assume a same-state reduction hypothesis for (d, D). Let $\mathbf{A}$ and $\mathbf{B}$ be n-site chains with common bond dimension D, with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then the combined tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.*

Proof. ✓ Proof This is exactly the defining property of the same-state reduction hypothesis. ☐

Theorem 16.20 (Fundamental Theorem for injective chains from same state). Lean ✓ Stmt *Assume a same-state reduction hypothesis for (d, D). Let $\mathbf{A}$ and $\mathbf{B}$ be n-site chains with common bond dimension D, with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.*

Proof. ✓ Proof Apply Lemma 16.19 to convert the fixed-length chain-state hypothesis into equality of the combined-tensor MPV families, then invoke Theorem 16.15. ☐

Remark 16.21 (Relationship to the translation-invariant all-sizes theorem). Theorem 16.15 treats a fixed chain length and assumes equality of the combined-tensor MPV families. The single-block Fundamental Theorem (Theorem 3.13) instead studies one tensor at all system sizes and concludes exact gauge equivalence $B^i = XA^iX^{-1}$. The chain theorem is recovered by converting the chain into one combined tensor; the price is that the hypothesis is all-length MPV equality for that combined tensor rather than equality of the chain state at one fixed length.

16.7.2 Finite-length MPV agreement

Definition 16.22 (Finite-length MPV agreement). Lean ✓ Stmt Two tensors A and B of the same bond dimension satisfy *finite-length MPV agreement from N_0 onward* if they generate the same MPV family for all system sizes $N \geq N_0$, meaning that for every $N \geq N_0$ and $\sigma \in [d]^N$,

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma$$

Theorem 16.23 (Finite-length agreement implies full agreement). Lean ✓ Stmt *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B generate the same MPV family for every system size $N \geq N_0$, then they generate the same MPV family for every system size.*

Proof. ✓ Proof Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show that the length-N_0 word span of B is all of $M_D(\mathbb{C})$ via the linear relation between the length-N_0 word-evaluation maps of A and B induced by trace agreement at length $N_0 + 1$, together with a finrank transfer argument.

2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\mathrm{tr}(M(S - 1)) = 0$.

3. Downward induction on $k = N_0 - |w|$: decompose $\mathrm{tr}(w_A)$ and $\mathrm{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

□

Theorem 16.24 (Strengthened single-block FT (finite-length)). Lean ✓ Stmt *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold N_0), then A and B are gauge equivalent.*

Proof. ✓ Proof Apply Theorem 16.23 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 3.13). □

16.8 Corollaries

16.8.1 Translation-invariant chains

Theorem 16.25 (Single gauge for translation-invariant chains). Lean ✓ Stmt *Let A and B be MPS tensors of bond dimension D, and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that*

$$B^i = X\, A^i\, X^{-1}$$

for every physical index i.

Proof. ✓ Proof Bundle the constant chains into one tensor. Injectivity of A implies injectivity of the combined tensor by Lemma 16.11, and the combined-tensor MPV equality is exactly the hypothesis needed for the single-block Fundamental Theorem. The resulting gauge matrix is independent of the site because every site carries the same local tensor. □

Corollary 16.26 (Translation-invariant collapse to one gauge). Lean ✓ Stmt *Let A and B be MPS tensors of bond dimension D, and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exist an invertible matrix $Z \in \mathrm{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that*

$$B^i = \lambda\, Z^{-1}\, A^i\, Z$$

for every physical index i.

Proof. ✓ Proof Apply Theorem 16.25 to obtain $B^i = XA^iX^{-1}$. Set $Z = X^{-1}$ and $\lambda = 1$. Then $\lambda^n = 1$ and the displayed identity is exactly the same gauge relation rewritten in the form used by the corollary. □

16.8.2 Non-translation-invariant descriptions of translation-invariant states

Corollary 16.27 (Translation-invariant states from nonuniform chains). *Not ready* *Assume a same-state reduction hypothesis for (d, D). Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an injective non-translation-invariant chain of length $n \geq 3$. Suppose the state generated by $\mathbf{A}$ is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain $(A_1, \dots, A_{n-1}, A_0)$. Then $\mathbf{A}$ admits a translation-invariant description with the same bond dimension: there exists an injective tensor B such that the constant chain $(B, \dots, B)$ generates the same state as $\mathbf{A}$. Explicitly, $B^i = A_0^i Z_0^{-1}$ (right multiplication,* not *conjugation), where Z_0 is extracted from the cyclic shift.*

Proof. Apply Theorem 16.20 to $\mathbf{A}$ and its cyclic shift. This gives invertible gauges $Z_0, \dots, Z_{n-1}$ with $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$. Iterating these identities yields $A_k^i = L_k^{-1} A_0^i R_k$, where the products L_k, R_k satisfy $R_k L_{k+1}^{-1} = Z_0^{-1}$. Substituting into the chain trace shows that the same state is generated by the constant tensor $B^i := A_0^i Z_0^{-1}$. Since A_0 is injective and Z_0 is invertible, B is injective. □

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 14.

16.8.3 Blocked chains

Lemma 16.28 (Block injectivity and blocked-tensor injectivity). *Lean* *✓ Stmt* *A tensor A is L-block injective if and only if its L-blocked tensor $A^{[L]}$ is injective.*

Proof. ✓ Proof The blocked physical alphabet is just a reindexing of words of length L. Decoding blocked indices therefore identifies the two spanning families, so the two injectivity conditions are equivalent. □

Definition 16.29 (Constant blocked chain). Lean ✓ Stmt For an MPS tensor A, a blocking length L, and a chain length n, the *blocked chain* is the constant n-site chain whose local tensor is the blocked tensor $A^{[L]}$.

Lemma 16.30 (Blocked chains are injective). *Lean* *✓ Stmt* *If A is L-block injective, then the constant blocked chain built from $A^{[L]}$ is injective at every site.*

Proof. ✓ Proof By Lemma 16.28, the local blocked tensor $A^{[L]}$ is injective, and therefore every site of the constant blocked chain is injective. □

Theorem 16.31 (Fundamental Theorem for blocked chains). *Lean* *✓ Stmt* *Let A and B be MPS tensors of bond dimension D, let L be a blocking length, and let n be a chain length. Assume that A is L-block injective and that the combined tensors of the two constant blocked chains built from $A^{[L]}$ and $B^{[L]}$ generate the same MPV family. Then the two blocked chains are gauge equivalent.*

Proof. ✓ Proof By Lemma 16.30, the blocked chain built from $A^{[L]}$ is injective. Apply Theorem 16.15 to the two constant blocked chains. □

Remark 16.32. The theorem above is the blocked-chain result established here. Recovering site-wise gauges for the original unblocked tensors gives the normal-MPS corollary of [MGRPG+18]; that step is outside this blocked-chain statement.

16.9 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 16.15) handles non-translation-invariant periodic chains. A parallel line of argument addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions D_k and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k A_k^i X_k^{-1}$ for each k independently, and a global block-permutation plus inner-automorphism decomposition. Throughout this section, assume $D_k \geq 1$ for every block k.

16.9.1 Per-block linear extension

Definition 16.33 (Per-block linear extension). Lean ✓ Stmt Let A_k, B_k be MPS tensors of bond dimension D_k with A_k injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \ldots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \to M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$-linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index i.

Lemma 16.34 (Per-block multiplicativity). Lean ✓ Stmt *For each block k, the per-block linear extension T_k is multiplicative: $T_k(MN) = T_k(M)\, T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.*

Proof. ✓ Proof Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. □

Lemma 16.35 (Per-block bijectivity). Lean ✓ Stmt *Each T_k is bijective.*

Proof. ✓ Proof A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of T_k is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. □

16.9.2 Product algebra automorphism

Definition 16.36 (Product algebra automorphism). Lean ✓ Stmt The *product algebra automorphism* is the $\mathbb{C}$-algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each T_k is the per-block linear extension. It is an algebra automorphism since each T_k is a multiplicative, unital, bijective linear map. Diagrammatically, this is the componentwise map on the product algebra obtained by applying the corresponding extension to each summand.

Theorem 16.37 (Block-permutation decomposition). Lean ✓ Stmt *The product algebra automorphism T decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all k) and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that*

$$(T(M))_{\sigma(k)} = X_k\, M\, X_k^{-1}$$

for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block k) and all k. Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.

Proof. ✓ Proof The automorphism T permutes the block ideals of $\prod_k M_{D_k}(\mathbb{C})$ (each block ideal is simple, and an automorphism sends simple ideals to simple ideals of the same dimension). This gives the permutation σ. On each block, the restricted map is a $\mathbb{C}$-algebra automorphism of $M_{D_k}(\mathbb{C})$; by Skolem–Noether (Theorem 3.10), it is inner. Thus T splits algebraically into the permutation of block ideals followed by inner conjugation on each surviving block. □

16.9.3 Nondegeneracy of the product trace pairing

Lemma 16.38 (Nondegeneracy of the product trace pairing). Lean ✓ Stmt *Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \operatorname{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.*

Proof. ✓ Proof Choose N supported on a single block k (all other components zero). The hypothesis reduces to $\operatorname{tr}(M_k N_k) = 0$ for all N_k, so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since k was arbitrary, $M = 0$. □

Lemma 16.39 (Injectivity of the product Gram map). Lean ✓ Stmt *Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \operatorname{tr}(M_k \cdot A_k^i))$ is injective.*

Proof. ✓ Proof Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \operatorname{tr}(M_k \cdot A_k^i))$, which holds by injectivity of A_k (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). □

16.10 Gauge equivalence for direct sums

The first lemmas in this section are elementary facts about block-diagonal tensors: they assemble already-known block gauges, and they record the easy consequences of applying the injective single-block theorem to each block separately. They do not derive the block matching appearing in [CPGSV17a, Theorem II.1]. In the source proof, that matching is obtained from BNT overlap, independence, and coefficient comparison; the corresponding global gauge assembly is Theorem 12.57.

Definition 16.40 (Flattened block-diagonal gauge). Lean Lean ✓ Stmt Given per-block gauges $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$, first form the dependent block-diagonal element $\bigoplus_k X_k$ and then reindex the dependent block-coordinate index to the standard index $\{0, \dots, \sum_k D_k - 1\}$ of the assembled tensor. The resulting element of $\mathrm{GL}_{\sum_k D_k}(\mathbb{C})$ is the global gauge used in the conjugation formula for $\bigoplus_k \mu_k A_k$.

Lemma 16.41 (Explicit direct-sum conjugation by the flattened gauge). Lean Lean ✓ Stmt *If each block satisfies $B_k^i = X_k A_k^i X_k^{-1}$, then the weighted direct sums satisfy, for every physical index i,*

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1},$$

where X is the flattened block-diagonal gauge of Definition 16.40. Consequently the two assembled tensors are gauge equivalent with this explicit witness.

Proof. ✓ Proof The block-diagonal matrix product is computed blockwise. The scalar weight μ_k commutes with the per-block conjugation, so each diagonal block of the right-hand side is $\mu_k B_k^i$. Reindexing from the dependent block coordinates to the standard bond index preserves multiplication and inverses. The gauge equivalence follows directly from the conjugation identity. □

Lemma 16.42 (Gauge equivalence of direct sums from per-block gauge equivalence). Lean ✓ Stmt *If each pair (A_k, B_k) is gauge equivalent, then the weighted block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. ✓ Proof Choose per-block gauge matrices via the hypothesis; then apply Lemma 16.41. □

Lemma 16.43 (Gauge equivalence of direct sums with phase-weight absorption). Lean ✓ Stmt *Let A_k, B_k be block families, and let $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ and scalars $\zeta_k \in \mathbb{C}$ satisfy $B_k^i = \zeta_k \cdot X_k A_k^i X_k^{-1}$ for every block k and physical index i. If the weights satisfy $\mu_A(k) = \mu_B(k)\,\zeta_k$, then the weighted block-diagonal tensors $\bigoplus_k \mu_A(k) A_k$ and $\bigoplus_k \mu_B(k) B_k$ are gauge equivalent.*

Proof. ✓ Proof Per-block, the weight identity $\mu_A(k) = \mu_B(k)\,\zeta_k$ rewrites the scaled relation as $\mu_B(k)\,B_k^i = X_k\,(\mu_A(k)\,A_k^i)\,X_k^{-1}$. Each weighted block then satisfies an ordinary conjugation, and the block-diagonal conjugation formula assembles the per-block X_k into the flattened global gauge $\bigoplus_k X_k$ relating $\bigoplus_k \mu_A(k) A_k$ to $\bigoplus_k \mu_B(k) B_k$. □

16.10.1 Same-index direct-sum lemmas from the single-block theorem

Remark 16.44. See Definition 2.39 for the formalized block-diagonal tensor construction. The results in this subsection assume that the block correspondence has already been fixed; they do not prove [CPGSV17a, Theorem II.1]. They are therefore retained here as same-index assembly corollaries, not among the BNT preparation steps used in the source proof or in the after-blocking canonical-form reduction.

Lemma 16.45 (Per-block application of the single-block theorem). Lean ✓ Stmt *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. ✓ Proof Apply the single-block Fundamental Theorem (Theorem 3.13) to each block independently. □

Remark 16.46 (Per-block restatement, not the multi-block paper theorem). This is a block-sum reformulation of the per-block injective single-block FT: the block pairing (A_k, B_k) is assumed already given and the blocks already matched. It is not the multi-block Fundamental Theorem of [CPGSV17a, Theorem II.1, lines 349–352], which derives the block matching and permutation from only the global same-MPV hypothesis.

Lemma 16.47 (Per-block same MPV implies global same MPV). Lean ✓ Stmt *If A_k and B_k generate the same MPV family for each block k, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. ✓ Proof By the MPV decomposition (Theorem 2.41), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. □

Remark 16.48 (Direct-sum congruence). This is a direct-sum congruence statement: per-block MPV equality lifts to global MPV equality via the block-diagonal decomposition formula.

Lemma 16.49 (Gauge equivalence for same-index direct sums). Lean ✓ Stmt *If each block A_k is injective and each pair (A_k, B_k) generates the same MPV family, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. ✓ Proof Apply the per-block single-block Fundamental Theorem (Lemma 16.45) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors. □

Remark 16.50 (Block-diagonal gauge assembly, not the FT block-matching theorem)*.* This combines per-block gauges into a block-diagonal gauge for the global assembled tensor, given pre-matched same-index blocks. It is not the FT block-matching theorem: it does not derive the block pairing or permutation from a global same-MPV hypothesis; the block correspondence is assumed in advance.

Lemma 16.51 (Gauge equivalence of direct sums from per-block MPV equality)**.** Lean ✓ Stmt *Let* $\{A_k\}_{k=0}^{r-1}$ *and* $\{B_k\}_{k=0}^{r-1}$ *be two families of MPS tensors with block dimensions* D_k*, each* A_k *injective, and* $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ *for all* k*. Then:*

1. *Per-block gauge equivalence:* $B_k^i = X_k\, A_k^i\, X_k^{-1}$ *for some* $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$*, for every* k *and* i*.*
2. *Global gauge equivalence: the block-diagonal tensors* $\bigoplus_k \mu_k A_k$ *and* $\bigoplus_k \mu_k B_k$ *are gauge equivalent.*

Explicitly, for $X = \bigoplus_k X_k$*,*

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1}.$$

Proof. ✓ Proof Per-block gauge equivalence follows from applying the single-block Fundamental Theorem to each block, using injectivity of A_k and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. With $X = \bigoplus_k X_k$, the displayed identity follows by multiplying block-diagonal matrices block by block. □

Lemma 16.52 (Explicit per-block gauges)**.** Lean ✓ Stmt *Under the hypotheses of Lemma 16.51, there exist invertible matrices* $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ *such that*

$$B_k^i = X_k\, A_k^i\, X_k^{-1}$$

for every block k *and every physical index* i*.*

Proof. ✓ Proof Extract the gauge matrix from the per-block part of Lemma 16.51 for each block. □

Lemma 16.53 (Product-algebra decomposition of per-block MPV equality)**.** Lean ✓ Stmt *Assume that every block dimension is nonzero. Under the hypotheses of Lemma 16.51, the product-algebra automorphism determined by the per-block linear extensions decomposes into a block permutation together with inner conjugation on each block: there exist a permutation* σ *of the blocks, compatible block-dimension equalities, and invertible matrices* X_k *such that, after reindexing the target block* $\sigma(k)$ *back to block* k*, the action on the* k*th matrix block is* $M \mapsto X_k M X_k^{-1}$*.*

Proof. ✓ Proof Apply Theorem 16.37 to the product-algebra automorphism built from the per-block linear extensions. □

16.10.2 MPV invariance under reindexing

Lemma 16.54 (MPV invariance under block permutation)**.** Lean ✓ Stmt *Permuting the block index set and transporting the weights correspondingly does not change the* $\mathrm{SameMPV}_2$ *family of the weighted block-diagonal tensor.*

Proof. ✓ Proof The sum of per-block MPV contributions is invariant under rearrangement of the summands. □

Lemma 16.55 (MPV invariance under dimension cast). Lean ✓ Stmt *If two block-dimension families agree pointwise, the corresponding weighted block-diagonal tensors generate the same* $\mathrm{SameMPV}_2$ *family after transporting the blocks across the dimension equalities.*

Proof. ✓ Proof Substituting the equality of dimensions makes the two tensor families identical. □

16.10.3 Converse: gauge equivalence implies same MPV

Lemma 16.56 (Gauge equivalence of direct sums implies same MPV). Lean ✓ Stmt *If the weighted block-diagonal tensors* $\bigoplus_k \mu_k A_k$ *and* $\bigoplus_k \mu_k B_k$ *are gauge equivalent, then they generate the same MPV family.*

Proof. ✓ Proof Gauge equivalence preserves all traces of word evaluations, hence all MPV coefficients. □

16.10.4 Canonical-form tensor as a weighted direct sum

Lemma 16.57 (Canonical-form tensor agrees with block sum). Lean ✓ Stmt *For a canonical-form record* C*, the assembled tensor* T_C *is definitionally equal to the weighted block-diagonal tensor* $\bigoplus_k \mu_k^C A_k^C$*, where* μ_k^C *are the block weights and* A_k^C *are the block tensors stored in* C*.*

Proof. ✓ Proof The equality holds by definition; T_C is defined as the weighted block-diagonal tensor formed from the weights and block tensors of C. □

Theorem 16.58 (Fundamental theorem for canonical forms with the same block structure). Lean ✓ Stmt *Let* C *be a canonical-form record with block weights* μ_k^C *and block tensors* A_k^C*. Let* $\{B_k\}$ *be a family of MPS tensors on the same block dimensions, and assume that each* A_k^C *is injective and generates the same MPV family as* B_k*. Then* T_C *and* $\bigoplus_k \mu_k^C B_k$ *are gauge equivalent.*

Proof. ✓ Proof Rewrite T_C as a weighted block-diagonal tensor via Lemma 16.57, then apply the multi-block global gauge-equivalence lemma (Lemma 16.49). □

Lemma 16.59 (Per-block equality of MPV families and per-block gauge equivalence). Lean ✓ Stmt *Let* $\{A_k\}$ *be a family of injective block tensors. Then* $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ *for all* k *if and only if* A_k *and* B_k *are gauge equivalent for all* k*.*

Proof. ✓ Proof The forward direction applies the single-block Fundamental Theorem to each block. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. □

16.11 Block separation from ordered canonical-form data

Lemma 16.51 assumes $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k. In practice, one starts from the weaker global hypothesis that the block-diagonal tensors satisfy the same MPV family: $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$. Block separation extracts the per-block identities from the global one under canonical-form normalization plus a strict separation of the weight moduli. The first step is the weighted block-sum identity

$$\sum_k \mu_k^N V^{(N)}(A_k)_\sigma = \sum_k \mu_k^N V^{(N)}(B_k)_\sigma,$$

which is then combined with overlap estimates and the leading-block peel.

The bundled canonical-form predicate and its same-structure block-separation theorem are recorded once, in Chapter 11 (Definition 11.12 and Lemma 11.19). This chapter only records the separated-data refinements used by the algebraic fundamental-theorem statements below.

Remark 16.60. See Definition 9.24 for the normal canonical-form data statement.

Lemma 16.61 (Block separation from weighted word identities). Lean ✓ Stmt *Let $\{A_k\}$ and $\{B_k\}$ be block families with strictly ordered, nonzero weights μ_k. Assume that every A_k and B_k is injective and left-canonical, that each self-overlap $O_{A_k A_k}(N)$ tends to 1, and that for every system size N and every word σ one has the weighted identity*

$$\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0.$$

Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k.

Proof. ✓ Proof The proof is the peeling argument described below for canonical-form data equipped with a separate strict ordering of the weight moduli. The strict weight ordering isolates the leading block, the overlap bounds show that the tail decays geometrically after dividing by the dominant weight, and one then iterates on the remaining blocks. □

Lemma 16.62 (Block separation from separated canonical data). Lean ✓ Stmt *Suppose the weights are nonzero with strictly decreasing norms, the A-blocks are injective and left-canonical with normalized self-overlaps, the B-blocks are injective and left-canonical, and the block-diagonal tensors generate the same MPV family. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k.*

Proof. ✓ Proof Equality of the block-diagonal MPV families gives the weighted word identity required by Lemma 16.61. The separated hypotheses supply exactly the injectivity, left-canonicality, strict-weight, and self-overlap assumptions needed there. □

Lemma 16.63 (Gauge comparison from separated canonical data). Lean ✓ Stmt *Under the hypotheses of Lemma 16.62, every block pair A_k, B_k is gauge equivalent, and the block-diagonal tensors are globally gauge equivalent.*

Proof. ✓ Proof First apply Lemma 16.62 to get $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k. Then apply Lemma 16.51. □

Lemma 16.64 (Explicit gauge from separated canonical data). Lean ✓ Stmt *Under the same hypotheses, there exist invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that*

$$B_k^i = X_k A_k^i X_k^{-1}$$

for every block k and physical index i.

Proof. ✓ Proof Combine Lemma 16.62 with the matrices X_k from Lemma 16.52. □

Lemma 16.65 (Explicit gauge from canonical-form data and strict weights). Lean ✓ Stmt *Under the canonical-form hypotheses of Definition 11.12, together with strict decrease of the weight moduli and the corresponding injective left-canonical comparison family, there exist invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that $B_k^i = X_k A_k^i X_k^{-1}$ for every block k and physical index i.*

Proof. ✓ Proof Apply Lemma 16.64 with the strict-ordering data from the separated hypotheses. □

16.12 Equal-case block matching with differing bond dimensions

Lemma 16.66 (Equal-case block matching with differing bond dimensions). Lean ✓ Stmt *Let P and Q be two BNT canonical forms (Definition 12.30), each admitting a unit-modulus copy weight in every basis sector, possibly with different sector counts and different bond dimensions before matching. If their block-diagonal tensors generate the same MPV family (Definition 2.12), then the basis sectors are bijectively matched by a permutation β, each matched basis pair is gauge-phase equivalent (Definition 2.23), the matched copy multiplicities agree, and the copy weights agree up to a unit-modulus phase ζ and a copy-index reindexing. This is the sector-data form of the equal-case theorem [CPGSV17a, Corollary II.2, lines 354–361 and appendix 1172–1192]; the corresponding global gauge corollary is recorded in Chapter 13.*

Proof. ✓ Proof On the BNT canonical form, the proof proceeds in three steps: bijective block matching from non-decaying cross-overlaps, coefficient identity via the global-gauge phase ζ, and multiset comparison of the scalar power sums recovering matched copy multiplicities and weights up to ζ^{-1}. □

16.13 Translation-invariant reduction

Theorem 16.67 (Translation-invariant reduction corollary). Lean ✓ Stmt *Let A and B be MPS tensors of bond dimension D with A injective. Consider the two constant translation-invariant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n \geq 1$. If the combined tensors of the two chains generate the same MPV family, then:*

1. *B is gauge equivalent to A: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ with $B^i = X\, A^i\, X^{-1}$ for all i.*
2. *B is injective.*

Proof. ✓ Proof Theorem 16.25 gives a single gauge matrix X with $B^i = XA^iX^{-1}$ for all i. Since gauge equivalence preserves injectivity, B is injective as well. □

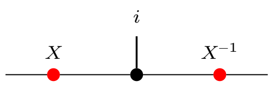
Theorem 16.68 (Translation-invariant reduction from same state). Lean ✓ Stmt *Assume a same-state reduction hypothesis for (d, D). Let A and B be MPS tensors of bond dimension D, and let $n \geq 3$. If the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same chain state, with A injective, then the same conclusion as in Theorem 16.67 holds: B is gauge equivalent to A, and B is injective.*

Proof. ✓ Proof The same-state reduction hypothesis upgrades equality of the two constant chain states to equality of their combined-tensor MPV families. Then Theorem 16.67 applies. □

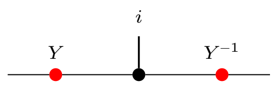
16.13.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 14.15) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

Lemma 16.69 (Gauge ratio commutes with tensor matrices). Lean ✓ Stmt *Diagrammatically, the hypothesis is that the two local gauges*



and



produce the same target tensor B^i from the same source tensor A^i. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = XA^iX^{-1} = YA^iY^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i.

Proof. ✓ Proof Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. □

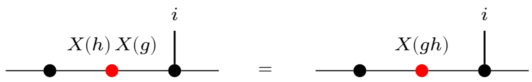
Lemma 16.70 (Gauge ratio is scalar for injective tensors). Lean ✓ Stmt *Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.*

Proof. ✓ Proof The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma 16.69, $Y^{-1}X$ commutes with all A^i. Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 16.12, it is a scalar matrix. □

Theorem 16.71 (Projective multiplication law for gauge matrices). Lean ✓ Stmt *Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \mathrm{GL}_D(\mathbb{C})$ satisfy $\widetilde{A}^i_g = X(g)\, A^i\, X(g)^{-1}$ for all i. Then for all $g, h \in G$, there exists $c(g,h) \in \mathbb{C}$ such that*

$$X(h)\, X(g) = c(g,h)\, X(g \cdot h).$$

In fact $c(g,h) \neq 0$ since both sides are invertible, so $c(g,h) \in \mathbb{C}^\times$. Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:



and both sides conjugate A^i to the same twisted tensor $\widetilde{A}^i_{gh}$.

Proof. ✓ Proof By the composition law $\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g$ (Lemma 14.5), both $X(h)\,X(g)$ and $X(gh)$ conjugate A to $\widetilde{A}_{gh}$. Lemma 16.70 then gives the scalar factor. □

16.14 Closing remarks

Remark 16.72. Appendix A of [MGRPG+18] strengthens the normal-MPS bound from $n \geq 3L$ to $n \geq 2L+1$ by a more refined blocking argument. The injective PEPS theorem of Chapter 17 and the normal PEPS theorem on arbitrary graphs both rest on the same two core lemmas (Lemma 16.13 and Lemma 16.14); the graph geometry enters only through the blocking step that reduces to a three-site chain.

Chapter 17

Fundamental Theorems for Injective and Normal PEPS

The following definitions and results extend the Fundamental Theorem to injective and normal PEPS, following Theorems 2 and 3 of [MGRPG+18].

Definition 17.1 (Graph edge for PEPS). Lean ✓ Stmt For a finite simple graph G, an *edge* is an ordered pair (u, v) with $u < v$ and u adjacent to v. The ordering is only used to avoid double counting.

Definition 17.2 (Incident edges at a vertex). Lean ✓ Stmt For a vertex v, an *incident edge* is an edge of G having v as one of its endpoints.

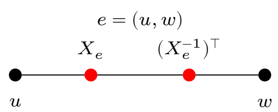
Definition 17.3 (PEPS tensor on a graph). Lean ✓ Stmt A PEPS tensor on G with physical dimension d consists of a bond dimension D_e for every edge e, together with a local tensor at each vertex v whose virtual indices are indexed by the edges incident to v and whose physical index lies in $\{0, \dots, d-1\}$.

Definition 17.4 (Virtual configuration). Lean ✓ Stmt A *virtual configuration* assigns to every edge e one basis index in the corresponding bond space of dimension D_e.

Definition 17.5 (PEPS state coefficient). Lean ✓ Stmt For a physical configuration σ on the vertices, the PEPS state coefficient is obtained by summing over all virtual configurations and, for each such configuration, multiplying the local tensor entries over all vertices.

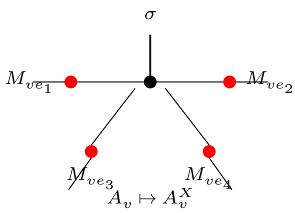
Definition 17.6 (Equality of PEPS states). Lean ✓ Stmt Two PEPS tensors represent the same state if all their state coefficients agree for every physical configuration.

Definition 17.7 (Edge gauge at a vertex). Lean ✓ Stmt For an edge $e = (u, w)$ with $u < w$ and gauge $X_e \in \mathrm{GL}(D_e)$, the gauge matrix applied at vertex v is X_e if $v = u$ and $(X_e^{-1})^\top$ if $v = w$. Diagrammatically, the ordered edge carries opposite endpoint actions:



Definition 17.8 (Gauge vertex action). Lean ✓ Stmt The gauged tensor at vertex v is A_v^X. For each incident edge ie, write M_{ie} for the matrix specified by Definition 17.7, applied at the vertex v

to the gauge on the underlying edge of ie. Then $A_v^X(\eta,\sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie),\eta'(ie)))A_v(\eta',\sigma)$. In tensor-network notation, one inserts the appropriate gauge matrix on each incident virtual leg of the local tensor:



Definition 17.9 (Apply gauge). Lean ✓ Stmt The PEPS tensor obtained by applying edge-gauge matrices to A, with vertex tensors A_v^X.

Definition 17.10 (Absorb edge gauges). Lean ✓ Stmt Given a tensor family B and an oriented gauge matrix X_e on every edge, the absorbed tensor family is $\widetilde{B} = B^X$; at each vertex, its local tensor is obtained by inserting the oriented endpoint matrices on the incident virtual legs.

Theorem 17.11 (Absorbed gauges keep the bond spaces). Lean ✓ Stmt *The absorbed tensor family has $D_{\widetilde{B}}(e) = D_B(e)$ for every edge e.*

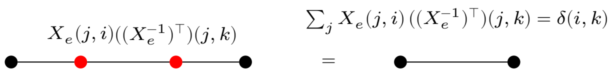
Proof. ✓ Proof Since $\widetilde{B} = B^X$ is defined with the same bond-dimension function as B, one has $D_{\widetilde{B}}(e) = D_{B^X}(e) = D_B(e)$ for every edge e. $\square$

Theorem 17.12 (Local formula for absorbed gauges). Lean ✓ Stmt *If M_{ie} is the oriented endpoint matrix attached to an incident edge ie at v, then $\widetilde{B}_v(\eta,\sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie),\eta'(ie)))B_v(\eta',\sigma)$.*

Proof. ✓ Proof Expanding $\widetilde{B} = B^X$ at a vertex gives $\widetilde{B}_v(\eta,\sigma) = (B^X)_v(\eta,\sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie),\eta'(ie)))B_v(\eta',\sigma)$. $\square$

Definition 17.13 (Gauge equivalence for PEPS). Lean ✓ Stmt Two PEPS tensors A, B are gauge-equivalent if they have the same bond dimensions and B is obtained from A by applying invertible gauge matrices on each edge.

Theorem 17.14 (Gauge invariance of PEPS state). Lean ✓ Stmt *Applying a gauge to a PEPS tensor preserves the state coefficients. Along each internal edge, the two endpoint insertions cancel:*



Theorem 17.15 (Gauge equivalence implies same state). Lean ✓ Stmt *Gauge equivalence implies that the two PEPS tensors represent the same state.*

Definition 17.16 (Local tensor evaluation). Lean ✓ Stmt The local tensor evaluated at vertex v with virtual-index weighting f, computing $\sum_\eta(\prod_{ie} f(ie)(\eta(ie))) \cdot A_v(\eta,\sigma)$. This is multilinear in the components of f, not linear in the full tuple.

Definition 17.17 (Vertex injectivity). Lean ✓ Stmt A PEPS tensor A is *vertex-injective* if, at every vertex v, the family of physical vectors

$$(A_v(\eta,\cdot) \in \mathbb{C}^d)_\eta$$

indexed by virtual configurations η on the edges incident to v is linearly independent in $\mathbb{C}^d$. Equivalently, extending A_v by linearity to a map from the free vector space on virtual configurations into the physical space $\mathbb{C}^d$ gives a linear map with trivial kernel. This matches the injectivity formulation of [MGRPG+18, Section 3], where the local tensor is interpreted as a map from virtual configurations to the physical space.

Definition 17.18 (Local tensor map). Lean ✓ Stmt For a fixed vertex v, the family of local physical vectors $\eta \mapsto A_v(\eta, \cdot)$ extends by linearity to a map from the coefficient space on local virtual configurations at v into $\mathbb{C}^d$.

Definition 17.19 (Local left inverse). Lean ✓ Stmt If A is vertex-injective, one may choose a linear map Ψ_v from $\mathbb{C}^d$ to the coefficient space on local virtual configurations at v such that $\Psi_v \circ \Phi_v = \mathbb{1}$.

Definition 17.20 (Physical realization of a local virtual operator). Lean ✓ Stmt For an endomorphism T of the coefficient space on local virtual configurations at v, define the local physical operator

$$O_T = \Phi_v \circ T \circ \Psi_v.$$

Theorem 17.21 (Local virtual operations on the image). *Lean* *✓ Stmt* *For every coefficient function c on the local virtual configurations at v,*

$$O_T(\Phi_v(c)) = \Phi_v(Tc).$$

Proof. ✓ Proof This is immediate from $\Psi_v \circ \Phi_v = \mathbb{1}$. □

Theorem 17.22 (Injectivity of local physical realization). *Lean* *✓ Stmt* *If two virtual operations have the same canonical physical realization, then they agree.*

Proof. ✓ Proof Evaluate both physical realizations on vectors in the image of the local tensor map and use vertex injectivity. □

Definition 17.23 (Local projector). Lean ✓ Stmt The local projector is the physical realization of the identity virtual operator:

$$P_v = \Phi_v \circ \Psi_v.$$

Definition 17.24 (Virtual pullback of a local physical operator). Lean ✓ Stmt For a physical operator O on the local physical space at v, define its virtual pullback by

$$\Psi_v \circ O \circ \Phi_v.$$

This is the local injectivity step used in the physical-to-virtual direction of the insertion-algebra argument in [MGRPG+18, Section 3].

Theorem 17.25 (Virtual pullback realizes the projected physical action). *Lean* *✓ Stmt* *For every coefficient function c, applying the local tensor map after the virtual pullback gives the projection of $O(\Phi_v(c))$ onto the image of Φ_v.*

Proof. ✓ Proof Expand the definitions. The chosen left inverse gives $\Psi_v \Phi_v = \mathbb{1}$, and the remaining map is the local projector. □

Theorem 17.26 (Virtual pullback for image-preserving physical actions). *Lean* *✓ Stmt* *If O sends every vector in the image of Φ_v back into that image, then its virtual pullback realizes exactly the action of O on $\Phi_v(c)$.*

Proof. ✓ Proof Use the preceding theorem and the image-preservation hypothesis. □

Theorem 17.27 (Recovery of a virtual operation from a physical realization). Lean ✓ Stmt *If a physical operator O realizes a virtual operation T on every vector in the image of Φ_v, then pulling back O recovers T.*

Proof. ✓ Proof After applying the local tensor map, both virtual operations have the same image. Vertex injectivity makes the local tensor map injective, so the virtual operations agree. □

Theorem 17.28 (Extensionality of physical-to-virtual pullback). Lean ✓ Stmt *Two physical endpoint operations have the same virtual pullback exactly when their projected actions agree on every vector in the image of the local tensor map:*

$$T_O = T_{O'} \iff P_v O \Phi_v(c) = P_v O' \Phi_v(c) \quad \textit{for all } c.$$

Proof. ✓ Proof Apply the local tensor map to the two virtual pullbacks and use the pullback specification. The converse follows from vertex injectivity; the forward direction is obtained by rewriting the two virtual pullbacks. □

Theorem 17.29 (Pullback of the canonical physical realization). Lean ✓ Stmt *Pulling back the canonical physical realization of a virtual operation recovers the original virtual operation.*

Proof. ✓ Proof The canonical physical realization agrees with the virtual operation on the image of the local tensor map. Apply the preceding recovery theorem. □

Theorem 17.30 (Physical realization of a pulled-back physical operator). Lean ✓ Stmt *Let $P_v = \Phi_v \circ \Psi_v$ be the local projector, and let $T_O = \Psi_v \circ O \circ \Phi_v$ be the virtual pullback of a physical operator O. Then*

$$\Phi_v \circ T_O \circ \Psi_v = P_v \circ O \circ P_v.$$

Proof. ✓ Proof Expand the definitions of the physical realization, the virtual pullback, and the local projector. □

Theorem 17.31 (Projected local recovery). Lean ✓ Stmt *Suppose that the compressed physical action $P_v O P_v$ agrees with the canonical physical realization of a virtual operation T. Then the virtual pullback of O is T.*

Proof. ✓ Proof The physical realization of the virtual pullback of O is $P_v O P_v$. By hypothesis this is the physical realization of T. Injectivity of the canonical physical realization gives equality of the virtual operations. □

Theorem 17.32 (Projected local recovery equivalence). Lean ✓ Stmt *The compressed physical action $P_v O P_v$ is the canonical physical realization of a virtual operation T if and only if the virtual pullback of O is T.*

Proof. ✓ Proof One implication follows by realizing both virtual operations physically and using the formula for the physical realization of a pulled-back operator. The converse is the preceding recovery theorem. □

Definition 17.33 (Matrix action on one incident virtual edge). Lean ✓ Stmt A matrix on one edge incident to v acts on the coefficient space of local virtual configurations by applying the matrix to that edge coordinate and summing over the input edge index while keeping the residual virtual configuration fixed.

Theorem 17.34 (One-edge matrix action on a basis configuration). Lean ✓ Stmt *If the coefficient function is the unit basis function at one local virtual configuration, and the matrix acts on the distinguished incident edge, the result is the sum over the new distinguished edge index with the residual local boundary held fixed.*

Proof. ✓ Proof Evaluate both sides on a local virtual configuration and split into the two cases according to whether the residual boundary agrees with the fixed residual boundary. □

Theorem 17.35 (Local tensor form of a one-edge basis action). Lean ✓ Stmt *Applying the local tensor map after a one-edge matrix action on a basis virtual configuration gives the corresponding linear combination of local tensor components.*

Proof. ✓ Proof Use the preceding coefficient identity and linearity of the local tensor map. □

Theorem 17.36 (Physical realization of one-edge virtual matrix action). Lean ✓ Stmt *If A is vertex-injective, then a matrix acting on one edge incident to v is realized by a physical operator on the physical space at v.*

Proof. ✓ Proof Apply the physical realization of local virtual operations to the one-edge matrix action. □

Definition 17.37 (Split at one incident edge). Lean ✓ Stmt Choosing one incident edge at v identifies the set of local virtual configurations with the product of the bond index on that edge and the residual indices on the remaining incident edges.

Definition 17.38 (Middle region of an edge). Lean ✓ Stmt For an edge $e = (u, v)$, the middle region is the complement $V \setminus \{u, v\}$.

Theorem 17.39 (Edge-centered vertex partition). Lean ✓ Stmt *For an edge $e = (u, v)$, the three regions $\{u\}$, $V \setminus \{u, v\}$, and $\{v\}$ cover the whole vertex set.*

Proof. ✓ Proof Every vertex is either u, v, or different from both. □

Theorem 17.40 (Nonempty middle region from a cardinal bound). Lean ✓ Stmt *If $2 < |V|$, then for every edge $e = (u, v)$ the middle region $V \setminus \{u, v\}$ is nonempty.*

Proof. ✓ Proof The middle region has cardinality $|V| - 2$. Hence $2 < |V|$ gives $|V| - 2 > 0$. □

Theorem 17.41 (Product splitting at an edge). Lean ✓ Stmt *For any edge $e = (u, v)$ and any function f on the vertex set with values in a commutative monoid M,*

$$\prod_{x \in V} f(x) = f(u) \Big(\prod_{x \in V \setminus \{u,v\}} f(x) \Big) f(v).$$

Proof. ✓ Proof Isolate the two distinguished endpoints in the product over V and group the remaining factors into the middle region. □

Theorem 17.42 (PEPS coefficient splitting at an edge). Lean ✓ Stmt *For any edge $e = (u, v)$, each summand in the PEPS coefficient factors into the tensor contribution at u, the product over the middle region $V \setminus \{u, v\}$, and the tensor contribution at v.*

Proof. ✓ Proof Expand the state coefficient and apply the product splitting at the chosen edge to the per-vertex product inside each summand. □

Definition 17.43 (Edge-centered boundary data). Lean ✓ Stmt For an edge $e = (u, v)$, the boundary data consist of the virtual index on e, the remaining virtual indices incident to u, and the remaining virtual indices incident to v. These are the indices over which the endpoint tensors couple to the blocked middle region.

Definition 17.44 (Boundary data read from a virtual configuration). Lean ✓ Stmt A global virtual configuration determines edge-centered boundary data by reading the index on the distinguished edge and the residual endpoint indices.

Definition 17.45 (Boundary compatibility at an edge). Lean ✓ Stmt A global virtual configuration is compatible with fixed edge-centered boundary data if it has the prescribed index on the distinguished edge and the prescribed residual indices at both endpoint stars.

Definition 17.46 (Middle configurations over fixed boundary data). Lean ✓ Stmt For fixed boundary data β, the middle configurations are the subtype of global virtual configurations satisfying the compatibility condition with β.

Definition 17.47 (Virtual configurations fibred by edge boundary data). Lean ✓ Stmt A global virtual configuration is equivalently a choice of edge-centered boundary data together with a middle configuration in the corresponding fibre.

Definition 17.48 (Left endpoint local configuration from boundary data). Lean ✓ Stmt The left endpoint local virtual configuration is reconstructed from the edge index and the residual data on the left endpoint star.

Definition 17.49 (Right endpoint local configuration from boundary data). Lean ✓ Stmt The right endpoint local virtual configuration is reconstructed from the edge index and the residual data on the right endpoint star.

Theorem 17.50 (Left endpoint reconstruction for compatible boundary data). *Lean* *✓ Stmt* *If a global virtual configuration is compatible with boundary data, then its restriction to the left endpoint star is the local configuration reconstructed from those data.*

Theorem 17.51 (Right endpoint reconstruction for compatible boundary data). *Lean* *✓ Stmt* *If a global virtual configuration is compatible with boundary data, then its restriction to the right endpoint star is the local configuration reconstructed from those data.*

Definition 17.52 (Boundary data with an inserted edge matrix). Lean ✓ Stmt For an edge $e = (u, v)$, inserted-edge boundary data consist of two independent indices on the bond e, together with the residual virtual indices incident to u and to v. These are the boundary indices used when a matrix is inserted on the distinguished bond.

Definition 17.53 (Diagonal inserted-edge boundary data). Lean ✓ Stmt Ordinary edge-centered boundary data determine inserted-edge boundary data by using the same bond index at both endpoints. This is the diagonal boundary datum selected by the identity matrix in the edge-insertion coefficient.

Definition 17.54 (Right-endpoint inserted-boundary reindexing). Lean ✓ Stmt Ordinary edge-boundary data together with an independent right endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the left bond index and the two residual endpoint boundaries.

Definition 17.55 (Left-endpoint inserted-boundary reindexing). Lean ✓ Stmt Ordinary edge-boundary data together with an independent left endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the right bond index and the two residual endpoint boundaries.

Definition 17.56 (Diagonal inserted-boundary configurations). Lean ✓ Stmt The diagonal inserted-boundary configurations are those inserted-edge boundary data for which the two distinguished bond indices agree. This is the support of the identity matrix inside the inserted-boundary sum.

Definition 17.57 (Diagonal boundary reindexing). Lean ✓ Stmt Ordinary edge-boundary data are equivalent to diagonal inserted-boundary data. This is the finite reindexing map used after the identity matrix has removed all off-diagonal inserted-boundary summands.

Definition 17.58 (Open middle tensor at an edge). Lean ✓ Stmt Fixing the residual endpoint data but leaving the bond e out of the middle region, the open middle tensor is the contraction over all virtual indices away from e that are compatible with those endpoint residual data.

Definition 17.59 (Middle physical configurations at an edge). Lean Lean ✓ Stmt For an edge $e = (u, v)$, the middle physical configurations are the physical indices on the vertices in $V \setminus \{u, v\}$. A global physical configuration restricts to such a middle configuration.

Definition 17.60 (Middle tensor with middle physical index). Lean Lean ✓ Stmt The ordinary and open middle contractions may be written using only the middle physical configuration, rather than a physical configuration on all vertices. This is the middle tensor in the three-site chain obtained after the edge blocking in arXiv:1804.04964, Section 3.

Theorem 17.61 (Middle restriction of the blocked tensor). *Lean* *Lean* *✓ Stmt* *The middle contractions written with a global physical configuration are the corresponding middle-indexed contractions evaluated on the restricted middle physical configuration.*

Proof. ✓ Proof Rewrite the product over $V \setminus \{u, v\}$ as the product over the corresponding finite set of middle vertices. □

Definition 17.62 (Restricting middle configurations away from the edge). Lean ✓ Stmt An ordinary middle configuration for the edge-blocked contraction restricts to a virtual configuration on all edges except the distinguished edge. This is the first reindexing map needed to compare the identity insertion with the ordinary edge-blocked middle tensor.

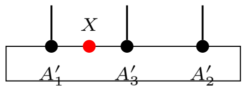
Definition 17.63 (Restoring the distinguished edge index). Lean ✓ Stmt Conversely, a compatible open middle configuration becomes an ordinary middle configuration after the fixed edge-boundary index is restored on the distinguished edge. Together with the restriction map, this gives the finite reindexing underlying the identity-insertion specialization.

Definition 17.64 (Open-middle reindexing equivalence). Lean ✓ Stmt For fixed edge-boundary data, ordinary middle configurations are equivalent to compatible open-middle configurations. This is the finite reindexing used when the inserted matrix is the identity.

Theorem 17.65 (Middle-indexed open tensor reindexing). *Lean* *✓ Stmt* *The ordinary middle tensor and the open middle tensor agree after restoring the fixed distinguished-edge index and reindexing the finite sum, with the physical index already restricted to the middle block.*

Proof. ✓ Proof Reindex the ordinary middle configurations by the equivalence with compatible open-middle configurations, then compare each local tensor entry. □

Definition 17.66 (Edge insertion coefficient). Lean ✓ Stmt For an edge $e = (u, v)$ and a matrix X acting on the bond space of e, the edge insertion coefficient is the contraction obtained by inserting X between the endpoint tensors and the open middle tensor:



Theorem 17.67 (Middle tensor reindexing for an opened edge). Lean ✓ Stmt *Restoring the fixed distinguished-edge index identifies the ordinary blocked middle tensor with the open middle tensor. This is the finite reindexing used before the identity matrix collapses the two inserted bond indices to the diagonal.*

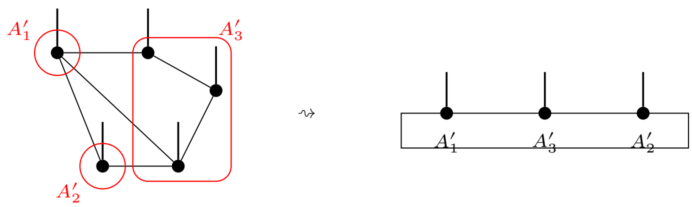
Theorem 17.68 (Diagonal summand of the identity insertion). Lean ✓ Stmt *On the diagonal boundary datum selected by the identity matrix, the inserted-edge summand is the ordinary edge-blocked summand. Thus the remaining identity-insertion step is purely a finite summation statement: the off-diagonal inserted boundary data vanish and the diagonal data reindex the ordinary edge-boundary sum.*

Theorem 17.69 (Off-diagonal summands of the identity insertion). Lean ✓ Stmt *If the two inserted bond indices are distinct, then the corresponding summand in the identity insertion is zero. This is the pointwise off-diagonal vanishing used before the inserted-boundary sum is restricted to its diagonal part.*

Theorem 17.70 (Identity insertion recovers the edge-blocked coefficient). Lean ✓ Stmt *If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the ordinary edge-blocked coefficient.*

Proof. ✓ Proof Split the inserted-boundary sum into the diagonal part, where the two inserted bond indices agree, and the off-diagonal part. The off-diagonal summands vanish for the identity matrix, while the diagonal summands reindex through the equivalence with ordinary edge-boundary data. □

Definition 17.71 (Blocked middle tensor at an edge). Lean ✓ Stmt Fixing the edge-centered boundary data, the blocked middle tensor is the sum over global virtual configurations compatible with those data of the product of local tensor entries over the middle region $V \setminus \{u, v\}$. The graph-local contraction is read as a three-site chain:



Definition 17.72 (Edge-blocked PEPS coefficient). Lean ✓ Stmt The edge-blocked coefficient is the three-block contraction obtained from the left endpoint tensor, the blocked middle tensor, and the right endpoint tensor at the chosen edge.

Theorem 17.73 (Edge-blocked coefficient equals the PEPS coefficient). Lean ✓ Stmt *For every edge e, the edge-blocked three-block coefficient is exactly the original PEPS state coefficient.*

Proof. ✓ Proof Write $\mathrm{Coeff}_A(\sigma)$ for the PEPS state coefficient and $C_A(e;\sigma)$ for the edge-blocked coefficient, and put $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. The equivalence $\omega \leftrightarrow (\beta, \omega_{M_e})$ between global virtual configurations and an edge boundary configuration with a compatible middle configuration rewrites the PEPS coefficient as

$$\sum_{\omega} A_u(\omega|_u, \sigma_u) \left(\prod_{x \in M_e} A_x(\omega|_x, \sigma_x) \right) A_v(\omega|_v, \sigma_v) = \sum_{\beta} A_u(\beta_u, \sigma_u)\, C_{A,M_e}(\beta, \sigma|_{M_e})\, A_v(\beta_v, \sigma_v).$$

Under the equivalence, $\omega|_u = \beta_u$ and $\omega|_v = \beta_v$; the inner sum over ω_{M_e} is exactly $C_{A,M_e}(\beta, \sigma|_{M_e})$. Thus $C_A(e;\sigma) = \mathrm{Coeff}_A(\sigma)$. □

Theorem 17.74 (Identity insertion equals the PEPS coefficient). Lean ✓ Stmt *If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the original PEPS state coefficient.*

Proof. ✓ Proof Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. With $\mathbb{1}$ inserted on e, the inserted-boundary sum is

$$C_A(e, \mathbb{1}; \sigma) = \sum_{a,b,\lambda,\rho} \delta_{a,b}\, A_u(a, \lambda; \sigma_u)\, C^{\text{open}}_{A,M_e}(\lambda, \rho; \sigma|_{M_e})\, A_v(b, \rho; \sigma_v).$$

Here a, b are the two copies of the distinguished bond index, while λ, ρ are the remaining virtual configurations at u and v. The factor $\delta_{a,b}$ leaves only the diagonal $a = b$, and the open-middle reindexing gives $C^{\text{open}}_{A,M_e}(\lambda, \rho; \sigma|_{M_e}) = C_{A,M_e}(a, \lambda, \rho; \sigma|_{M_e})$. Hence $C_A(e, \mathbb{1}; \sigma) = C_A(e; \sigma) = \mathrm{Coeff}_A(\sigma)$. □

Theorem 17.75 (Same state gives the same edge-blocked coefficients). Lean ✓ Stmt *If two PEPS tensors have the same state, then their edge-blocked coefficients agree at every edge and every physical configuration.*

Proof. ✓ Proof For every physical configuration σ,

$$C_A(e; \sigma) = \mathrm{Coeff}_A(\sigma) = \mathrm{Coeff}_B(\sigma) = C_B(e; \sigma).$$

□

Theorem 17.76 (Same state gives the same identity insertions). Lean ✓ Stmt *If two PEPS tensors have the same state, then their edge-centered identity insertions agree at every edge and every physical configuration.*

Proof. ✓ Proof For every physical configuration σ,

$$C_A(e, \mathbb{1}; \sigma) = \mathrm{Coeff}_A(\sigma) = \mathrm{Coeff}_B(\sigma) = C_B(e, \mathbb{1}; \sigma).$$

□

Definition 17.77 (Middle tensor family at an edge). Lean Lean ✓ Stmt For the edge-blocked three-site chain at $e = (u, v)$, the middle tensor is the family obtained by fixing the two residual endpoint boundaries and evaluating the open middle contraction on the middle physical index. Its injectivity is the middle-block part of the assertion following the edge-blocking diagram.

Definition 17.78 (Edge-blocked three-site injectivity). Lean ✓ Stmt For a chosen edge $e = (u, v)$, the edge-blocked three-site chain is injective when the left endpoint tensor map, the middle tensor family, and the right endpoint tensor map are all injective.

Definition 17.79 (Singleton blocked region). Lean Lean Lean Lean ✓ Stmt For a singleton block $\{v\}$, the blocked physical index is the original physical index and the blocked tensor is

$$T_{A,\{v\}}(\eta, \sigma) = A_v(\eta, \sigma).$$

Its injectivity is $\mathrm{inj}(T_{A,\{v\}})$. The comparison with a region-injectivity predicate κ records the implication

$$\mathrm{inj}(T_{A,\{v\}}) \Longrightarrow \kappa(\{v\}).$$

Theorem 17.80 (Vertex injectivity gives singleton-block injectivity). Lean ✓ Stmt *If A is vertex-injective, then every singleton blocked tensor is injective. For every vertex v,*

$$\mathrm{inj}(T_{A,\{v\}}).$$

Proof. ✓ Proof For $R = \{v\}$, the contraction defining $T_{A,R}$ has one factor. Hence

$$T_{A,\{v\}}(\eta, \sigma) = A_v(\eta, \sigma).$$

Thus $\mathrm{inj}(A_v)$ is exactly $\mathrm{inj}(T_{A,\{v\}})$. Vertex injectivity gives $\forall v, \mathrm{inj}(A_v)$, and therefore $\forall v, \mathrm{inj}(T_{A,\{v\}})$. For a fixed v, this is the displayed assertion. □

Theorem 17.81 (Singleton comparison gives the region base case). Lean Lean ✓ Stmt *Assume that a region-injectivity predicate κ satisfies, for every vertex v,*

$$\mathrm{inj}(T_{A,\{v\}}) \implies \kappa(\{v\}).$$

Then, for every vertex v,

$$A \textit{ vertex-injective} \implies \kappa(\{v\}).$$

Proof. ✓ Proof For each v, the preceding theorem gives A vertex-injective $\implies \mathrm{inj}(T_{A,\{v\}})$. The comparison hypothesis gives $\mathrm{inj}(T_{A,\{v\}}) \implies \kappa(\{v\})$. Their composition is the claimed implication. □

Theorem 17.82 (Finite regions from singleton regions and union closure). Lean ✓ Stmt *Assume that κ satisfies singleton comparison and union closure: $\mathrm{inj}(T_{A,\{v\}}) \implies \kappa(\{v\})$ for every vertex v, and $\kappa(R) \wedge \kappa(S) \implies \kappa(R \cup S)$. If A is vertex-injective, then every nonempty finite region R satisfies $\kappa(R)$.*

Proof. ✓ Proof For $R \neq \emptyset$, write $R = \bigcup_{v \in R}\{v\}$. The preceding theorem turns singleton comparison and vertex injectivity into $\kappa(\{v\})$ for every $v \in R$. Applying the finite-union consequence of union closure to these singleton regions gives $\kappa(R)$. □

Definition 17.83 (Region injectivity supplies the edge-middle tensor). Lean ✓ Stmt A comparison from finite-region injectivity to the edge-middle tensor records the following assertion: if the middle vertex region $V \setminus \{u, v\}$ is injective after blocking, then the middle tensor in the edge-blocked three-site chain is injective. This isolates the finite-region contraction claim in [MGRPG+18, Section 3].

Theorem 17.84 (Endpoint injectivity in the edge-blocked chain). Lean ✓ Stmt *If the original PEPS tensor is vertex-injective, then the two endpoint tensor maps in the edge-blocked three-site chain are injective.*

Proof. ✓ Proof If $e = (u, v)$, vertex injectivity gives $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$, which are the left and right endpoint tensor maps of the edge-blocked chain. □

Theorem 17.85 (Endpoint injectivity at all edges). Lean ✓ Stmt *If the original PEPS tensor is vertex-injective, then the two endpoint tensor maps in the edge-blocked three-site chain are injective at every edge.*

Proof. ✓ Proof For each $e = (u, v)$, Theorem 17.84 gives $\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,v})$. □

Theorem 17.86 (Reducing edge-blocked injectivity to the middle block). Lean ✓ Stmt *For a vertex-injective PEPS tensor, injectivity of the edge-blocked middle tensor gives the full injectivity statement for the edge-blocked three-site chain.*

Proof. ✓ Proof Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. The endpoint theorem supplies $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$, while the hypothesis is $\mathrm{inj}(T_{A,M_e})$. Hence

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v}).$$

□

Theorem 17.87 (All edge-blocked chains from middle-tensor injectivity)**.** Lean ✓ Stmt *If every edge-middle tensor is injective, then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at every edge.*

Proof. ✓ Proof For each $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. Vertex injectivity gives $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$, while the hypothesis gives $\mathrm{inj}(T_{A,M_e})$. Hence

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v})$$

is the injectivity assertion for the edge-blocked three-site chain at e. □

Theorem 17.88 (Region injectivity gives edge-blocked injectivity)**.** Lean ✓ Stmt *Suppose the middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose this region-injectivity assertion has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at the edge $e = (u, v)$.*

Proof. ✓ Proof Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. The comparison gives $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$. Vertex injectivity supplies $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$. The preceding reduction then gives

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v}).$$

□

Theorem 17.89 (Middle tensors from all middle regions)**.** Lean ✓ Stmt *Suppose every edge-middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then every edge-middle tensor family is injective.*

Proof. ✓ Proof For each edge $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. The hypotheses give $\kappa(M_e)$. The comparison implication $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$ yields $\mathrm{inj}(T_{A,M_e})$ for every edge. □

Theorem 17.90 (All edge-blocked chains from middle-region injectivity)**.** Lean ✓ Stmt *Suppose every edge-middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at every edge.*

Proof. ✓ Proof For every edge $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. The region hypothesis gives $\kappa(M_e)$, and the comparison gives $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$. Vertex injectivity gives $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$. Thus

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v}).$$

□

Theorem 17.91 (Edge-middle tensor from singleton regions, nonempty middle case)**.** Lean ✓ Stmt *Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. Assume $M_e \neq \emptyset$. Assume also singleton comparison, union closure for κ, and the comparison $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$. If A is vertex-injective, then the edge-middle tensor family T_{A,M_e} is injective.*

Proof. ✓ Proof The finite-region theorem applied to $M_e = \bigcup_{w \in M_e} \{w\}$ gives $\kappa(M_e)$. The edge-middle comparison sends this to $\mathrm{inj}(T_{A,M_e})$. $\square$

Theorem 17.92 (One edge-blocked chain from singleton regions, nonempty middle case)**.** Lean ✓ Stmt *Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. Under the hypotheses of the preceding theorem, vertex injectivity of A gives injectivity of the edge-blocked three-site chain at e.*

Proof. ✓ Proof The preceding theorem gives $\mathrm{inj}(T_{A,M_e})$. The endpoint reduction combines this middle-tensor injectivity with vertex injectivity at u and v. $\square$

Theorem 17.93 (All edge-blocked chains from singleton regions, nonempty middle case)**.** Lean ✓ Stmt *If $M_e = V \setminus \{u, v\}$ is nonempty for every edge $e = (u, v)$, then the preceding one-edge statement applies at every edge. Thus vertex injectivity of A gives injectivity of every edge-blocked three-site chain.*

Proof. ✓ Proof Apply the one-edge statement separately to each edge e. $\square$

Theorem 17.94 (All edge-blocked chains from singleton regions, cardinal form)**.** Lean Lean ✓ Stmt *Assume singleton comparison, union closure for κ, the comparison $\kappa(M_e) \Longrightarrow \mathrm{inj}(T_{A,M_e})$, and vertex injectivity of A. If $2 < |V|$, then $V \setminus \{u, v\} \neq \emptyset$ for every edge $e = (u, v)$, and these hypotheses give $\mathrm{inj}(T_{A,V\setminus\{u,v\}})$ and edge-blocked three-site injectivity at every edge.*

Proof. ✓ Proof For each $e = (u, v)$, Theorem 17.40 gives $V \setminus \{u, v\} \neq \emptyset$. Apply the preceding all-edge singleton theorem. $\square$

Definition 17.95 (Finite kernel descent datum)**.** Lean ✓ Stmt Let V be a vertex type with decidable equality. A finite kernel descent datum is a predicate K on finite subsets $S \subseteq V$ together with implications

$$j \in S \quad \Longrightarrow \quad K(S) \Longrightarrow K(S \setminus \{j\}).$$

In the edge-blocking proof, $K(S)$ is the assertion that the boundary coefficients c_ρ, with the virtual indices exposed at that stage, give zero after the tensors in S have been contracted.

Lemma 17.96 (Finite descent of kernel conditions)**.** Lean Lean ✓ Stmt *For a finite kernel descent datum and any finite $R \subseteq V$,*

$$K(R) \Longrightarrow K(\emptyset).$$

More generally, if $K(\emptyset) \Longrightarrow Q$, then $K(R) \Longrightarrow Q$.

Proof. ✓ Proof Induct on the finite set R. If $R = \emptyset$, there is nothing to prove. Otherwise write $R = S \cup \{j\}$ with $j \notin S$. The deletion implication gives $K(R) \Longrightarrow K(S)$, and the induction hypothesis gives $K(S) \Longrightarrow K(\emptyset)$. $\square$

Definition 17.97 (Edge-middle kernel descent data)**.** Lean Lean ✓ Stmt Let $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. An edge-middle kernel descent datum is indexed by boundary labels ρ consisting of the two residual endpoint configurations. It assigns to every finitely supported coefficient family $c = (c_\rho)_\rho$ a finite kernel descent datum $K_c(S)$ such that

$$\left(\forall \tau, \ \sum_\rho c_\rho T^\rho_{A,M_e}(\tau) = 0 \right) \Longrightarrow K_c(M_e),$$

and

$$K_c(\emptyset) \Longrightarrow \forall \rho, \ c_\rho = 0.$$

Theorem 17.98 (Kernel descent gives edge-middle injectivity). Lean Lean ✓ Stmt *Let $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. If an edge-middle kernel descent datum is given, then T_{A,M_e} is injective. Consequently, if A is vertex-injective, the edge-blocked three-site chain at e is injective.*

Proof. ✓ Proof Let $c = (c_\rho)_\rho$ be a finite linear relation among the middle tensor vectors:

$$\forall \tau, \ \sum_\rho c_\rho T^\rho_{A,M_e}(\tau) = 0.$$

The initial part of the descent datum gives $K_c(M_e)$. The finite descent lemma gives $K_c(M_e) \Longrightarrow K_c(\emptyset)$, and the terminal part gives $\forall \rho, \ c_\rho = 0$. Thus every finite relation is trivial, which is $\mathrm{inj}(T_{A,M_e})$. Vertex injectivity at u and v gives $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$; the edge-blocked three-site injectivity at e follows. □

Theorem 17.99 (Edge blocking preserves injectivity). Lean ✓ Stmt *For a tensor map or blocked tensor family T, write $\mathrm{inj}(T)$ for the assertion that T is injective. If A is vertex-injective, then for every edge $e = (u, v)$,*

$$\mathrm{inj}(T_{A,u}), \qquad \mathrm{inj}(T_{A,V\setminus\{u,v\}}), \qquad \mathrm{inj}(T_{A,v}).$$

Here $T_{A,u}$ and $T_{A,v}$ are the two endpoint tensor maps, and $T_{A,V\setminus\{u,v\}}$ is the blocked middle tensor family. This is the precise assertion after the edge-blocking diagram in [MGRPG+18, Section 3].

Proof. Fix $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. From vertex injectivity, construct the edge-middle descent datum required by Theorem 17.98: for each finite relation

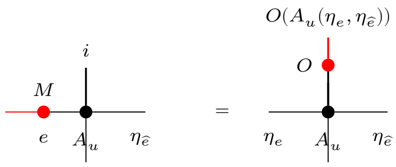
$$\forall \tau, \ \sum_\rho c_\rho T^\rho_{A,M_e}(\tau) = 0,$$

the associated conditions $K_c(S)$ satisfy $K_c(M_e)$, $j \in S \Longrightarrow K_c(S) \Longrightarrow K_c(S \setminus \{j\})$, and $K_c(\emptyset) \Longrightarrow \forall \rho, \ c_\rho = 0$. The descent implication is the local contraction identity: for $j \in S$, vertex injectivity at j gives a left inverse L_j, and applying L_j to the relation contracts away that tensor,

$$j \in S, \ K_c(S) \ \overset{L_j}{\Longrightarrow} \ K_c(S \setminus \{j\}).$$

The kernel descent theorem gives $\mathrm{inj}(T_{A,M_e})$. The same theorem, together with vertex injectivity at u and v, gives $\mathrm{inj}(T_{A,u})$, $\mathrm{inj}(T_{A,M_e})$, and $\mathrm{inj}(T_{A,v})$ as the edge-blocked three-site injectivity assertion. □

Theorem 17.100 (Local virtual insertion realized physically on either endpoint). Lean ✓ Stmt *Let A be vertex-injective and let $e = (u, v)$ be an edge. For every matrix M on the bond space of e, the one-leg virtual action induced by M on the local tensor at u, and likewise at v, is realized by a physical operator on that endpoint's physical space.*



Proof. ✓ Proof Apply the one-edge realization theorem to the two endpoint incidences of the distinguished edge. □

Theorem 17.101 (Left-endpoint projected recovery). Lean ✓ Stmt *Let $e = (u, v)$. A physical operator at the left endpoint pulls back to the one-edge virtual action of $M^\top$ if and only if its compressed physical action is the canonical realization of that one-edge virtual action.*

Proof. ✓ Proof This is the projected local recovery equivalence applied to the left incident edge of e, with the transpose convention used by the left-endpoint coefficient identity. □

Theorem 17.102 (Right-endpoint projected recovery). Lean ✓ Stmt *Let $e = (u, v)$. A physical operator at the right endpoint pulls back to the one-edge virtual action of M if and only if its compressed physical action is the canonical realization of that one-edge virtual action.*

Proof. ✓ Proof This is the projected local recovery equivalence applied to the right incident edge of e. □

Theorem 17.103 (Endpoint recovery of a common virtual insertion). Lean ✓ Stmt *Let $e = (u, v)$. If $P_u O_1 P_u = \Phi_u T_{u,e}(M^\top)\Psi_u$ and $P_v O_2 P_v = \Phi_v T_{v,e}(M)\Psi_v$, then $\Psi_u O_1 \Phi_u = T_{u,e}(M^\top)$ and $\Psi_v O_2 \Phi_v = T_{v,e}(M)$. Here $T_{w,e}(N)$ denotes the virtual operator on the legs incident to w which acts by N on the leg belonging to e and by the identity on the other incident legs.*

Proof. ✓ Proof The endpoint equivalences give $P_u O_1 P_u = \Phi_u T_{u,e}(M^\top)\Psi_u \iff \Psi_u O_1 \Phi_u = T_{u,e}(M^\top)$ and $P_v O_2 P_v = \Phi_v T_{v,e}(M)\Psi_v \iff \Psi_v O_2 \Phi_v = T_{v,e}(M)$. The two hypotheses are the left sides of these equivalences. □

Theorem 17.104 (Right-endpoint physical realization of the inserted coefficient). Lean ✓ Stmt *Let $e = (u, v)$ and let σ be a physical configuration. If a physical operator on v realizes the one-edge virtual matrix action on the distinguished incident edge, then the inserted-edge coefficient at σ is obtained by leaving the left endpoint and open middle contraction explicit and applying that physical operator to the right endpoint tensor.*

Proof. ✓ Proof Expand the physical realization on a basis local configuration at the right endpoint. The resulting sum over the right bond index is then reindexed together with the ordinary edge-boundary data. □

Theorem 17.105 (Left-endpoint physical realization of the inserted coefficient). Lean ✓ Stmt *Let $e = (u, v)$ and let σ be a physical configuration. If a physical operator on u realizes the one-edge virtual action of $M^\top$ on the distinguished incident edge, then the inserted-edge coefficient for M at σ is obtained by applying that physical operator to the left endpoint tensor and leaving the open middle contraction and the right endpoint explicit.*

Proof. ✓ Proof Expand the physical realization on a basis local configuration at the left endpoint. The transpose converts the ordinary right bond index and the new left bond index into the coefficient M_{ij}; the resulting finite sum is then reindexed with the ordinary edge-boundary data. □

Theorem 17.106 (Endpoint physical realizations of the inserted coefficient). Lean ✓ Stmt *If the PEPS tensor is vertex-injective, then for every physical configuration σ, each inserted edge matrix is realized by physical operators at the endpoint tensors: the left endpoint realizes $M^\top$, the right endpoint realizes M, and both realizations reproduce the same inserted-edge coefficient at σ in the edge-centered three-site decomposition.*

Proof. ✓ Proof Apply the local physical-realization theorem at the two endpoints and then use the two coefficient identities above. □

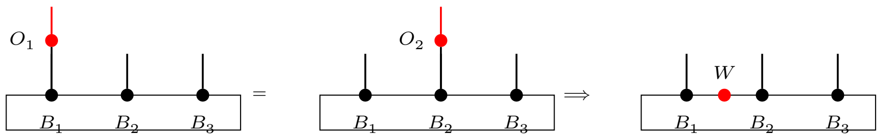
Theorem 17.107 (Equal neighboring physical actions recover a virtual insertion). Lean ✓ Stmt *Let $e = (u, v)$. Assume that the edge-blocked three-site chain at e is injective. Let O_1, O_2 be physical operators at the endpoint tensors, and write Φ_u, Φ_v for the two local tensor maps. Suppose that, for every physical configuration σ,*

$$\sum_{\beta} O_1(A_u(\beta_u))_{\sigma_u}\, C_{\mathrm{mid}}(\sigma; \beta_{\mathrm{L}}, \beta_{\mathrm{R}})\, A_v(\beta_v)_{\sigma_v} = \sum_{\beta} A_u(\beta_u)_{\sigma_u}\, C_{\mathrm{mid}}(\sigma; \beta_{\mathrm{L}}, \beta_{\mathrm{R}})\, O_2(A_v(\beta_v))_{\sigma_v}.$$

Then

$$\exists M \in M_{D(e)}(\mathbb{C}), \qquad O_1\Phi_u = \Phi_u T_{u,e}(M^{\mathsf{T}}), \qquad O_2\Phi_v = \Phi_v T_{v,e}(M).$$

The same implication is represented diagrammatically by the $O_1, O_2 \mapsto W$ step:



This is the converse part of the insertion-algebra lemma in [MGRPG+18, Section 3], where the recovered matrix is denoted W.

Definition 17.108 (Edge-blocked insertion algebra correspondence). Lean ✓ Stmt For two PEPS tensors A and B and an edge e, this property is

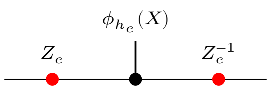
$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\mathrm{alg}} M_{D_B(e)}(\mathbb{C}), \qquad \forall \sigma,\ \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

Theorem 17.109 (Edge-blocked insertion algebra isomorphism). Lean ✓ Stmt *Suppose the two edge-blocked three-site chains associated to A and B at the edge e are injective, and suppose the two PEPS tensors define the same state. Then*

$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\mathrm{alg}} M_{D_B(e)}(\mathbb{C}), \qquad \forall \sigma,\ \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

This is the edge-blocked form of the algebra isomorphism $X \mapsto Y$ in [MGRPG+18, Section 3]:

Theorem 17.110 (Edge gauge from the insertion algebra isomorphism). Lean ✓ Stmt *Suppose the edge insertion correspondence is a $\mathbb{C}$-algebra isomorphism between the two full matrix algebras on the chosen bond. Then the two bond dimensions on that edge are equal. If $h_e : D_A(e) = D_B(e)$ is this equality, and if $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \to M_{D_B(e)}(\mathbb{C})$ is the induced algebra isomorphism, then the insertion correspondence is $Y = Z_e\, \phi_{h_e}(X)\, Z_e^{-1}$ for an invertible edge matrix Z_e.*



This is the finite-dimensional matrix-algebra step in [MGRPG+18, Section 3].

Proof. ✓ Proof Apply Theorem 3.11 to the full matrix algebras determined by the two edge bond spaces. ☐

Definition 17.111 (Factorized local gauge datum). Lean ✓ Stmt Fix A, B, a bond-dimension equality, and a vertex v. Let $\mathcal{T}_v^A$ be the local tensor map of A, let L_v^A be the chosen left inverse of $\mathcal{T}_v^A$, and let $\Pi_v^A = \mathcal{T}_v^A L_v^A$. The factorized local gauge datum consists of the following two identities for local virtual configurations η, η':

$$\Pi_v^A B_v(\eta, -) = B_v(\eta, -), \qquad L_v^A B_v(\eta, -)(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e),$$

with $X_e \in \mathrm{GL}(D_A(e))$ on each incident edge. This is the local output required from the blocked-middle reduction in [MGRPG+18, Section 3].

Definition 17.112 (Blocked-middle local gauge formula). Lean ✓ Stmt The blocked-middle local gauge formula at v is the assertion that there are matrices $X_e \in \mathrm{GL}(D_A(e))$, one on each incident edge, such that for all local virtual configurations η and physical indices σ,

$$B_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Theorem 17.113 (Blocked-middle formula implies factorized local gauge). Lean ✓ Stmt *If the blocked-middle local gauge formula holds at v with matrices X_e, then the factorized local gauge datum holds at v with the same matrices.*

Proof. ✓ Proof For fixed η, set $c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$. The displayed blocked-middle formula is precisely $B_v(\eta, -) = \mathcal{T}_v^A c_\eta$. Hence

$$\Pi_v^A B_v(\eta, -) = \Pi_v^A \mathcal{T}_v^A c_\eta = \mathcal{T}_v^A c_\eta = B_v(\eta, -),$$

and

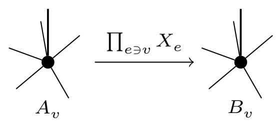
$$L_v^A B_v(\eta, -)(\eta') = L_v^A \mathcal{T}_v^A c_\eta(\eta') = c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e).$$

□

Theorem 17.114 (Blocked-middle formula gives a local gauge formula). Lean ✓ Stmt *Suppose that the edge-centred three-site reduction supplies invertible matrices X_e on the incident edges of v such that, for all local virtual configurations η and physical indices σ,*

$$B_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Then the same family X_e is a local gauge at v:

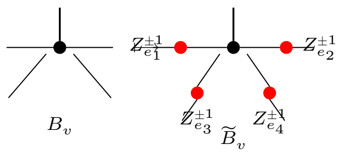


Proof. ✓ Proof By the preceding theorem, $L_v^A B_v(\eta, -)(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$. Since $\Pi_v^A B_v(\eta, -) = B_v(\eta, -)$ and $\Pi_v^A = \mathcal{T}_v^A L_v^A$,

$$B_v(\eta, \sigma) = \sum_{\eta'} L_v^A B_v(\eta, -)(\eta') A_v(\eta', \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

□

Theorem 17.115 (Absorbing edge gauges into the second PEPS). Lean ✓ Stmt *Given edge gauges X_e, set $\widetilde{B} = B^X$. Then $D_{\widetilde{B}} = D_B$ and, for every vertex v, $\widetilde{B}_v(\eta, \sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie), \eta'(ie)))B_v(\eta', \sigma)$, where $M_{ie} = X_e$ or $M_{ie} = (X_e^{-1})^t$ according to the orientation of the endpoint ie.*



After this absorption, arbitrary insertions of the same matrix on the same edge are compared in the first PEPS and the modified second PEPS.

Proof. ✓ Proof With $\widetilde{B} = B^X$, Theorem 17.11 gives $D_{\widetilde{B}} = D_B$, and Theorem 17.12 gives $\widetilde{B}_v(\eta, \sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie), \eta'(ie)))B_v(\eta', \sigma)$. $\square$

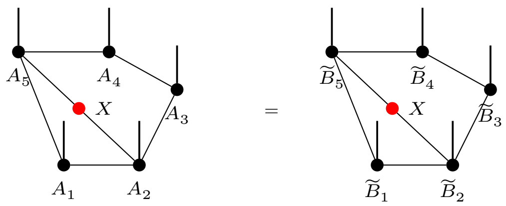
Definition 17.116 (Post-absorption inserted-edge comparison). Lean ✓ Stmt For tensor families $A, \widetilde{B}$, the post-absorption comparison holds when $h : D_A = D_{\widetilde{B}}$ and the following equality holds. For each edge e, write $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \to M_{D_{\widetilde{B}}(e)}(\mathbb{C})$ for the matrix-algebra transport induced by $h_e : D_A(e) = D_{\widetilde{B}}(e)$. Then, for every matrix $X \in M_{D_A(e)}(\mathbb{C})$ and every physical configuration σ,

$$C_A(e, X; \sigma) = C_{\widetilde{B}}(e, \phi_{h_e}(X); \sigma).$$

Theorem 17.117 (Post-absorption edge insertion equality). Lean ✓ Stmt *Let A and B be vertex-injective PEPS with the same state, and assume the bond-dimension equality $h : D_A = D_B$. After the edge gauges have been absorbed into the second tensor family, there are gauges Z_e and $\widetilde{B} = B^Z$ such that $D_{\widetilde{B}} = D_A$ and, for every edge e, write $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \to M_{D_{\widetilde{B}}(e)}(\mathbb{C})$ for the matrix-algebra transport induced by the corresponding equality $h_e : D_A(e) = D_{\widetilde{B}}(e)$. Then, for every matrix $X \in M_{D_A(e)}(\mathbb{C})$ and every physical configuration σ,*

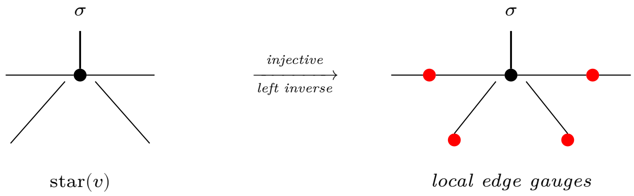
$$C_A(e, X; \sigma) = C_{\widetilde{B}}(e, \phi_{h_e}(X); \sigma),$$

Diagrammatically, this is the equality



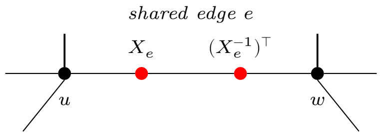
from [MGRPG+ 18, Section 3, lines 1037–1065].

Theorem 17.118 (Local gauge extraction). Lean Lean ✓ Stmt *Under the factorized local gauge hypothesis, the local gauge formula follows. What remains open is to upgrade equality of the edge-blocked coefficients to the blocked-middle formula using the corresponding three-site injective-chain argument. The physical-realization and physical-to-virtual steps in that argument are recorded separately as Theorems 17.100 and 17.107. After Theorem 17.117 supplies the factorized local gauge hypothesis, injectivity on the star neighborhood provides a left inverse that isolates the local edge gauges:*



Proof. ✓ Proof The factorized local gauge datum already supplies the incident-edge invertible matrices and the factorized local coefficient identity. □

Theorem 17.119 (Gauge consistency)**.** Lean Not ready *Choose an edge $e = (u, v)$ and block all other vertices into one middle tensor. The three resulting tensors form an injective chain, so the one-dimensional injective comparison assigns an invertible matrix to the edge e. Repeating this for every edge and absorbing these matrices into the second tensor family gives the post-absorption insertion equality of [MGRPG⁺18, Section 3]: for every edge and every matrix, inserting that matrix on the edge in the first PEPS gives the same state as inserting it on the same edge in the modified second PEPS. Applying the local extraction at each vertex then gives one edge gauge family whose endpoint actions transform the first tensor family into the second:*



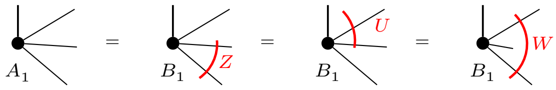
Remark 17.120. The PEPS diagrams in this section record the gauge steps used in [MGRPG⁺18, Section 3]. The edge-orientation and vertex-action diagrams fix the endpoint convention for the edge gauges obtained by blocking the PEPS to a three-site injective MPS. The cancellation diagram is the converse contracted identity: inserting opposite endpoint actions on an internal edge leaves the PEPS coefficient unchanged. The edge-gauge absorption diagram records the formation of $\widetilde{B}$ before the post-absorption insertion equality. The local-extraction diagram summarizes the comparison of one-site-enlarged injective regions with injective complements, after the edge gauges have been absorbed into $\widetilde{B}$. The global-consistency diagram summarizes repeating the edge blocking over all edges, so the two endpoint actions on a shared edge are one edge gauge. The two-injective-tensor diagram records the two-block comparison from [MGRPG⁺18, Section 3], and the final one-vertex-complement diagram records its application to one vertex and its complement. These pictures use the blueprint tensor-dot convention, but each one is attached to the same contraction, insertion, or blocking step as the corresponding argument in the paper.

Definition 17.121 (Left-identity three-leg residual form)**.** Lean ✓ Stmt For three virtual leg spaces α, β, γ and an endomorphism Z of $\beta \otimes \gamma$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as the identity on the α leg and as Z on the remaining two legs. This is the $I_\alpha \otimes Z$ form appearing after the first tensor is inverted in [MGRPG⁺18, Section 3].

Definition 17.122 (Right-identity three-leg residual form)**.** Lean ✓ Stmt For an endomorphism U of $\alpha \otimes \beta$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as U on the first two legs and as the identity on the γ leg. This is the $U \otimes I_\gamma$ form in [MGRPG⁺18, Section 3].

Definition 17.123 (Middle-identity three-leg residual form)**.** Lean ✓ Stmt For an endomorphism W of $\alpha \otimes \gamma$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as W on the outer two legs and as the identity on the middle β leg. This is the third residual form in [MGRPG⁺18, Section 3].

Theorem 17.124 (Scalar reduction of the residual two-tensor gauges). Lean ✓ Stmt *In the proof of the generalized two-injective-tensor comparison, applying the inverse of the second injective tensor leaves three possible virtual operators on the exposed legs of the first tensor:*



Inverting the first injective tensor then shows that the corresponding residual operator on the three exposed virtual legs has the three decompositions

$$\sum_i I \otimes Z_i^{(1)} \otimes Z_i^{(2)} = \sum_i U_i^{(1)} \otimes U_i^{(2)} \otimes I = \sum_i W_i^{(1)} \otimes I \otimes W_i^{(2)}$$

Equivalently, for nonzero virtual leg spaces α, β, γ, if one endomorphism of $\alpha \otimes \beta \otimes \gamma$ lies simultaneously in the three subspaces $I_\alpha \otimes \mathrm{End}(\beta \otimes \gamma)$, $\mathrm{End}(\alpha \otimes \beta) \otimes I_\gamma$, and the analogous subspace with I_β in the middle leg, then it is a scalar multiple of the identity. Thus the residual virtual operators are scalar multiples of the identity. This is the scalar step in the proof of [MGRPG+18, Section 3].

Proof. ✓ Proof Write the common operator as T. The equality $T = I_\alpha \otimes Z = U \otimes I_\gamma$ gives, for all indices,

$$\delta_{aa'} Z_{bc,b'c'} = U_{ab,a'b'} \delta_{cc'}.$$

If $c \neq c'$, the right hand side is zero; choosing $a = a'$ gives $Z_{bc,b'c'} = 0$. Hence Z is diagonal in the γ-index. Similarly, the equality $T = I_\alpha \otimes Z = W_{\alpha\gamma} \otimes I_\beta$ gives

$$\delta_{aa'} Z_{bc,b'c'} = W_{ac,a'c'} \delta_{bb'},$$

so $Z_{bc,b'c'} = 0$ whenever $b \neq b'$. Thus $Z_{bc,b'c'} = 0$ unless $b = b'$ and $c = c'$.

It remains to compare the surviving diagonal entries. From $\delta_{aa'} Z_{bc,bc} = U_{ab,a'b}$, the value $Z_{bc,bc}$ is independent of c. From the middle-identity equality it is also independent of b. Therefore $Z = \lambda I_{\beta\otimes\gamma}$. Substitution into

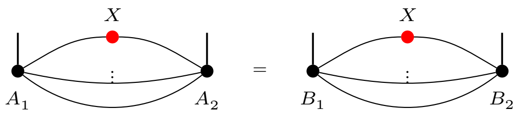
$$I_\alpha \otimes Z = U \otimes I_\gamma = W_{\alpha\gamma} \otimes I_\beta$$

gives $U = \lambda I_{\alpha\otimes\beta}$ and $W_{\alpha\gamma} = \lambda I_{\alpha\otimes\gamma}$. □

Theorem 17.125 (Generalized two-injective-tensor comparison). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let A_1, A_2, B_1, B_2 be injective tensors with the same shared virtual bonds and nonempty external boundary spaces, and assume that there is at least one shared virtual bond. For the set $\mathcal{B}$ of shared bonds, a bond $b \in \mathcal{B}$, and $X \in M_{D_b}(\mathbb{C})$, set*

$$C_{A_1,A_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2) = \sum_{\substack{\mu,\nu \\ \mu|_{\mathcal{B}\setminus\{b\}} = \nu|_{\mathcal{B}\setminus\{b\}}}} X_{\mu_b,\nu_b} A_1(\eta_1, \mu, \sigma_1) A_2(\eta_2, \nu, \sigma_2).$$

Assume $C_{A_1,A_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2) = C_{B_1,B_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2)$ for every $b, X, \eta_1, \eta_2, \sigma_1, \sigma_2$:



Then

$$\exists \lambda \in \mathbb{C}^{\times}, \qquad A_1 = \lambda B_1, \qquad A_2 = \lambda^{-1} B_2.$$

This is the blueprint form of the two-injective tensor comparison in [MGRPG+18, Section 3].

Proof. The proof is the comparison lemma of [MGRPG+18, Section 3]. After grouping the relevant virtual spaces, reduce to three nonzero virtual legs α, β, γ. Applying the inverse of A_2 to the insertion equalities obtained by freeing, respectively, the (β, γ), (α, β), and (α, γ) legs gives endomorphisms Z, U, and W such that, for every external index η, shared-boundary index μ, and physical index σ of the first block,

$$A_1(\eta, \mu, \sigma) = \sum_{\nu} B_1(\eta, \nu, \sigma)(I_\alpha \otimes Z)_{\nu,\mu} = \sum_{\nu} B_1(\eta, \nu, \sigma)(U \otimes I_\gamma)_{\nu,\mu} = \sum_{\nu} B_1(\eta, \nu, \sigma)(W_{\alpha\gamma} \otimes I_\beta)_{\nu,\mu}.$$

Injectivity of B_1 gives the residual equality

$$I_\alpha \otimes Z = U \otimes I_\gamma = W_{\alpha\gamma} \otimes I_\beta.$$

Write $Z = \sum_i Z_i^{(1)} \otimes Z_i^{(2)}$, $U = \sum_i U_i^{(1)} \otimes U_i^{(2)}$, and $W = \sum_i W_i^{(1)} \otimes W_i^{(2)}$. The previous display is exactly

$$\sum_i I \otimes Z_i^{(1)} \otimes Z_i^{(2)} = \sum_i U_i^{(1)} \otimes U_i^{(2)} \otimes I = \sum_i W_i^{(1)} \otimes I \otimes W_i^{(2)}.$$

Theorem 17.124 gives $Z = \lambda I_{\beta\otimes\gamma}$, $U = \lambda I_{\alpha\otimes\beta}$, and $W_{\alpha\gamma} = \lambda I_{\alpha\otimes\gamma}$. Hence $A_1(\eta, \mu, \sigma) = \lambda B_1(\eta, \mu, \sigma)$. If $\lambda = 0$, then $A_1 = 0$, contradicting injectivity of A_1; hence $\lambda \neq 0$.

Put $X = I_{D_b}$ in the insertion identity. Since $A_1 = \lambda B_1$, for every boundary and physical index one has

$$\sum_{\mu} B_1(\eta_1, \mu, \sigma_1)(A_2(\eta_2, \mu, \sigma_2) - \lambda^{-1} B_2(\eta_2, \mu, \sigma_2)) = 0.$$

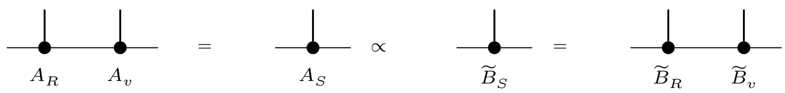
Injectivity of B_1 gives $A_2(\eta_2, \mu, \sigma_2) = \lambda^{-1} B_2(\eta_2, \mu, \sigma_2)$. □

Theorem 17.126 (One-vertex versus complement comparison)**.** Lean ✓ Stmt *Let the shared bond type and the two external boundary spaces be nonempty. After the edge gauges have been absorbed into the second tensor family $\widetilde{B}$, block one vertex v against its complement. Assume that $A_v, A_{v^c}, \widetilde{B}_v, \widetilde{B}_{v^c}$ are injective as two-block tensors. The post-absorption insertion equality gives, for every shared bond b, every matrix $X \in M_{D_b}(\mathbb{C})$, and all external and physical indices,*

$$C_{A_v, A_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{\widetilde{B}_v, \widetilde{B}_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Then

$$\exists \lambda_v \in \mathbb{C}^{\times}, \qquad A_v = \lambda_v \widetilde{B}_v.$$



This is the final local comparison following the post-absorption insertion equality in [MGRPG+18, Section 3].

Proof. ✓ Proof With $A_1 = A_v$, $A_2 = A_{v^c}$, $B_1 = \widetilde{B}_v$, and $B_2 = \widetilde{B}_{v^c}$, the post-absorption equality is precisely

$$\forall b, X, \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}, \quad C_{A_1, A_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{B_1, B_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Theorem 17.125 gives

$$\exists \lambda_v \in \mathbb{C}^\times, \qquad A_v = \lambda_v \widetilde{B}_v, \qquad A_{v^c} = \lambda_v^{-1} \widetilde{B}_{v^c}.$$

The desired local proportionality is the first equality. □

Remark 17.127 (Normal-region form). The normal-region application in [MGRPG⁺18, Section 3] uses the same scalar comparison after comparing two injective regions $R \subset S = R \cup \{v\}$. If the region comparisons give $A_R = \lambda_R \widetilde{B}_R$ and $A_S = \lambda_S \widetilde{B}_S$, then the source-paper diagram reads

$$A_R A_v = A_S = \lambda_S \widetilde{B}_S = \lambda_S \widetilde{B}_R \widetilde{B}_v.$$

Applying the left inverse supplied by injectivity of the block $A_R = \lambda_R \widetilde{B}_R$ gives $A_v = (\lambda_S / \lambda_R) \widetilde{B}_v$.

Theorem 17.128 (Fundamental Theorem for injective PEPS, conditional on bond-dimension equality)**.** Lean Not ready *Suppose that the corresponding virtual edge spaces of A and B are already identified. If A and B are vertex-injective and give the same PEPS state, then there are invertible matrices X_e, one for each edge, such that at every vertex the tensor B_v is obtained from A_v by the oriented endpoint action of the incident matrices.*

Theorem 17.129 (Fundamental Theorem for injective PEPS)**.** Lean Not ready *Let A and B be vertex-injective PEPS tensors on a finite simple graph. If they give the same PEPS state, then their corresponding virtual edge spaces have the same dimensions and, after identifying these spaces, there are invertible matrices X_e, one for each edge, such that B_v is obtained from A_v by the oriented endpoint action of the matrices incident to v. This is Theorem 2 of [MGRPG⁺18].*

17.1 Normal PEPS

Definition 17.130 (Four-region decomposition for a union of regions)**.** Lean Lean Lean Lean Lean ✓ Stmt Let V be a finite vertex set. For two regions $A, B \subseteq V$, define the four regions $A \setminus B$, $A \cap B$, $B \setminus A$, and $V \setminus (A \cup B)$. These are the four regions used in the proof of Lemma 17.135. They also form an indexed family over $\{0, 1, 2, 3\}$ in the order $A \setminus B$, $A \cap B$, $B \setminus A$, $V \setminus (A \cup B)$.

Lemma 17.131 (Identities for the four-region decomposition)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let V be a finite vertex set and let $A, B \subseteq V$. The left-only and overlap regions reconstruct A, the overlap and right-only regions reconstruct B, the three inside regions reconstruct $A \cup B$, and all four regions reconstruct V. The same identities also hold in the indexed four-region family: pieces 0 and 1 reconstruct A, pieces 1 and 2 reconstruct B, and pieces 0, 1, 2 reconstruct $A \cup B$. These same reconstructions are available as finite unions over the corresponding subfamilies of the indexed four-region family. Piece 3 is the complement of $A \cup B$, equivalently the complement of the three inside indexed pieces, and also $A^c \cap B^c$. The singleton subfamily $\{3\}$ gives the outside block. Pieces 2, 3 reconstruct A^c, equivalently the complement of pieces 0, 1, and pieces 0, 3 reconstruct B^c, equivalently the complement of pieces 1, 2, again both as explicit unions and as finite unions over the indexed subfamilies. Conversely, pieces 0, 1 are the complement of pieces 2, 3, pieces 1, 2 are the complement of pieces 0, 3, and adding all four pieces reconstructs V. Equivalently, each vertex lies in one of the four regions, and the indexed four-region family has union V. In fact, every vertex belongs to exactly one indexed piece.*

Proof. ✓ Proof Check membership of an arbitrary vertex in A and in B. □

Lemma 17.132 (Disjointness in the four-region decomposition). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *In the four-region decomposition, the left-only and right-only parts are disjoint from the overlap and from each other. Each inside part, and hence the union of the three inside regions, is disjoint from the outside region. In indexed form, the pieces forming A are disjoint from the pieces forming A^c, and the pieces forming B are disjoint from the pieces forming B^c, both as explicit binary unions and as finite unions over the corresponding selected indexed subfamilies. The indexed four-region family is pairwise disjoint.*

Proof. ✓ Proof Use the standard disjointness identities for set difference and intersection, together with the reconstruction of $A \cup B$. □

Definition 17.133 (Contraction chain for the union injectivity proof). Lean ✓ Stmt Let K, E_{012}, E_2, and E_{12} be complex vector spaces. A contraction chain for the proof of [MGRPG+18, Section 3] consists of three linear maps $\mathcal{C}_{\{0,1,2\}} : K \to E_{012}$, $\mathcal{C}_{\{2\}} : K \to E_2$, and $\mathcal{C}_{\{1,2\}} : K \to E_{12}$, together with the three implications used in Lemma 17.134.

Lemma 17.134 (Kernel criterion for the union contraction chain). Lean Lean ✓ Stmt *Let $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$, and $P_3 = (A \cup B)^c$. Let K be the boundary vector space attached to P_3, and let $\mathcal{C}_{\{0,1,2\}} : K \to E_{012}$, $\mathcal{C}_{\{2\}} : K \to E_2$, and $\mathcal{C}_{\{1,2\}} : K \to E_{12}$ be the three contraction maps in the proof of [MGRPG+18, Section 3]. If*

$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{2\}}(X) = 0, \qquad \mathcal{C}_{\{2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{1,2\}}(X) = 0,$$

and $\mathcal{C}_{\{1,2\}}(X) = 0 \Longrightarrow X = 0$, then $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$.

Proof. ✓ Proof For $X \in K$ with $\mathcal{C}_{\{0,1,2\}}(X) = 0$, apply the three implications in sequence:

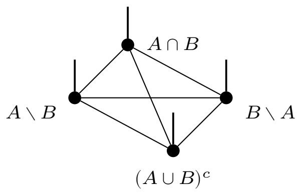
$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{1,2\}}(X) = 0 \Longrightarrow X = 0.$$

□

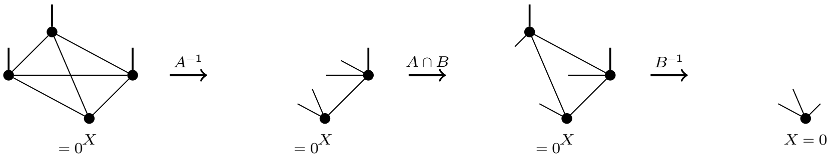
Lemma 17.135 (Union of injective regions). Not ready *Let V be a finite vertex set. Write $\mathrm{inj}(R)$ for the assertion that the PEPS tensor blocked to the region $R \subseteq V$ is injective. For two possibly overlapping regions $A, B \subseteq V$,*

$$\mathrm{inj}(A) \wedge \mathrm{inj}(B) \Longrightarrow \mathrm{inj}(A \cup B).$$

The proof in [MGRPG+18, Section 3] blocks the network into four parts $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$, and applies the two available left inverses in sequence:



Taking a boundary tensor X on $(A \cup B)^c$ with $\mathcal{C}_{\{0,1,2\}}(X) = 0$, injectivity of A gives $\mathcal{C}_{\{2\}}(X) = 0$; reinserting $A \cap B$ and applying injectivity of B then gives $X = 0$:



Proof. Put $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$, and $P_3 = (A \cup B)^c$. Then $P_0 \cup P_1 = A$, $P_1 \cup P_2 = B$, and $P_0 \cup P_1 \cup P_2 = A \cup B$. Let $\mathcal{C}_I$ denote contraction of the blocked tensor over the region $\bigcup_{i \in I} P_i$. To prove injectivity of $A \cup B$, take a boundary tensor X on P_3 such that $\mathcal{C}_{\{0,1,2\}}(X) = 0$. Since $P_0 \cup P_1 = A$ is injective, its left inverse gives

$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \quad \Longrightarrow \quad \mathcal{C}_{\{2\}}(X) = 0.$$

The remaining two steps are

$$\begin{aligned} \mathcal{C}_{\{2\}}(X) = 0 \quad &\Longrightarrow \quad \mathcal{C}_{\{1,2\}}(X) = 0, \\ \mathcal{C}_{\{1,2\}}(X) = 0 \quad &\Longrightarrow \quad X = 0. \end{aligned}$$

The first implication is obtained by contracting with the tensor on P_1, and the second is the left inverse for the injective region $P_1 \cup P_2 = B$. Thus $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$, and therefore $P_0 \cup P_1 \cup P_2 = A \cup B$ is injective. □

Definition 17.136 (Injectivity predicate for blocked regions). Lean Lean ✓ Stmt A region-injectivity hypothesis assigns to each finite vertex region the assertion that the tensor obtained by blocking that region is injective. This predicate is used for regions such as R, S, T, and their complements in the normal PEPS proof. The union-closure hypothesis records the conclusion supplied by the source union-of-injective-regions lemma: the union of two injective regions is injective.

Lemma 17.137 (Finite unions of injective regions). *Lean* *✓ Stmt* *Under the union-closure hypothesis for region injectivity, any nonempty finite union of injective finite regions is injective.*

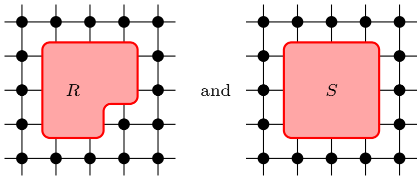
Proof. ✓ Proof Induct on the nonempty finite indexing set. The singleton case is the given injectivity hypothesis, and the induction step is one application of binary union closure. □

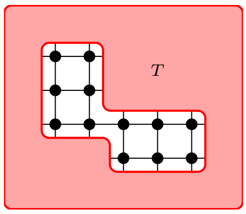
Lemma 17.138 (Indexed inside union under closure). *Lean* *Lean* *✓ Stmt* *Under the union-closure hypothesis for region injectivity, if the indexed subfamilies $\{0, 1\}$ and $\{1, 2\}$ in the four-region decomposition are injective, then the inside subfamily $\{0, 1, 2\}$ is injective. Equivalently, injectivity of the two source-paper regions $(A \setminus B) \cup (A \cap B)$ and $(A \cap B) \cup (B \setminus A)$ gives injectivity of $(A \setminus B) \cup (A \cap B) \cup (B \setminus A) = A \cup B$.*

Proof. ✓ Proof Rewrite the three selected indexed unions as A, B, and $A \cup B$, then apply binary union closure. □

Definition 17.139 (Complement of a finite region). Lean Lean ✓ Stmt For a finite vertex region $R \subseteq V$, its complement is $V \setminus R$. Equivalently, a vertex lies in the complement exactly when it does not lie in R.

Definition 17.140 (Square-lattice coordinate regions). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The finite $n \times m$ square lattice is represented by pairs of coordinates. The corresponding square-lattice graph has horizontal and vertical nearest-neighbor edges; the two elementary adjacency lemmas record the right-neighbor and upper-neighbor cases used for translated edge blockings, and the corresponding coordinate edges name those two nearest-neighbor edges as elements of the edge set. These named right and upward edges have the expected horizontal and vertical orientations. Every edge of this graph is horizontal or vertical, but not both. Since edges are represented by ordered endpoint pairs, a horizontal edge is oriented from left to right and is exactly the coordinate right edge from its left endpoint; similarly, a vertical edge is oriented from bottom to top and is exactly the coordinate upward edge from its lower endpoint. A coordinate rectangle is a product of a horizontal coordinate set and a vertical coordinate set. The 2×3 and 3×2 coordinate-product predicates assert only that these two coordinate sets have cardinalities $2, 3$ and $3, 2$, respectively. Contiguous coordinate rectangles are products of contiguous coordinate intervals, and the contiguous 2×3 and 3×2 predicates name the rectangular blocks used in the normal square-lattice theorem. The displayed region R is the union of a 2×3 contiguous rectangle with the 3×2 contiguous rectangle occupying its upper two rows and extending one column to the right. The displayed region S is the union of two 2×3 contiguous rectangles shifted by one horizontal coordinate, hence the full 3×3 square spanned by them. The displayed local-window model for T is the complement, inside a local 5×6 coordinate window, of the vertical 2×3 edge block and the horizontal 3×2 edge block shown in the source picture. The finite-lattice complementary block is the whole finite square lattice with those two edge blocks removed; this is the block denoted A_3 in the proof of [MGRPG+18, Theorem 3]. The source shapes are:





Lemma 17.141 (Basic identities for square-lattice coordinate regions)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Membership in a coordinate rectangle is coordinatewise membership, the contiguous coordinate interval has the expected coordinate inequalities, membership in a contiguous coordinate rectangle is the conjunction of the four bounding inequalities, and membership in R is the disjunction of membership in the 2×3 lower-left piece or in the shifted 3×2 upper piece. Membership in S is the disjunction of membership in the left 2×3 piece or in the horizontally shifted 2×3 piece. Membership in the local-window T region is membership in the local 5×6 window together with nonmembership in the union of the two displayed edge blocks. Bounded contiguous coordinate rectangles of sizes 2×3 and 3×2 satisfy, respectively, the predicates for contiguous 2×3 and contiguous 3×2 square-lattice rectangles. The cardinality of a coordinate rectangle is the product of the two coordinate cardinalities, and the full coordinate rectangle is the whole finite vertex set.*

Proof. ✓ Proof The coordinate-rectangle statements follow from membership and cardinality identities for Cartesian products of finite coordinate sets. The contiguous-rectangle statement then unfolds the two coordinate intervals. The membership statements for R, S, and the two edge blocks removed from T additionally unfold membership in a union of two such rectangles, while membership in T also unfolds the set difference from the local 5×6 window. The shape predicates follow by using the displayed lower-left corners as witnesses. □

Lemma 17.142 (The normal regions R and S are unions of contiguous rectangles). Lean Lean ✓ Stmt *If the displayed 3×3 window containing R and S lies in the finite square lattice, then R is the union of a contiguous 2×3 rectangle and a contiguous 3×2 rectangle, while S is the union of two contiguous 2×3 rectangles.*

Proof. ✓ Proof This is the defining description of R and S, together with the coordinate bounds which make the displayed rectangles contiguous 2×3 and 3×2 rectangles. □

Lemma 17.143 (The edge blocks removed from T are contiguous rectangles). Lean Lean Lean ✓ Stmt *If the two edge blocks cut out of the displayed T-region lie in the finite square lattice, then they are respectively a contiguous 2×3 rectangle and a contiguous 3×2 rectangle.*

Proof. ✓ Proof This is the defining description of the two removed edge blocks, together with the coordinate bounds which make them valid contiguous rectangles. □

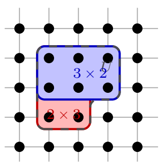
Lemma 17.144 (The local T-region and its removed edge blocks fill the window). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *The two edge blocks removed from the displayed local T-region lie inside the local 5×6 coordinate window. They are disjoint from each other and from this local T region, and their union with it is exactly that local window.*

Proof. ✓ Proof This follows by unfolding the definition of T as the set-theoretic complement of the two displayed edge blocks inside the local window. □

Lemma 17.145 (The edge-complementary block fills the finite lattice). Lean Lean Lean Lean ✓ Stmt *The finite-lattice complementary block around the edge is disjoint from the two displayed edge blocks, and its union with them is the full finite square lattice. The local-window T region is contained in this finite-lattice complementary block.*

Proof. ✓ Proof This is the set-theoretic complement of the two displayed edge blocks in the full finite square lattice. The containment of the local-window T region follows because its points also avoid the two removed edge blocks. □

Definition 17.146 (Rectangular covers of square-lattice regions). Lean Lean Lean Lean Lean ✓ Stmt A rectangular cover of a square-lattice region is a nonempty finite family of regions whose union is exactly the given region, and each member is a contiguous 2×3 or 3×2 rectangle. The two cases used in the proof are the cover of the local displayed T-region and the cover of the finite-lattice complementary block A_3.



$U = \bigcup_j C_j, \quad C_j \cong 2 \times 3$ or 3×2

Lemma 17.147 (The present local T-window and A_3 cover obstructions). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *The present origin-based local-window model for the displayed T-region admits no nonempty finite cover by contained contiguous 2×3 and 3×2 rectangles whenever the ambient lattice contains the 5×6 window. The origin-based rotated local-window model used for vertical edges has the analogous obstruction whenever the ambient lattice contains the 6×5 window. The normalized 5×6 and 6×5 cases are recorded as corollaries. The present origin-based horizontal edge-complement model has the analogous obstruction whenever the ambient lattice contains the 5×7 frame; the normalized 5×7 case is again recorded as* *a corollary. The present origin-based vertical edge-complement model has the corresponding obstruction whenever the ambient lattice contains the* 7×5 *frame; the normalized* 7×5 *case is recorded as a corollary.*

Proof. ✓ Proof The lower-left point of the local window belongs to the present T model. Any contiguous 2×3 rectangle containing it also contains a point of the removed vertical edge block, while any contiguous 3×2 rectangle containing it also contains a point of the removed horizontal edge block. The rotated statement is the same coordinate argument with the two shapes interchanged. The horizontal edge-complement statement uses the same lower-left point, and so does the normalized vertical edge-complement statement. These cases are all instances of the same point-forcing obstruction to an exact rectangular cover. □

Lemma 17.148 (Injectivity from a rectangular cover)**.** Lean ✓ Stmt *If a square-lattice region has a nonempty finite cover by contiguous* 2×3 *and* 3×2 *rectangles, then rectangular injectivity and finite union closure imply that the region is injective.*

Proof. ✓ Proof Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. □

Lemma 17.149 (The normal regions R and S differ in one site)**.** Lean Lean ✓ Stmt *For the displayed regions* R *and* S *with the same lower-left corner,* R *is contained in* S*. The difference* $S \setminus R$ *is exactly the lower-right site of the* 3×3 *square: in coordinates, the site one obtains by shifting two steps horizontally and zero steps vertically from the lower-left corner.*

Proof. ✓ Proof Expand the two displayed unions. The only point of the second 2×3 piece of S which is not already in the union defining R is its lower-right point. □

Definition 17.150 (Normal rectangular injectivity hypotheses)**.** Lean Lean ✓ Stmt In the square-lattice normal theorem, every 2×3 and 3×2 rectangular region is assumed injective. The corresponding abstract hypotheses specify which finite regions have these two shapes and assert injectivity for all of them. In the coordinate square-lattice specialization, these two families are the contiguous coordinate 2×3 and 3×2 rectangles.

Lemma 17.151 (Coordinate rectangular injectivity gives abstract rectangular injectivity)**.** Lean ✓ Stmt *The coordinate square-lattice rectangular injectivity hypotheses determine abstract rectangular injectivity hypotheses, with the two abstract rectangular families taken to be the contiguous coordinate* 2×3 *and* 3×2 *rectangles.*

Proof. ✓ Proof This is the direct specialization of the abstract rectangular families to the two contiguous coordinate-rectangle predicates. □

Lemma 17.152 (Coordinate rectangular injectivity gives injective rectangles)**.** Lean Lean ✓ Stmt *Under the coordinate square-lattice rectangular injectivity hypotheses, every bounded contiguous* 2×3 *rectangle and every bounded contiguous* 3×2 *rectangle is injective.*

Proof. ✓ Proof This is the defining rectangular injectivity hypothesis applied to the coordinate witnesses for the two rectangle shapes. □

Lemma 17.153 (Rectangular injectivity gives injective T-edge blocks)**.** Lean Lean Lean ✓ Stmt *For the displayed local* T*-region inside its* 5×6 *coordinate window, if the smaller edge-block bounds* $x_0 + 5 \leq n$ *and* $y_0 + 5 \leq m$ *hold in the finite* $n \times m$ *square lattice, then the coordinate square-lattice rectangular injectivity hypotheses imply that the vertical* 2×3 *block and horizontal* 3×2 *block removed from the window are injective. If, in addition, region injectivity is closed under unions, then the union of these two removed edge blocks is injective.*

Proof. ✓ Proof Apply the rectangular injectivity hypotheses to the already identified contiguous rectangle shapes of the two edge blocks. The final assertion is one application of union closure to these two injective blocks. ☐

Lemma 17.154 (Injectivity from a rectangular cover of the local T-region)**.** Lean Lean Lean ✓ Stmt *If the displayed local T-region has a nonempty finite cover by contiguous 2×3 and 3×2 rectangles, then rectangular injectivity and finite union closure imply that T is injective. The rotated vertical local T-region has the same conditional injectivity criterion.*

Proof. ✓ Proof Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. ☐

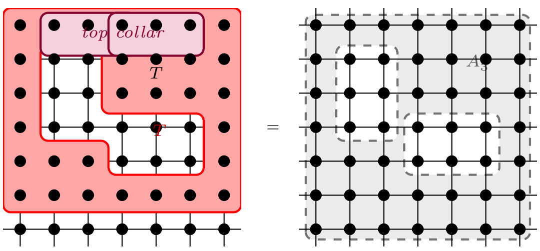
Lemma 17.155 (Injectivity from a rectangular cover of the edge complement)**.** Lean Lean ✓ Stmt *If the finite-lattice complementary block A_3 has a nonempty finite cover by contiguous 2×3 and 3×2 rectangles, then rectangular injectivity and finite union closure imply that A_3 is injective. In the normalized horizontal-edge 5×7 frame, it is enough to give such a cover of the local T-region, since adjoining the two top-collar 3×2 rectangles gives a cover of A_3.*

Proof. ✓ Proof Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. For the normalized 5×7 edge complement, first extend the local T-cover by the two top-collar rectangles. ☐

Lemma 17.156 (A cover of T extends to a cover of the edge complement)**.** Lean ✓ Stmt *In the normalized horizontal-edge 5×7 frame, any rectangular cover of the local T-region extends to a rectangular cover of the edge-complementary block A_3 by adjoining the two top-collar contiguous 3×2 rectangles.*

Proof. ✓ Proof Index the new cover by the old cover indices together with two extra indices for the top-collar rectangles. The set equality is exactly the top-collar decomposition of the edge-complementary block. ☐

Lemma 17.157 (The normalized edge complements have collars)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *In the normalized horizontal-edge 5×7 frame, the finite-lattice complementary block A_3 is the local-window T-region together with two top-collar contiguous 3×2 rectangles. Consequently, if the local T-region is injective, then rectangular injectivity and union closure imply that A_3 is injective. There is also a transposed 7×5 horizontal-frame collar identity with two contiguous 2×3 right-collar rectangles. For the rotated vertical edge-blocking regions, the corresponding complement is the rotated local T-region together with the same two right-collar rectangles; hence rotated T-injectivity, and hence a rectangular cover of rotated T, implies injectivity of the vertical complementary block.*



Proof. ✓ Proof The two set identities follow by expanding the coordinate inequalities. The injectivity statements apply union closure first to the local T-region and the first collar rectangle, and then to the second collar rectangle. ☐

Lemma 17.158 (Normalized edge-blocking data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *In the normalized 5×7 horizontal-edge frame, the red block is the vertical 2×3 block, the blue block is the horizontal 3×2 block, and the complementary block is the finite-lattice edge complement A_3. The distinguished edge is a horizontal edge of the square-lattice graph. The left endpoint of the distinguished edge lies in the red block, the right endpoint lies in the blue block, the three regions are pairwise disjoint, and their union is the whole finite square lattice. If the local T-region is injective, rectangular injectivity and union closure give injectivity of all three blocks; in particular a rectangular cover of T suffices. The normalized 7×5 vertical-edge frame has the analogous red and blue rectangular blocks, a distinguished vertical edge, and the same set-theoretic blocking data; at this stage its complementary-block injectivity is either kept as an explicit hypothesis or obtained from rotated T-injectivity, and therefore from a rectangular cover of rotated T, by the vertical collar lemma. In both cases the named blocking datum supplies the corresponding three injective regions for the edge-blocked chain.*

Proof. ✓ Proof The endpoint memberships are coordinate checks. Rectangular injectivity gives the rectangular red and blue blocks. For the horizontal edge, the collar lemma gives the complement, and the rectangular cover criterion removes the direct T-injectivity input when such a cover is supplied. For the vertical edge, complement injectivity is an explicit input or is obtained from the rotated T-cover. The disjointness and cover statements are finite-set complement identities. □

Lemma 17.159 (Translated horizontal edge-blocking data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *For any coordinate origin whose horizontal blocking window lies in the finite square lattice, there is a distinguished horizontal edge with the same relative position as in the normalized picture. The translated red region is the 2×3 block, the translated blue region is the 3×2 block, and the complementary region is the finite-lattice complement of their union. If this complementary region is injective, these three regions form the one-edge blocking datum; if it has a rectangular cover, rectangular injectivity and union closure supply that complementary injectivity. In both forms the datum supplies the three injective regions used in the edge-blocked chain. The distinguished edge is the corresponding coordinate right edge. At coordinate origin $0, 0$, the translated edge and the three translated regions specialize to the normalized horizontal picture.*

Proof. ✓ Proof The endpoint memberships and horizontal orientation are coordinate checks. Rectangular injectivity gives the red and blue blocks. The complementary block is supplied either as an input or from a rectangular cover, and the disjointness and covering identities follow from the finite-set complement formulas. □

Lemma 17.160 (Translated vertical edge-blocking data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *For any coordinate origin whose vertical blocking window lies in the finite square lattice, there is a distinguished vertical edge with the same relative position as in the normalized vertical picture. The translated red region is the 3×2 block, the translated blue region is the 2×3 block, and the complementary region is the finite-lattice complement of their union. If this complementary region is injective, these three regions form the one-edge blocking datum; if it has a rectangular cover, rectangular injectivity and union closure supply that complementary injectivity. In both forms the datum supplies the three injective regions used in the edge-blocked chain. The distinguished edge is the corresponding coordinate upward edge. At coordinate origin $0, 0$, the translated edge and the three translated regions specialize to the normalized vertical picture.*

Proof. ✓ Proof The endpoint memberships and vertical orientation are coordinate checks. Rectangular injectivity gives the red and blue blocks. The complementary block is supplied either as an input or from a rectangular cover, and the disjointness and covering identities follow from the finite-set complement formulas. □

Definition 17.161 (Translated edge windows). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt A translated edge window records that a chosen edge is one of the translated horizontal or vertical edge pictures, together with a rectangular cover of the complementary block. Such a window gives the one-edge blocking datum for that edge, hence the three injective regions used in the edge-blocked chain. The coordinate right and upward edges obtain such windows directly from the corresponding translated picture. Equivalently, an actual coordinate right or upward edge obtains such a window when it has the necessary room around it for the translated 5×7 or 7×5 frame and the complementary cover is supplied. The same criterion applies to an arbitrary horizontal or vertical square-lattice edge after identifying it with its coordinate right or upward representative. The oriented margin-and-cover datum is the same criterion packaged as per-edge data in the rectangular coordinate model; it directly supplies the one-edge blocking datum, its three injective regions, and the identities $u_e \in R_e$, $v_e \in B_e$, $R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset$, and $R_e \cup B_e \cup C_e = V$. The boundary obstruction lemmas show that these translated-window and margin-cover criteria are interior criteria, not the final every-edge construction in the open square-lattice model.

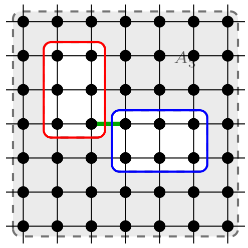
Theorem 17.162 (Construction from translated edge windows). *Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Given rectangular injectivity hypotheses, finite union closure, and a translated edge window at every edge of the finite square lattice, the corresponding one-edge blockings determine the normal edge-blocking hypotheses. This is the conditional construction; constructing such windows for all edges of the 7×7 lattice remains separate; the current open rectangular translated-window criterion does not by itself supply such a family. The one-edge datum recovered from the resulting hypotheses is the datum supplied by the corresponding translated window, and the resulting hypotheses give the three injective regions at every edge, together with $u_e \in R_e$, $v_e \in B_e$, $R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset$, and $R_e \cup B_e \cup C_e = V$. Equivalently, it is enough to supply the oriented margin-and-cover data at every edge; the resulting hypotheses recover the corresponding datum, give the same three injectivity conclusions, and give the same endpoint, disjointness, and covering conclusions. The obstruction theorem records that this interior margin-cover criterion does not by itself supply data at every edge of the open 7×7 lattice.*

Proof. ✓ Proof Apply the general construction for one-edge blocking data to the datum supplied by the chosen translated window at each edge. □

Lemma 17.163 (Conditional injectivity of the normal R and S regions). *Lean Lean ✓ Stmt If region injectivity is closed under unions, then the coordinate square-lattice rectangular injectivity hypotheses imply that the displayed regions R and S are injective.*

Proof. ✓ Proof Region R is the union of one injective 2×3 rectangle and one injective 3×2 rectangle. Region S is the union of two injective 2×3 rectangles. Apply union closure in each case. □

Definition 17.164 (Normal edge-blocking hypotheses). Lean Lean ✓ Stmt Around one edge, the normal proof uses a blocking into three injective regions: a red region containing one endpoint, a blue region containing the other endpoint, and an injective complementary region. The three regions are pairwise disjoint and cover the whole vertex set. The edge-blocking hypotheses are precisely this one-edge datum at every edge; the red, blue, and complementary regions are read off from it. The local blocking picture used in the proof of [MGRPG+18, Theorem 3]:



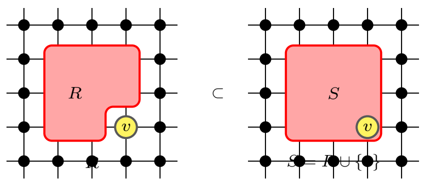
Theorem 17.165 (Injectivity supplied by normal edge blocking). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *A family of one-edge blocking data is precisely the normal edge-blocking hypotheses. If H is assembled from the family d_e, then the one-edge datum that H associates with e is d_e. Write the endpoints of e as $u_e < v_e$ in the canonical vertex ordering. If the three regions are R_e, B_e, and C_e, then*

$$u_e \in R_e, \qquad v_e \in B_e, \qquad R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset, \qquad R_e \cup B_e \cup C_e = V,$$

and the underlying PEPS tensor is injective on R_e, B_e, and C_e.

Proof. ✓ Proof By construction, the one-edge datum at each edge e is d_e. The displayed endpoint, disjointness, coverage, and injectivity statements are exactly the corresponding assumptions of that one-edge datum. □

Definition 17.166 (Normal one-site separation hypotheses). Lean ✓ Stmt For each site, the normal theorem assumes two injective regions with injective complements that differ exactly at that site. This is the one-site comparison used after the edge gauges have been obtained. Diagrammatically, in the square-lattice proof this is the comparison between the source regions R and $S = R \cup \{v\}$:

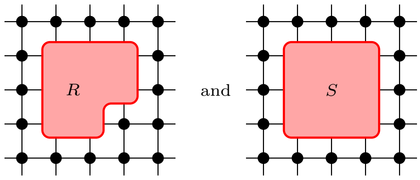


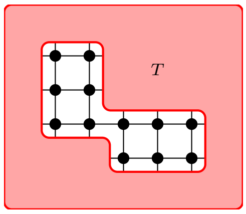
Theorem 17.167 (Membership in a one-site separated region). Lean ✓ Stmt *For the two regions associated to a site v, a vertex w belongs to the region containing v if and only if either $w = v$, or w belongs to the comparison region omitting v.*

Proof. ✓ Proof If $w = v$, use the distinguished-site membership and nonmembership assumptions. If $w \neq v$, use the equality of the two regions away from the distinguished site. □

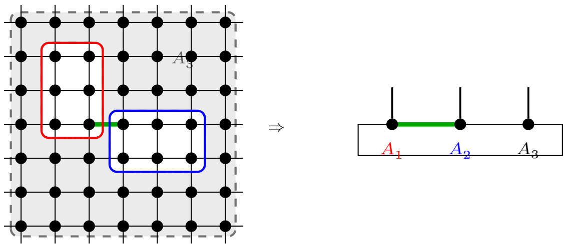
Definition 17.168 (Normal square-lattice blocking regions). Lean Lean ✓ Stmt In the translationally invariant square-lattice proof, the source paper assumes that every 2×3 and 3×2 region is injective. The formal statement here is the corresponding abstract, scope-restricted hypothesis bundle: it records the rectangular injectivity assumptions, the size bound $n, m \geq 7$

with $|V| = nm$, and three finite regions R, S, and T whose injectivity is assumed directly. It does not by itself assert that these regions are the displayed square-lattice unions, nor that their translated variants give the edge-centred blocking around every edge. Conversely, for the coordinate square lattice with $n, m \geq 7$, the rectangular hypotheses, union closure, and a rectangular cover of the displayed T-region yield this abstract R, S, T structure, with R and S obtained from their rectangular decompositions and T obtained from the supplied cover. The source regions are:

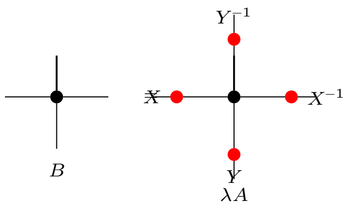




Theorem 17.169 (Normal edge blocking to a three-site injective chain)**.** Not ready *Under the square-lattice normality hypotheses of [MGRPG+18, Theorem 3], the regions R, S, and T, together with their translated variants, give an edge-centred blocking into red, blue, and complementary injective regions. These are the regions on which the three-site injective-chain insertion argument is applied:*

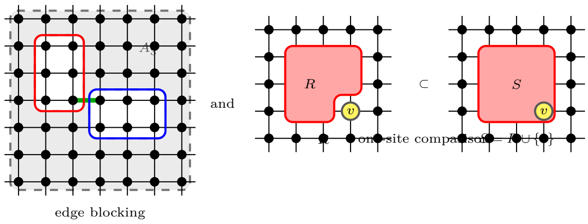


Theorem 17.170 (Translationally invariant normal gauge absorption)**.** Not ready *In the translationally invariant normal square-lattice case, the edge gauges obtained after blocking are represented by one horizontal matrix X and one vertical matrix Y, unique up to scalar. Absorbing these matrices into B gives a tensor $\widetilde{B}$, equivalently the final local gauge formula of [MGRPG+18, Theorem 3]:*



Theorem 17.171 (Fundamental Theorem for normal square-lattice PEPS). Not ready *Let A and B be translationally invariant normal square-lattice PEPS tensors such that every 2×3 and 3×2 region is injective. If they generate the same state on an $n \times m$ region with $n, m \geq 7$, then they are related by horizontal and vertical gauges X, Y, up to a scalar λ satisfying $\lambda^{nm} = 1$. This is [MGRPG+18, Theorem 3].*

Definition 17.172 (Normal PEPS blocking hypotheses). Lean ✓ Stmt The general normal theorem assumes both the edge-centred blockings into three injective regions and the one-site comparison regions with injective complements. These two assumptions are collected as the normal PEPS blocking hypotheses.



Theorem 17.173 (Square-lattice normal blocking hypotheses from margin covers). Lean Lean Lean Lean ✓ Stmt *Given rectangular injectivity hypotheses, finite union closure, oriented margin-and-cover hypotheses at every edge, and one-site separation, the finite square lattice satisfies the general normal blocking hypotheses. The edge-blocking hypotheses are exactly the margin-cover construction, the one-edge datum at an edge is the datum supplied there, and the one-site separation is exactly the supplied one-site separation.*

Proof. ✓ Proof Combine the edge-blocking hypotheses constructed from the margin-cover hypotheses with the given one-site separation hypotheses. □

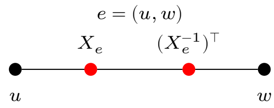
Theorem 17.174 (Fundamental Theorem for normal PEPS). Not ready *Suppose two normal PEPS generate the same state, can be blocked into three-site injective chains around every edge, and for every site admit two injective regions with injective complements that differ exactly at that site. Then their defining tensors are related by a local gauge, and the gauges are unique up to the scalar freedom allowed by the graph. This is the general normal-PEPS theorem stated after [MGRPG+18, Theorem 3].*

Definition 17.175 (Oriented endpoint scalar). Lean ✓ Stmt An edge-scalar family $c = (c_e)_{e \in E}$ determines the endpoint scalar $c_{v,e}$ by $c_{v,e} = c_e$ at the lower endpoint of e and $c_{v,e} = c_e^{-1}$ at the upper endpoint.

Definition 17.176 (Vertex-balanced edge scalars). Lean ✓ Stmt An edge-scalar family is vertex-balanced when $\prod_{e \ni v} c_{v,e} = 1$ for every vertex v.

Definition 17.177 (Gauge equivalence modulo balanced edge scalars). Lean ✓ Stmt Two gauge families are equivalent modulo balanced edge scalars if there is a nonzero scalar c_e on each edge such that the corresponding endpoint actions satisfy $X_{v,e} = c_{v,e} Y_{v,e}$, and $\prod_{e \ni v} c_{v,e} = 1$ for every vertex v.

Theorem 17.178 (Multiplication of oriented edge scalars). Lean ✓ Stmt *If two edge-scalar families are multiplied, then their endpoint scalars satisfy $(cd)_{v,e} = c_{v,e} d_{v,e}$ on every oriented half-edge.*



Proof. ✓ Proof At the lower endpoint this is ordinary multiplication of the edge scalars. At the upper endpoint, $(c_e d_e)^{-1} = c_e^{-1} d_e^{-1}$ because the edge scalars lie in $\mathbb{C}^\times$. □

Theorem 17.179 (Inversion of oriented edge scalars). Lean ✓ Stmt *Inverting an edge-scalar family gives* $(c^{-1})_{v,e} = c_{v,e}^{-1}$ *on every oriented endpoint.*

Proof. ✓ Proof At the lower endpoint, $(c^{-1})_{v,e} = c_e^{-1} = c_{v,e}^{-1}$. At the upper endpoint, $(c^{-1})_{v,e} = (c_e^{-1})^{-1} = c_{v,e}^{-1}$. □

Theorem 17.180 (Nonzero oriented edge scalars). Lean ✓ Stmt *Every oriented endpoint scalar satisfies* $c_{v,e} \neq 0$.

Proof. ✓ Proof Edge scalars are units, and the inverse of a unit is again a unit. □

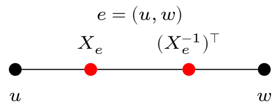
Theorem 17.181 (Product of balanced edge scalars). Lean ✓ Stmt *The pointwise product of two vertex-balanced edge-scalar families is again vertex-balanced.*

Proof. ✓ Proof For every vertex v, $\prod_{e \ni v} (cd)_{v,e} = (\prod_{e \ni v} c_{v,e})(\prod_{e \ni v} d_{v,e}) = 1 \cdot 1 = 1$. □

Theorem 17.182 (Inverse of balanced edge scalars). Lean ✓ Stmt *The pointwise inverse of a vertex-balanced edge-scalar family is again vertex-balanced.*

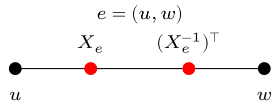
Proof. ✓ Proof For every vertex v, $\prod_{e \ni v} (c^{-1})_{v,e} = (\prod_{e \ni v} c_{v,e})^{-1} = 1^{-1} = 1$. □

Theorem 17.183 (Reflexivity of balanced edge-scalar equivalence). Lean ✓ Stmt *Every edge-gauge family is equivalent to itself modulo balanced edge scalars, witnessed by* $c_e = 1$ *for every edge* e.



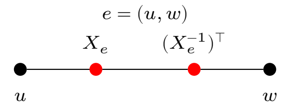
Proof. ✓ Proof If $c_e = 1$, then $c_{v,e} = 1$, $X_{v,e} = c_{v,e} X_{v,e}$, and $\prod_{e \ni v} c_{v,e} = 1$. □

Theorem 17.184 (Symmetry of balanced edge-scalar equivalence). Lean ✓ Stmt *If one gauge family differs from another by balanced edge scalars, then the second differs from the first by the inverse balanced edge-scalar family.*



Proof. ✓ Proof If $X_{v,e} = c_{v,e} Y_{v,e}$ with $c_{v,e} \neq 0$, then $Y_{v,e} = c_{v,e}^{-1} X_{v,e}$. The inverse scalar family is balanced by Theorem 17.182. □

Theorem 17.185 (Transitivity of balanced edge-scalar equivalence). Lean ✓ Stmt *If one gauge family differs from a second by balanced edge scalars, and the second differs from a third by balanced edge scalars, then the first differs from the third by the pointwise product of the two scalar families.*



Proof. ✓ Proof From $X_{v,e} = c_{v,e} Y_{v,e}$ and $Y_{v,e} = d_{v,e} Z_{v,e}$ one obtains $X_{v,e} = (c_{v,e} d_{v,e}) Z_{v,e}$. The product scalar family is balanced by Theorem 17.181. □

Theorem 17.186 (The balanced edge-scalar quotient is an equivalence relation)**.** Lean ✓ Stmt *Gauge equivalence modulo balanced edge scalars is reflexive, symmetric, and transitive.*

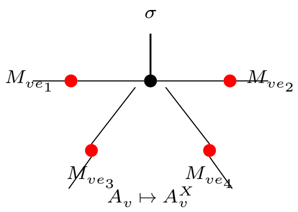
Proof. ✓ Proof Reflexivity is witnessed by 1, symmetry by c^{-1}, and transitivity by cd. □

Definition 17.187 (The balanced edge-scalar quotient relation)**.** Lean ✓ Stmt The preceding equivalence relation is the quotient relation on edge-gauge families modulo vertex-balanced edge scalars.

Theorem 17.188 (Balanced edge scalars preserve the local gauge action)**.** Lean ✓ Stmt *If two gauge families differ by a balanced edge-scalar reweighting, then they produce the same gauged tensor at every vertex.*

Proof. ✓ Proof In each summand of the gauged tensor at v, the edge scalars contribute the factor $\prod_{e \ni v} c_{v,e} = 1$. Hence the summand, and therefore the whole local tensor, is unchanged. □

Theorem 17.189 (Balanced edge scalars preserve the gauged tensor)**.** Lean ✓ Stmt *If two gauge families differ by a balanced edge-scalar reweighting, then applying them to the same PEPS tensor produces the same gauged tensor.*



Proof. ✓ Proof For every vertex v, virtual index η, and physical index σ, $(A^X)_v(\eta, \sigma) = (A^Y)_v(\eta, \sigma)$ for the two gauged tensors by Theorem 17.188. Hence the two gauged tensors are equal. □

Theorem 17.190 (Balanced edge scalars preserve the gauged state)**.** Lean ✓ Stmt *If two gauge families differ by a balanced edge-scalar reweighting, then the PEPS tensors obtained by applying those gauge families represent the same state.*

Proof. ✓ Proof The two gauged tensors are equal, hence all of their state coefficients are equal. □

Theorem 17.191 (Gauge uniqueness modulo balanced edge scalars)**.** Lean Not ready *If two gauge families relate the same vertex-injective PEPS tensor to the same target tensor, then their edge gauges differ by nonzero scalars c_e whose oriented product at every vertex is* 1*. Thus the correct graph quotient is by balanced edge scalars.*

Remark 17.192. The local left inverse Ψ_v, the realization map $T \mapsto O_T$, the split at one incident edge, and the partition $V = \{u\} \sqcup (V \setminus \{u, v\}) \sqcup \{v\}$ isolate the local indices and maps needed for the edge-centered reduction of [MGRPG+18, Section 3]. The middle-region blocked coefficient is constructed by the preceding definitions; the remaining local-gauge step is the comparison with the corresponding three-site injective chain. The corrected uniqueness statement is a quotient by balanced edge scalars; its proof still requires the same blocking construction together with the corresponding tensor-factor uniqueness statement at a single vertex.

Chapter 18

Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07], with the N-site Hilbert space modelled as coefficient functions

$$\{0, \dots, d-1\}^N \to \mathbb{C},$$

rather than as an explicit tensor product.

Definition 18.1 (N-site coefficient space)**.** Lean ✓ Stmt For local dimension d and chain length N, the ambient coefficient space is

$$\mathscr{H}_N^{(d)} := \{0, \dots, d-1\}^N \to \mathbb{C}.$$

This is the concrete model used throughout the chapter for periodic-chain states.

18.1 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \cdots A^{i_L}$. The local ground-space map is

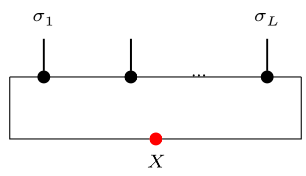
$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}), \qquad \Gamma_L(X)(\sigma) := \mathrm{tr}(A^\sigma X),$$

where A^σ abbreviates $A^{\sigma(0)} \cdots A^{\sigma(L-1)}$.

Definition 18.2 (Ground-space map)**.** Lean ✓ Stmt The map Γ_L above is a $\mathbb{C}$-linear map

$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower red node denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The chain length L is fixed by the surrounding formula, not by an extra label inside the diagram.

Lemma 18.3 (Evaluation formula). Lean ✓ Stmt *For every boundary matrix $X \in M_D(\mathbb{C})$ and every length-L word σ,*

$$\Gamma_L(X)(\sigma) = \operatorname{tr}(A^\sigma X).$$

Definition 18.4 (Local ground space). Lean ✓ Stmt The *local ground space* is the image

$$G_L(A) := \operatorname{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

Lemma 18.5 (Dimension bound). Lean ✓ Stmt *For every L,*

$$\dim G_L(A) \leq D^2.$$

Proof. ✓ Proof Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2.$$

□

Lemma 18.6 (Ambient-space dimension). Lean ✓ Stmt *The coefficient-function model satisfies*

$$\dim(\{0, \dots, d-1\}^L \to \mathbb{C}) = d^L.$$

Lemma 18.7 (Nontriviality criterion). Lean ✓ Stmt *If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:*

$$G_L(A) \neq (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

Proof. ✓ Proof If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 18.6; Lemma 18.5 gives the contradiction $d^L \leq D^2$. □

18.2 Parent interaction and chain Hamiltonian

The parent interaction at range L is the orthogonal projector onto $G_L(A)^\perp$ in the L-site Euclidean space.

Definition 18.8 (Euclidean transport of the local ground space). Lean ✓ Stmt Via the canonical identification between coefficient functions and the Euclidean space $\ell^2(\{0, \dots, d-1\}^L)$, one transports the local ground space to a Hilbert-space subspace

$$G_L^{\mathrm{ES}}(A) \subseteq \ell^2(\{0, \dots, d-1\}^L).$$

Lemma 18.9 (Membership in the transported local ground space). Lean ✓ Stmt *A Euclidean vector lies in $G_L^{\mathrm{ES}}(A)$ exactly when transporting it back through the canonical coefficient-space equivalence gives an element of the original local ground space $G_L(A)$.*

Proof. ✓ Proof This is just the definition of $G_L^{\mathrm{ES}}(A)$ as the image of $G_L(A)$ under the inverse of the canonical equivalence. □

Definition 18.10 (Parent interaction). Lean ✓ Stmt The parent interaction at range L is the orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp},$$

on the local L-site space.

Lemma 18.11 (Parent interaction kills ground-space elements). Lean ✓ Stmt *For any $v \in G_L(A)$, we have $h_L(A)\, v = 0$.*

Proof. ✓ Proof Since $h_L(A)$ is the orthogonal projector onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. □

Definition 18.12 (Periodic window extraction). Lean ✓ Stmt For a configuration $\sigma \in \{0, \dots, d-1\}^N$ on the periodic chain and a starting site i, the extracted word

$$\mathrm{extractWindow}_{L,i}(\sigma) \in \{0, \dots, d-1\}^L$$

reads the entries of σ on the cyclic interval $i, i+1, \dots, i+L-1$ modulo N.

Definition 18.13 (Periodic window replacement). Lean ✓ Stmt If $L \le N$, the configuration

$$\mathrm{replaceWindow}_{L,i}(\sigma, \tau)$$

is obtained from σ by replacing the cyclic window starting at i by the length-L word τ.

Lemma 18.14 (Extracting a replaced window). Lean ✓ Stmt *If $L \le N$, then extracting the same cyclic window after replacement recovers the inserted word:*

$$\mathrm{extractWindow}_{L,i}(\mathrm{replaceWindow}_{L,i}(\sigma, \tau)) = \tau.$$

Lemma 18.15 (Replacing an extracted window). Lean ✓ Stmt *If $L \le N$, then replacing the cyclic window at i in σ by the values extracted from that same σ leaves σ unchanged:*

$$\mathrm{replaceWindow}_{L,i}(\sigma, \mathrm{extractWindow}_{L,i}(\sigma)) = \sigma.$$

Proof. ✓ Proof Case-split on whether the offset $(k-i+N) \bmod N$ lies in $[0, L)$: inside the window the value comes from extractWindow, which returns σ at the same position; outside the window both sides already agree with σ. □

For a periodic chain of length N, the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 18.16 (Translated local term). Lean ✓ Stmt For each site index $i \in \{0, \dots, N-1\}$, the translated local term h_i embeds the parent interaction into the full N-site space at position i (with periodic boundary conditions). Thus h_i acts on the cyclic interval $[i, i+L-1]$ and as the identity outside that interval.

Definition 18.17 (Parent Hamiltonian). Lean ✓ Stmt The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

Definition 18.18 (Frustration-free condition). Lean ✓ Stmt A state ψ is frustration-free if each local term annihilates it:

$$\forall\, i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

Lemma 18.19 (MPV window membership). Lean ✓ Stmt *For any window of L consecutive sites starting at position i in an N-site periodic chain ($L \le N$), the restriction of the MPV state to that window lies in $G_L(A)$.*

Proof. ✓ Proof The witness is the product of A-matrices on the complement sites. Trace cyclicity rotates the full N-site product so that window indices come first, matching the ground-space map definition. □

Lemma 18.20 (Local term annihilates the MPV). Lean ✓ Stmt *For each site i, $h_i V^{(N)}(A) = 0$.*

Proof. ✓ Proof Combines MPV window membership with the fact that the parent interaction kills ground-space elements. □

Lemma 18.21 (Parent Hamiltonian annihilates the MPV). Lean ✓ Stmt *For every N, the MPV state satisfies*

$$H_N(A, L)\, V^{(N)}(A) = 0.$$

Proof. ✓ Proof Sum of zeros, since each local term annihilates the MPV. □

Lemma 18.22 (Frustration-freeness of the MPV). Lean ✓ Stmt *The MPV state is frustration-free for the parent model.*

Proof. ✓ Proof Immediate from Lemma 18.20. □

18.3 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

18.3.1 Restriction maps

Given a state ψ on $L + 1$ sites, we define two restriction maps.

Remark 18.23 (Word and restriction auxiliary formulas). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction operators are the associated linear maps on coefficient spaces, together with their pointwise formulas.

Definition 18.24 (Left restriction). Lean ✓ Stmt Fix the last physical index j:

$$\sigma \longmapsto \psi(\sigma_1, \dots, \sigma_L, j).$$

Definition 18.25 (Right restriction). Lean ✓ Stmt Fix the first physical index i:

$$\sigma \longmapsto \psi(i, \sigma_1, \dots, \sigma_L).$$

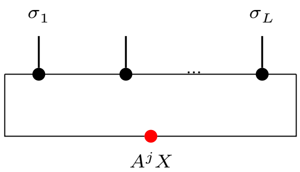
Definition 18.26 (Left ground membership). Lean ✓ Stmt A state ψ on $L+1$ sites satisfies the left ground condition if for every j, the state obtained by fixing the last index to j lies in $G_L(A)$.

Definition 18.27 (Right ground membership). Lean ✓ Stmt A state ψ on $L + 1$ sites satisfies the right ground condition if for every i, the state obtained by fixing the first index to i lies in $G_L(A)$.

18.3.2 Forward direction: ground space restricts

Lemma 18.28 (Left restriction). Lean ✓ Stmt *If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$ for some X, then for each j, the state obtained by fixing the last index to j lies in $G_L(A)$ with witness $A^j X$.*

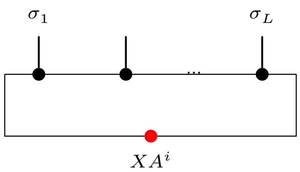
Proof. ✓ Proof The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation: $\psi(\sigma_1, \dots, \sigma_L, j) = \mathrm{tr}(A^{\sigma_1} \cdots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. $\square$

Lemma 18.29 (Right restriction). Lean ✓ Stmt *If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$, then for each i, the state obtained by fixing the first index to i lies in $G_L(A)$ with witness XA^i.*

Proof. ✓ Proof On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity: $\psi(i, \sigma_1, \dots, \sigma_L) = \mathrm{tr}(A^i A^\sigma X) = \mathrm{tr}(A^\sigma \cdot XA^i) = \Gamma_L(XA^i)(\sigma)$. $\square$

18.3.3 Injectivity of the ground-space map

Theorem 18.30 (Ground-space map injectivity). Lean ✓ Stmt *If A is injective and $L \geq 1$, then Γ_L is injective.*

Proof. ✓ Proof If $\Gamma_L(X) = 0$ then $\mathrm{tr}(A^\sigma X) = 0$ for every length-L word σ. Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. $\square$

Theorem 18.31 (Ground-space map injectivity from word span). Lean ✓ Stmt *If the length-L products $\{A^\sigma : |\sigma| = L\}$ span $M_D(\mathbb{C})$, then Γ_L is injective.*

Proof. ✓ Proof If $\Gamma_L(X) = 0$, then $\mathrm{tr}(A^\sigma X) = 0$ for every word σ of length L. The spanning hypothesis turns this into $\mathrm{tr}(MX) = 0$ for every $M \in M_D(\mathbb{C})$, and the trace pairing gives $X = 0$. $\square$

Lemma 18.32 (Ground-space map injectivity for block-injective tensors). Lean ✓ Stmt *If A is L_0-block-injective, then Γ_{L_0} is injective.*

Proof. ✓ Proof By definition, L_0-block injectivity says precisely that the length-L_0 products span $M_D(\mathbb{C})$. Apply Theorem 18.31. $\square$

Theorem 18.33 (Ground-space dimension). Lean ✓ Stmt *If A is injective and $L \geq 1$, then* $\dim G_L(A) = D^2$.

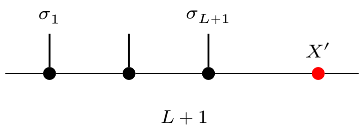
Proof. ✓ Proof The upper bound $\dim G_L(A) \leq D^2$ is Lemma 18.5. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. $\square$

18.3.4 The intersection property

Theorem 18.34 (Intersection property). Lean ✓ Stmt *If A is injective and $L \geq 2$, then a state ψ on $L+1$ sites satisfying both the left and right ground conditions lies in $G_{L+1}(A)$:*

$$\psi \in G_L^{\text{left}} \cap G_L^{\text{right}} \implies \psi \in G_{L+1}(A).$$

Proof. ✓ Proof The "invert-and-regrow" argument compares the left and right restrictions on their common $(L-1)$-site overlap and then reconstructs the full $(L+1)$-site word from a single boundary matrix:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of Γ_L) such that $\psi(i,\sigma) = \mathrm{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each j there exists a unique Z_j with $\psi(\sigma, j) = \mathrm{tr}(A^\sigma Z_j)$.
3. Matching on the $(L-1)$-site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all i, j.
4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.28), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Then $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$, where $X' := \sum_j c_j Z_j$.
5. By trace cyclicity, $\psi(\sigma) = \mathrm{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

□

Theorem 18.35 (Intersection characterization). Lean ✓ Stmt *For injective A and $L \geq 2$: $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$.*

Proof. ✓ Proof Immediate from the two forward-direction lemmas and Theorem 18.34. □

18.3.5 Unique ground state

Contiguous and cyclic window operators

Remark 18.36 (Contiguous window operators). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For a non-wrapping interval $[s, s+M-1] \subseteq \{0, \ldots, N-1\}$, one can insert a length-M word into that interval and keep the complementary sites fixed. The associated restriction map evaluates an N-site state on those configurations. The auxiliary lemmas state the full-window case $s = 0$, $M = N$, the compatibility of this restriction with deleting the last or first site, and the fact that fixing the first K entries of a contiguous $(K+L)$-window leaves the contiguous L-window beginning at $s+K$.

Remark 18.37 (Cyclic window operators). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt
On the periodic chain, a cyclic window starting at i reads the sites $i, i+1, \dots, i+L-1$ modulo N. Deleting the final site of a cyclic $(L+1)$-window reads the value at cyclic offset L in the outside configuration and gives the L-window starting at the same site. When this window does not wrap around the ring, the cyclic construction agrees with the contiguous one, and so do the two restriction maps.

Lemma 18.38 (Iterated contiguous-window intersection). Lean ✓ Stmt *If an N-site state satisfies the L-site ground condition on every non-wrapping contiguous window (with $L \geq 2$), then it lies in $G_N(A)$.*

Proof. ✓ Proof The induction repeatedly merges two neighbouring admissible windows $[i, i+L-1]$ and $[i+1, i+L]$ into the larger window $[i, i+L]$ by applying Theorem 18.34 to their common overlap $[i+1, i+L-1]$. Strong induction on window size, combining adjacent windows via Theorem 18.34 at each step. ☐

Definition 18.39 (Suffix restriction). Lean ✓ Stmt Fix a prefix $u : \{0, \dots, K-1\} \to \{0, \dots, d-1\}$. The suffix restriction of an $(K+L)$-site state ψ is the L-site state given by

$$(\mathrm{tailRestrict}_u\, \psi)(\sigma) = \psi(u, \sigma).$$

Definition 18.40 (Suffix ground membership). Lean ✓ Stmt A state ψ on $K+L$ sites lies in the suffix ground condition if every fixed prefix u gives a suffix state in $G_L(A)$.

Lemma 18.41 (Ground-space suffix restriction). Lean Lean Lean ✓ Stmt *If $\psi \in G_{K+L}(A)$, then every fixed-prefix suffix restriction of ψ lies in $G_L(A)$.*

Proof. ✓ Proof Write $\psi(\rho) = \mathrm{tr}(A^\rho X)$ for a boundary matrix X. Fixing a prefix u gives

$$\psi(u, \sigma) = \mathrm{tr}(A^u A^\sigma X) = \mathrm{tr}(A^\sigma X A^u),$$

so the suffix restriction again has the form of a ground-space vector. ☐

Lemma 18.42 (Left-tail compatibility). Lean ✓ Stmt *Assume ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every fixed prefix of length K. Then there exist matrices $(Z_j)_j$ and $(Y_u)_u$ such that*

$$\psi(\sigma, j) = \mathrm{tr}(A^\sigma Z_j), \qquad \psi(u, \tau) = \mathrm{tr}(A^\tau Y_u),$$

and for all j and u,

$$Z_j A^u = A_j Y_u.$$

Proof. ✓ Proof Choose Z_j from the long left-window condition and Y_u from the suffix ground condition. Restricting the suffix state at its last site gives the boundary matrix $A_j Y_u$, while first fixing the last site and then taking the suffix gives $Z_j A^u$. The two restrictions are the same because appending the fixed prefix and then the last letter gives the same configuration as appending the last letter to the suffix and then placing the prefix in front. Injectivity of Γ_{L_0}, obtained from L_0-block injectivity, yields $Z_j A^u = A_j Y_u$. ☐

Theorem 18.43 (Common right factor from word compatibility). Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$. If matrices $(Z_j)_j$ satisfy*

$$Z_j A^\sigma = A_j Y_\sigma$$

for every word σ of length K, then there exists a matrix X such that $Z_j = A_j X$ for all j.

Proof. ✓ Proof For $K = 1$, the hypothesis gives $Z_j A_i = A_j Y_i$ for every letter i. Every blocked word of length L_0 is nonempty, so this identity extends to the coefficients of the blocked tensor $A^{[L_0]}$. Since $A^{[L_0]}$ is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields $Z_j = A_j X$ for a common matrix X. For larger K, strip the first letter: for each i, the matrices $Z_j A_i$ satisfy the same compatibility hypothesis with words of length $K - 1$, so the induction hypothesis reduces the problem to the case $K = 1$. □

Theorem 18.44 (Open-chain right extension at the injectivity length). Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$. If an element ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every prefix of length K, then $\psi \in G_{K+L_0+1}(A)$.*

Proof. ✓ Proof Lemma 18.42 produces matrices $(Z_j)_j$ and $(Y_u)_u$ with $Z_j A^u = A_j Y_u$ for every prefix word u of length K. Theorem 18.43 gives a common right factor X such that $Z_j = A_j X$ for all j. Therefore each last-site restriction of ψ agrees with the corresponding last-site restriction of the ground- space vector determined by X. Since every configuration is obtained by adjoining its final letter to its initial $(K + L_0)$-tuple, the two states coincide, so ψ lies in $G_{K+L_0+1}(A)$. □

Theorem 18.45 (Open-chain normal range reduction). Lean ✓ Stmt *Let A be L_0-block injective with $D \geq 1$ and $L_0 > 0$. If an N-site state satisfies the ground-space constraint on every non-wrapping contiguous window of length $L_0 + 1$, for every choice τ of the complementary outside configuration, with $N \geq L_0 + 1$, then it lies in $G_N(A)$.*

Proof. ✓ Proof Induct on the extra length beyond $L_0 + 1$. The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 18.39 and the contiguous-window transport above identify every fixed-prefix suffix with one of the assumed length-$L_0 + 1$ windows. Theorem 18.44 then regrows the rightmost site. □

Lemma 18.46 (Cyclic-window range antitonicity). Lean Lean ✓ Stmt *On a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if $L' \leq L \leq N$, then $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$.*

Proof. ✓ Proof Fix a cyclic window of length $L+1$ and restrict its last site. The resulting length-L vector is again in the ground space, so one-site peeling gives the inclusion for consecutive ranges. Iteration gives the general antitonicity. □

Theorem 18.47 (Cyclic-window normal open-chain reduction). Lean ✓ Stmt *Let A be L_0-block injective with $L_0 > 0$ and $D \geq 1$. If an N-site periodic-chain state satisfies all cyclic ground-space constraints at some range $L_0 + 1 \leq L \leq N$, then it lies in the open-chain ground space $G_N(A)$.*

Proof. ✓ Proof Lemma 18.46 reduces the cyclic hypotheses to range $L_0 + 1$. On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 18.45 applies. □

Lemma 18.48 (Wrapped-window one-sided compatibility). Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$ and $M \geq L_0$. If the reduced wrapped window at the last site has boundary matrices Y_τ, then for every physical letter j one has*

$$C_\tau^+ A^j X = Y_\tau A^j,$$

where C_τ^+ is the complementary word seen by that wrapped window.

Proof. ✓ Proof The trace identity for the wrapped window is tested against all words of length L_0. Injectivity of Γ_{L_0}, which follows from L_0-block injectivity, strips the test word and gives the displayed matrix identity. □

Lemma 18.49 (Mirror wrapped-window one-sided compatibility). Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$ and $M \geq L_0$. If the opposite reduced wrapped window has boundary matrices Y_τ, then*

$$XA^jC_\tau^- = A^jY_\tau$$

for every physical letter j, where C_τ^- is the complementary word seen from the opposite boundary-crossing position.

Proof. ✓ Proof Factor the cyclic word at the opposite wrapped position, rotate the trace, and use injectivity of Γ_{L_0} to remove the length-L_0 test word. □

Theorem 18.50 (Reduced wrapped-boundary compatibilities). Lean ✓ Stmt *Let A be L_0-block injective with $L_0 > 0$ and $D \geq 1$. Suppose $\psi \in \mathcal{G}_{N,L}(A)$, $N \geq 2$, $L_0 < L \leq N$, and $\psi = \Gamma_N(X)$ for a boundary matrix X. Then there are two families of boundary matrices Y_τ^+ and Y_τ^-, indexed by outside configurations τ, such that for every physical letter j,*

$$C_\tau^+A^jX = Y_\tau^+A^j, \qquad XA^jC_\tau^- = A^jY_\tau^-,$$

where C_τ^+ and C_τ^- are the complementary words exposed by the two opposite wrapped windows of reduced length $L_0 + 1$.

Proof. ✓ Proof First use Lemma 18.46 to reduce the cyclic constraints from range L to range $L_0 + 1$. Then apply the wrapped window and mirror wrapped window compatibility lemmas at the two boundary-crossing positions to obtain the displayed one-sided identities. □

Lemma 18.51 (Middle-word backgrounds for the two wrapped complements). Lean Lean Lean Lean ✓ Stmt *Fix a dummy physical letter η and a middle word μ of length $N - (L_0 + 1)$. Write $\tau_\eta^+(\mu)$ for the last-site wrapped background and $\tau_\eta^-(\mu)$ for the opposite wrapped background. Their exposed complements are both exactly μ.*

Proof. ✓ Proof With the same dummy letter η on the remaining sites, fill the physical sites $L_0, \ldots, N-2$ with μ for the last-site wrapped window, and fill the sites $1, \ldots, N - L_0 - 1$ with μ for the opposite wrapped window. The two complement-extraction identities are then immediate from the index arithmetic. □

Lemma 18.52 (Same-witness identities from compared wrapped witnesses). Lean ✓ Stmt *Fix a dummy physical letter η. Suppose the wrapped and mirror one-sided witnesses Y^+ and Y^- have been extracted. If, for every middle word μ, the witnesses agree on the backgrounds built from η,*

$$Y^+_{\tau_\eta^+(\mu)} = Y^-_{\tau_\eta^-(\mu)},$$

then there is a single family Y_μ satisfying

$$A^\mu A^b X = Y_\mu A^b, \qquad XA^aA^\mu = A^aY_\mu$$

for all letters a, b.

Proof. ✓ Proof Define Y_μ to be the wrapped witness evaluated on the background $\tau_\eta^+(\mu)$. Lemma 18.51 rewrites the wrapped complement as μ, while the assumed comparison at $\tau_\eta^-(\mu)$ rewrites the mirror witness to the same Y_μ. □

Lemma 18.53 (Same-witness common-middle commutation). Lean ✓ Stmt *Suppose a single family of boundary matrices Y_μ, indexed by middle words μ, satisfies the two one-sided identities*

$$A^\mu A^b X = Y_\mu A^b, \qquad X A^a A^\mu = A^a Y_\mu$$

for every pair of physical letters a, b. Then X commutes with every word product $A^a A^\mu A^b$ obtained by adjoining one letter on each side of the common middle.

Proof. ✓ Proof Multiply the second identity on the right by A^b and the first identity on the left by A^a:

$$X A^a A^\mu A^b = A^a Y_\mu A^b = A^a A^\mu A^b X.$$

□

Theorem 18.54 (Boundary matrix commutation from the wrapping window). Lean ✓ Stmt *Let A be injective with $D \geq 1$, and let $\psi = \Gamma_N(X)$ for some boundary matrix $X \in M_D(\mathbb{C})$. If every cyclic restriction of length L of ψ lies in $G_L(A)$, with $N \geq 2$ and $1 < L \leq N$, then, for every $j = 0, \dots, d-1$,*

$$X A^j = A^j X$$

Proof. ✓ Proof The wrapping window across the boundary produces matrix identities of the form $XMW = MY$ for arbitrary test matrices M and complementary words W. Spanning first over the length-$(L-1)$ tails and then over the complementary words, injectivity forces $(XM - MX)W = 0$ for every W, hence $XM = MX$ for every matrix M. In particular, X commutes with each letter A^j. □

Definition 18.55 (MPV submodule). Lean ✓ Stmt The subspace spanned by the MPV state $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^{\sigma_0} \cdots A^{\sigma_{N-1}})$.

Lemma 18.56 (The MPV is the ground-space map at the identity). Lean ✓ Stmt *For every chain length N,*

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

Lemma 18.57 (MPV lies in the ground space). Lean ✓ Stmt $V^{(N)}(A) \in G_N(A)$ *for every N, with boundary matrix $X = I$.*

Proof. ✓ Proof $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^\sigma) = \mathrm{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. □

Definition 18.58 (Unique ground state). Lean ✓ Stmt A submodule S has a unique ground state if $\dim S = 1$.

Theorem 18.59 (One-dimensionality criterion). Lean ✓ Stmt *For finite-dimensional S, $\dim S = 1$ iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c\,\psi_0$.*

Proof. ✓ Proof Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. □

Definition 18.60 (Periodic chain ground space). Lean ✓ Stmt For $N > 0$ and $L \leq N$, define the *periodic chain ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d{-}1\}^N \to \mathbb{C}) : \forall i \in \{0, \dots, N{-}1\},\ \psi|_{[i,i+L-1]} \in G_L(A)\},$$

where the condition $\psi|_{[i,i+L-1]} \in G_L(A)$ means that for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \top$ by convention.

Lemma 18.61 (The MPV satisfies every cyclic window constraint). Lean ✓ Stmt *If* $0 < N$ *and* $L \le N$*, then*

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

Theorem 18.62 (MPV nonvanishing for block-injective tensors). Lean ✓ Stmt *If* A *is* L_0*-block-injective with* $L_0 > 0$*,* $D \ge 1$*, and* $N \ge L_0 + 1$*, then* $V^{(N)}(A) \ne 0$ *in the* N*-site space.*

Proof. ✓ Proof Suppose for contradiction that $V^{(N)}(A) = 0$, i.e. $\mathrm{tr}(A^\sigma) = 0$ for every word σ of length N. For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\mathrm{tr}(A^w A^u) = 0$. Since A is L_0-block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeating the argument with words of length $N - 2L_0$ (since $|w'| + L_0 = N - L_0$, the product $A^{w'} A^{u_1}$ has length $N - L_0$, so $A^{w'} A^{u_1} = 0$ by the previous step) and iterating, we eventually reach $\mathrm{tr}(A^u) = 0$ for $|u| = L_0$, which gives $I = 0$, a contradiction. □

18.3.6 Block-word stripping

Theorem 18.63 (Common right factor from block-word compatibility). Lean ✓ Stmt *Assume* A *is* L_0*-block-injective with* $L_0 > 0$*. Let* $\{Z_u\}_{|u|=L_0}$ *be a family indexed by words of length* L_0*. If for some* $K \ge 0$ *and every suffix word* w *of length* K *there exists* $Y_w \in M_D(\mathbb{C})$ *such that, for every word* u *of length* L_0*,*

$$Z_u A^w = A^u Y_w$$

then there exists $X \in M_D(\mathbb{C})$ *with* $Z_u = A^u X$ *for every word* u *of length* L_0*.*

Proof. ✓ Proof Induct on K. For $K = 0$ the empty suffix gives $Z_u = A^u Y_\emptyset$. For the successor step, fix the first physical index and apply the induction hypothesis to the shortened suffix. This reduces the statement to letter compatibility on the right. Extending that letter compatibility to all length-L_0 words and decomposing $\mathbb{1}$ in the spanning family $\{A^u : |u| = L_0\}$ yields a common factor X. □

Theorem 18.64 (Block-word commutation from long-word commutation). Lean ✓ Stmt *If* A *is* L_0*-block-injective with* $L_0 > 0$*,* $m \ge L_0$*, and* $X \in M_D(\mathbb{C})$ *commutes with every length-*m *product* A^ω*, then* X *already commutes with every length-*L_0 *product* A^u*.*

Proof. ✓ Proof Write each length-m word as a concatenation uw with $|u| = L_0$ and $|w| = m - L_0$. The hypothesis gives

$$XA^u A^w = A^u A^w X = A^u(A^w X).$$

Thus the family $Z_u := XA^u$ satisfies the hypothesis of Theorem 18.63. One obtains $XA^u = A^u R$ for all $|u| = L_0$. Decomposing $\mathbb{1}$ in the span of the length-L_0 words shows $R = X$, hence $XA^u = A^u X$. □

Theorem 18.65 (Centrality from long-word commutation). Lean ✓ Stmt *If* A *is* L_0*-block-injective with* $L_0 > 0$*,* $m \ge L_0$*, and* $X \in M_D(\mathbb{C})$ *commutes with every length-*m *word product, then* X *commutes with every matrix in* $M_D(\mathbb{C})$*.*

Proof. ✓ Proof By Theorem 18.64, X commutes with every length-L_0 word product. These products span $M_D(\mathbb{C})$ by block injectivity, so linearity gives $XM = MX$ for every $M \in M_D(\mathbb{C})$. □

Lemma 18.66 (Amplifying fixed-length word commutation). Lean ✓ Stmt *If a boundary matrix* X *commutes with every word product of length* m*, then it commutes with every word product whose length is a multiple* $q\,m$*.*

Proof. ✓ Proof Proceed by induction on q. For $q = 0$, the claim is trivial. For the inductive step, let a word of length $(q+1)m$ be given and write it as a length-m prefix followed by a length-$q\,m$ suffix. The prefix commutes with X by hypothesis, and the suffix commutes by the induction hypothesis. □

Lemma 18.67 (Padded complement annihilation). Lean ✓ Stmt *If A is L_0-block-injective and $q \geq 1$, then an operator Z that annihilates every length-k word product on the left is zero whenever $k \leq qL_0$:*

$$ZA^w = 0 \quad (|w| = k) \qquad \Longrightarrow \qquad Z = 0.$$

Proof. ✓ Proof The hypothesis $\operatorname{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$ implies the same spanning statement at every positive multiple qL_0. Each length-qL_0 word factors as a length-k prefix followed by padding, so Z annihilates all generators of the full span and hence annihilates the identity. □

Lemma 18.68 (Generator commutation from long-word commutation). Lean ✓ Stmt *Under the hypotheses of Theorem 18.65, one has $XA^i = A^iX$ for every physical index i.*

Proof. ✓ Proof Apply Theorem 18.65 with $M = A^i$. □

Theorem 18.69 (Boundary contraction in the MPV span from long-word commutation). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and let $m \geq L_0$. If a boundary matrix X commutes with every length-m word product, then $\Gamma_N(X)$ lies in the MPV line $\operatorname{span}\{V^{(N)}(A)\}$.*

Proof. ✓ Proof Theorem 18.65 makes X commute with the whole matrix algebra. The center of $M_D(\mathbb{C})$ consists of scalar matrices, so $X = c\mathbb{1}$, and therefore $\Gamma_N(X) = cV^{(N)}(A)$. □

Theorem 18.70 (Boundary contraction in the MPV span from positive-length commutation). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$. If X commutes with every word product of some positive length m, then $\Gamma_N(X)$ lies in the MPV line.*

Proof. ✓ Proof Lemma 18.66 amplifies the length-m commutation hypothesis to length $L_0\,m \geq L_0$. Then Theorem 18.69 applies. □

Theorem 18.71 (Boundary contraction in the MPV span from a common-middle witness). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$. If the common-middle same-witness identities of Lemma 18.53 hold for X, then $\Gamma_N(X)$ lies in the MPV line.*

Proof. ✓ Proof The common-middle identities give commutation with all words of length $|\mu| + 2$, a positive length. The positive-length containment theorem then applies. □

Theorem 18.72 (Boundary contraction in the MPV span from wrapped-witness comparison). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and fix a dummy physical letter η. If the wrapped and mirror one-sided witnesses satisfy*

$$Y^+_{\tau^+_\eta(\mu)} = Y^-_{\tau^-_\eta(\mu)}$$

for every middle word μ, then $\Gamma_N(X)$ lies in the MPV line.

Proof. ✓ Proof Lemma 18.52 uses the comparison at the two η-backgrounds to turn the wrapped witnesses into a single same-witness common-middle family. Then Theorem 18.71 applies. □

Theorem 18.73 (Conditional normal-range reduction from witness comparison). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$. Assume $N \geq 2$ and $L_0 < L \leq N$. Suppose that, for every dummy letter η and every boundary matrix obtained from an N-site chain ground state, the wrapped and mirror witnesses supplied by Theorem 18.50 agree at $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$ for every middle word μ. Then the range-L chain ground space is contained in the MPV line.*

Proof. ✓ Proof The cyclic-to-open-chain reduction writes any chain ground state as $\Gamma_N(X)$. The reduced wrapped-boundary compatibility theorem supplies the two one-sided witness families for this X. The comparison hypothesis, applied to a dummy letter η, says that these witnesses agree at $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$ for every middle word μ. Theorem 18.72 finishes. □

Theorem 18.74 (Chain ground space equals the MPV line in the injective case). Lean ✓ Stmt *If A is injective, $D \geq 1$, $N \geq 2$, and $1 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) = \operatorname{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof Let $\psi \in \mathcal{G}_{N,L}(A)$. The non-wrapping window conditions and Lemma 18.38 imply $\psi \in G_N(A)$, so $\psi = \Gamma_N(X)$ for some boundary matrix X. The wrapping window condition now applies Theorem 18.54 and forces X to commute with every letter A^j. Since A is injective, the letters generate the full matrix algebra, hence X is scalar. Therefore ψ is proportional to $V^{(N)}(A)$, while the reverse inclusion is Lemma 18.61. □

Theorem 18.75 (Normal-range reduction: containment in the MPV line). Lean Not ready *If A is normal, $D \geq 1$, L_0-block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) \subseteq \operatorname{span}\{V^{(N)}(A)\}.$$

Proof. Not ready The closure property needed here is the one from [CPGSV21, Section 4.3, lines 2049–2094]. Theorem 18.50 supplies the two one-sided wrapped-boundary identities for the open-chain boundary matrix X. The padded-annihilation lemma states that exact complement-span length mismatches are no longer the obstruction. The new conditional reduction theorem shows that the proof would finish as soon as those actual wrapped-window witnesses are compared after reindexing their complements to a common middle word. What is still missing is precisely that comparison of the witnesses themselves. □

Theorem 18.76 (Chain ground space equals the MPV line in the normal case). Lean Not ready *If A is normal, $D \geq 1$, L_0-block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) = \operatorname{span}\{V^{(N)}(A)\}.$$

Theorem 18.77 (Unique ground state on the periodic chain). Lean ✓ Stmt *For an injective tensor A with $D \geq 1$, if $N \geq 2$ and $1 < L \leq N$, then the periodic chain ground space is one-dimensional.*

Proof. ✓ Proof By Theorem 18.74, the ground space is the line $\operatorname{span}\{V^{(N)}(A)\}$. It remains to show that the MPV is nonzero. If $V^{(N)}(A) = 0$, then Lemma 18.56 gives

$$\Gamma_N(\mathbb{1}) = \Gamma_N(0).$$

Since A is injective and $N > 0$, Theorem 18.30 forces $\mathbb{1} = 0$, a contradiction. Therefore the spanning line is one-dimensional. □

Theorem 18.78 (Unique ground state for block-injective tensors). Lean Not ready *If A is L_0-block-injective with $L_0 > 0$ and $D \geq 1$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on the periodic chain.*

Theorem 18.79 (Optimal unique ground state for normal tensors). Lean Not ready *If A is normal (hence L_0-block-injective for some $L_0 > 0$), $D \geq 1$, and in normal form, the interaction range can be reduced to $L_0 + 1$.*

18.4 Commuting parent Hamiltonians

A parent Hamiltonian has *commuting* local terms when all translated interaction projectors commute with each other.

Definition 18.80 (Commuting parent Hamiltonian). Lean ✓ Stmt The parent Hamiltonian with block length L on N sites has *commuting* local terms if $h_i h_j = h_j h_i$ for all $i, j \in \{0, \dots, N-1\}$. See Ref. [CPGSV16, Definition 3.9].

Definition 18.81 (Nearest-neighbor commuting parent Hamiltonian). Lean ✓ Stmt A *nearest-neighbor commuting parent Hamiltonian* (NNCPH) is a commuting parent Hamiltonian with block length $L = 2$. See Ref. [CPGSV16, Definition 3.9].

Remark 18.82 (Elementary consequences of the commuting definition). Lean Lean Lean ✓ Stmt The nearest-neighbor condition is simply the length-2 specialization of the general commuting condition; the latter is symmetric in the two sites, and it implies that the full parent Hamiltonian commutes with each local interaction term.

Theorem 18.83 (RFP implies NNCPH). Lean ✓ Stmt *If A is a normal renormalization fixed point, then for every chain length $N \geq 2$ the parent Hamiltonian with $L = 2$ has commuting local terms. This implication is cited here from the product-of-entangled-pairs description in Appendix B of Theorem 3.10 in [CPGSV16].*

Remark 18.84 (Product-pair data for NNCPH). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt On an even chain, one can extract the p-th adjacent two-site word of a configuration and compare the MPS coefficients with a repeated two-site pair state. The one-pair case recovers the original two-site amplitude. For an Appendix B structural witness, the gauge-removed tensor ΛU_i has the same MPV coefficients as A, and the structural two-site amplitude is exactly the two-site MPV coefficient. Hence the coefficient part of the extraction reduces to proving the repeated-pair identity for this gauge-removed tensor. The corresponding data also include a family of idempotent chain-level projectors whose values are the nearest-neighbor local terms. The final theorem in these data shows that these local terms commute. Consequently the same data yields the nearest-neighbor commuting condition on every finite chain. The Appendix B construction extracts the structural decomposition from normal left-canonical RFP data. The missing mathematical input is the product-pair extraction through the two-site amplitude determined by that structural witness.

Theorem 18.85 (NNCPH implies RFP). Lean ✓ Stmt *If A is normal, in left-canonical form, and its parent Hamiltonian with $L = 2$ has commuting local terms for every chain length $N \geq 2$, then A is a renormalization fixed point. This implication is cited here from Beigi's ground-space characterization of one-dimensional commuting nearest-neighbor Hamiltonians; the left-canonical hypothesis fixes the normalization of the transfer map.*

18.5 Decorrelation and commuting parent Hamiltonians

This section develops the abstract operator-algebraic formulation of decorrelation and its connection to commuting parent Hamiltonians, following [CPGSV16, Appendix D, Section D.2].

Definition 18.86 (Decorrelation). Lean ✓ Stmt Given an idempotent endomorphism P_K and sets of observables $\mathcal{O}_A$, $\mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A$, $O_B \in \mathcal{O}_B$.

Definition 18.87 (Commuting parent Hamiltonian structure). Lean ✓ Stmt A subspace K (given by its idempotent endomorphism P_K) has a *commuting parent Hamiltonian structure* if there exist idempotent endomorphisms P_{AX} and P_{XB} that commute and satisfy $P_{AX} \circ P_{XB} = P_K$.

Theorem 18.88 (Idempotence gives a trivial commuting parent Hamiltonian). Lean ✓ Stmt *If*

$$P_K \circ P_K = P_K,$$

then P_K has a commuting parent Hamiltonian structure.

Proof. ✓ Proof Take $P_{AX} = P_{XB} = P_K$. Their commutativity is immediate, and the product identity is exactly the assumed idempotence of P_K. □

Lemma 18.89 (Cancellation for commuting idempotents). Lean ✓ Stmt *If P and Q are commuting idempotents, then*

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

Proof. ✓ Proof Expand the middle factor, use commutativity to rewrite $QPQP$ as $QPPQ$, and then apply idempotence of P and Q. □

Theorem 18.90 (Idempotence of the product projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_K = P_K.$$

Proof. ✓ Proof Since P_{AX} and P_{XB} are idempotent and commute, their product $P_{AX} \circ P_{XB}$ is idempotent. Substituting $P_K = P_{AX} \circ P_{XB}$ gives the claim. □

Theorem 18.91 (Absorption by the left projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{AX} \circ P_K = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{AX},

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 18.92 (Absorption by the right projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{XB} = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{XB},

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 18.93 (Cross-absorption by the right projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{XB} \circ P_K = P_K.$$

Proof. ✓ Proof By commutativity, $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$, and idempotence of P_{XB} yields $P_{XB} \circ P_K = P_K$. □

Theorem 18.94 (Cross-absorption by the left projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{AX} = P_K.$$

Proof. ✓ Proof By commutativity, $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$, and idempotence of P_{AX} yields $P_K \circ P_{AX} = P_K$. □

Theorem 18.95 (Reverse product identity). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{XB} \circ P_{AX} = P_K.$$

Proof. ✓ Proof This is immediate from commutativity of P_{AX} and P_{XB} together with the identity $P_{AX} \circ P_{XB} = P_K$. □

Theorem 18.96 (Commuting complementary projectors). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition and*

$$Q_{AX} = 1 - P_{AX}, \qquad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

Proof. ✓ Proof Expanding both products and using $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ shows that the complementary projectors commute. □

Theorem 18.97 (Commuting parent Hamiltonian implies decorrelation). Lean ✓ Stmt *If a subspace K has a commuting parent Hamiltonian decomposition $P_K = P_{AX} \circ P_{XB}$ with $[P_{AX}, P_{XB}] = 0$, and A-observables commute with P_{XB} while B-observables commute with P_{AX}, then regions A and B are decorrelated w.r.t. K. This is the backward direction of [CPGSV16, Appendix D, Section D.2, Proposition D.3].*

Proof. ✓ Proof The key identity is $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Using the commutation hypotheses to slide O_A past P_{XB} and O_B past P_{AX}, the five-fold composition factors through this zero term. □

Theorem 18.98 (Commuting parent Hamiltonian implies decorrelation). Lean ✓ Stmt *Corollary of Theorem 18.97: if P_K has a commuting parent Hamiltonian decomposition and observables respect locality, then regions A and B are decorrelated w.r.t. K.*

Lemma 18.99 (Frustration-free identity). Lean ✓ Stmt *For any endomorphisms P and Q,*

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

Proof. ✓ Proof Expanding the left-hand side and cancelling gives the result. □

Theorem 18.100 (Commuting parent Hamiltonian identity). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where $Q_{AX} = 1 - P_{AX}$ and $Q_{XB} = 1 - P_{XB}$ are the complementary projectors. This is the frustration-free form of the parent Hamiltonian from [CPGSV16, Appendix D, Section D.2].

Proof. ✓ Proof Applying Lemma 18.99 to P_{AX} and P_{XB}, and substituting $P_{AX} \circ P_{XB} = P_K$. □

Theorem 18.101 (Ground-space membership). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K v = v \quad \Longleftrightarrow \quad P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic form of equation (D.2) from [CPGSV16]: $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Proof. ✓ Proof ($\Rightarrow$) If $P_K v = v$, then $P_{AX}(P_K v) = (P_{AX} \circ P_K) v = P_K v = v$ by left absorption, and similarly for P_{XB}.

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then $P_K v = (P_{AX} \circ P_{XB}) v = P_{AX}(P_{XB} v) = P_{AX} v = v$. □

18.6 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length N. For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [KL18].

Definition 18.102 (Transported parent ground space). Lean ✓ Stmt Denote by $\mathcal{G}^{\mathrm{ES}}_{N,L}(A)$ the parent ground space after identifying the coefficient space with its Euclidean ℓ^2-model.

Definition 18.103 (Transported parent Hamiltonian). Lean ✓ Stmt Denote by $H^{\mathrm{ES}}_{N,L}(A)$ the conjugate of $H_N(A, L)$ by the canonical identification between the coefficient space and its Euclidean avatar.

Theorem 18.104 (Euclidean transport of the parent ground space). Lean ✓ Stmt *Under the canonical ℓ^2 transport of the N-site state space, the transported parent ground space is exactly the kernel of the transported parent Hamiltonian.*

Proof. ✓ Proof This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and its Euclidean-space avatar, so the transported ground space is precisely the transported kernel. □

Lemma 18.105 (Membership in the transported parent ground space). Lean ✓ Stmt *A vector belongs to the transported parent ground space exactly when the transported parent Hamiltonian annihilates it.*

Remark 18.106 (Euclidean local projector ingredients). Lean Lean Lean Lean ✓ Stmt Let $P_L^{\mathrm{ES}}(A)$ denote the orthogonal projector onto $(G_L^{\mathrm{ES}}(A))^\perp$ on the Euclidean L-site space. For a cyclic window starting at i and an outside configuration τ, the transported restriction map $R_{i,\tau}$ evaluates an N-site Euclidean vector on that window. The transported local term $h_{i,\mathrm{ES}}$ is the corresponding N-site local projector, and $R_{i,\tau}^\dagger P_L^{\mathrm{ES}}(A) R_{i,\tau}$ is the basic positive summand in its cyclic-averaging formula.

Lemma 18.107 (Positivity of the Euclidean local summands)**.** Lean Lean Lean ✓ Stmt *The Euclidean local projector $P_L^{\mathrm{ES}}(A)$ is positive. Consequently, every summand $R_{i,\tau}^\dagger P_L^{\mathrm{ES}}(A) R_{i,\tau}$ is positive, and so is the finite sum over τ.*

Proof. ✓ Proof Orthogonal projectors are positive. Conjugation by $R_{i,\tau}$ preserves positivity, and finite sums of positive operators remain positive. □

Lemma 18.108 (Cyclic averaging for the transported ES local terms)**.** Lean Lean ✓ Stmt *The transported parent Hamiltonian is the sum of the transported local terms $H_{N,\mathrm{ES}} = \sum_i h_{i,\mathrm{ES}}$, and each transported local term admits the exact cyclic-averaging formula*

$$h_{i,\mathrm{ES}} = d^{-L} \sum_\tau R_{i,\tau}^\dagger P_L^{\mathrm{ES}}(A) R_{i,\tau}.$$

Proof. ✓ Proof Unfold the transported local term and compare it configuration-by-configuration with the cyclic restrictions. The d^{-L} factor compensates for the d^L choices of the window coordinates in the ambient configuration parameter τ. □

Lemma 18.109 (Positivity of the transported ES Hamiltonian)**.** Lean Lean ✓ Stmt *Each transported local term $h_{i,\mathrm{ES}}$ is positive, hence the full transported parent Hamiltonian $H_{N,\mathrm{ES}}$ is positive.*

Proof. ✓ Proof Apply the averaging formula above: $h_{i,\mathrm{ES}}$ is a nonnegative scalar multiple of a finite sum of positive conjugates of $P_L^{\mathrm{ES}}(A)$, hence is itself positive. Summing over i yields positivity of $H_{N,\mathrm{ES}}$. □

Lemma 18.110 (Symmetric-projection structure of transported ES local terms)**.** Lean Lean ✓ Stmt *The Euclidean local projector $P_L^{\mathrm{ES}}(A)$ is a symmetric projection, and every transported local term $h_{i,\mathrm{ES}}$ on the N-site chain is a symmetric projection.*

Proof. ✓ Proof The local operator $P_L^{\mathrm{ES}}(A)$ is an orthogonal projection onto $(G_L^{\mathrm{ES}}(A))^\perp$. For $L \le N$, restricting $h_{i,\mathrm{ES}}v$ to a cyclic window gives $P_L^{\mathrm{ES}}(A)$ applied to the corresponding restriction of v; applying $h_{i,\mathrm{ES}}$ a second time therefore applies $(P_L^{\mathrm{ES}}(A))^2 = P_L^{\mathrm{ES}}(A)$ on that window. The positivity result above supplies symmetry, and the case $L > N$ is the zero projection by definition. □

Lemma 18.111 (Kernel of the Euclidean parent interaction)**.** Lean ✓ Stmt *The Euclidean local interaction $P_L^{\mathrm{ES}}(A)$ annihilates exactly the Euclidean local ground space:*

$$P_L^{\mathrm{ES}}(A)v = 0 \iff v \in G_L^{\mathrm{ES}}(A).$$

Proof. ✓ Proof Since $P_L^{\mathrm{ES}}(A)$ is the orthogonal projector onto $(G_L^{\mathrm{ES}}(A))^\perp$, its kernel is the original subspace $G_L^{\mathrm{ES}}(A)$. □

Lemma 18.112 (Local-term kernel via cyclic restrictions)**.** Lean Lean ✓ Stmt *For $L \le N$, a transported local term $h_{i,\mathrm{ES}}$ annihilates a vector v iff every boundary-filled restriction of v to the cyclic L-window starting at i lies in $G_L^{\mathrm{ES}}(A)$.*

Proof. ✓ Proof Restricting $h_{i,\mathrm{ES}}v$ to the same cyclic window gives $P_L^{\mathrm{ES}}(A)$ applied to the corresponding restriction of v. Lemma 18.111 identifies the vanishing of this local projector with membership in $G_L^{\mathrm{ES}}(A)$. □

Theorem 18.113 (Adjacent local kernels realize the intersection property). Lean Lean Lean ✓ Stmt *If A is injective and $L \geq 2$, then on an $(L+1)$-site chain the kernels of the two adjacent transported L-site local terms are exactly the transported $(L+1)$-site MPS ground space:*

$$h_{0,\mathrm{ES}}v = 0 \quad \text{and} \quad h_{1,\mathrm{ES}}v = 0 \quad \iff \quad v \in G_{L+1}^{\mathrm{ES}}(A).$$

Proof. ✓ Proof Lemma 18.112 translates the two local kernel assumptions into the left and right L-site ground conditions. The open-chain intersection property, Theorem 18.34, then regrows the shared $(L-1)$-site overlap to the $(L+1)$-site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel–restriction characterization. □

Lemma 18.114 (Non-overlap positivity for transported ES local terms). Lean Lean Lean Lean Lean ✓ Stmt *Let $L \leq N$ and let $h_{i,\mathrm{ES}}, h_{j,\mathrm{ES}}$ be transported local terms whose cyclic L-windows are site-disjoint: no site of the periodic chain has cyclic offset $< L$ from both i and j. Then the terms commute pointwise: for every vector v,*

$$h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v) = h_{j,\mathrm{ES}}(h_{i,\mathrm{ES}}v),$$

and their ordered cross term is nonnegative:

$$0 \leq \mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle.$$

Proof. ✓ Proof In coordinates, $h_{i,\mathrm{ES}}$ applies $P_L^{\mathrm{ES}}(A)$ only to the i-window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the i-window does not change the extracted j-window, and the i- and j-window replacements commute. Therefore the two embedded local operators commute. The preceding projection-geometry identity for commuting symmetric projections gives the nonnegative ordered cross term. □

Lemma 18.115 (Quadratic form to norm bound). Lean ✓ Stmt *Let H be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all v,*

$$\gamma\,\langle v, Hv\rangle \leq \langle Hv, Hv\rangle$$

i.e. $H^2 \geq \gamma H$ as a quadratic form, for some $\gamma > 0$. Then for every v in the orthogonal complement of $\ker H$,

$$\gamma\,\|v\| \leq \|Hv\|.$$

In particular, every positive eigenvalue of H is at least γ. The positive-semidefiniteness hypothesis is essential: without it, $H = -\mathrm{Id}$ with $\gamma = 2$ satisfies the quadratic-form inequality vacuously but violates the norm bound.

Proof. ✓ Proof By the finite-dimensional spectral theorem, H diagonalises in an orthonormal eigenbasis with real eigenvalues λ_k. For a unit eigenvector ψ_k with eigenvalue λ_k the hypothesis gives $\gamma\,\lambda_k \leq \lambda_k^2$, whence $\lambda_k \leq 0$ or $\lambda_k \geq \gamma$. Since H is positive semidefinite (its quadratic form is ≥ 0), the negative case is excluded and every nonzero eigenvalue satisfies $\lambda_k \geq \gamma$. Writing $v = \sum_k c_k\psi_k$ and projecting onto $(\ker H)^\perp$ keeps only terms with $\lambda_k \geq \gamma$, so $\|Hv\|^2 = \sum_k |c_k|^2\lambda_k^2 \geq \gamma^2\sum_k |c_k|^2 = \gamma^2\|v\|^2$. □

Lemma 18.116 (Parent-Hamiltonian quadratic-form reduction). Lean Lean ✓ Stmt *Let $H^{\mathrm{ES}}_{N,L}(A)$ be the transported parent Hamiltonian. If a constant $\gamma > 0$ satisfies the uniform quadratic-form estimate*

$$\gamma \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}(A)v, v\rangle \le \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}(A)v, H^{\mathrm{ES}}_{N,L}(A)v\rangle$$

for every $N \ge 2L$ and every vector v, then

$$\gamma \|v\| \le \|H^{\mathrm{ES}}_{N,L}(A)v\|$$

for every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$. In particular, with $L > 1$ and $\gamma = 1/(4L)$, the explicit gap-bound conclusion follows once this same quadratic-form estimate is supplied.

Proof. ✓ Proof The transported Hamiltonian is positive by Lemma 18.109, and its transported ground space is its kernel by Theorem 18.104. Therefore the abstract spectral-theorem conversion of Lemma 18.115 applies directly. The explicit constant is positive because $L > 1$ implies $4L > 0$. □

Remark 18.117 (Projection-geometry identities for row-sum arguments). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity $\operatorname{Re}\langle Pv, Pv\rangle = \operatorname{Re}\langle Pv, v\rangle$ for symmetric projections; nonnegativity of this quadratic form; nonnegativity of $\operatorname{Re}\langle Pv, Qv\rangle$ for commuting symmetric projections P, Q; the Cauchy–Schwarz conversion from $\|PQv\| \le c\|Pv\|$ to $\operatorname{Re}\langle Pv, Qv\rangle \ge -c \operatorname{Re}\langle Pv, v\rangle$; finite-sum expansions of $\operatorname{Re}\langle(\sum_i P_i)v, v\rangle$ and $\operatorname{Re}\langle(\sum_i P_i)v, (\sum_i P_i)v\rangle$; the diagonal/off-diagonal split of the ordered double sum; and the aggregate off-diagonal estimate obtained from row sums $\sum_{j\ne i} c_{ij} \le 1$.

Lemma 18.118 (Ordered projection row-sum quadratic-form reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. If the ordered off-diagonal forms obey*

$$\operatorname{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)c_{ij} \operatorname{Re}\langle P_i v, v\rangle \qquad (i \ne j),$$

with row sums $\sum_{j\ne i} c_{ij} \le 1$ and $\gamma \le 1$, then $H^2 \ge \gamma H$ as a quadratic form.

Proof. ✓ Proof Expand $\operatorname{Re}\langle Hv, Hv\rangle$ as an ordered double sum. Symmetric idempotence identifies each diagonal term with $\operatorname{Re}\langle P_i v, v\rangle$, while the row-summable ordered cross-term bounds control the off-diagonal contribution by $-(1-\gamma)\sum_i \operatorname{Re}\langle P_i v, v\rangle$. □

Lemma 18.119 (Finite-overlap projection row-sum reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $\gamma \le 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row for some positive integer m. If noninteracting ordered cross terms are nonnegative and, on each interacting pair,*

$$\operatorname{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)\frac{1}{m} \operatorname{Re}\langle P_i v, v\rangle,$$

then $H = \sum_i P_i$ satisfies $H^2 \ge \gamma H$ as a quadratic form.

Proof. ✓ Proof Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row cardinality bound gives $\sum_{j\ne i} c_{ij} \le 1$; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 18.118. □

Lemma 18.120 (Finite-overlap projection norm-compression reduction). Lean Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $\gamma \le 1$. Suppose an interaction predicate has at most $m > 0$ off-diagonal neighbours in each row, noninteracting ordered cross terms are nonnegative, and interacting pairs satisfy the norm-compression estimate*

$$\|P_i P_j v\| \le (1-\gamma)\frac{1}{m}\|P_i v\|.$$

Then $H = \sum_i P_i$ satisfies $H^2 \ge \gamma H$ as a quadratic form. The same conclusion holds from a separate compression coefficient η whenever $\eta \le (1-\gamma)/m$.

Proof. ✓ Proof Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into $\operatorname{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)m^{-1}\operatorname{Re}\langle P_i v, v\rangle$. If a separate coefficient η is used, the assumption $\eta \le (1-\gamma)/m$ first weakens that estimate to the displayed one. The finite-overlap row-sum reduction then applies. □

Lemma 18.121 (Parent-Hamiltonian ordered row-sum quadratic-form reduction). Lean ✓ Stmt *For a fixed chain length N, suppose the transported local terms $h_{i,\mathrm{ES}}$ satisfy the ordered row-sum estimate*

$$\operatorname{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \ge -(1-\gamma)c_{ij}\operatorname{Re}\langle h_{i,\mathrm{ES}}v, v\rangle \qquad (i \ne j),$$

with $\sum_{j\ne i} c_{ij} \le 1$ and $\gamma \le 1$. Then the transported parent Hamiltonian satisfies

$$\gamma \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}v, v\rangle \le \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}v, H^{\mathrm{ES}}_{N,L}v\rangle$$

for every vector v.

Proof. ✓ Proof Lemma 18.110 supplies the symmetric-projection hypothesis for the family $P_i = h_{i,\mathrm{ES}}$. Apply Lemma 18.118 to this family and use $H^{\mathrm{ES}}_{N,L} = \sum_i h_{i,\mathrm{ES}}$. □

Lemma 18.122 (Parent-Hamiltonian gap from ordered local row bounds). Lean ✓ Stmt *Assume uniformly for every $N \ge 2L$ that the transported local terms satisfy the ordered row-sum estimate above with $\gamma = 1/(4L)$. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$,*

$$\frac{1}{4L}\|v\| \le \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof The fixed-chain row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N. The positivity and kernel identification of the transported Hamiltonian then allow Lemma 18.116 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. □

Lemma 18.123 (Parent-Hamiltonian finite-overlap Friedrichs quadratic reduction). Lean ✓ Stmt *For a fixed chain length N, let m be a positive integer and let $\gamma \le 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row. If non-overlapping local terms have nonnegative ordered cross terms, overlapping local terms satisfy the finite-overlap Friedrichs estimate*

$$\operatorname{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \ge -(1-\gamma)\frac{1}{m}\operatorname{Re}\langle h_{i,\mathrm{ES}}v, v\rangle,$$

and the transported local terms are symmetric projections, then the transported Hamiltonian satisfies ${H^{\mathrm{ES}}_{N,L}}^2 \ge \gamma H^{\mathrm{ES}}_{N,L}$ as a quadratic form.

Proof. ✓ Proof Apply the finite-overlap projection reduction with $P_i = h_{i,\mathrm{ES}}$. □

Lemma 18.124 (Finite-overlap norm-compression reduction). Lean Lean ✓ Stmt *For a fixed chain length N, let $m > 0$ and $\gamma \le 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row, non-overlapping local terms have nonnegative ordered cross terms, and overlapping local terms satisfy*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \le (1-\gamma)\frac{1}{m}\|h_{i,\mathrm{ES}}v\|.$$

Then $H^{\mathrm{ES}}_{N,L}{}^2 \ge \gamma H^{\mathrm{ES}}_{N,L}$ as a quadratic form. The same conclusion holds from a separate coefficient η satisfying $\eta \le (1-\gamma)/m$.

Proof. ✓ Proof Apply the finite-overlap norm-compression projection reduction to the family $P_i = h_{i,\mathrm{ES}}$. If a separate coefficient η is used, first weaken the compression estimate using $\eta \le (1-\gamma)/m$. □

Definition 18.125 (Cyclic-window support). Lean ✓ Stmt For a length-L cyclic window on $\{0, \dots, N-1\}$ starting at i, its support is

$$S_i = \{i, i+1, \dots, i+L-1\} \pmod N,$$

with repeated visits counted once.

Lemma 18.126 (Cyclic-window support offsets). Lean ✓ Stmt *If $L \le N$, then a site k belongs to the cyclic support S_i exactly when its cyclic offset from i is below L:*

$$k \in S_i \iff (k-i) \bmod N < L.$$

Proof. ✓ Proof Write $k = i + r \pmod N$ for a representative offset r. Membership in S_i gives an offset $r < L$, and conversely an offset below L gives the corresponding point of the support. □

Definition 18.127 (Cyclic-window overlap). Lean ✓ Stmt Two length-L cyclic windows overlap, written $i \sim_L j$, when their supports meet: $S_i \cap S_j \ne \emptyset$.

Lemma 18.128 (Concrete non-overlap positivity for cyclic supports). Lean Lean ✓ Stmt *If $L \le N$ and the cyclic supports S_i and S_j do not meet, then the transported local ES terms have nonnegative ordered cross term:*

$$0 \le \mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle.$$

Proof. ✓ Proof The support-offset characterization converts failure of $S_i \cap S_j \ne \emptyset$ into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies. □

Lemma 18.129 (Cyclic-window overlap row cardinality). Lean ✓ Stmt *If $2L \le N$ and $L > 1$, then for every $i \in \{0, \dots, N-1\}$,*

$$\#\{j \ne i : i \sim_L j\} \le 2(L-1).$$

Proof. ✓ Proof If S_i and S_j meet, choose offsets $a, b < L$ with $i + a \equiv j + b \pmod N$. Since $2L \le N$, the offset from i to j is either the clockwise distance $a - b \in \{1, \dots, L-1\}$ or the counterclockwise distance $b - a \in \{1, \dots, L-1\}$; the zero distance is excluded by $j \ne i$. Thus every overlapping off-diagonal start lies among the $L-1$ clockwise starts or the $L-1$ counterclockwise starts. □

Lemma 18.130 (Parent-Hamiltonian gap from finite-overlap Friedrichs data). Lean ✓ Stmt *Fix $L > 1$ and set $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ that the hypotheses of the fixed-chain finite-overlap Friedrichs reduction hold with $\gamma = 1/(4L)$ and this value of m. Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$.*

Proof. ✓ Proof The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible N. Lemma 18.116 converts that estimate to the norm lower bound. □

Lemma 18.131 (Parent-Hamiltonian gap from cyclic-window Friedrichs data). Lean ✓ Stmt *Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length-L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \geq -(1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\,\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle.$$

Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$.

Proof. ✓ Proof Lemma 18.129 gives the row-cardinality bound $m = 2(L-1)$ for the concrete cyclic-window overlap relation, and Lemma 18.110 supplies the symmetric projection structure of each transported local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 18.128 gives the required nonnegative ordered cross term. Apply Lemma 18.130 with these facts and the displayed overlapping-window estimate. □

Lemma 18.132 (Parent-Hamiltonian gap from overlapping cyclic Friedrichs data). Lean ✓ Stmt *Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length-L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \geq -(1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\,\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$.

Proof. ✓ Proof Apply Lemma 18.131. Its non-overlap positivity and finite-row hypotheses are already internal to that cyclic-window reduction; the displayed overlapping estimate supplies its remaining ordered cross-term condition. □

Lemma 18.133 (Parent-Hamiltonian gap from overlapping cyclic norm-compression data). Lean ✓ Stmt *Fix $L > 1$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq (1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\|h_{i,\mathrm{ES}}v\|.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$.

Proof. ✓ Proof The norm-compression condition is converted to the ordered Friedrichs lower bound by the projection-geometry Cauchy–Schwarz lemma. The preceding overlapping-cyclic Friedrichs reduction then gives the norm lower bound. □

Lemma 18.134 (Parent-Hamiltonian gap from a cyclic overlap compression constant). Lean Lean ✓ Stmt *Fix $L > 1$, and let $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq \eta\,\|h_{i,\mathrm{ES}}v\|.$$

If $\gamma > 0$, $\gamma \leq 1$, and $\eta \leq (1-\gamma)/m$, then $0 < \gamma$ and, for every admissible N and every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$\gamma\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

The assertion $0 < \gamma$ repeats the hypothesis, in the same shape as the strict statement below. In particular, if $\eta \geq 0$ and $\eta m < 1$, then $0 < 1 - \eta m$ and, for every admissible N and every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$(1-\eta m)\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof The cyclic-window cardinality lemma gives at most $m = 2(L-1)$ overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The finite-overlap norm-compression reduction gives the quadratic-form estimate with the chosen γ, and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with $\gamma = 1 - \eta m$. □

Theorem 18.135 (Martingale criterion for spectral gap)**.** Not ready *Consider a frustration-free Hamiltonian*

$$H = \sum_{i=1}^{n} h_i$$

with projector interaction terms ($h_i^2 = h_i$). Suppose that for every pair of overlapping terms h_i, h_j there exist constants $c_{ij} \geq 0$ and $\gamma > 0$ satisfying

$$h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j),$$

and suppose further that $c_{ij} = c_{ji}$ for every overlapping pair (i,j) and that $\sum_{j:\, j \neq i} c_{ij} \leq 1$ for every i. Then the smallest positive eigenvalue of H satisfies $\lambda^{+}_{\min}(H) \geq \gamma$.

Proof. Not ready The relevant geometry is a family of translated length-L projector windows on the ring; two terms h_i and h_j interact only when their supports overlap, equivalently when the cyclic distance $d_N(i,j)$ is smaller than L. From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i<j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i<j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is ≥ 0. For the overlapping sum, the martingale condition gives $h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j)$. Summing over unordered pairs $\{i,j\}$:

$$\sum_{\{i,j\},\text{overlap}} (h_i h_j + h_j h_i) \geq -(1-\gamma) \sum_{\{i,j\},\text{overlap}} c_{ij}(h_i + h_j).$$

Collecting terms for each h_i and using the symmetry assumption $c_{ij} = c_{ji}$:

$$\sum_{\{i,j\},\text{overlap}} c_{ij}(h_i + h_j) = \sum_i \Big(\sum_{\substack{j \neq i \\ \text{overlap}}} c_{ij} \Big) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis $\sum_{j \neq i} c_{ij} \leq 1$. Combining, $H^2 \geq H - (1-\gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. □

Lemma 18.136 (Martingale condition for MPS). Not ready *Let A be injective and fix an interaction range $L \geq 2$ as in Definition 18.10. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the L-site parent interaction, and for each chain length and site i let h_i denote its translate as in Definition 18.16. Then for every pair of overlapping terms h_i and h_j with $0 < d_N(i,j) < L$, where $d_N(i,j) := \min(|i-j|, N-|i-j|)$ is the cyclic distance on the ring (i.e. whose L-site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 18.135 holds with constants c_{ij} and $\gamma > 0$ depending only on A (not on N).* Note: *The proof below fully establishes the martingale constants only for $L = 2$; for $L > 2$ the existence of suitable constants is asserted without complete proof (see the remark at the end of the proof). Only the $L = 2$ case is used in the spectral-gap theorem below. (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale argument.)*

Proof. Not ready Two terms h_i, h_j with $0 < d_N(i,j) < L$ have supports overlapping in $L - d_N(i,j)$ sites (where d_N is the cyclic distance on the N-site ring). Since A is injective, the intersection property (Theorem 18.34) gives the qualitative identity $\ker(h_i) \cap \ker(h_j) = G_{[i,j+L-1]}(A)$. Because the local Hilbert space is finite-dimensional, the subspaces $\ker(h_i)$ and $\ker(h_j)$ live in a fixed finite-dimensional space. The intersection identity shows that $\ker(h_i) \cap \ker(h_j)$ is exactly $G_{[i,j+L-1]}(A)$; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle $\theta_{d_N(i,j)}$ between $\ker(h_i)$ and $\ker(h_j)$ is strictly positive (for the fixed injective tensor A). Because the angle depends only on the overlap pattern $d_N(i,j) \in \{1, \dots, L-1\}$ and not on the absolute position, the number of distinct angles is at most $L - 1$. Let $\alpha := \max_{1 \leq s < L} \cos\theta_s < 1$ be the worst-case cosine of the Friedrichs angles.

Row-sum bound. Since $L \geq 2$, each site i overlaps with at most $2(L-1)$ neighbours (those j with $0 < d_N(i,j) < L$). Set $c_{ij} := \frac{1}{2(L-1)}$ for every overlapping pair and $c_{ij} := 0$ otherwise. These constants depend only on the cyclic distance $d_N(i,j)$ and therefore satisfy $c_{ij} = c_{ji}$. Then $\sum_{j \neq i} c_{ij} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 18.135.

Positivity of γ. The Friedrichs angle bound gives, for each overlapping pair, $h_i h_j + h_j h_i \geq -\alpha\,(h_i + h_j)$. The martingale inequality holds with $\gamma = 1 - 2(L-1)\alpha$, which requires $c_{ij}(1-\gamma) = \frac{1}{2(L-1)} \cdot 2(L-1)\alpha = \alpha$ (matching the Friedrichs bound exactly). Since $\alpha < 1$ (the Friedrichs angle is strictly positive), it remains to show $\gamma > 0$, i.e. $\alpha < \frac{1}{2(L-1)}$. In the blocked picture used in the spectral-gap theorem below, $L = 2$ and each site overlaps with at most 2 neighbours, so the condition reduces to $\alpha < \frac{1}{2}$. For $L = 2$ with an injective tensor, the two projectors h_i and h_{i+1} act on a finite-dimensional space, and the intersection property (Theorem 18.34) guarantees that $\ker(h_i) \cap \ker(h_{i+1})$ is exactly $G_{[i,i+2]}(A)$; hence $\|P_i P_{i+1}\| < 1$ (the Friedrichs cosine is strictly less than 1). On a finite-dimensional space with two projectors, the Friedrichs cosine satisfies the quantitative bound $\alpha \leq 1 - c_{\min}$ where $c_{\min} > 0$ depends on the minimal principal angle between the two kernels. The bound $\alpha < \frac{1}{2}$ then follows from the Nachtergaele finite-size criterion [Nac96], which establishes that the overlap $\|P_i P_{i+1}\|$ for injective MPS is bounded away from 1 by a constant depending only on the tensor A. For general $L > 2$, the uniform constants $c_{ij} = \frac{1}{2(L-1)}$ make the condition $\alpha < \frac{1}{2(L-1)}$ increasingly restrictive as L grows. A more refined choice of separation-dependent constants $c_{ij} = c_{d_N(i,j)}$ that weight small-separation (large-overlap) pairs more heavily can exploit the exponential decay of the Friedrichs cosine α_s in s to satisfy the row-sum constraint. However, this refinement is not needed for the main result: the spectral-gap argument below is applied only with $L = 2$ in the blocked picture. In all cases, the constants depend only on A and are independent of N [FNW92, KL18]. ☐

Theorem 18.137 (Explicit Friedrichs-angle gap bound). Lean Not ready *For an injective tensor A and a range $L > 1$, the theorem asserts the analytic estimate with the explicit constant*

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, it asserts that if $N \geq 2L$ and $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$, then

$$\gamma_L \|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. Not ready The local averaging, positivity, and symmetric-projection statements are now established in Lemmas 18.108, 18.109, and 18.110. Lemma 18.116 reduces the present theorem to the remaining MPS-specific task: deriving the uniform quadratic-form estimate. Lemma 18.122 establishes the finite-sum algebra from ordered cross-term row bounds, and Lemmas 18.123 and 18.130 state the common finite-overlap specialization with $m = 2(L-1)$. Lemma 18.114 supplies the non-overlap positivity statement for site-disjoint windows, and Lemma 18.128 translates this to the concrete support-disjoint cyclic windows. Lemma 18.129 gives the finite-overlap row-cardinality bound. Lemma 18.131 combines these ingredients with the finite-overlap reduction, so the remaining analytic task is the Friedrichs-angle estimate for overlapping cyclic windows. Lemma 18.132 states the same overlap-only formulation, and Lemma 18.133 states a sufficient norm-compression formulation. □

Theorem 18.138 (Spectral gap for MPS parent Hamiltonians). Lean ✓ Stmt *For an injective tensor A and a range $L > 1$, there exists $\gamma > 0$ such that for every $N \geq 2L$ and every vector $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$ one has*

$$\gamma \|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

By Lemma 18.115, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

Proof. Not ready This follows immediately from Theorem 18.137 by taking as γ the explicit constant produced there, which yields the required existential spectral-gap constant. □

18.7 Ground space of non-injective MPS

For a general (non-injective) MPS tensor in block-diagonal canonical form, the ground space is no longer one-dimensional. Instead, it is spanned by the MPVs of the individual normal blocks — the basis of normal tensors (BNT) from Chapter 12 [FNW92, PGVWC07].

Definition 18.139 (Assembled parent ground space). Lean ✓ Stmt For a family of blocks $A = (A_j)_{j=1}^{r}$ and weights μ, write $\mathrm{ParentGround}_{N,L}(A)$ for the periodic chain ground space of the block-diagonal tensor assembled from the blocks.

Lemma 18.140 (Definition by assembled chain ground space). Lean ✓ Stmt *By definition, this subspace is exactly the chain ground space of the assembled tensor.*

Definition 18.141 (Span of BNT block states). Lean ✓ Stmt The BNT span is the linear span of the periodic MPV states of the individual blocks:

$$\mathrm{BNTSpan}_N(A) := \mathrm{span}\{V^{(N)}(A_j) : j = 1, \ldots, r\}.$$

Definition 18.142 (Route B use of the product-word span condition). Lean ✓ Stmt For a fixed length m, the Route B product-word span condition is the finite-length spanning condition already stated in Remark 25.42: the simultaneous block-word evaluations

$$\omega \longmapsto (A_j^{\omega})_j$$

span the whole product algebra $\prod_j M_{D_j}(\mathbb{C})$. The remaining mathematical step is to derive this condition for some positive length from the separated canonical-form/BNT hypotheses. This is the block-injective span requirement in [CPGSV21, Section 4.3, lines 2049–2094]: after blocking, the parent ground space is generated by the basis of normal tensors, with the improved bound coming from block-diagonal boundary conditions.

Lemma 18.143 (Each BNT MPV lies in the assembled parent ground space). Lean ✓ Stmt *If $1 < L$ and $N \geq L+1$, then for each block A_j of a canonical-form/BNT decomposition, the corresponding MPV state lies in the assembled parent ground space.*

Proof. ✓ Proof One first applies Lemma 18.61 to the individual block. The proof then inserts the resulting boundary matrix into the j-th diagonal block of the assembled tensor, with the compensating scalar factor coming from the nonzero weight μ_j. □

Lemma 18.144 (Periodic block lift). Lean ✓ Stmt *For each block index j with $\mu_j \neq 0$, the chain ground space of the block A_j embeds into the chain ground space of the assembled tensor:*

$$\mathrm{ChainGround}_{N,L}(A_j) \subseteq \mathrm{ChainGround}_{N,L}(\mathrm{Assemble}(\mu, A)).$$

Proof. ✓ Proof The cyclic-window definition of the chain ground space reduces the inclusion to the local block-embedding statement that each ground vector of A_j on the window of length L lifts to the assembled tensor by inserting the block-diagonal boundary matrix with the compensating scalar factor μ_j^{-L}. □

Lemma 18.145 (Periodic block decomposition: forward inclusion). Lean ✓ Stmt *Assume $\mu_j \neq 0$ for every block index j. The supremum of the blockwise periodic-chain ground spaces is contained in the assembled tensor's periodic-chain ground space:*

$$\bigsqcup_j \mathrm{ChainGround}_{N,L}(A_j) \subseteq \mathrm{ChainGround}_{N,L}(\mathrm{Assemble}(\mu, A)),$$

where j ranges over all block indices.

Proof. ✓ Proof Aggregate the block lift over all block indices. □

Lemma 18.146 (Forward inclusion for assembled parent ground spaces). Lean ✓ Stmt *For a canonical-form/BNT family, the supremum of the blockwise periodic-chain ground spaces is contained in the assembled parent ground space:*

$$\bigsqcup_j \mathrm{ChainGround}_{N,L}(A_j) \subseteq \mathrm{ParentGround}_{N,L}(A).$$

Proof. ✓ Proof The canonical-form/BNT hypothesis supplies $\mu_j \neq 0$ for every j, and the assembled parent ground space is the chain ground space of the assembled tensor by definition. □

Lemma 18.147 (Block-diagonal commutant reduction from span extension). Lean Lean Lean ✓ Stmt *Suppose the virtual sector projections P_j of a dependent direct-sum bond space lie in the span of a family of matrices. Any boundary matrix commuting with that family commutes with each P_j, hence is block diagonal. Applied after pulling an assembled tensor back along the direct-sum reindexing, length-m word commutation plus the corresponding projection-span hypothesis forces all off-block entries of the pulled-back boundary matrix to vanish.*

Proof. ✓ Proof Commutation is linear in the matrix from the spanning family, so it extends from the generators to their span. Since each P_j lies in that span, the boundary matrix commutes with every sector projection. Lemma 2.38 then gives block diagonality, and Lemma 2.37 gives the entrywise off-block-zero formulation. □

Lemma 18.148 (Block projections from product-algebra span). Lean ✓ Stmt *Let $T_a = (T_a(i))_i$ be a family of tuples spanning the full product algebra $\prod_i M_{n_i}(\mathbb{C})$, and let $c_i \neq 0$ be nonzero scalars. Then, for every sector k, the sector projection P_k lies in the span of the scaled dependent block-diagonal matrices*

$$\bigoplus_i c_i T_a(i).$$

Proof. ✓ Proof Apply the linear map $(M_i)_i \mapsto \bigoplus_i c_i M_i$ to the full-span product-algebra element with $(c_k)^{-1}I$ in the k-th component and zero in every other component. Its image is exactly P_k. □

Lemma 18.149 (Projection span from product-word span). Lean Lean Lean Lean ✓ Stmt *If the simultaneous length-m block-word tuples*

$$\omega \longmapsto (A_j^\omega)_j$$

span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$ and every μ_j is nonzero, then the pulled-back length-m word products of Assemble(μ, A) *span every virtual sector projection P_j. Consequently, any boundary matrix commuting with all those assembled words has block-diagonal pulled-back form, and its off-block entries vanish.*

Proof. ✓ Proof First embed the product algebra linearly as dependent block-diagonal matrices, scaling the j-th block by μ_j^m. Since $\mu_j^m \neq 0$, the product-algebra span contains the tuple with $(\mu_j^m)^{-1}I$ in one component and zero elsewhere; its scaled diagonal image is exactly P_j. The word-evaluation formula for the assembled tensor identifies these scaled diagonal matrices with the pulled-back assembled word products, and the commutant conclusion follows from Lemma 18.147. □

Definition 18.150 (Pairwise block-separating word polynomials). Lean Lean ✓ Stmt A length-S partial selector for block k on a finite target set T is a tuple in the span of simultaneous length-S block-word evaluations which is I on block k and 0 on every block in T. The pairwise version asks for such a partial selector for each ordered pair of distinct blocks, with $T = \{j\}$.

Definition 18.151 (Pair trace-separation criterion). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For two blocks A and B, the length-S pair word tuple is $w \mapsto (A^w, B^w)$. The pair product-span condition says that these tuples span $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$. Equivalently, in the trace-dual formulation, the only pair of test matrices (Δ_A, Δ_B) whose trace pairing with every length-S tuple vanishes is $(0,0)$. The cumulative variant replaces one fixed length by all word lengths at most S, and the all-length variant allows every finite word.

Lemma 18.152 (Three-block trace reduction from a nonzero middle matrix). Lean Lean ✓ Stmt *Let A be L-block injective, and let $\Delta_A \neq 0$. Suppose that, for all words u, v, t of length L,*

$$\operatorname{tr}(\Delta_A A_v A_t A_u) + \operatorname{tr}(\Delta_B B_v B_t B_u) = 0.$$

Then for every $Z \in M_{D_A}(\mathbb{C})$ there is a matrix $W \in M_{D_B}(\mathbb{C})$ such that, for all words t of length L,

$$\operatorname{tr}(ZA_t) + \operatorname{tr}(WB_t) = 0.$$

This is the trace-dual algebraic step in the two-block case of the direct-sum lemma of [PGVWC07].

Proof. ✓ Proof Since $\Delta_A \neq 0$, Lemma 8.11 writes Z as a linear combination of matrices $A_u \Delta_A A_v$. For one such term, cyclicity of the trace rewrites the three-block relation as

$$\operatorname{tr}(A_u \Delta_A A_v A_t) + \operatorname{tr}(B_u \Delta_B B_v B_t) = 0.$$

Taking the same linear combination of the matrices $B_u \Delta_B B_v$ gives the required W, and linearity gives the general case. □

Lemma 18.153 (Finite-chain image inclusion from a three-block relation). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Under the hypotheses of Lemma 18.152, the length-L image spaces satisfy*

$$\mathcal{G}_L^A \subseteq \mathcal{G}_L^B.$$

If the corresponding nonzero middle matrix exists on both sides, then $\mathcal{G}_L^A = \mathcal{G}_L^B$.

Proof. ✓ Proof Write a vector in $\mathcal{G}_L^A$ as $t \mapsto \operatorname{tr}(A_t Z)$. The preceding lemma supplies W with $\operatorname{tr}(ZA_t) + \operatorname{tr}(WB_t) = 0$ for every word t. Cyclicity of the trace gives

$$\operatorname{tr}(A_t Z) = \operatorname{tr}(ZA_t) = -\operatorname{tr}(WB_t) = \operatorname{tr}(B_t(-W)),$$

so the vector lies in $\mathcal{G}_L^B$. The equality statement follows by applying the same argument with the two blocks exchanged. □

Lemma 18.154 (Dimension step for two-block direct-sum image spaces). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If A and B are length-L block-injective, the map $Z \mapsto (t \mapsto \operatorname{tr}(A_t Z))$ has image dimension D_A^2, and similarly for B. Therefore an inclusion $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$, together with $D_B \leq D_A$, forces $D_A = D_B$ and $\mathcal{G}_L^A = \mathcal{G}_L^B$. In particular, the three-block trace relation from Lemma 18.153 gives this equality whenever the source-paper size ordering is available. If the inclusion comes from the strictly larger block, $D_B < D_A$, then the same dimension step already contradicts the existence of the three-block trace relation. More generally, if an external source input rules out the simultaneous conclusion $D_A = D_B$ and $\mathcal{G}_L^A = \mathcal{G}_L^B$, then the three-block image spaces $\mathcal{G}_{3L}^A$ and $\mathcal{G}_{3L}^B$ have zero intersection. With injectivity at the three-block length, the strict-size branch itself already gives homogeneous pair trace separation at length $3L$. One such source input is non-proportionality of the periodic MPV states at a sufficiently long chain length, equivalently distinctness of the corresponding periodic MPV lines: equality of the local image spaces gives equality of the periodic-chain constraint spaces, and injective uniqueness identifies those spaces with the two MPV lines. For same-dimensional separated BNT blocks, the non-equivalence condition supplies such non-proportional MPV states at arbitrarily large chain lengths; the direct-sum injectivity hypotheses remain explicit. Zero intersection of the two three-block image spaces also gives homogeneous pair trace separation at* *that length once the boundary-to-chain maps at the three-block length are injective. Combining the equal-dimension and unequal-dimension branches over all ordered block pairs gives one common three-block trace-separation length, and hence the positive-length product-word span required by the selector construction. Since canonical-form/BNT blocks are one-site injective, one may take* $L = 2$*; injectivity at lengths* 2 *and* 6 *then follows by passing from a fixed injective length to positive multiples. This length-*6 *trace separation is equivalently a full homogeneous pair span; concatenating one-letter words propagates fullness to all later homogeneous lengths, so the direct-sum witness also gives a uniform finite pair-span period window. Combining this window with the conditional period-window homogenization theorem gives the selector-format homogeneous word span without separately assuming homogeneous identity pairs. The same conclusion applies to normal-CF-BNT data once one-site injectivity is supplied explicitly, since the normal-CF-BNT predicate carries the irreducibility and normalization hypotheses used in the direct-sum comparison.*

Proof. ✓ Proof Block injectivity makes the boundary-to-chain map injective, so $\dim \mathcal{G}_L^A = D_A^2$ and $\dim \mathcal{G}_L^B = D_B^2$. Inclusion gives $D_A^2 \le D_B^2$, while $D_B \le D_A$ gives the reverse inequality. Hence $D_A = D_B$, and equality of dimensions inside the inclusion gives equality of the image spaces. In the strict-size branch this contradicts $D_B < D_A$. For the conditional directness statement, a vector in the intersection of the two three-block image spaces has two boundary representations. If the first boundary matrix is zero, the vector is zero. Otherwise the difference of the two coefficient formulas is exactly the homogeneous three-block trace relation, so the dimension step gives the forbidden collapse. In the equal-size branch, equal local image spaces impose the same cyclic local constraints on a periodic chain. The injective uniqueness theorem then identifies both cyclic ground spaces with the spans of their MPV states, contradicting non-proportionality of those states. □

Lemma 18.155 (All-length pair trace separation has finite cumulative length). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If no nonzero trace functional vanishes on all finite pair words, then there is a finite* S *such that vanishing on every word of length at most* S *already forces the test pair to be zero. For a finite family of ordered block pairs, these bounds can be replaced by one common bound.*

Proof. ✓ Proof The trace condition is dual to the span of all finite pair words. If that span is the whole product algebra, every vector lies in some cumulative span. Since the product algebra is finite-dimensional, the increasing sequence of cumulative spans stabilizes; at the stabilization index it is already the whole product algebra. Applying the finite trace-duality criterion gives the stated finite bound. This proves the cumulative finite-dimensional step; the remaining BNT step is to convert it to a single homogeneous length. Cumulative span-top alone is not identity padding, because the length-zero word already contributes the identity. □

Lemma 18.156 (Pair algebra projections are full). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If one of the two blocks is injective, then the corresponding coordinate projection of the pair algebra generated by the simultaneous one-letter pairs is the full matrix algebra. Moreover, every finite pair word lies in this pair algebra, and the all-word linear span is the submodule generated by the pair algebra. If the pair algebra is the graph of an algebra automorphism of the matrix algebra, Skolem–Noether turns this graph case into gauge-phase equivalence, so under non-gauge-equivalence the graph branch is impossible. For two blocks of unequal bond dimensions, the graph branch is also impossible because it would identify matrix algebras with different finite dimensions over* $\mathbb{C}$*.*

Proof. ✓ Proof Projecting the pair generators to one coordinate gives exactly the one-block generator family. Injectivity says that the linear span of this family is the full matrix algebra, and the linear span is contained in the unital algebra it generates. The pair-word membership follows by induction on the word, using closure of the generated algebra under multiplication. Conversely, the all-word linear span is closed under componentwise multiplication, so it contains the algebra generated by the pair generators. In the graph case, each generator satisfies $B_i = \varphi(A_i)$; applying Skolem–Noether to φ gives an ordinary gauge, hence a gauge-phase equivalence with scalar 1. $\square$

Lemma 18.157 (Conditional padding from exact homogeneous identity witnesses)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Homogeneous identity-padding witnesses multiply by concatenation of words. Therefore, if one has a positive period and a full residue window of homogeneous pair spans, then the simultaneous identity pair lies in all sufficiently large homogeneous pair spans. Combined with all-length pair trace separation, this period-window input gives one homogeneous pair trace-separation length. This does not prove the direct-sum input of [PGVWC07] by itself: fullness of spans up to a bounded length is weaker than the exact homogeneous input used here.*

Proof. ✓ Proof The length-zero word gives the identity only at length zero, so it is not a positive-length padding theorem. The useful padding facts are homogeneous: if the identity pair is in the span at lengths m and n, componentwise multiplication puts it in the span at length $m + n$. A period witness and a complete residue window therefore give homogeneous padding at every sufficiently large length, which is the input needed by the all-length-to-fixed-length implication. $\square$

Lemma 18.158 (Pair trace separation gives pairwise selectors)**.** Lean Lean Lean Lean Lean Lean ✓ Stmt *At a fixed length S, the pair trace-separation criterion is dual to full pair product-span. Therefore a pair trace-separation witness supplies a word polynomial that is identity on the first block and zero on the second. A common witness for all ordered distinct pairs gives pairwise block-separating words at the same length.*

Proof. ✓ Proof If the pair span were proper, a nonzero linear functional would vanish on it. Nondegeneracy of the matrix trace pairing represents that functional as (Δ_A, Δ_B), contradicting pair trace separation. Once the pair span is the whole product algebra, the tuple $(I, 0)$ is in the span; using the same coefficients on the ambient block family gives the required partial selector. $\square$

Lemma 18.159 (Block selectors from pairwise block separation)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If every ordered pair of distinct blocks admits a common-length pairwise separator, then full block-selector words exist. More precisely, multiplying the pairwise separators for a fixed block k against all other blocks gives a selector of length $(r-1)S$ for k. Hence common block injectivity plus pairwise separators implies the finite product-word span in the block-injectivity proposition in [CPGSV17a, lines 340–345] and the block-injective canonical-form trace-separation condition.*

Proof. ✓ Proof The empty word gives the identity tuple, selecting on the empty target set. The span of word tuples is closed under pointwise multiplication because the product of a length-L word and a length-S word is the concatenated length-$(L + S)$ word. Inducting over the finite set of blocks different from k and multiplying the corresponding pairwise separators keeps the value I on k and introduces one additional zero block at each step. Extracting coefficients from the final tuple-span membership gives the coefficient form of block-selector words. $\square$

Lemma 18.160 (Product-word span from block-injective selector words)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let (A_j) be a finite family of block-injective tensors. Suppose there are length-S block-selector words: for each sector k, a linear combination of length-S words evaluates to I on A_k and to 0 on every other block. Then the simultaneous length-$(1+S)$ block-word evaluations span $\prod_j M_{D_j}(\mathbb{C})$, with positive length. In particular, a canonical form with BNT separation supplies the required block injectivity, but the selector-word conclusion does not require the full canonical-form/BNT data. It is enough, by Lemma 18.159, to construct pairwise separators for every ordered pair of distinct blocks. Lemma 18.158, a common pair trace-separation length is one sufficient finite-dimensional witness. This witness remains homogeneous: either it is supplied directly as fixed-length pair separation, or it is obtained from an explicit period-window hypothesis as in Lemma 18.157. If the assembly weights are all nonzero, then the pulled-back words at the resulting length for* $\mathrm{Assemble}(\mu, A)$ *span every virtual sector projection, and the corresponding commutant reduction is available.*

Proof. ✓ Proof Block injectivity writes each target sector matrix as a fixed-length linear combination of word evaluations. For a canonical-form/BNT family, Definition 12.1 gives one-site injectivity of each block, hence 1-block injectivity by Lemma 2.30. For a target tuple $(M_j)_j$, write M_k as a one-letter linear combination in block k, multiply by the selector for k, and sum over k. The selector kills all off-target blocks, while it is the identity on block k, so the length-$(1+S)$ word tuples span the full product algebra. If one starts with pairwise separators instead, first use Lemma 18.159 to multiply them into full selectors; if one starts from the pair trace-separation criterion, first use Lemma 18.158. The projection-span and commutant consequences then follow from Lemma 18.149. ☐

Lemma 18.161 (Conditional boundary-matrix block split from projection span)**.** Lean Lean Lean Lean ✓ Stmt *Let A be in canonical-form/BNT, and assume $1 < L$, $N \geq L+1$, and $0 < m$. Assume every virtual sector projection lies in the pulled-back span of the length-m word products of the assembled tensor. If a boundary matrix for the assembled tensor commutes with all those length-m words, then the vector obtained by applying the assembled ground-space map to that boundary matrix lies in $\bigsqcup_j \mathcal{G}_{N,L}(A_j)$. Consequently, if every assembled periodic-chain ground vector admits such a commuting boundary matrix representation (the result of the wrapping-window comparison), then*

$$\mathcal{G}_{N,L}(\mathrm{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j),$$

and together with the already-proved forward inclusion this gives the corresponding conditional equality. Composing the reverse inclusion with blockwise injective uniqueness gives the conditional containment of the assembled parent ground space in the BNT span; this lemma keeps the projection-span and boundary-representation hypotheses explicit.

Proof. ✓ Proof The commutant reduction makes the pulled-back boundary matrix block diagonal. Restricting the same word-commutation identities to a diagonal block and using the injectivity of that block shows, by the scalar-commutant lemma, that each diagonal block is scalar. Expanding the assembled ground-space map on a block-diagonal boundary matrix gives a finite sum of scalar multiples of the block MPV states, and each block MPV lies in its own periodic chain ground space. The reverse-inclusion, equality, and BNT-span forms are then reformulations of this boundary-matrix calculation, combined with the forward inclusion and the blockwise uniqueness theorem. ☐

Lemma 18.162 (Conditional Route B block split from product-word span). Lean Lean Lean ✓ Stmt *Let A be in canonical-form/BNT, and assume $1 < L$, $N \geq L + 1$, and $0 < m$. If the length-m product-word tuples span $\prod_j M_{D_j}(\mathbb{C})$ and every assembled periodic-chain ground vector comes with a boundary matrix representation commuting with all length-m assembled words, then*

$$\mathcal{G}_{N,L}(\mathrm{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Combined with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\mathrm{BNTSpan}_N(A)$*. The product-word span and boundary representation remain hypotheses here; they are not derived by this composition lemma.*

Proof. ✓ Proof The product-word span witness gives the sector-projection span by Lemma 18.149; the nonzero weights required there are supplied by the canonical-form/BNT hypothesis. The conditional boundary-matrix block split from Lemma 18.161 then gives the reverse inclusion. The equality and BNT-span forms are the corresponding projection-span results after this product-word-to-projection-span reduction, followed by the blockwise uniqueness argument already built into the projection-span theorem. □

Lemma 18.163 (Conditional Route B block split from BNT selector words). Lean Lean Lean ✓ Stmt *Let A be in canonical-form/BNT, assume $1 < L$ and $N \geq L + 1$, and suppose length-S block-selector words are given. If every assembled periodic-chain ground vector has a boundary matrix representation commuting with all length-$(1 + S)$ assembled words, then*

$$\mathcal{G}_{N,L}(\mathrm{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Together with the forward inclusion, this gives the corresponding conditional block-splitting equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\mathrm{BNTSpan}_N(A)$*. The selector-word hypothesis and the boundary representation remain explicit: this lemma does not derive selector existence from BNT separation, nor the wrapping/open-chain boundary representation. It is the formal composition matching the block-injective ground-space route of [CPGSV21, Section 4.3, lines 2049–2094], where the regrowing argument is restricted to block-diagonal boundary conditions.*

Proof. ✓ Proof Lemma 18.160 turns the canonical-form/BNT hypotheses plus length-S selector words into the product-word span condition at the positive length $1 + S$. Lemma 18.162 then applies Route B with the boundary matrix commuting with the same length-$(1 + S)$ assembled words. The equality and BNT-span containment are exactly the corresponding product-word-span consequences after this selector-to-product-span replacement. □

Theorem 18.164 (Containment in the BNT span by block decomposition). Lean Not ready *If A is in canonical-form/BNT and $1 < L$ and $N \geq L + 1$, every vector in the assembled parent ground space should decompose into blockwise chain ground states, and therefore lie in the BNT span.*

Proof. Not ready The remaining structural step is the periodic block decomposition of the assembled chain ground space. Lemma 18.159 reduces finite selector words to pairwise block-separating word polynomials. Lemma 18.163 then gives the conditional Route B composition once those selector words and the wrapping-window commuting boundary matrix are available. What remains open is to derive the needed separation data and to supply the compatible boundary representation. With the block decomposition available, one applies the blockwise injective uniqueness theorem to each block separately and recovers the BNT span. □

Theorem 18.165 (Ground space equals span of BNT). Not ready Lean *If $1 < L$ and $N \geq L+1$, then for a canonical-form/BNT family the assembled parent ground space equals the BNT span:*

$$\mathrm{ParentGround}_{N,L}(A) = \mathrm{BNTSpan}_N(A).$$

Proof. Not ready We prove the two inclusions separately. The inclusion

$$\mathrm{ParentGround}_{N,L}(A) \subseteq \mathrm{BNTSpan}_N(A)$$

is exactly Theorem 18.164. For the reverse inclusion, Lemma 18.143 shows that each BNT MPV lies in $\mathrm{ParentGround}_{N,L}(A)$. Since the parent ground space is a linear subspace, it therefore contains every linear combination of such vectors, and hence contains $\mathrm{BNTSpan}_N(A)$. Thus

$$\mathrm{BNTSpan}_N(A) \subseteq \mathrm{ParentGround}_{N,L}(A),$$

and the two spaces are equal. □

Remark. A separate kernel-identification statement for these assembled block families would convert the chain-ground-space statement into the corresponding statement for $\ker(H_N)$.

Chapter 19

Exponential Decay of Correlations

This chapter derives the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 18), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \le 1$, the strict transfer-operator gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 7. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

19.1 Connected correlators from the transfer map

Definition 19.1 (One-point expectation). Lean ✓ Stmt Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R. Under the hypotheses of Theorem 6.11, that fixed point exists and is unique up to scaling; we normalize it so that $\mathrm{tr}(\rho^R) = 1$. For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \mathrm{tr}(X\rho^R).$$

Definition 19.2 (Two-point expectation at distance n). Lean ✓ Stmt For observables $X, Y \in M_D(\mathbb{C})$ and $n \ge 0$,

$$\langle X_0 Y_n \rangle := \mathrm{tr}\big(Y\, \mathcal{E}_A^n(X\rho^R)\big)\,.$$

Equivalently: insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.

Definition 19.3 (Connected correlator). Lean ✓ Stmt The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

19.2 Spectral expansion and exponential decay

Theorem 19.4 (Sum of exponentials (spectral expansion interface)). Lean ✓ Stmt *If coefficients $c_j \in \mathbb{C}$ and eigenvalues $\lambda_j \in \mathbb{C}$ ($j = 1, \dots, D^2 - 1$) satisfy, for all $n \geq 0$,*

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \lambda_j^n,$$

then the connected correlator equals the stated sum of exponentials. The source [CPGSV21, Sec. 2.3] states that the connected correlator is a sum of $D^2 - 1$ pure exponentials $C(X, Y; n) = \sum_{j\geq 2}^{D^2} c_{XY}(j)\lambda_j^n$, which always exists for a normal MPS via the transfer-map spectral decomposition.

Proof. ✓ Proof Given the expansion hypothesis, the equality is immediate. The source [CPGSV21, Sec. 2.3] derives this expansion from the spectral decomposition of $\mathcal{E}_A^n$ restricted to the complement of the fixed-point eigenspace: the leading eigenvalue $\lambda_1 = 1$ contributes $\langle X_0\rangle\langle Y_0\rangle$, which is subtracted in the connected correlator, leaving the sum over subleading eigenvalues. ☐

Theorem 19.5 (Exponential decay bound (interface)). Lean ✓ Stmt *If a constant $C_{XY} \in \mathbb{R}$ and $\lambda_2 \in \mathbb{C}$ satisfy, for all $n \geq 0$,*

$$|C(X, Y; n)| \leq C_{XY}\, |\lambda_2|^n,$$

then the connected correlator satisfies that exponential decay bound. The source [CPGSV21, Sec. 2.3] obtains this by combining the sum-of-exponentials expansion with the triangle inequality, and Sec. 4 provides the transfer-map gap condition $|\lambda_j| < 1$ for the subleading eigenvalues.

Proof. ✓ Proof Given the bound hypothesis, the estimate is immediate. The source [CPGSV21, Sec. 2.3] obtains this bound by applying the triangle inequality to the sum-of-exponentials expansion and using the transfer-map gap condition $|\lambda_j| \leq |\lambda_2| < 1$ for all subleading eigenvalues. ☐

Remark 19.6 (Transfer-map spectral hypothesis). The hypothesis "$|\lambda_2| < 1$" in Theorem 19.5 is a transfer-map gap condition: all non-leading eigenvalues of $\mathcal{E}_A$ lie strictly inside the unit disk. It is distinct from the Hamiltonian spectral gap of Chapter 18. The complementary transfer-map gap $\rho(\mathcal{E}_A - P) < 1$ is established for primitive channels (Theorem 4.51, Chapter 4). The mixed-transfer gap needed for block separation is supplied for irreducible trace-preserving tensors (Theorem 7.29), and quantitative single-tensor transfer-map bounds for injective tensors are recorded in Theorem 19.11. Thus the analytical input needed to make Theorems 19.4 and 19.5 unconditional is already present; what remains is extracting the spectral expansion coefficients c_j, λ_j from the eigendecomposition of $\mathcal{E}_A$ (Issue #1447).

Definition 19.7 (Correlation length). Lean ✓ Stmt The correlation length associated with a subleading eigenvalue λ_2 of the transfer map ($|\lambda_2| < 1$) is

$$\xi := -\frac{1}{\log|\lambda_2|}.$$

Lemma 19.8 (Correlation length is positive). Lean ✓ Stmt *If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.*

Proof. ✓ Proof Since $0 < |\lambda_2| < 1$, we have $\log|\lambda_2| < 0$, so $\xi = -1/\log|\lambda_2| > 0$. ☐

19.3 Quantitative transfer-map gap bounds

The transfer-map gap theorems in Chapter 4 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section gives the *quantitative* refinements with explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$.

Theorem 19.9 (Exponential convergence of injective primitive channels). Lean ✓ Stmt *Fix an injective, trace-preserving MPS tensor A with positive definite fixed point ρ. Let*

$$P(X) = \frac{\operatorname{tr}(X)}{\operatorname{tr}(\rho)}\rho$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \le 1$ such that for all $n \ge 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \;\le\; C\,(1-\delta)^n\,\|X\|.$$

The transfer-map gap δ exists by primitivity (Theorem 4.51, the complementary transfer-map gap).

Proof. ✓ Proof Injectivity implies primitivity of the transfer map. The complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ then follows from the primitive transfer-map gap theorem for the complementary map. The Gelfand formula yields a geometric bound $\|\,(\mathcal{E}_A - P)^n\| \le C\,(1-\delta)^n$ for suitable constants, giving the claimed estimate. □

Theorem 19.10 (Correlation length bound). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \ge 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e. $\operatorname{tr}(X) = 0$),*

$$\|\mathcal{E}_A^n(X)\| \;\le\; C\,e^{-n/\xi}\,\|X\|.$$

Traceless matrices lie in $\ker P$, where P is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate $\xi = -1/\log\rho(\mathcal{E}_A - P)$.

Proof. ✓ Proof For traceless X we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. The exponential convergence theorem gives a geometric bound $C\,(1-\delta)^n\,\|X\|$. Rewriting $(1-\delta)^n = e^{-n/\xi}$ with $\xi = -1/\log(1-\delta)$ yields the claimed estimate. □

Theorem 19.11 (Transfer-map gap from injectivity). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \le 1 - \delta$.*

Proof. ✓ Proof Injectivity implies eventual full Kraus rank. This yields primitivity, hence a complementary transfer-map gap. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. □

19.4 Relation to parent Hamiltonians

Remark 19.12. When the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 18.17) whose ground state is the matrix product vector (Lemma 18.21), the exponential decay bound (Theorem 19.5) shows that ground-state correlations decay with a finite correlation length ξ.

19.5 Renormalization fixed points and zero correlation length

Definition 19.13 (Renormalization fixed point). Lean ✓ Stmt An MPS tensor A is a *renormalization fixed point* (RFP) when its transfer map is idempotent, $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. See [CPGSV21, Definition 3.2].

Theorem 19.14 (Kraus-isometry criterion for RFP). Lean ✓ Stmt *Suppose there exists an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1,i_2),j}$, such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1,i_2),j} A^j$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. Then A is a renormalization fixed point. This is the backward direction of [CPGSV21, Theorem 3.1].

Proof. ✓ Proof Expanding $\mathcal{E}_A^2$ gives a double Kraus sum indexed by pairs (i_1, i_2). Substituting the decomposition above and using the isometry relation $V^\dagger V = 1$ collapses the double sum to the single Kraus family $\{A^j\}$, hence $\mathcal{E}_A^2 = \mathcal{E}_A$. ☐

Definition 19.15 (Local orthogonality). Lean ✓ Stmt A single BNT block is *locally orthogonal* when its self-transfer map is idempotent. For a single tensor this coincides with the RFP condition (Definition 19.13); in the full multi-block BNT setting, local orthogonality additionally requires that mixed transfer operators vanish. See [CPGSV21, Definition 3.5].

Theorem 19.16 (RFP normal tensor is injective). Lean ✓ Stmt *If A is a nonzero normal RFP tensor, then A is injective.*

Proof. ✓ Proof The RFP condition $\mathcal{E}_A^2 = \mathcal{E}_A$ implies that the cumulative span $\operatorname{span}\{A^w : |w| \le n\}$ stabilises after one step. Since A is normal (hence has an injective blocked form), the one-step span already equals $M_D(\mathbb{C})$, so A is injective. ☐

Theorem 19.17 (Left-canonical normal RFP tensor is injective). Lean ✓ Stmt *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then A is injective. This is the left-canonical injectivity step in [CPGSV21, Appendix B].

Proof. ✓ Proof The normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ forces at least one matrix A^i to be nonzero, since otherwise the left-hand side would vanish. Hence A is a nonzero normal renormalization fixed point, and Theorem 19.16 applies. ☐

Theorem 19.18 (Unitary diagonal fixed point for a left-canonical normal RFP). Lean ✓ Stmt *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \qquad \text{and} \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [CPGSV21, Appendix B].

Proof. ✓ Proof By Theorem 19.17, the tensor A is injective. Therefore its transfer map is irreducible, and hence A is irreducible as a tensor. Applying Theorem 9.19 gives a unitary conjugate B for which the trace-preserving normalization remains valid and a diagonal positive definite matrix Λ with $\mathcal{E}_B(\Lambda) = \Lambda$. □

Theorem 19.19 (Rank-one classification for RFP transfer maps). Lean ✓ Stmt *Let A be an injective left-canonical RFP tensor with positive-definite fixed point ρ of the transfer map $\mathcal{E}_A$. Then $\mathcal{E}_A = P_\rho$, where $P_\rho(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$ is the rank-one fixed-point projection.*

Proof. ✓ Proof The exponential convergence bound gives $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \le C(1-\delta)^n\|X\|$. Idempotence of $\mathcal{E}_A$ implies $\mathcal{E}_A^{n+1} = \mathcal{E}_A$ for all $n \ge 0$, so $\|\mathcal{E}_A(X) - P_\rho(X)\| \le C(1-\delta)^{n+1}\|X\|$ for all n. Taking $n \to \infty$ gives $\mathcal{E}_A = P_\rho$. □

Theorem 19.20 (Full structural form for RFP tensors). Lean ✓ Stmt *A normal tensor A in canonical form II that is a renormalization fixed point admits a decomposition $A^i = X\Lambda U^i X^{-1}$, where Λ is diagonal positive and the family $U = (U^i)$ is an isometry in the physical indices.*

Proof. ✓ Proof Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a canonical Kraus form, applies Kraus freedom to identify the physical-index family with an isometry U, and transports the decomposition back through the conjugation. □

Theorem 19.21 (Canonical-form blocks are injective). Lean ✓ Stmt *For a family of tensors in canonical form, every block is injective. This is a direct projection of the injectivity field of the canonical-form predicate; see [CPGSV16, Section 2.3].*

Proof. ✓ Proof Immediate from the `block_injective` field of `IsCanonicalForm`. □

Theorem 19.22 (BNT basis blocks are injective). Lean ✓ Stmt *Let P be a sector decomposition in BNT canonical form. Then every basis block P_j is injective as an MPS tensor. This is the basis-block injectivity part of [CPGSV16, Section III] (`thm:charact-MPS`, lines 543–555; basis-block normality established at lines 271–301 and 317–344).*

Proof. ✓ Proof Direct projection of the `basis_injective` field of `IsBNTCanonicalForm`. The result holds for arbitrary copy multiplicities, since the statement concerns only BNT basis blocks and not copy weights. □

Theorem 19.23 (RG flow convergence for canonical-form tensors). Lean ✓ Stmt *For a tensor in canonical form with blocks (A_k) and weights (μ_k), the iterated blocking $\mathcal{E}_{A_k}^{2^n}$ converges pointwise to an idempotent linear map $\mathcal{E}_\infty$ as $n \to \infty$. That is, there exists $\mathcal{E}_\infty$ with $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \to \mathcal{E}_\infty(\rho)$ for all density matrices ρ. This is [CPGSV21, Appendix B].*

Proof. ✓ Proof Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0. The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \le C(1-\delta)^n\|X\|$ (Theorem 19.9) then gives pointwise convergence $\mathcal{E}^n(X) \to P(X)$, and composing with the subsequence $2^n \to \infty$ yields the result. The witness $\mathcal{E}_\infty = P$ is the rank-one fixed-point projection, which is idempotent. □

Remark 19.24. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \ge 1$. Informally, this corresponds to zero correlation length ($\xi = 0$). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

Definition 19.25 (Correlations independent of distance). Lean ✓ Stmt For a positive definite right fixed point $\rho^R > 0$, an MPS tensor has *correlations independent of distance* (CID) when for all local observables X and Y and all separations $n, m \geq 1$,

$$C(X, Y; n) = C(X, Y; m).$$

Definition 19.26 (Zero correlation length). Lean ✓ Stmt An MPS tensor has *zero correlation length* when it is both locally orthogonal and correlations are independent of distance. See [CPGSV21, Definition 3.6].

Theorem 19.27 (CID implies RFP). *Lean* *✓ Stmt* *If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV21, Theorem 3.8].*

Proof. ✓ Proof Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations $n = 2$ and $n = 1$. □

Theorem 19.28 (Zero correlation length iff idempotent transfer). *Lean* *✓ Stmt* *An MPS tensor has zero correlation length (i.e. is both locally orthogonal and CID) if and only if its transfer map is idempotent. This is [CPGSV21, Theorem 3.8].*

Proof. ✓ Proof The forward direction extracts local orthogonality (which equals idempotence by definition). The reverse uses $\mathcal{E}^n = \mathcal{E}$ for $n \geq 1$ to show the connected correlator is separation-independent. □

Theorem 19.29 (Canonical-form blocks satisfy RFP iff ZCL). *Lean* *✓ Stmt* *For a tensor in canonical form, each block is a renormalization fixed point if and only if it has zero correlation length. This is the RFP–ZCL part of [CPGSV21, Theorem 3.10].*

Proof. ✓ Proof Apply Theorem 19.28 to the chosen canonical-form block. □

Chapter 20

Concrete Examples

This chapter collects concrete MPS tensors from [CPGSV21, Section III]. Each example is specified by its physical dimension d, bond dimension D, and explicit matrix family $\{A^i\}$. Key structural properties (injectivity, RFP status, symmetry) are stated for each tensor.

20.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

Definition 20.1 (GHZ tensor). Lean ✓ Stmt Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \mathrm{diag}(1, 0), \qquad A^1 = |1\rangle\langle 1| = \mathrm{diag}(0, 1).$$

For a system of N sites the resulting MPV is the GHZ state

$$|\mathrm{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Theorem 20.2 (GHZ transfer map is idempotent). Lean ✓ Stmt *The transfer map of the GHZ tensor satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$.*

Proof. ✓ Proof The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij} X_{ij}$ (extraction of the diagonal). This operation is idempotent. □

Theorem 20.3 (GHZ tensor is an RFP). Lean ✓ Stmt *The GHZ tensor is a renormalization fixed point.*

Proof. ✓ Proof Immediate from Theorem 20.2. □

Theorem 20.4 (GHZ tensor is not injective). Lean ✓ Stmt *The GHZ tensor is not injective:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\}$ *is the two-dimensional space of diagonal matrices in* $M_2(\mathbb{C})$*, which does not equal* $M_2(\mathbb{C})$*.*

Proof. ✓ Proof Every element of $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. □

Theorem 20.5 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor). Lean ✓ Stmt *Let $\mathbb{Z}_2 = \{1, \sigma_x\}$ act on $\mathbb{C}^2$ by the Pauli-X matrix σ_x. The GHZ tensor is on-site symmetric under this action: for each $g \in \mathbb{Z}_2$, the twisted tensor $\widetilde{A}_g$ defines the same MPV family as A, i.e. $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$.*

Proof. ✓ Proof The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\widetilde{A}_g^i = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. □

20.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

Definition 20.6 (AKLT tensor). Lean ✓ Stmt Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}}\sigma_z, \qquad A^1 = \frac{\sqrt{2}}{\sqrt{3}}|0\rangle\langle 1|, \qquad A^2 = -\frac{\sqrt{2}}{\sqrt{3}}|1\rangle\langle 0|,$$

where $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}}\sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}}\sigma_-$ for $\sigma_+ = \sqrt{2}\,|0\rangle\langle 1|$ and $\sigma_- = \sqrt{2}\,|1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin-$\frac{1}{2}$ to spin-$\frac{1}{2}$.

Theorem 20.7 (AKLT tensor is not injective). Lean ✓ Stmt *The AKLT tensor is not injective at a single site:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ *is the three-dimensional space of traceless matrices* $\mathfrak{sl}(2, \mathbb{C})$*, which is a proper subspace of* $M_2(\mathbb{C})$*.*

Theorem 20.8 (AKLT tensor is normal). Lean ✓ Stmt *The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.*

Theorem 20.9 ($\mathrm{SO}(3)$ on-site symmetry of the AKLT tensor). Not ready *The AKLT tensor is on-site symmetric under the spin-1 representation of* $\mathrm{SO}(3)$ *(the spin-1 representation of* $\mathrm{SU}(2)$ *factors through* $\mathrm{SO}(3)$*), with the virtual representation being the projective spin-$\frac{1}{2}$ representation. The projective phase is the non-trivial element of $H^2(\mathrm{SO}(3), \mathrm{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase.*

20.3 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

Definition 20.10 (Cluster-state tensor). Not ready Set $d = D = 2$. The cluster-state tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

Theorem 20.11 (Cluster-state tensor is not injective). Not ready *The cluster-state tensor is not injective:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\} = \mathrm{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ *has dimension* 2*, not* $\dim M_2(\mathbb{C}) = 4$*.*

Proof. Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. □

Remark 20.12. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \tfrac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\tfrac{1}{\sqrt{2}}|-\rangle\langle 1|.$$

Since $\{|+\rangle, |-\rangle\}$ and $\{\langle 0|, \langle 1|\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 20.13 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor). Not ready *The length-*2 *blocked cluster-state tensor is on-site symmetric under* $\mathbb{Z}_2 \times \mathbb{Z}_2$*. The two generators act on the* 4*-dimensional blocked physical space* $(\mathbb{C}^2)^{\otimes 2}$ *by* $\sigma_x \otimes I$ *and* $I \otimes \sigma_x$*, which commute and each square to the identity, defining a linear representation of* $\mathbb{Z}_2 \times \mathbb{Z}_2$*. The corresponding virtual matrices are* $V_1 = \sigma_z$ *and* $V_2 = \sigma_x$*, which anticommute (*$V_1 V_2 = -V_2 V_1$*). This projective phase is the non-trivial element of* $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1)) \cong \mathbb{Z}_2$*, witnessing a non-trivial SPT phase.*

Chapter 21

Channel Representations and Normal Forms

This chapter collects the representation material for finite-dimensional quantum channels from [Wol12, Chapter 2]: the Choi–Jamiołkowski isomorphism, Kraus representation theorems, Stinespring and Naimark dilations, ordered completely positive maps, Radon–Nikodym and open-system representations, trace-pairing expansions, SVD and Lorentz normal forms, and the channel determinant. These results are important for the later channel theory, but they are not prerequisites for the matrix-product-state Fundamental Theorem.

21.1 Representations

The following definitions support the Choi–Jamiolkowski isomorphism (Theorem 21.6) and the representation theorems of [Wol12, Chapter 2].

Definition 21.1 (Partial traces). Lean Lean ✓ Stmt Let $X \in M_{d \cdot d'}(\mathbb{C})$ be a bipartite matrix indexed by

$$(\{0, \dots, d-1\} \times \{0, \dots, d'-1\})^2.$$

The *left partial trace* $\mathrm{tr}_A(X)$ and *right partial trace* $\mathrm{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\mathrm{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \qquad (\mathrm{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

Definition 21.2 (Maximally entangled state). Lean ✓ Stmt The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1,i_2),(j_1,j_2)} = \frac{1}{d}\delta_{i_1 j_1}\delta_{i_2 j_2}$ and $\mathrm{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 21.3 (Tensor product of a map with identity). Lean ✓ Stmt Given a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *tensor extension* $T \otimes \mathrm{id}$ acts on bipartite matrices $X \in M_{D\times D}(\mathbb{C})$ by applying T to each "slice":

$$(T \otimes \mathrm{id})(X)_{(i_1,i_2),(j_1,j_2)} = (T(X^{(i_2,j_2)}))_{i_1 j_1},$$

where $X^{(i_2,j_2)}_{ab} = X_{(a,i_2),(b,j_2)}$ is the bipartite slice.

21.2 Choi–Jamiołkowski isomorphism

Definition 21.4 (Choi matrix). Lean ✓ Stmt The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is

$$\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \in M_{D\times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1,i_2),(j_1,j_2)} = \frac{1}{D}(T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiolkowski convention of [Wol12, Proposition 2.1].

Theorem 21.5 (Choi matrix of the identity map). Lean ✓ Stmt *The identity channel has Choi matrix* $|\Omega\rangle\langle\Omega|$.

Proof. ✓ Proof Apply the identity map in the definition of the Choi matrix. □

Theorem 21.6 (Complete positivity iff the Choi matrix is positive semidefinite). Lean ✓ Stmt *A linear map* $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ *is completely positive (in the Kraus sense) if and only if its Choi matrix* $\tau \geq 0$.

Proof. ✓ Proof ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c\,K_i(a,b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a,b) = v_m(a,b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. □

Theorem 21.7 (Trace preservation implies the partial-trace condition). Lean ✓ Stmt *If T is trace-preserving, then* $\mathrm{tr}_A(\tau) = \frac{1}{D}\mathbb{1}_D$.

Proof. ✓ Proof $(\mathrm{tr}_A(\tau))_{ij} = \mathrm{tr}(T(\Omega^{(ij)})) = \mathrm{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D}E_{ij}$ is the (i,j)-slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D}\delta_{ij}$. □

Theorem 21.8 (Hermiticity preservation and the Choi matrix). Lean ✓ Stmt *The Choi matrix τ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$.*

Proof. ✓ Proof The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D}(T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B. □

Theorem 21.9 (Trace of the Choi matrix of a trace-preserving map). Lean ✓ Stmt *If T is trace-preserving, then* $\mathrm{tr}(\tau) = 1$.

Proof. ✓ Proof $\mathrm{tr}(\tau) = \mathrm{tr}(\mathrm{tr}_A(\tau)) = \mathrm{tr}(\frac{1}{D}\mathbb{1}_D) = 1$. □

21.3 Representation corollaries for channel decompositions

The next three corollaries follow from the Choi/Kraus/Stinespring representations already developed above. They correspond to [Wol12, Propositions 2.2–2.4].

Theorem 21.10 (Polarization of sesquilinear sandwiches). Lean ✓ Stmt *For any* $A, B, X \in M_D(\mathbb{C})$,

$$4\,AXB^\dagger = (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger + i\,(A+iB)X(A+iB)^\dagger - i\,(A-iB)X(A-iB)^\dagger.$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

Proof. ✓ Proof Working entrywise, reduces to the scalar polarization identity $4\,\alpha\,\overline{\delta} = \sum_{k=0}^{3} i^k(\alpha + i^k\beta)\overline{(\gamma + i^k\delta)}$ in $\mathbb{C}$, which holds after substituting $i^2 = -1$. ☐

Theorem 21.11 (CP decomposition of sandwich sums). Lean ✓ Stmt *For any finite Kraus-like families $\{A_i\}, \{B_i\}$, the map $T(X) = \sum_i A_i X B_i^\dagger$ satisfies $4\,T = T_1 - T_2 + i\,T_3 - i\,T_4$ with each T_k a CP map given explicitly by a single-side Kraus sum of the combined families.*

Proof. ✓ Proof Apply the polarization identity summand-by-summand. ☐

Theorem 21.12 (No information without disturbance). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be linear. If $T(|v\rangle\langle v|) = |v\rangle\langle v|$ for every $v \in \mathbb{C}^D$, then $T = \mathrm{id}$. In particular, a quantum channel preserving every pure state equals the identity.*

Proof. ✓ Proof Rank-one outer products $|u\rangle\langle v|$ polarize into rank-one self-outer-products:

$$4\,|u\rangle\langle v| = \sum_{k=0}^{3} i^k |u + i^k v\rangle\langle u + i^k v|,$$

so T preserves every $|u\rangle\langle v|$. Since the matrix units $E_{ij} = |e_i\rangle\langle e_j|$ span $M_D(\mathbb{C})$, T equals the identity on a spanning set, hence everywhere. ☐

Definition 21.13 (Pure-state ensemble density). Lean ✓ Stmt Given a finite family $\{\psi_i\}_{i\in\iota}$ of (unnormalized) vectors in $\mathbb{C}^D$, its *pure-ensemble density* is $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$. Weights $p_i \geq 0$ with $\sum p_i = 1$ can be absorbed by replacing $\psi_i \mapsto \sqrt{p_i}\psi_i$.

Theorem 21.14 (Isometric mixing preserves the density). Lean ✓ Stmt *If two pure-state ensembles $\{\psi_i\}_{i\in\iota_1}$ and $\{\phi_j\}_{j\in\iota_2}$ are related by an isometric mixing matrix $V \in \mathbb{C}^{\iota_1\times\iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij}\phi_j$, then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 21.15.*

Proof. ✓ Proof Expanding and applying the orthogonality relation $\sum_i V_{ij}\overline{V_{ij'}} = \delta_{jj'}$ from $V^\dagger V = \mathbb{1}$:

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left(\sum_i V_{ij}\overline{V_{ij'}}\right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

☐

Theorem 21.15 (Hughston–Jozsa–Wootters converse). Lean ✓ Stmt *If two pure-state ensembles $\{\psi_i\}_{i\in\iota_1}$ and $\{\phi_j\}_{j\in\iota_2}$ induce the same pure-ensemble density operator and $|\iota_2| \leq |\iota_1|$, then there exists a tall isometric mixing matrix $V \in \mathbb{C}^{\iota_1\times\iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij}\phi_j$. The cardinality hypothesis is what makes V a tall isometry; the symmetric case $|\iota_1| \leq |\iota_2|$ follows by swapping the roles of the ensembles.*

Proof. ✓ Proof Embed each vector ψ_i, ϕ_j as the 0-th column of a $D \times D$ matrix with zeros elsewhere; call these K_i, $L_j \in M_D(\mathbb{C})$. A direct entry-wise computation shows that for any $X \in M_D(\mathbb{C})$ both $\sum_i K_i X K_i^\dagger$ and $\sum_j L_j X L_j^\dagger$ collapse to $X_{00}\cdot\rho$, so the density equality $\rho_\psi = \rho_\phi$ forces the two Kraus families to define the same CP map. Theorem 21.26 then supplies an isometry V with $V^\dagger V = \mathbb{1}$ and $K_i = \sum_j V_{ij}\,L_j$; reading the equation off at column 0 recovers the vector relation $\psi_i = \sum_j V_{ij}\phi_j$. ☐

Theorem 21.16 (Hughston–Jozsa–Wootters equivalence). Lean ✓ Stmt *Under the cardinality hypothesis $|\iota_2| \leq |\iota_1|$, two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.*

Proof. ✓ Proof Combine the two directions above. ☐

21.4 Kraus representation theorem

Theorem 21.17 (Trace preservation implies Kraus normalization). Lean ✓ Stmt *If* $T(X) = \sum_i K_i X K_i^\dagger$ *is trace-preserving, then* $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Proof. ✓ Proof For any N, $\mathrm{tr}((\sum_i K_i^\dagger K_i)\,N) = \sum_i \mathrm{tr}(K_i^\dagger K_i N) = \sum_i \mathrm{tr}(K_i N K_i^\dagger) = \mathrm{tr}(T(N)) = \mathrm{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. □

Theorem 21.18 (Kraus normalization implies trace preservation). Lean ✓ Stmt *If* $\sum_i K_i^\dagger K_i = \mathbb{1}$, *then* $T(X) = \sum_i K_i X K_i^\dagger$ *is trace-preserving.*

Proof. ✓ Proof $\mathrm{tr}(T(X)) = \sum_i \mathrm{tr}(K_i X K_i^\dagger) = \sum_i \mathrm{tr}(K_i^\dagger K_i X) = \mathrm{tr}((\sum_i K_i^\dagger K_i)\,X) = \mathrm{tr}(X)$. □

Theorem 21.19 (Unitality implies Kraus unital normalization). Lean ✓ Stmt *If* $T(X) = \sum_i K_i X K_i^\dagger$ *satisfies* $T(\mathbb{1}) = \mathbb{1}$, *then*

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

Proof. ✓ Proof Evaluate the Kraus formula at the identity matrix. □

Theorem 21.20 (Unitary freedom in Kraus operators). Lean ✓ Stmt *Let* U *be a unitary* $r \times r$ *matrix* ($U^\dagger U = \mathbb{1}$) *and suppose* $K_j = \sum_\ell U_{j\ell} \widetilde{K}_\ell$. *Then* $\{K_j\}$ *and* $\{\widetilde{K}_\ell\}$ *define the same Kraus map: for every* $X \in M_D(\mathbb{C})$,

$$\sum_j K_j X K_j^\dagger = \sum_\ell \widetilde{K}_\ell X \widetilde{K}_\ell^\dagger.$$

Proof. ✓ Proof Substitute the combination formula, expand the triple sum over j, ℓ, ℓ', swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. □

Theorem 21.21 (Isometric mixing preserves the Kraus map). Lean ✓ Stmt *Let* W *be an isometry between two Kraus index spaces, so* $W^\dagger W = \mathbb{1}$. *If*

$$K_j = \sum_\ell W_{j\ell} \widetilde{K}_\ell,$$

then the Kraus families $\{K_j\}$ *and* $\{\widetilde{K}_\ell\}$ *define the same completely positive map.*

Proof. ✓ Proof Expand the Kraus sums, interchange the finite summations, and use $W^\dagger W = \mathbb{1}$ to collapse the coefficient matrix. □

Theorem 21.22 (The transition matrix of orthonormal Kraus families is unitary). Lean ✓ Stmt *Let* $\{K_j\}_{j=0}^{r-1}$ *and* $\{K'_j\}_{j=0}^{r-1}$ *be two Hilbert–Schmidt orthonormal Kraus families, and suppose*

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell}\, K'_\ell.$$

Then the transition matrix U *is unitary:* $U^\dagger U = \mathbb{1}_r$.

Proof. ✓ Proof Expand $\mathrm{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. □

Theorem 21.23 (Dual map equality from primal map equality). Lean ✓ Stmt *If two Kraus families satisfy $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X, then $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ for all Y.*

Proof. ✓ Proof Use the trace pairing: for all X, $\operatorname{tr}(X^\dagger(\sum B_\alpha^\dagger Y B_\alpha)) = \operatorname{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger)Y) = \operatorname{tr}((\sum A_j X^\dagger A_j^\dagger)Y) = \operatorname{tr}(X^\dagger(\sum A_j^\dagger Y A_j))$. Nondegeneracy of the trace pairing gives the result. □

Theorem 21.24 (Equal Stinespring Gramians). Lean ✓ Stmt *If two Kraus families define the same CPM, then $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$.*

Proof. ✓ Proof Specialise Theorem 21.23 at $Y = \mathbb{1}$. □

Theorem 21.25 (Rectangular Kraus freedom). Lean ✓ Stmt *If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X and $r_2 \le r_1$ (where $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) such that $B_\alpha = \sum_j V_{\alpha j} A_j$.*

Proof. ✓ Proof Pad the smaller family to size r_1, establish Gram matrix equality $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$ from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry V with $V^\dagger V = \mathbb{1}_{r_2}$. □

Theorem 21.26 (Rectangular Kraus freedom with general finite index types). Lean ✓ Stmt *Variant of Theorem 21.25 with general finite index sets ι_1, ι_2 in place of standard index sets of cardinalities r_1 and r_2.*

Proof. ✓ Proof Reindex by choosing enumerations of those finite index sets and apply Theorem 21.25. □

Theorem 21.27 (Isometric freedom of Kraus representations). Lean ✓ Stmt *Let $\{B_\alpha\}_{\alpha\in\iota_1}$ and $\{A_j\}_{j\in\iota_2}$ be finite Kraus families with $|\iota_2| \le |\iota_1|$. Then the following are equivalent:*

1. *$\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.*
2. *There exists a matrix $V = (V_{\alpha j})$ with $V^\dagger V = \mathbb{1}$ such that $B_\alpha = \sum_j V_{\alpha j} A_j$ for every α.*

Proof. ✓ Proof The implication (1) ⇒ (2) is Theorem 21.26. Conversely, expand $\sum_\alpha B_\alpha X B_\alpha^\dagger$ using $B_\alpha = \sum_j V_{\alpha j} A_j$, interchange the finite sums, and use $V^\dagger V = \mathbb{1}$ to collapse the coefficient matrix to the diagonal. □

Theorem 21.28 (Unitary freedom of Kraus representations). Lean ✓ Stmt *Let $\{B_\alpha\}_{\alpha\in\iota}$ and $\{A_j\}_{j\in\iota}$ be two Kraus families with the same finite index set. Then the following are equivalent:*

1. *$\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.*
2. *There exists a unitary matrix $U = (U_{\alpha j})$ such that $B_\alpha = \sum_j U_{\alpha j} A_j$ for every α.*

Proof. ✓ Proof Apply Theorem 21.27 with $|\iota| = |\iota|$. In the square case an isometry matrix is unitary, and the converse is immediate. □

Remark 21.29 (Choi rank and minimal Kraus cardinality). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt If a completely positive map admits a Kraus representation with exactly r operators, then its Choi matrix is a sum of r rank-one outer products. Consequently,

$$\operatorname{rank}(\tau_E) \le r.$$

Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly $\mathrm{rank}(\tau_E)$ operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

Proof. ✓ Proof For the forward implication, expand the Choi matrix using the Kraus formula and write $\tau_E = \sum_j v_j v_j^\dagger$ with one vector v_j for each Kraus operator. The column space of this sum lies in the span of the r vectors $\{v_j\}$, so the rank is at most r. For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators. □

21.5 Stinespring dilation

Definition 21.30 (Stinespring isometry). Lean ✓ Stmt Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

Lemma 21.31 (Entry formula for the Stinespring isometry). Lean ✓ Stmt *For indices $i, k \in \{0, \dots, D-1\}$ and $j \in \{0, \dots, r-1\}$,*

$$V_{(i,j),k} = (K_j)_{ik}.$$

Theorem 21.32 (Stinespring Gram identity). Lean ✓ Stmt *The Stinespring isometry satisfies*

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

Theorem 21.33 (Stinespring dual representation). Lean ✓ Stmt *For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,*

$$V^\dagger \,(A \otimes \mathbb{1}_r)\, V = \sum_j K_j^\dagger A\, K_j = T^*(A).$$

This is the Stinespring representation of the dual *(Heisenberg picture) map T^*.*

Proof. ✓ Proof Compute $(V^\dagger(A \otimes \mathbb{1}_r)V)_{ab}$ by summing over the product index (i,j); the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l, giving $\sum_j K_j^\dagger A K_j$. □

Theorem 21.34 (The Stinespring isometry condition iff trace preservation). Lean ✓ Stmt $V^\dagger V = \mathbb{1}_D$ *if and only if* $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. *In particular, V is an isometry precisely when the Kraus map is trace-preserving.*

Proof. ✓ Proof $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}}\, V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}}\, (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. □

Theorem 21.35 (Stinespring Schrödinger representation). Lean ✓ Stmt *The Kraus map* $T(\rho) = \sum_j K_j \rho K_j^\dagger$ *equals the partial trace over the dilation space:*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \mathrm{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

Proof. ✓ Proof Expand $(V\rho V^\dagger)_{(i,k),(j,k)}$ and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. □

Theorem 21.36 (Existence of a Stinespring dilation). Lean ✓ Stmt *Every completely positive map admits an ancilla dimension r, Kraus operators $\{K_j\}_{j=0}^{r-1}$, and the concrete representation $\pi(A) = A \otimes \mathbb{1}_r$ such that*

$$E(A) = V^\dagger \pi(A) V.$$

Proof. ✓ Proof Choose a Kraus representation of E and apply the explicit Heisenberg-picture Stinespring formula. □

Theorem 21.37 (CPTP maps admit isometric Stinespring dilations). Lean ✓ Stmt *Every quantum channel admits a Stinespring dilation whose matrix V is an isometry, equivalently $V^\dagger V = \mathbb{1}$.*

Proof. ✓ Proof Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition $V^\dagger V = \mathbb{1}$. □

21.6 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Theorems 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

Definition 21.38 (CP partial order). Lean ✓ Stmt For linear maps $S, T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, write $T \le S$ (in the CP order) when $S - T$ is completely positive.

Definition 21.39 (Block-top contraction). Lean ✓ Stmt For natural numbers r, s, let $C = C_{r,s} \in \mathbb{C}^{r\times(r+s)}$ be the rectangular $r \times (r+s)$ matrix whose rows are the first r rows of the identity on $\mathbb{C}^{r+s}$: $C_{ij} = 1$ if $j = i < r$ and 0 otherwise.

Lemma 21.40 (C is a co-isometry). Lean ✓ Stmt *The block-top matrix satisfies $CC^\dagger = \mathbb{1}_r$.*

Proof. ✓ Proof Direct entrywise computation: $(CC^\dagger)_{ii'} = \sum_j C_{ij}\overline{C_{i'j}}$ collapses to $\delta_{ii'}$. □

Lemma 21.41 (C is a contraction). Lean ✓ Stmt *The block-top matrix satisfies $C^\dagger C \le \mathbb{1}_{r+s}$.*

Proof. ✓ Proof The matrix $C^\dagger C$ is diagonal with a 1 on the first r coordinates and a 0 on the last s, so $\mathbb{1} - C^\dagger C$ is positive semidefinite. □

Lemma 21.42 (Intertwining of Stinespring isometries). Lean ✓ Stmt *Let $K : \{0, \dots, r-1\} \to M_D(\mathbb{C})$ and $L : \{0, \dots, s-1\} \to M_D(\mathbb{C})$ be two Kraus families. Write $K +\!\!+ L$ for their concatenation as a family indexed by $\{0, \dots, r+s-1\}$. Then*

$$V_K = (\mathbb{1}_D \otimes C_{r,s}) V_{K+\!\!+L}.$$

Proof. ✓ Proof Entrywise expansion of both sides: the Kronecker product with $C_{r,s}$ picks out the first r coordinates of the dilation space, matching the action of $V_{K+\!\!+L}$ on those coordinates, which is precisely V_K. □

Theorem 21.43 (Ordered CP maps via Stinespring contractions). Lean ✓ Stmt *Let $T_1, T_2 : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be CP maps with $T_1 \le T_2$. Then there exist an ancilla dimension m, Heisenberg-form Stinespring matrices V_1, V_2 realizing T_1, T_2 via $T_i(A) = V_i^\dagger (A \otimes \mathbb{1}) V_i$, and a contraction $\widetilde{C}$ on the dilation space with $\widetilde{C}^\dagger \widetilde{C} \le \mathbb{1}$ such that*

$$V_1 = (\mathbb{1}_D \otimes \widetilde{C})\, V_2.$$

This is [Wol12, Theorem 2.3].

Proof. ✓ Proof Choose a Heisenberg-form Kraus family K_1 of T_1 with Stinespring matrix $V_1 = V_{K_1}$, and Kraus operators L of the CP map $T_2 - T_1$ in Schrödinger orientation. Let $L^\dagger$ denote the conjugate-transposed family. Set $m = r_1 + s$ and let $K_2 = K_1 +\!\!+ L^\dagger$. Then $T_2(A) = T_1(A) + (T_2 - T_1)(A) = \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger$, and the latter equals $\sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger$ which is the Heisenberg form for K_2, yielding $T_2(A) = V_{K_2}^\dagger (A \otimes \mathbb{1}) V_{K_2}$. Taking $\widetilde{C} = C_{r_1,s}$ the intertwining $V_{K_1} = (\mathbb{1}_D \otimes \widetilde{C}) V_{K_2}$ is Lemma 21.42, and $\widetilde{C}^\dagger \widetilde{C} \le \mathbb{1}$ is Lemma 21.41. □

Definition 21.44 (Block-diagonal projectors on the dilation space). Lean Lean ✓ Stmt For r, s, let $P_{\text{top}} = C_{r,s}^\dagger C_{r,s}$ and $P_{\text{bot}} = \mathbb{1} - P_{\text{top}}$. Both are PSD (for P_{bot}, by Lemma 21.41) and $P_{\text{top}} + P_{\text{bot}} = \mathbb{1}$.

Theorem 21.45 (Radon–Nikodym theorem for CP maps). Lean ✓ Stmt *Let $T_1, T_2 : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be completely positive. There exist an ancilla dimension m, a Kraus family K with Stinespring matrix $V = V_K$, and positive semidefinite operators $P_1, P_2 : \mathbb{C}^m \to \mathbb{C}^m$ with $P_1 + P_2 = \mathbb{1}_m$ such that, for $i = 1, 2$ and every $A \in M_D(\mathbb{C})$,*

$$T_i(A) = V^\dagger (A \otimes P_i) V.$$

This is [Wol12, Theorem 2.4].

Proof. ✓ Proof Take Heisenberg-form Kraus families K_1 for T_1 and Schrödinger-form L for T_2, and form $K = K_1 +\!\!+ L^\dagger$ on $\mathbb{C}^{r_1+s}$. Choose $P_1 = P_{\text{top}}$, $P_2 = P_{\text{bot}}$ as in Definition 21.44. The Kronecker identity $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger (A \otimes \mathbb{1})(\mathbb{1} \otimes C)$ rewrites $V^\dagger (A \otimes P_{\text{top}}) V$ as $((\mathbb{1} \otimes C)V)^\dagger (A \otimes \mathbb{1})((\mathbb{1} \otimes C)V)$, which equals $V_{K_1}^\dagger (A \otimes \mathbb{1}) V_{K_1} = T_1(A)$ by the intertwining Lemma 21.42. For the complementary block, $A \otimes P_{\text{bot}} = A \otimes \mathbb{1} - A \otimes P_{\text{top}}$ and sandwiching by V yields $(T_1 + T_2)(A) - T_1(A) = T_2(A)$. □

Theorem 21.46 (Open-system representation of quantum channels). Lean ✓ Stmt *Every quantum channel $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits a Stinespring isometry V on an ancilla space $\mathbb{C}^r$ such that*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \operatorname{tr}_r(V \rho V^\dagger).$$

Proof. ✓ Proof Apply the existence of an isometric Stinespring dilation (Theorem 21.37) and the Schrödinger-picture identity (Theorem 21.35). □

Theorem 21.47 (Open-system representation, partial-trace form). Lean ✓ Stmt *Every quantum channel T admits an isometric Stinespring dilation V such that $T(\rho) = \operatorname{tr}_E(V \rho V^\dagger)$.*

Proof. ✓ Proof Rewrite the componentwise identity of Theorem 21.46 as a partial trace over the ancilla factor. □

Remark 21.48 (Unitary form). The fully unitary open-system form $T(\rho) = \operatorname{tr}_E(U(\rho \otimes |\phi\rangle\langle\phi|)U^\dagger)$ of [Wol12, Theorem 2.5] follows from the isometric form by extending the Stinespring isometry V to a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^r$; this uses orthonormal basis extension and is left to a later theorem.

21.7 POVMs and Naimark dilation

Definition 21.49 (Positive operator-valued measure). Lean ✓ Stmt A *positive operator-valued measure* with n outcomes on $\mathbb{C}^D$ is a family $\{E_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^D$ satisfying the resolution of identity

$$\sum_{i=0}^{n-1} E_i = \mathbb{1}_D.$$

Definition 21.50 (Naimark Kraus square roots). Lean Lean ✓ Stmt For a POVM $\{E_i\}$, each effect $E_i \geq 0$ admits a square-root factorisation $E_i = M_i^\dagger M_i$ with $M_i \in M_D(\mathbb{C})$. The operators M_i are the *Naimark Kraus square roots.*

Theorem 21.51 (Naimark square roots satisfy the Kraus normalization). Lean ✓ Stmt *For a POVM $\{E_i\}$ with square roots $E_i = M_i^\dagger M_i$, one has*

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

Proof. ✓ Proof Sum the identities $E_i = M_i^\dagger M_i$ and use the defining resolution of the identity for the POVM. □

Definition 21.52 (Naimark isometry). Lean ✓ Stmt The *Naimark isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^n$ of a POVM is the Stinespring-type construction

$$V = \sum_{i=0}^{n-1} M_i \otimes |i\rangle, \qquad V_{(a,i),\,k} = (M_i)_{a,k}.$$

Definition 21.53 (Naimark projective measurement). Lean Lean ✓ Stmt The *Naimark projectors* on $\mathbb{C}^D \otimes \mathbb{C}^n$ are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \qquad (P_i)_{(a,b),(c,d)} = \delta_{a,c}\delta_{b,i}\delta_{d,i}.$$

Theorem 21.54 (Naimark isometry condition). Lean ✓ Stmt *The Naimark isometry satisfies $V^\dagger V = \mathbb{1}_D$.*

Proof. ✓ Proof The Stinespring Gram identity gives $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$ by the resolution of identity. □

Theorem 21.55 (Naimark projectors form a projective measurement). Lean Lean Lean Lean ✓ Stmt *The Naimark projectors satisfy $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_iP_j = 0$ for $i \neq j$, and $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$.*

Proof. ✓ Proof Each identity follows from direct entrywise computation using the delta-function form of $(P_i)_{(a,b),(c,d)}$. □

Theorem 21.56 (Naimark dilation). Lean ✓ Stmt *Every POVM $\{E_i\}_{i=0}^{n-1}$ arises as a projective measurement on a dilation: for the isometry V and projectors P_i above,*

$$E_i = V^\dagger P_i V.$$

Proof. ✓ Proof Computing entrywise, $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}}\,(M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$. □

Theorem 21.57 (Existential Naimark dilation). Lean ✓ Stmt *For every POVM $\{E_i\}$ on $\mathbb{C}^D$ there exist a dilation dimension r, an isometry $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$, and a projective measurement $\{P_i\}$ on the dilation satisfying $E_i = V^\dagger P_i V$.*

Proof. ✓ Proof Take $r = n$ and the explicit witnesses $V = \sum_i M_i \otimes |i\rangle$ and $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. □

Definition 21.58 (Naimark dilation as a structure). Lean ✓ Stmt A *Naimark dilation* of a POVM $\{E_i\}$ consists of an isometry V into a larger Hilbert space together with a projective measurement $\{P_i\}$ there such that $E_i = V^\dagger P_i V$ for all i.

Theorem 21.59 (Canonical dilation satisfies the Naimark dilation axioms). Lean ✓ Stmt *The explicit isometry $V = \sum_i M_i \otimes |i\rangle$ and projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ satisfy the defining axioms of a Naimark dilation.*

Proof. ✓ Proof Combine the previously proved identities $V^\dagger V = \mathbb{1}_D$, $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, $\sum_i P_i = \mathbb{1}$, and $V^\dagger P_i V = E_i$. □

Theorem 21.60 (Concrete uniqueness for canonical projectors). Lean ✓ Stmt *Let $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^n$ satisfy $V^\dagger P_i V = E_i$ for the canonical projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. Then there exists an isometry W on the dilated space such that $V = WV_0$, where V_0 is the canonical Naimark isometry of $\{E_i\}$.*

Proof. ✓ Proof For each outcome i, the i-th block of V has Gram matrix E_i. Comparing with the canonical square root M_i gives $V_i^\dagger V_i = M_i^\dagger M_i$, hence $V_i = U_i M_i$ for a unitary U_i on $\mathbb{C}^D$. Assemble the U_i block-diagonally to obtain an isometry $W = \bigoplus_i U_i$ on $\mathbb{C}^D \otimes \mathbb{C}^n$, and then $V = W \sum_i M_i \otimes |i\rangle = WV_0$. □

Definition 21.61 (POVM from a PSD resolution of identity on a dilation). Lean ✓ Stmt Given an isometry $V : \mathbb{C}^D \to \mathbb{C}^{d'}$ with $V^\dagger V = \mathbb{1}_D$ and a family $\{P_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^{d'}$ summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

Definition 21.62 (Quantum instrument). Lean ✓ Stmt A *quantum instrument* with n outcomes is a family $\{\Phi_i\}_{i=0}^{n-1}$ of completely positive maps on $M_D(\mathbb{C})$ whose sum $\sum_i \Phi_i$ is trace-preserving.

Definition 21.63 (Instrument operations). Lean Lean Lean Lean ✓ Stmt Associated to an instrument are the total channel $\sum_i \Phi_i$, the unnormalized update $\rho \mapsto \Phi_i(\rho)$ for each outcome i, the outcome probability $p_i(\rho) = \mathrm{tr}(\Phi_i(\rho))$, and the normalized posterior state $\Phi_i(\rho)/p_i(\rho)$ whenever $p_i(\rho) \neq 0$.

Theorem 21.64 (Instrument total map is a channel). Lean ✓ Stmt *The total map $\sum_i \Phi_i$ of an instrument is a quantum channel.*

Proof. ✓ Proof Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition. □

Theorem 21.65 (Conservation of probability for instruments). Lean ✓ Stmt *For every state $\rho \in M_D(\mathbb{C})$, the outcome probabilities $p_i(\rho) := \mathrm{tr}(\Phi_i(\rho))$ sum to $\mathrm{tr}(\rho)$.*

Proof. ✓ Proof Linearity of trace and trace preservation of $\sum_i \Phi_i$ give $\sum_i \mathrm{tr}(\Phi_i(\rho)) = \mathrm{tr}((\sum_i \Phi_i)(\rho)) = \mathrm{tr}(\rho)$. ☐

Theorem 21.66 (Non-negativity of instrument probabilities). Lean ✓ Stmt *If $\rho \in M_D(\mathbb{C})$ is positive semidefinite, then each outcome probability $p_i(\rho) = \mathrm{tr}(\Phi_i(\rho))$ is a non-negative real number.*

Proof. ✓ Proof Each Φ_i is completely positive, hence positive, so $\Phi_i(\rho)$ is positive semidefinite and its trace is a non-negative real. ☐

21.8 Trace-pairing expansion in transfer-matrix form

Definition 21.67 (Trace-self-dual basis). Lean ✓ Stmt A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \mathrm{tr}(\sigma_i X)\sigma_i.$$

Definition 21.68 (Trace-pairing coefficients). Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \mathrm{tr}(\sigma_i T(\sigma_j)).$$

Theorem 21.69 (Trace-pairing expansion of a linear map). Lean ✓ Stmt *If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits the expansion*

$$T(\rho) = \sum_i \sum_j t_{ij}\,\mathrm{tr}(\sigma_j \rho)\sigma_i, \qquad t_{ij} = \mathrm{tr}(\sigma_i T(\sigma_j)).$$

Proof. ✓ Proof Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. ☐

21.9 SVD normal form (existence)

[Wol12, Section 2.3] discusses two families of normal forms for the transfer-matrix representation of a quantum channel: the *singular value decomposition* (SVD) and, for qubit channels, the *Lorentz normal form* obtained by $\mathrm{SL}(2,\mathbb{C})$ filtering operations. The SVD normal form is proved below. The Lorentz normal form theorem (Theorem 21.81) requires the compactness and minimisation argument for $\mathrm{SL}(2,\mathbb{C})$ filterings in [Wol12, Propositions 2.9 and 2.11]; see Section 21.10.

Theorem 21.70 (SVD for positive semidefinite matrices). Lean ✓ Stmt *Every positive semidefinite matrix $M \in M_D(\mathbb{C})$ admits a decomposition*

$$M = U\,\mathrm{diag}(\sigma)U^\dagger,$$

where $U \in \mathcal{U}(D)$ is unitary and $\sigma : \{0, \dots, D-1\} \to \mathbb{R}_{\geq 0}$ has non-negative entries.

Proof. ✓ Proof The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semidefiniteness forces those eigenvalues to be non-negative, and the decomposition is written in SVD form with U the eigenvector unitary and σ the eigenvalue sequence. ☐

Theorem 21.71 (SVD existence for invertible matrices). Lean ✓ Stmt *Every invertible complex square matrix $M \in M_D(\mathbb{C})$ admits a singular value decomposition*

$$M = U\,\mathrm{diag}(\sigma)V^\dagger,$$

with $U, V \in \mathcal{U}(D)$ unitary and $\sigma_i > 0$ for all i.

Proof. ✓ Proof Apply the spectral theorem to the positive definite matrix $M^\dagger M$ to obtain $M^\dagger M = V\,\mathrm{diag}(\lambda)V^\dagger$ with $\lambda_i > 0$. Set $\sigma_i := \sqrt{\lambda_i}$ and $\Sigma := \mathrm{diag}(\sigma)$; since Σ is a real positive diagonal, it is self-adjoint and invertible. Define $U := MV\Sigma^{-1}$. Then $U^\dagger U = \Sigma^{-1}V^\dagger(M^\dagger M)V\Sigma^{-1} = \Sigma^{-1}\,\mathrm{diag}(\lambda)\Sigma^{-1} = \mathbb{1}$, so U is unitary, and $U\Sigma V^\dagger = MV(\Sigma^{-1}\Sigma)V^\dagger = MVV^\dagger = M$. □

Theorem 21.72 (SVD representation of a transfer matrix). Lean ✓ Stmt *Every invertible transfer matrix $\widehat{T}$ of a linear super-operator $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits a singular value decomposition $\widehat{T} = U\,\mathrm{diag}(\sigma)V^\dagger$ with U, V unitary on $\mathbb{C}^D \otimes \mathbb{C}^D$ and $\sigma_i > 0$.*

Proof. ✓ Proof Direct specialisation of Theorem 21.71 to the index type $\{0, \dots, D-1\} \times \{0, \dots, D-1\}$ used by the transfer-matrix representation. □

Remark 21.73 (Sorted / unique singular values). Theorem 21.71 produces the singular values as an unordered family, without the usual convention that σ_i are sorted in non-increasing order or the statement that they are uniquely determined by M. Downstream applications in [Wol12, Section 2.3], such as trace-norm identities, polar decomposition, and the Lorentz normal form, will typically require the sorted / unique variant; that refinement is future work.

21.10 Lorentz normal form

This section states the existence theorems from [Wol12, Section 2.3].

Definition 21.74 (SL-filtering operation). An *SL-filtering* for $D \times D$ matrices is a completely positive map of the form $\Phi(X) = SXS^\dagger$ where $S \in M_D(\mathbb{C})$ satisfies $\det S = 1$. Such maps are invertible and have Kraus rank 1.

Definition 21.75 (Doubly-stochastic map). A linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *doubly-stochastic* if $T(\mathbb{1}) \propto \mathbb{1}$ and the reduced density matrix $\mathrm{tr}_1[\tau]$ of its Choi matrix $\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)$ is proportional to the identity. This is the normal form in [Wol12, Proposition 2.8].

Lemma 21.76 (Infimum attainment for SL-filterings). *For a positive-definite Choi matrix τ, the infimum of $\mathrm{tr}[(S_2 \otimes_k S_1)\tau(S_2 \otimes_k S_1)^\dagger]$ over $S_1, S_2 \in M_D(\mathbb{C})$ with $\det S_1 = \det S_2 = 1$ is attained. **Not yet proved**—requires compactness of bounded subsets of $\mathrm{SL}(D, \mathbb{C})$ and the extreme-value theorem.*

Theorem 21.77 (Generic normal form for CP maps). *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a completely positive map whose Choi matrix is positive-definite (equivalently, T has full Kraus rank). Then there exist SL-filterings Φ_1, Φ_2 such that $\Phi_2 \circ T \circ \Phi_1$ is doubly-stochastic. **Not yet proved**—proof blocked on Lemma 21.76.*

Definition 21.78 (Diagonal Lorentz normal form). For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ for its matrix in the normalized Pauli basis $\{\sigma_0/\sqrt{2}, \sigma_1/\sqrt{2}, \sigma_2/\sqrt{2}, \sigma_3/\sqrt{2}\}$. The channel is in *diagonal Lorentz normal form* when it is unital and every off-diagonal entry of $\widehat{T'}$ is zero.

Definition 21.79 (Non-diagonal Lorentz normal form)**.** For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ in the normalized Pauli basis. The channel is in *non-diagonal Lorentz normal form* when, for some $x \in [0, 1]$,

$$\widehat{T'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x/\sqrt{3} & 0 & 0 \\ 0 & 0 & x/\sqrt{3} & 0 \\ 2/3 & 0 & 0 & 1/3 \end{pmatrix}.$$

Trace preservation supplies the first row of the displayed matrix.

Definition 21.80 (Singular Lorentz normal form)**.** For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ in the normalized Pauli basis. The channel is in *singular Lorentz normal form* when

$$\widehat{T'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

equivalently, every input state is mapped to $(1 + \sigma_3)/2$. Trace preservation supplies the first row of the displayed matrix.

Theorem 21.81 (Lorentz normal form for qubit channels)**.** *For every qubit channel $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, there exist $\mathrm{SL}(2, \mathbb{C})$-filterings Φ_1, Φ_2 such that the filtered channel $T' = \Phi_2 \circ T \circ \Phi_1$ is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular.* ***Not yet proved****—proof blocked on Lemma 21.76 and the Lorentz group classification of $\mathrm{SL}(2, \mathbb{C})$ orbits.*

21.11 Determinant of a quantum channel

Definition 21.82 (Channel determinant)**.** Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, Section 6.1.1].

Remark 21.83 (Channel determinant as an operator determinant)**.** Lean ✓ Stmt The channel determinant agrees with the ordinary determinant of the linear endomorphism $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$.

Definition 21.84 (Unitary channel)**.** Lean Lean ✓ Stmt For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$. It is automatically a quantum channel.

Theorem 21.85 (Unitary channels have determinant one)**.** *Lean* *Lean* *✓ Stmt* *If $T(\rho) = U\rho U^\dagger$ is a unitary channel, then $\det T = 1$ and hence $|\det T| = 1$.*

Proof. ✓ Proof Vectorization identifies T with a Kronecker product $\overline{U} \otimes U$, whose determinant is $\overline{\det U}^D (\det U)^D = 1$. □

Theorem 21.86 (Determinant bound for positive trace-preserving maps)**.** *Lean* *Lean* *✓ Stmt* *For any positive trace-preserving map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, $|\det T| \leq 1$. This is [Wol12, Theorem 6.1(1)].*

Proof. ✓ Proof Every eigenvalue μ of T satisfies $|\mu| \leq 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \leq 1$ follows by induction. □

Theorem 21.87 (Determinant one iff the channel is unitary)**.** Lean Lean ✓ Stmt *For a CPTP map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$,*

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Theorem 6.1(2)].

Proof. ✓ Proof Reverse direction is Theorem 21.85. Forward direction: $|\det T| = 1$ together with $|\mu| \leq 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Hence T is bijective. A bijective CPTP map is a $*$-automorphism: the Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$ is unital (since T is TP), so the Kadison–Schwarz inequality $T^*(Y^\dagger Y) \geq T^*(Y)^\dagger T^*(Y)$ becomes an equality for every Y. Hence T^*, and therefore T, preserves products. By the Skolem–Noether theorem, every $*$-automorphism of $M_D(\mathbb{C})$ has the form $T(\rho) = U\rho U^\dagger$ for some unitary U. Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. □

Chapter 22

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Chapter 7].

Section 22.1 establishes definitions and the basic exponential form (Proposition 22.14). Section 22.2 gives the Duhamel formula and perturbation bound. Section 22.3 develops the generator theory and proves the GKSL theorem.

22.1 Dynamical semigroups and the exponential form

Definition 22.1 (Dynamical semigroup). Lean ✓ Stmt A family of linear maps $T : \mathbb{R} \to (M_D(\mathbb{C}) \to_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T_{t+s} = T_t \circ T_s$ for all $t, s \geq 0$; and
2. (initial condition) $T_0 = \mathrm{id}$.

This is [Wol12, Equation (7.1)].

Definition 22.2 (Continuous dynamical semigroup). Lean ✓ Stmt A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T_t$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 22.3 (Exponential semigroup). Lean Lean ✓ Stmt Given a generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

Theorem 22.4 (Semigroup law for the exponential). Lean Lean ✓ Stmt *For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.*

Proof. ✓ Proof Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. □

Theorem 22.5 (Exponential semigroup at time 0). Lean ✓ Stmt *The continuous-linear-map exponential satisfies $e^{0L} = \mathbb{1}$.*

Proof. ✓ Proof The exponential of the zero endomorphism is the identity. □

Theorem 22.6 (Continuity of the exponential semigroup). Lean Lean ✓ Stmt *The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.*

Proof. ✓ Proof The exponential is analytic (convergence radius $= \infty$), hence continuous. □

Theorem 22.7 (Derivative of the exponential semigroup). Lean ✓ Stmt *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

This is [Wol12, Equation (7.2)].

Proof. ✓ Proof Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \mathrm{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \mathrm{End}(M_D(\mathbb{C}))$. □

Theorem 22.8 (Derivative at the origin). Lean ✓ Stmt *One has*

$$\frac{d}{dt}e^{tL}\bigg|_{t=0} = L.$$

Proof. ✓ Proof This is Theorem 22.7 at $t = 0$ together with Theorem 22.5. □

Theorem 22.9 (Generator uniqueness). Lean ✓ Stmt *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

Proof. ✓ Proof Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. □

Theorem 22.10 (CLM and linear-map forms coincide). Lean ✓ Stmt *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

Proof. ✓ Proof This is a direct unpacking of the definition of the linear-map exponential semigroup. □

Theorem 22.11 (Pointwise derivative formula). Lean ✓ Stmt *For every $X \in M_D(\mathbb{C})$ and every $t \in \mathbb{R}$,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

Proof. ✓ Proof Evaluate the derivative of the continuous-linear-map semigroup at X and transport the statement through Theorem 22.10. □

Theorem 22.12 (Linear-map exponential semigroup at time 0). Lean ✓ Stmt *The linear-map exponential semigroup satisfies $e^{0L} = \mathbb{1}$.*

Proof. ✓ Proof Transport Theorem 22.5 through Theorem 22.10. □

Theorem 22.13 (Exponential semigroup as a dynamical semigroup). Lean ✓ Stmt *The family $t \mapsto e^{tL}$ satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

Proof. ✓ Proof Transport the continuous-linear-map semigroup law from Theorem 22.4 through Theorem 22.10; the initial condition is Theorem 22.12. □

Theorem 22.14 (Every continuous semigroup is exponential). Lean ✓ Stmt *Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form*

$$T_t = e^{tL}$$

for some generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [Wol12, Proposition 7.1].

Proof. ✓ Proof Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s\, ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. □

22.2 Perturbation theory

Theorem 22.15 (Duhamel formula). Lean ✓ Stmt *Let $\Delta = L' - L$. For $t \geq 0$,*

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta\, e^{sL'}\, ds.$$

This is [Wol12, Lem. 7.1].

Proof. ✓ Proof Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta\, e^{sL'}$ (Theorem 22.7), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s)\, ds$. □

Theorem 22.16 (Perturbation bound). Lean ✓ Stmt *For $t \geq 0$,*

$$\|e^{tL'} - e^{tL}\| \leq t\, \|\Delta\| \cdot \sup_{s\in[0,t]} \|e^{sL}\| \cdot \sup_{s\in[0,t]} \|e^{sL'}\|,$$

where $\Delta = L' - L$. This is [Wol12, Corollary 7.1].

Proof. ✓ Proof Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 22.15). □

Corollary 22.17 (Perturbation bound under unit-norm control). Lean ✓ Stmt *For $t \geq 0$, if $\|e^{sL}\| \leq 1$ and $\|e^{sL'}\| \leq 1$ for every $s \in [0, t]$, then*

$$\|e^{tL'} - e^{tL}\| \leq t\, \|L' - L\|.$$

Proof. ✓ Proof Apply Theorem 22.16 and bound both suprema by 1. □

Lemma 22.18 (Dyson-series summability from factorial majorant). Lean ✓ Stmt *Assume a factorial majorant of Dyson–Phillips iterates:*

$$\|D_n(t)\| \leq M\, \frac{(t\, \|L' - L\|\, M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

Proof. ✓ Proof This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. □

Theorem 22.19 (Factorial norm bound for Dyson iterates). Lean ✓ Stmt *For $s \in [0, t]$ and $M = \sup_{u\in[0,t]} \|e^{uL}\|$,*

$$\|\widetilde{T}_s^{(n)}\| \leq M\, \frac{(s\, \|\Delta\|\, M)^n}{n!}.$$

Proof. ✓ Proof Induction on n. The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$ and integrates $\int_0^s u^n\,du = s^{n+1}/(n+1)$, recovering the factorial. ☐

Corollary 22.20 (Dyson series summability). Lean ✓ Stmt *The Dyson–Phillips series $\sum_n \widetilde{T}_t^{(n)}$ converges in operator norm for every $t \geq 0$.*

Proof. ✓ Proof Compose the factorial norm bound (Theorem 22.19) with the Weierstrass M-test step (Lemma 22.18). ☐

Corollary 22.21 (Factorial norm bound at the final time). Lean ✓ Stmt *For $t \geq 0$, setting $s = t$ in Theorem 22.19 gives*

$$\|\widetilde{T}_t^{(n)}\| \leq M\,\frac{(t\,\|\Delta\|\,M)^n}{n!},$$

where $M = \sup_{u\in[0,t]} \|e^{uL}\|$.

Proof. ✓ Proof This is the special case $s = t$ of Theorem 22.19. ☐

Theorem 22.22 (Continuity of Dyson iterates). Lean ✓ Stmt *Each Dyson iterate $t \mapsto \widetilde{T}_t^{(n)}$ is continuous in the time parameter.*

Proof. ✓ Proof By induction on n. The base case is continuity of the matrix exponential. For the inductive step, the parametric integral $t \mapsto \int_0^t e^{(t-s)L}\,\Delta\,\widetilde{T}_s^{(n)}\,ds$ is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of $\mathbb{R}$ since $\widetilde{T}_t^{(n+1)} = 0$ for $t \leq 0$. ☐

Theorem 22.23 (Factorial bound on the Dyson remainder). Lean ✓ Stmt *For $s \in [0,t]$, $M = \sup_{u\in[0,t]} \|e^{uL}\|$, and $M' = \sup_{u\in[0,t]} \|e^{uL'}\|$:*

$$\|T'_s - \textstyle\sum_{n<N} \widetilde{T}_s^{(n)}\| \leq M'\,\frac{(s\,\|\Delta\|\,M)^N}{N!}.$$

Proof. ✓ Proof Induction on N. The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity $R_{N+1}(s) = \int_0^s e^{(s-u)L}\,\Delta\,R_N(u)\,du$ (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial. ☐

Theorem 22.24 (Dyson series identity [Wol12, Equation (7.13)]). Lean ✓ Stmt *For $t \geq 0$, the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \widetilde{T}_t^{(n)} = e^{tL'}.$$

Proof. ✓ Proof The partial-sum remainder $\|e^{tL'} - \sum_{n<N} \widetilde{T}_t^{(n)}\| \leq M'\,(t\|\Delta\|M)^N/N!$ tends to zero as $N \to \infty$, since $c^N/N! \to 0$ for any constant c. Together with the summability of the series (Corollary 22.20), this identifies the sum via uniqueness of limits. ☐

Corollary 22.25 (Tsum form of the Dyson series). Lean ✓ Stmt $\sum_{n=0}^{\infty} \widetilde{T}_t^{(n)} = e^{tL'}$ *(tsum form).*

Proof. ✓ Proof Immediate from Theorem 22.24. ☐

22.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 22.54), which shows that such generators have the standard Lindblad form.

22.3.1 Generator decomposition and conditional complete positivity

Definition 22.26 (Generator decomposition). Lean ✓ Stmt A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Equation (7.14)].

Definition 22.27 (Conditional complete positivity). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Proposition 7.2].

Definition 22.28 (Trace-annihilating map). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *trace-annihilating* if $\mathrm{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 22.29 (Trace constraint). Lean ✓ Stmt A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Equation (7.20)].

Theorem 22.30 (Trace constraint implies trace-annihilating). Lean ✓ Stmt *If (ϕ, κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.*

Proof. ✓ Proof Let $\phi(\rho) = \sum_i K_i\rho K_i^\dagger$. By the cyclic property of trace, $\mathrm{tr}(L(\rho)) = \mathrm{tr}((\sum_i K_i^\dagger K_i)\rho) - \mathrm{tr}(\kappa\rho) - \mathrm{tr}(\kappa^\dagger\rho) = \mathrm{tr}((\kappa + \kappa^\dagger)\rho) - \mathrm{tr}(\kappa\rho) - \mathrm{tr}(\kappa^\dagger\rho) = 0$. □

22.3.2 The Lindblad form

Definition 22.31 (Lindblad form). Lean ✓ Stmt A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \Big(L_j\rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \Big),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Equation (7.21)].

Theorem 22.32 (Lindblad form is trace-annihilating). Lean ✓ Stmt *Every Lindblad form defines a trace-annihilating linear map.*

Proof. ✓ Proof The Hamiltonian part satisfies $\mathrm{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\mathrm{tr}(L_j\rho L_j^\dagger) = \mathrm{tr}(L_j^\dagger L_j\rho) = \mathrm{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. □

Theorem 22.33 (Lindblad form equals generator decomposition). Lean ✓ Stmt *A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j\rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$. This is [Wol12, Equation (7.24)].*

Proof. ✓ Proof Compute $\kappa^\dagger = -iH + \frac{1}{2}\sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$. ☐

Theorem 22.34 (Lindblad form is CCP). Lean ✓ Stmt *Every Lindblad form defines a conditionally completely positive map.*

Proof. ✓ Proof By Theorem 22.33, the Lindblad form equals a generator decomposition with ϕ CP. ☐

Definition 22.35 (Commutator form of Lindblad equation). Lean ✓ Stmt The *commutator form* of the Lindblad equation writes the generator as

$$L(\rho) = i[\rho, H] + \frac{1}{2}\sum_j ([L_j,\, \rho L_j^\dagger] + [L_j\rho,\, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Equation (7.22)].

Theorem 22.36 (Commutator form equals standard Lindblad form). Lean ✓ Stmt *The commutator form (Definition 22.35) defines the same linear map as the standard Lindblad form (Definition 22.31).*

Proof. ✓ Proof Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j\rho, L_j^\dagger]$ yields the standard dissipator terms. ☐

22.3.3 Characterization of conditional complete positivity

Theorem 22.37 (Conditional complete positivity implies projected Choi positivity). Lean ✓ Stmt *If L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

This is [Wol12, Proposition 7.2].

Proof. ✓ Proof Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P, so the projected Choi matrix reduces to the projected Choi matrix of ϕ, which is positive. ☐

Theorem 22.38 (Projected Choi positivity implies conditional complete positivity). Lean ✓ Stmt *If L is Hermiticity-preserving and $P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Proposition 7.2].*

Proof. ✓ Proof Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiolkowski isomorphism applied to Q gives the CP map ϕ, and $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ. ☐

22.3.4 Completely positive semigroups and conditionally completely positive generators

Theorem 22.39 (CP semigroups have CCP generators). Lean ✓ Stmt *If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof From $(e^{tL} \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 22.38 gives CCP. □

Theorem 22.40 (CCP generators yield CP semigroups). Lean ✓ Stmt *If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \qquad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL}, and the cone of CP maps is closed. □

Theorem 22.41 (CP semigroups iff CCP generators). Lean ✓ Stmt *e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.*

Proof. ✓ Proof Combine Theorems 22.39 and 22.40. □

22.3.5 Freedom in generator representation

Theorem 22.42 (Traceless Kraus operators exist). Lean ✓ Stmt *Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K'_j = K_j + c_j\mathbb{1}$ is traceless. This is part of [Wol12, Proposition 7.4].*

Proof. ✓ Proof Set $c_j = -\operatorname{tr}(K_j)/d$. □

Theorem 22.43 (Shift invariance of generators). Lean ✓ Stmt *If $K'_j = K_j + c_j\mathbb{1}$ and κ' is adjusted per [Wol12, Equation (7.19)], then (ϕ', κ') and (ϕ, κ) define the same generator.*

Proof. ✓ Proof Direct algebraic expansion. □

22.3.6 Uniqueness of the traceless Lindblad form

Definition 22.44 (Traceless Kraus operators). Lean ✓ Stmt A Lindblad form has *traceless Kraus operators* if $\operatorname{tr}(L_j) = 0$ for every j.

Theorem 22.45 (Uniqueness of the completely positive part). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Proposition 7.4, item 2].*

Proof. ✓ Proof Compare projected Choi matrices and use Choi injectivity. □

Theorem 22.46 (Uniqueness of the drift term modulo phase). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Proposition 7.4, item 2].*

Proof. ✓ Proof After identifying $\phi = \phi'$ (Theorem 22.45), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ. Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ. The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

Theorem 22.47 (Combined uniqueness of the traceless Lindblad form). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

Proof. ✓ Proof Combine Theorems 22.45 and 22.46. □

22.3.7 Trace-annihilation and trace preservation

Theorem 22.48 (Trace-annihilating generators yield trace-preserving semigroups). Lean ✓ Stmt
If L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.

Proof. ✓ Proof For fixed ρ, the function $f(t) = \operatorname{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \operatorname{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\operatorname{tr}(e^{tL}(\rho)) = f(t) = f(0) = \operatorname{tr}(\rho)$. □

Theorem 22.49 (Trace-preserving semigroups have trace-annihilating generators). Lean ✓ Stmt
If e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.

Proof. ✓ Proof Differentiate the identity $\operatorname{tr}(e^{tL}(\rho)) = \operatorname{tr}(\rho)$ at $t = 0$ from the right. □

22.3.8 The GKSL/Lindblad theorem

Definition 22.50 (GKSL generator). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 22.51 (A trace-constrained generator decomposition gives a GKSL generator). Lean ✓ Stmt *If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Theorem 7.1, Equation (7.20)].*

Proof. ✓ Proof CP follows from Theorem 22.40. The trace constraint implies trace-annihilation by Theorem 22.30, so Theorem 22.48 gives trace preservation. □

Theorem 22.52 (GKSL generators admit a trace-constrained generator decomposition). Lean ✓ Stmt *If L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$.

Proof. ✓ Proof Apply Theorem 22.39 to obtain a CCP decomposition. Then apply Theorem 22.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. □

Theorem 22.53 (GKSL iff conditional complete positivity and trace annihilation). Lean ✓ Stmt
L is a GKSL generator if and only if L is CCP and trace-annihilating.

Proof. ✓ Proof Forward: apply Theorem 22.39 to the CP part and Theorem 22.49 to the TP part. Reverse: combine Theorem 22.40 with Theorem 22.48. □

Theorem 22.54 (GKSL iff Lindblad form). Lean ✓ Stmt *L is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \right)$$

with $H = H^\dagger$. This is [Wol12, Theorem 7.1].

Proof. ✓ Proof Forward: apply Theorem 22.52 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 22.42 and rewrite κ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 22.33 gives the Lindblad formula. Reverse: Theorems 22.34 and 22.32 show that every Lindblad form satisfies the right-hand side of Theorem 22.53. □

22.3.9 Kossakowski matrix form

Definition 22.55 (Kossakowski form). Lean ✓ Stmt A *Kossakowski form* consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \tfrac{1}{2} \sum_{k,l} C_{lk}([F_k, \rho F_l^\dagger] + [F_k\rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

Theorem 22.56 (Kossakowski form iff Lindblad form). Lean ✓ Stmt *A linear map admits a Kossakowski form iff it admits a Lindblad form.*

Proof. ✓ Proof Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. □

22.4 Primitivity and irreducibility of QDS

22.4.1 Auxiliary spectral and semigroup lemmas

Definition 22.57 (Quantum dynamical semigroup). Lean ✓ Stmt A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every $t \geq 0$.

Theorem 22.58 (Semigroup power identity). Lean ✓ Stmt *For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.*

Theorem 22.59 (Eigenvector propagation under powers). Lean ✓ Stmt *If $E(v) = \mu v$, then $E^n(v) = \mu^n v$ for all $n \in \mathbb{N}$.*

Lemma 22.60 (Density matrices are nonzero). Lean ✓ Stmt *Every density matrix is nonzero.*

Theorem 22.61 (Compression invariance is stable under powers). Lean ✓ Stmt *If a corner compression is invariant under E, it remains invariant under every power E^n.*

Theorem 22.62 (Generator eigenvectors along the exponential semigroup). Lean ✓ Stmt *If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.*

Theorem 22.63 (Spectral mapping for semigroup generators). Lean ✓ Stmt *If μ is in the spectrum of L, then $e^{t\mu}$ is in the spectrum of e^{tL}.*

Theorem 22.64 (Root-of-unity rigidity for phases). Lean ✓ Stmt *If $e^{it\theta}$ is a root of unity for every $t > 0$, then $\theta = 0$.*

Theorem 22.65 (Peripheral generator eigenvalues are purely imaginary). Lean ✓ Stmt *Peripheral spectral points of the generator have vanishing real part.*

Theorem 22.66 (Peripheral cardinality bound). Lean ✓ Stmt *The number of peripheral eigenvalues is bounded by* $\dim(M_D(\mathbb{C}))$.

Theorem 22.67 (Bounded order under peripheral power-closure). Lean ✓ Stmt *If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding* $\dim(M_D(\mathbb{C}))$ *equals* 1.

Theorem 22.68 (Peripheral powers stay peripheral under irreducibility). Lean ✓ Stmt *For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.*

Theorem 22.69 (Continuous-linear-map power evaluation). Lean ✓ Stmt *Evaluating powers of a linear map agrees with powers in its continuous linear realization.*

Theorem 22.70 (Spectral-radius criterion from eigenvalue bounds). Lean ✓ Stmt *If all eigenvalues satisfy $|\lambda| < 1$, then the spectral radius is < 1.*

Theorem 22.71 (Primitive powers annihilate trace-zero matrices). Lean ✓ Stmt *If E is an irreducible primitive channel with faithful fixed density σ, then $E^n(X) \to 0$ for every matrix X with $\mathrm{tr}(X) = 0$.*

Proof. ✓ Proof Decompose E^n into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus < 1, so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component. □

Theorem 22.72 (Fixed points of an irreducible channel are trace multiples). Lean ✓ Stmt *If E is an irreducible channel with faithful fixed density σ, then every nonzero fixed point V of E satisfies $V = \mathrm{tr}(V)\,\sigma$.*

Proof. ✓ Proof Subtract the trace multiple $\mathrm{tr}(V)\,\sigma$ and apply the vanishing theorem for trace-zero fixed points of an irreducible channel. □

Corollary 22.73 (Nonzero fixed points have nonzero trace). Lean ✓ Stmt *Every nonzero fixed point of an irreducible channel has nonzero trace.*

Proof. ✓ Proof Otherwise the preceding theorem would force the fixed point to vanish. □

Theorem 22.74 (Peripheral eigenvectors become periodic fixed points). Lean ✓ Stmt *For every positive time t, a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue μ of T_t admits a nonzero eigenvector V and a positive integer p such that $T_t(V) = \mu V$ and $T_{pt}(V) = V$.*

Proof. ✓ Proof Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives $\mu^p = 1$, and the semigroup power identity turns the eigenvector equation for T_t into a fixed-point equation for T_{pt}. □

Theorem 22.75 (Peripheral eigenvectors with nonzero trace). Lean ✓ Stmt *For every positive time t, under the same hypotheses every peripheral eigenvalue of T_t has a nonzero eigenvector with nonzero trace.*

Proof. ✓ Proof Apply Theorem 22.74 and then use the previous corollary at the irreducible time pt. □

Theorem 22.76 (Trace-preserving eigenvectors force eigenvalue 1). Lean ✓ Stmt *If E is trace-preserving, $E(V) = \mu V$, and $\mathrm{tr}(V) \neq 0$, then $\mu = 1$.*

Proof. ✓ Proof Take traces in $E(V) = \mu V$ and use trace preservation. □

Theorem 22.77 (Peripheral collapse under irreducibility at all times). Lean ✓ Stmt *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

Proof. ✓ Proof Combine Theorem 22.75 with Theorem 22.76. □

Theorem 22.78 (Primitivity from irreducibility at all times). Lean ✓ Stmt *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

Proof. ✓ Proof The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 22.77 collapses the peripheral set to $\{1\}$. □

Remark 22.79 (Semigroup irreducible/primitive equivalence). ✓ Stmt The semigroup analogue of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 22.95 and 22.96).

Remark 22.80 (Reduction to a smaller time step for semigroups). ✓ Stmt Starting from one irreducible time t_0, we construct a positive time step $u = t_0/N$ with $T_{t_0} = (T_u)^N$, prove peripheral eigenvalue collapse at time u, and deduce primitivity of T_u.

Theorem 22.81 (The stationary density is fixed for all times). Lean ✓ Stmt *If $t_0 > 0$, T_{t_0} is irreducible, and σ is its unique fixed density matrix, then $T_u(\sigma) = \sigma$ for every $u \geq 0$.*

Proof. ✓ Proof By semigroup commutativity, $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$, and uniqueness of the fixed density at time t_0 identifies $T_u(\sigma)$ with σ. □

Theorem 22.82 (Fixed points at the irreducible time are trace multiples). Lean ✓ Stmt *If T_{t_0} is irreducible with faithful fixed density σ, then every fixed point V of T_{t_0} satisfies $V = \operatorname{tr}(V)\,\sigma$.*

Proof. ✓ Proof This is the fixed-point trace-scalar theorem for the irreducible channel T_{t_0}. □

Definition 22.83 (Residual slice index). Lean ✓ Stmt For $u \geq 0$ and $s > 0$, the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

Definition 22.84 (Residual slice time). Lean ✓ Stmt For $u \geq 0$ and $s > 0$, the residual slice time is the remainder

$$r_n = s\,\mathrm{fract}\left(\frac{nu}{s}\right).$$

Theorem 22.85 (Residual slice decomposition). Lean Lean ✓ Stmt *For $u \geq 0$ and $s > 0$, one has $r_n \in [0, s]$ and*

$$nu = m_n s + r_n.$$

Proof. ✓ Proof This is the floor-plus-fraction decomposition of nu/s, multiplied by s. □

Theorem 22.86 (A residual slice can vanish). Lean ✓ Stmt *Suppose $u \geq 0$ and $s > 0$, and that $T_s(\delta) = \delta$ and $T_{nu}(\delta) \to 0$ as $n \to \infty$. Then there exists $a \in [0, s]$ such that $T_a(\delta) = 0$.*

Proof. ✓ Proof The residual slice times lie in the compact interval $[0, s]$, so a subsequence converges to some $a \in [0, s]$. The residual decomposition rewrites $T_{nu}(\delta)$ in terms of $T_{r_n}(\delta)$, and continuity of the semigroup identifies the limit with $T_a(\delta)$. □

Theorem 22.87 (Reduced-time eigenvector has nonzero trace). Lean ✓ Stmt *Under the reduced-time hypotheses $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$, if μ is a peripheral eigenvalue of T_u, then T_u admits a μ-eigenvector with nonzero trace.*

Theorem 22.88 (Peripheral collapse at the reduced time step). Lean ✓ Stmt *Under the same hypotheses, every peripheral eigenvalue of T_u equals 1.*

Theorem 22.89 (Primitivity at the reduced time step). Lean ✓ Stmt *Under the same hypotheses, T_u is primitive.*

Theorem 22.90 (Irreducibility descends to a reduced time step). Lean ✓ Stmt *If $T_{t_0} = (T_u)^N$ and T_{t_0} is irreducible, then T_u is irreducible.*

Proof. ✓ Proof Any invariant compression for T_u remains invariant under powers, hence also for T_{t_0}; irreducibility of T_{t_0} rules this out. □

Theorem 22.91 (Existence of an irreducible reduced time step). Lean ✓ Stmt *If $t_0 > 0$, T_{t_0} is irreducible, and σ is fixed for all times, then there exists a positive time step u such that*

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

the slice T_u is a channel, T_u is irreducible, and $T_u(\sigma) = \sigma$.

Proof. ✓ Proof Take $u = t_0/N$ with $N = (\dim M_D(\mathbb{C}))!$. The semigroup power identity gives the factorization of T_{t_0}, channelity comes from the semigroup hypotheses, and Theorem 22.90 gives irreducibility of T_u. □

Theorem 22.92 (Fixed points of the primitive reduced time step are trace-scalars). Lean ✓ Stmt *If T_u is primitive and σ is the common fixed density, then every fixed point of any positive-time slice T_s has the form $\tau = \mathrm{tr}(\tau)\,\sigma$.*

Theorem 22.93 (Existence of a primitive reduced time step). Lean ✓ Stmt *If T_{t_0} is irreducible for some $t_0 > 0$, then there exists a positive reduced time step u such that T_u is primitive.*

Theorem 22.94 (Irreducibility propagates to all positive times). Lean ✓ Stmt *If T_{t_0} is irreducible for some $t_0 > 0$, then every positive-time slice T_s is irreducible.*

Proof. ✓ Proof Let σ be the unique faithful fixed density of T_{t_0}. By Theorem 22.81, it is fixed for all times, and Theorem 22.82 makes the fixed-point space of T_{t_0} one-dimensional. Choose a primitive reduced time step by Theorem 22.93; then the trace-scalar description of fixed points from Theorem 22.92 gives the uniqueness criterion for irreducibility at every positive time. □

Theorem 22.95 (An irreducible semigroup is primitive). Lean ✓ Stmt *Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If T_{t_0} is irreducible for some $t_0 > 0$, then T_t is primitive (and irreducible) for every $t > 0$. This is [Wol12, Proposition 7.5].*

Proof. ✓ Proof First apply Theorem 22.94 to propagate irreducibility from the single time t_0 to every positive time. Then apply Theorem 22.78 to conclude primitivity of every positive-time slice. □

Theorem 22.96 (Irreducibility iff primitivity for a quantum dynamical semigroup). Lean ✓ Stmt *For a quantum dynamical semigroup $T_t = e^{tL}$:*

$$(\exists\, t_0 > 0,\ T_{t_0} \text{ irreducible}) \iff (\forall\, t > 0,\ T_t \text{ primitive and irreducible}).$$

This is [Wol12, Proposition 7.5].

Proof. ✓ Proof Forward: apply Theorem 22.95. Reverse: take $t_0 = 1$ (or any positive time); the right-hand side already includes irreducibility of T_t for every $t > 0$. □

22.5 Kernel of the adjoint Liouvillian

Definition 22.97 (Faithful stationary state). Lean ✓ Stmt A generator L has a faithful stationary state if there exists a positive definite density matrix ρ_0 with $L(\rho_0) = 0$.

Definition 22.98 (Adjoint dissipator). Lean ✓ Stmt For a Lindblad operator L_j, the adjoint dissipator on observables is

$$\mathcal{D}^*_{L_j}(A) = L_j^\dagger A L_j - \tfrac{1}{2} L_j^\dagger L_j A - \tfrac{1}{2} A L_j^\dagger L_j.$$

Definition 22.99 (Adjoint generator). Lean ✓ Stmt The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}^*_{L_j}(A).$$

Theorem 22.100 (Trace-pairing characterization of the adjoint generator). Lean ✓ Stmt *For every density matrix ρ and every observable A,*

$$\operatorname{tr}(\rho\, L^*(A)) = \operatorname{tr}(L(\rho)\, A).$$

Proof. ✓ Proof Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term. □

Definition 22.101 (Commutant of the Lindblad data). Lean ✓ Stmt The commutant of a Lindblad form is the set of matrices commuting with H, with every Lindblad operator L_j, and with every adjoint $L_j^\dagger$.

Definition 22.102 (Adjoint kernel). Lean ✓ Stmt The adjoint kernel is the set of observables A satisfying $L^*(A) = 0$.

Theorem 22.103 (Commutant lies in the adjoint kernel). Lean Lean ✓ Stmt *If an observable commutes with H, with every L_j, and with every $L_j^\dagger$, then it lies in* $\ker(L^*)$.

Proof. ✓ Proof The Hamiltonian commutator term vanishes, and each adjoint dissipator $\mathcal{D}^*_{L_j}(A)$ is zero under the commutation hypotheses. □

Theorem 22.104 (Adjoint kernel lies in the commutant). Lean ✓ Stmt *Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by $\operatorname{tr}(\rho\, L^*(A)) = \operatorname{tr}(L(\rho)\, A)$. If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any A:*

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \ \forall j.$$

That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$. This is [Wol12, Theorem 7.2].

Proof. ✓ Proof From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\operatorname{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j. Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. □

Theorem 22.105 (Adjoint kernel equals the commutant). Lean ✓ Stmt *Under the same hypotheses as Theorem 22.104,*

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Theorem 7.2].

Proof. ✓ Proof The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 22.104. □

22.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Proposition 7.6] are as follows.

Definition 22.106 (Nontrivial projection). Lean ✓ Stmt A projection is nontrivial if it is orthogonal and neither 0 nor $\mathbb{1}$.

Definition 22.107 (Rank-deficient fixed density). Lean ✓ Stmt Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 22.108 (Rank-deficient kernel element). Lean ✓ Stmt Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 22.109 (Invariant compression). Lean ✓ Stmt Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 22.110 (Block-upper-triangular Lindblad form). Lean ✓ Stmt Condition (4): there exists a nontrivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ for all j, where $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$.

Definition 22.111 (Generator-preserved compression). Lean ✓ Stmt For a projection P, the generator preserves the compression $PM_d(\mathbb{C})P$ if

$$P\,L(PXP)\,P = L(PXP)$$

for every $X \in M_d(\mathbb{C})$.

Definition 22.112 (Reducible quantum dynamical semigroup). Lean ✓ Stmt A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

Theorem 22.113 (Semigroup invariance implies generator invariance). Lean ✓ Stmt *If the semigroup preserves $PM_d(\mathbb{C})P$ for all nonnegative times, then the generator preserves the same compression.*

Proof. ✓ Proof Differentiate the identity $Pe^{tL}(PXP)P = e^{tL}(PXP)$ at $t = 0$. □

Theorem 22.114 (Generator invariance implies semigroup invariance). Lean ✓ Stmt *If the generator preserves $PM_d(\mathbb{C})P$, then so does the exponential semigroup e^{tL} for every $t \geq 0$.*

Proof. ✓ Proof Apply the compression map termwise to the exponential series. Every iterate of L preserves the compression, so the whole series does as well. □

Theorem 22.115 (Reducibility as a generator-level criterion). Lean ✓ Stmt *A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.*

Proof. ✓ Proof Combine Theorems 22.113 and 22.114. □

Theorem 22.116 (Rank-deficient fixed densities and kernel elements). Lean ✓ Stmt *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in* $\ker L$. *This is [Wol12, Proposition 7.6, (1)↔(2)].*

Proof. ✓ Proof The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator–semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. □

Theorem 22.117 (A rank-deficient fixed density yields an invariant compression). Lean ✓ Stmt *If L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Proposition 7.6, (1)→(3)].*

Proof. ✓ Proof Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0. Since ρ_0 is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. □

Theorem 22.118 (Invariant compression yields block-upper-triangular Lindblad data). Lean ✓ Stmt *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

Proof. ✓ Proof First pass from semigroup invariance to generator invariance by Theorem 22.113. The generator-level vanishing condition forces $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ through the sum-of-squares identity. □

Theorem 22.119 (Block-upper-triangular Lindblad data yield invariant compression). Lean ✓ Stmt *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

Proof. ✓ Proof The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 22.114. □

Theorem 22.120 (Invariant compression and triangular Lindblad data). Lean Lean ✓ Stmt *For a GKSL generator L, the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Proposition 7.6, (3)↔(4)].*

Proof. ✓ Proof Combine Theorems 22.119 and 22.118. □

Theorem 22.121 (Triangular Lindblad data give deficient kernels). Lean ✓ Stmt *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P, then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Proposition 7.6, (4)→(2)].*

Proof. ✓ Proof The block-upper-triangular structure implies L preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix ρ_m supported in $PM_d(\mathbb{C})P$ for each m. Passing to a subsequential limit gives a density matrix ρ_0 with $P\rho_0P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. □

Theorem 22.122 (Equivalent formulations of reducibility). Lean ✓ Stmt *For a GKSL generator $L : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the four conditions*

1. *rank-deficient fixed density,*
2. *rank-deficient kernel element,*
3. *invariant compression, and*
4. *block-upper-triangular Lindblad form*

are equivalent. This is [Wol12, Proposition 7.6].

Proof. ✓ Proof The cycle (1)↔(2), (1)→(3), (3)↔(4), and (4)→(2) closes the equivalence chain. □

22.6.1 Sufficient conditions for non-reducibility

Definition 22.123 (Lindblad span). Lean ✓ Stmt The $\mathbb{C}$-linear span of the Lindblad operators $\{L_j\}$.

Definition 22.124 (Hermitian-closed Lindblad span). Lean ✓ Stmt The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$-subspace of $M_d(\mathbb{C})$.

Definition 22.125 (Trivial Lindblad commutant). Lean ✓ Stmt The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_d(\mathbb{C})$.

Definition 22.126 (Kossakowski rank). Lean ✓ Stmt The minimal number of Lindblad operators across all GKSL representations of L, which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

Theorem 22.127 (κ block vanishing from generator compression). Lean ✓ Stmt *Let $(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_d(\mathbb{C})P$. If $(1-P)L_jP = 0$ for every j, then $(1-P)\kappa P = 0$.*

Proof. ✓ Proof From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(1-P)L(P) = 0$: the $(1-P)\varphi(P)$ term vanishes because each $(1-P)L_jP = 0$, and the $(1-P)(P\kappa^\dagger)$ term vanishes because $(1-P)P = 0$. What remains is $(1-P)\kappa P = 0$. □

Theorem 22.128 (Full algebra generation forbids block-upper-triangular form). Lean ✓ Stmt *Let $(H, \{L_j\})$ be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. ✓ Proof Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. By Theorem 22.127, $(1-P)\kappa P = 0$. Induction on elements of the subalgebra shows $(1-P)AP = 0$ for every generator. Since the generated algebra is all of $M_d(\mathbb{C})$, every matrix satisfies $(1-P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ — a contradiction. □

Theorem 22.129 (Hermitian span with trivial commutant forbids triangular form). Lean ✓ Stmt *If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. ✓ Proof Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. The block vanishing $(1-P)L_jP = 0$ and Hermitian closure give $PL_j(1-P) = 0$ as well, so $[P, L_j] = 0$ for all j. But then P lies in the commutant of the Lindblad span, contradicting triviality. □

Theorem 22.130 (Large Kossakowski rank forbids block-upper-triangular form). Lean ✓ Stmt *If the Kossakowski rank satisfies*

$$\operatorname{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

Proof. ✓ Proof A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most $d^2 - d$. The formal rank minimization argument then contradicts the large-rank bound. □

Theorem 22.131 (No block-upper-triangular Lindblad form implies non-reducibility). Lean ✓ Stmt *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

Proof. ✓ Proof A reducible semigroup would have an invariant compression, and Theorem 22.118 would then produce a block-upper-triangular Lindblad form. □

Theorem 22.132 (Full algebra generation implies non-reducibility). Lean ✓ Stmt *If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the QDS is not reducible. This is [Wol12, Corollary 7.2, condition 1].*

Proof. ✓ Proof By Theorem 22.128, no block-upper-triangular decomposition exists. Then apply Theorem 22.131. □

Theorem 22.133 (Hermitian span and trivial commutant imply non-reducibility). Lean ✓ Stmt *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Corollary 7.2, condition 2].*

Proof. ✓ Proof By Theorem 22.129, no block-upper-triangular decomposition exists. Then apply Theorem 22.131. □

Theorem 22.134 (Large Kossakowski rank implies non-reducibility). Lean ✓ Stmt *If* $\mathrm{rank}(C) > d^2 - d$*, then the QDS is not reducible. This is [Wol12, Corollary 7.2, condition 3].*

Proof. ✓ Proof The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 22.130. Then apply Theorem 22.131. □

Chapter 23

Operator Convexity and Jensen Inequalities

This chapter collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis ([Bha97, Chapter V]; [Wol12, Theorem 5.1]; Hansen–Pedersen's Jensen inequality [HP82]; and Lieb's concavity theorem [Lie73]). They are provided here as cited inputs, since the corresponding matrix-analysis proofs for real powers $x \mapsto x^p$ on the Loewner order and for the matrix logarithm are not developed elsewhere in this document.

23.1 Diagonal Jensen inequality

Lemma 23.1 (Diagonal Jensen inequality)**.** Lean ✓ Stmt *Let $f : \mathbb{R} \to \mathbb{R}$ be convex on $[0, \infty)$, let A be a positive semidefinite matrix on a finite-dimensional space indexed by n, and let $v \in \mathbb{C}^n$ satisfy $\langle v, v\rangle = 1$. Then*

$$f(\mathfrak{R}\langle v, Av\rangle) \le \mathfrak{R}\langle v, f(A)\, v\rangle,$$

where $f(A)$ is defined by the Hermitian continuous functional calculus.

Proof. ✓ Proof Write $A = U\, \mathrm{diag}(\mu)\, U^*$ by the spectral theorem and set $w = U^* v$. Since U is unitary, $p_i := |w_i|^2$ satisfies $\sum_i p_i = \|v\|^2 = 1$, so (p_i) is a probability distribution on n. The eigenvalues μ_i lie in $[0, \infty)$ because A is positive semidefinite. A direct computation yields $\mathfrak{R}\langle v, Av\rangle = \sum_i p_i \mu_i$ and $\mathfrak{R}\langle v, f(A)v\rangle = \sum_i p_i f(\mu_i)$, so the scalar Jensen inequality applied to the weights (p_i) and points (μ_i) gives the conclusion. See [Bha97, Chapter V]. □

23.2 Trace concavity and convexity of matrix powers

Theorem 23.2 (Trace concavity of real powers)**.** Lean Lean ✓ Stmt *For $p \in [0, 1]$, PSD matrices A_1, A_2, and $t \in [0, 1]$:*

$$t\, \mathfrak{R}\, \mathrm{tr}(A_1^p) + (1-t)\mathfrak{R}\, \mathrm{tr}(A_2^p) \le \mathfrak{R}\, \mathrm{tr}((tA_1 + (1-t)A_2)^p).$$

Follows from operator concavity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Let $A = tA_1 + (1-t)A_2$ and let $\{\psi_j\}$ with eigenvalues $\mu_j \geq 0$ be an orthonormal eigenbasis of A. A direct computation gives $\Re \operatorname{tr}(A^p) = \sum_j \mu_j^p$ and $\mu_j = t\, a_j + (1-t)\, b_j$ where $a_j = \Re\langle \psi_j, A_1\psi_j\rangle$, $b_j = \Re\langle \psi_j, A_2\psi_j\rangle$. Scalar concavity of $x \mapsto x^p$ on $[0,\infty)$ yields $\mu_j^p \geq t\, a_j^p + (1-t)\, b_j^p$, and the diagonal Jensen inequality (Lemma 23.1, concave form) gives $a_j^p \leq \Re\langle \psi_j, A_1^p \psi_j\rangle$ (and similarly for b_j). Summing over j and using $\sum_j \Re\langle \psi_j, M\psi_j\rangle = \Re \operatorname{tr} M$ yields the conclusion. See [Bha97, Chapter V]. □

Theorem 23.3 (Trace convexity of real powers). Lean Lean ✓ Stmt *For $p \in [1,2]$, PSD matrices A_1, A_2, and $t \in [0,1]$:*

$$\Re \operatorname{tr}((tA_1 + (1-t)A_2)^p) \leq t\,\Re \operatorname{tr}(A_1^p) + (1-t)\Re \operatorname{tr}(A_2^p).$$

Follows from operator convexity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Identical eigenbasis reduction as in Theorem 23.2, with scalar convexity (rather than concavity) of $x \mapsto x^p$ on $[0,\infty)$ for $p \geq 1$ and the convex form of the diagonal Jensen inequality. □

23.3 Operator Jensen inequality for positive maps

Definition 23.4 (Finite-POVM dilation construction). Lean Lean ✓ Stmt For a finite family of matrices $\{C_i\}_{i\in\iota}$ and a defect block S, one forms an isometric dilation whose rows are the adjoints of the matrices C_i together with the adjoint of S. One also forms the scalar block-diagonal matrix whose ι-blocks carry prescribed weights and whose defect block carries a single scalar.

Lemma 23.5 (Compression auxiliary for Jensen). Lean ✓ Stmt *Let Y be positive definite and let W be an isometry. Then*

$$(W^*YW)^{-1} \leq W^*Y^{-1}W.$$

Proof. ✓ Proof Apply the Schur-complement criterion to the block matrix $\begin{pmatrix} Y^{-1} & W \\ W^* & W^*YW \end{pmatrix}$. The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by W gives the stated inequality. □

Lemma 23.6 (Finite-POVM compression identities). Lean Lean Lean Lean Lean Lean ✓ Stmt *If the defect block satisfies*

$$SS^* = \mathbb{1} - \sum_{i\in\iota} C_i C_i^*,$$

then the dilation is an isometry, compressing the weighted scalar block-diagonal matrix gives

$$\sum_{i\in\iota} w_i\, C_i C_i^* + t\, SS^*,$$

the defect relation rewrites as

$$\sum_{i\in\iota} C_i C_i^* + SS^* = \mathbb{1},$$

and strict positivity of all weights together with $t > 0$ implies positive definiteness of the scalar block-diagonal matrix. If all weights are nonzero and the defect scalar is nonzero, the inverse diagonal has reciprocal weights, and compressing this inverse gives

$$\sum_{i\in\iota} w_i^{-1}\, C_i C_i^* + t^{-1}\, SS^*.$$

Proof. ✓ Proof Expand the block matrix multiplication entrywise. The defect identity gives the isometry relation after summing the ι-blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positive definiteness uses strict positivity of every diagonal entry, while the inverse formula uses the nonvanishing of every diagonal entry. □

Lemma 23.7 (Finite-POVM resolvent inequality). Lean ✓ Stmt *Let $w_i \geq 0$ and $t > 0$. If the defect block satisfies*

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*,$$

then

$$\left(\sum_{i \in \iota} w_i \, C_i C_i^* + t \, \mathbb{1} \right)^{-1} \leq \sum_{i \in \iota} (w_i + t)^{-1} \, C_i C_i^* + t^{-1} \, SS^*.$$

Proof. ✓ Proof Apply the compression inequality to the positive definite scalar block-diagonal matrix with entries $w_i + t$ and defect entry t. The compression identities rewrite the compressed matrix as $\sum_i w_i C_i C_i^* + t\mathbb{1}$ and its inverse compression as the displayed upper bound. □

Theorem 23.8 (Jensen for concave real powers). Lean Not ready *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [0, 1]$:*

$$T(A^p) \leq (TA)^p.$$

This is [Wol12, Theorem 5.1] for concave $f(x) = x^p$.

Theorem 23.9 (Map form of Jensen for concave real powers). Lean ✓ Stmt *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [0, 1]$, the inequality $T(A^p) \leq (TA)^p$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

Theorem 23.10 (Jensen for convex real powers). Lean Not ready *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [1, 2]$:*

$$(TA)^p \leq T(A^p).$$

This is [Wol12, Theorem 5.1] for convex $f(x) = x^p$.

Theorem 23.11 (Map form of Jensen for convex real powers). Lean ✓ Stmt *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [1, 2]$, the inequality $(TA)^p \leq T(A^p)$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

Theorem 23.12 (Jensen for concave log). Lean Not ready *For a positive* unital *map T and positive-definite A:*

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Theorem 5.1] for $f = \log$. Requires unitality ($T(\mathbb{1}) = \mathbb{1}$), not merely subunitality.

Theorem 23.13 (Map form of Jensen for the concave logarithm). Lean ✓ Stmt *For a positive unital map T and a positive-definite matrix A, the inequality $T(\log A) \leq \log(TA)$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

23.4 Lieb concavity theorem

Theorem 23.14 (Lieb concavity). Lean Not ready *For $s \in [0,1]$, any matrix K, and positive-definite matrices A_1, A_2, B_1, B_2, the map $(A,B) \mapsto \operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave:*

$$t\,\Re\operatorname{tr}(K^\dagger A_1^s K B_1^{1-s}) + (1-t)\Re\operatorname{tr}(K^\dagger A_2^s K B_2^{1-s}) \le \Re\operatorname{tr}(K^\dagger (tA_1+(1-t)A_2)^s K (tB_1+(1-t)B_2)^{1-s}).$$

See [Lie73, And79].

Theorem 23.15 (Map form of Lieb concavity). Lean ✓ Stmt *For $s \in [0,1]$, any matrix K, and positive-definite matrices A_1, A_2, B_1, B_2, the function $(A,B) \mapsto \Re\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave, with the same inequality as Theorem 23.14.*

Proof. ✓ Proof Direct application of the axiom. □

Corollary 23.16 (Lieb concavity, $K = \mathbb{1}$ case). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, and positive-definite matrices A_1, A_2, B_1, B_2, the map $(A,B) \mapsto \Re\operatorname{tr}(A^s B^{1-s})$ is jointly concave:*

$$t\,\Re\operatorname{tr}(A_1^s B_1^{1-s}) + (1-t)\Re\operatorname{tr}(A_2^s B_2^{1-s}) \le \Re\operatorname{tr}((tA_1 + (1-t)A_2)^s (tB_1 + (1-t)B_2)^{1-s}).$$

Obtained from Theorem 23.15 by taking $K = \mathbb{1}$.

Proof. ✓ Proof Specialize Theorem 23.15 at $K = \mathbb{1}$ and simplify $\mathbb{1}^\dagger = \mathbb{1}$ together with $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$. □

Corollary 23.17 (Concavity of Lieb's map in the first argument). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, matrix K, and positive-definite matrices A_1, A_2, B, the map $A \mapsto \Re\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:*

$$t\,\Re\operatorname{tr}(K^\dagger A_1^s K B^{1-s}) + (1-t)\Re\operatorname{tr}(K^\dagger A_2^s K B^{1-s}) \le \Re\operatorname{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K B^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 23.15 at $B_1 = B_2 = B$, so that the convex combination collapses: $tB + (1-t)B = B$. □

Corollary 23.18 (Concavity of Lieb's map in the second argument). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, matrix K, and positive-definite matrices A, B_1, B_2, the map $B \mapsto \Re\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:*

$$t\,\Re\operatorname{tr}(K^\dagger A^s K B_1^{1-s}) + (1-t)\Re\operatorname{tr}(K^\dagger A^s K B_2^{1-s}) \le \Re\operatorname{tr}(K^\dagger A^s K (tB_1 + (1-t)B_2)^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 23.15 at $A_1 = A_2 = A$, so that the convex combination collapses: $tA + (1-t)A = A$. □

Chapter 24

Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

24.1 Von Neumann entropy

Definition 24.1 (Von Neumann entropy). Lean ✓ Stmt For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \ldots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = -\sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where $0 \log 0 := 0$.

Theorem 24.2 (Entropy is non-negative). Lean ✓ Stmt *For any density matrix ρ, $S(\rho) \geq 0$.*

Proof. ✓ Proof Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. □

Lemma 24.3 (Eigenvalue sum). Lean ✓ Stmt *The eigenvalues of a density matrix sum to 1.*

Proof. ✓ Proof Follows from $\mathrm{tr}(\rho) = \sum_i \lambda_i = 1$. □

Theorem 24.4 (Eigenvalue bound). Lean ✓ Stmt *Each eigenvalue of a density matrix lies in $[0, 1]$.*

Proof. ✓ Proof Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. □

Theorem 24.5 (Entropy upper bound). Lean ✓ Stmt *For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$,*

$$S(\rho) \leq \log D.$$

Proof. ✓ Proof By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. □

24.2 Tripartite partial traces

Definition 24.6 (Partial trace over A). Lean ✓ Stmt For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\mathrm{tr}_A\, \rho_{ABC})_{(b_1,c_1)(b_2,c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a,b_1,c_1)(a,b_2,c_2)}.$$

Definition 24.7 (Partial trace over C). Lean ✓ Stmt The partial trace over C:

$$(\mathrm{tr}_C\, \rho_{ABC})_{(a_1,b_1)(a_2,b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1,b_1,c)(a_2,b_2,c)}.$$

Definition 24.8 (Partial trace over AC). Lean ✓ Stmt The partial trace over A and C:

$$(\mathrm{tr}_{AC}\, \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a,b_1,c)(a,b_2,c)}.$$

Lemma 24.9 (Partial trace preserves Hermiticity). *Lean* *✓ Stmt* *If ρ_{ABC} is Hermitian, then $\mathrm{tr}_A(\rho_{ABC})$, $\mathrm{tr}_C(\rho_{ABC})$, and $\mathrm{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\mathrm{tr}_A(\rho_{AB})$ and $\mathrm{tr}_B(\rho_{AB})$.*

Proof. ✓ Proof Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. □

24.3 Strong subadditivity

Theorem 24.10 (Strong subadditivity). *Lean* *Not ready* *For a tripartite density matrix ρ_{ABC},*

$$S(\rho_{ABC}) + S(\rho_B) \le S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 24.11 (SSA equality). Lean ✓ Stmt A tripartite density matrix ρ_{ABC} satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 24.12 (Quantum Markov decomposition). Lean ✓ Stmt A *Hayashi Markov decomposition* of a tripartite state ρ_{ABC} consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on B, a probability vector $(p_j)_j$, and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j\, \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

The terminology follows Hayashi's presentation of quantum Markov structure [Hay06]; the block decomposition used by the MPDO argument is the structure theorem of [HJPW04].

Theorem 24.13 (Hayashi SSA-equality characterization). Lean Not ready *For a tripartite density matrix ρ_{ABC},*

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B.

This equality criterion is cited here as an input for the later MPDO arguments. It is the equality criterion needed by Appendix C of arXiv:1606.00608. The equality conditions are reviewed in [Hay06, Rus02]; the structural block-decomposition formulation used for MPDOs appears in [HJPW04].

24.4 Mutual information

Definition 24.14 (Mutual information). Lean ✓ Stmt The *quantum mutual information* of a bipartite state ρ_{AB} is

$$I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

Theorem 24.15 (Mutual information is non-negative). Lean ✓ Stmt *For any bipartite density matrix ρ_{AB}, $I(A{:}B) \geq 0$.*

Proof. ✓ Proof Apply strong subadditivity (Theorem 24.10) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A{:}C) \geq 0$. □

24.5 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

Definition 24.16 (Entropy formulation of von Neumann entropy). Lean ✓ Stmt This formulation has the same value $S(\rho) = -\sum_i \lambda_i \log \lambda_i$ as in Definition 24.1.

Theorem 24.17 (Entropy formulation of strong subadditivity). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,*

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 24.10. The underlying axiom's proof in [LR73] is deferred to the umbrella entropy issue.

Definition 24.18 (Entropy formulation of quantum Markov decomposition). Lean ✓ Stmt This is the entropy formulation of the Hayashi Markov decomposition from Definition 24.12.

Theorem 24.19 (SSA equality iff quantum Markov decomposition). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC}, equality in strong subadditivity holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B.*

This formulation introduces no new axiom: it is the same equality criterion as Theorem 24.13.

Definition 24.20 (Entropy formulation of mutual information). Lean ✓ Stmt This formulation has the same value $I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ as in Definition 24.14.

Theorem 24.21 (Subadditivity from trivial-middle SSA). Lean ✓ Stmt *For a tripartite density matrix ρ_{ABC} with $\dim B = 1$,*

$$S(\rho_{ABC}) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Proof. ✓ Proof Apply Theorem 24.17 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 24.27, hence its entropy vanishes by Theorem 24.28. □

24.6 Mutual information: monotonicity and area-law bound

The two inequalities below are the downstream MPDO-facing consequences of the entropy inequalities in this chapter. The monotonicity inequality is the strong-subadditivity content underlying the MPDO mutual-information monotonicity $I_L \leq I_{L+1}$ (arXiv:1606.00608, Proposition C.1); the elementary area-law bound is the single-site entropy bound underlying the MPDO area-law bound $I_L \leq 4 \log D$ (arXiv:1606.00608, cited from the Wolf area-law bound). Both inequalities follow directly from strong subadditivity (taken as an axiom) and the single-system $S(\rho) \leq \log D$ bound; neither introduces a new axiom.

Theorem 24.22 (Entropy nonnegativity for finite index sets). Lean ✓ Stmt *For any PSD Hermitian matrix ρ with $\mathrm{tr}(\rho) = 1$ on an arbitrary finite index set, $S(\rho) \geq 0$. This is the analogue of Theorem 24.2 for arbitrary finite index sets: it does not require the index set to be $\{0, \dots, D-1\}$, so it applies to bipartite density matrices on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$.*

Proof. ✓ Proof The eigenvalues of a PSD matrix are non-negative, and eigenvalues of a trace-1 Hermitian matrix with non-negative eigenvalues are bounded above by 1 (each single eigenvalue is at most the total sum). Apply $x \log x \leq 0$ on $[0, 1]$ pointwise and sum. □

Theorem 24.23 (Mutual information is monotone under enlargement). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,*

$$I(A{:}B)_{\mathrm{tr}_C \rho_{ABC}} \;\leq\; I(A{:}BC)_{\rho_{ABC}},$$

where the left-hand side is the bipartite mutual information of the reduced state $\rho_{AB} = \mathrm{tr}_C(\rho_{ABC})$ and the right-hand side is evaluated by expanding $I(A{:}BC) = S(\rho_A) + S(\rho_{BC}) - S(\rho_{ABC})$.

Proof. ✓ Proof Expanding both sides in entropy form and cancelling the common $S(\rho_A)$ term, the inequality reduces to strong subadditivity $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$. The bipartite B-reduced state of ρ_{AB} matches the tripartite B-reduced state $\mathrm{tr}_{AC}(\rho_{ABC})$, ensuring the $S(\rho_B)$ term is the one supplied by Theorem 24.17. □

Theorem 24.24 (Elementary area-law bound on mutual information). Lean ✓ Stmt *For a bipartite density matrix ρ_{AB} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with $d_A, d_B \geq 1$ whose single-system reduced states are obtained by partial trace,*

$$I(A{:}B) \;\leq\; \log d_A + \log d_B.$$

Proof. ✓ Proof The reduced states are density matrices because partial trace preserves positivity and trace. Combine $S(\rho_A) \leq \log d_A$ and $S(\rho_B) \leq \log d_B$ (Theorem 24.5) with $S(\rho_{AB}) \geq 0$ (Theorem 24.22). □

24.7 Trivial-factor corollaries

The partial traces and von Neumann entropy are introduced in Chapter 24. This supplement collects elementary trace-preservation identities and the dimension-1 entropy bound that follow directly from those definitions. They are listed as separate results because each proof requires only unfolding definitions and re-indexing finite sums.

Theorem 24.25 (Partial trace over A preserves the full trace). Lean ✓ Stmt *For any tripartite matrix ρ_{ABC},*

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_A(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions,

$$\mathrm{tr}(\mathrm{tr}_A(\rho_{ABC})) = \sum_{b,c} \sum_{a} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

Theorem 24.26 (Partial trace over C preserves the full trace). Lean ✓ Stmt *For any tripartite matrix ρ_{ABC},*

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_C(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions,

$$\mathrm{tr}(\mathrm{tr}_C(\rho_{ABC})) = \sum_{a,b} \sum_{c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

Theorem 24.27 (Partial trace over AC preserves the full trace). Lean ✓ Stmt *For any tripartite matrix ρ_{ABC},*

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions and interchanging the outer sums,

$$\mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})) = \sum_{b} \sum_{a,c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

Theorem 24.28 (Entropy vanishes in dimension 1). Lean ✓ Stmt *If $\rho \in M_1(\mathbb{C})$ is Hermitian and $\mathrm{tr}(\rho) = 1$, then*

$$S(\rho) = 0.$$

Proof. ✓ Proof The unique eigenvalue equals the trace, hence equals 1, and $-1 \cdot \log 1 = 0$:

$$\lambda_0 = \mathrm{tr}(\rho) = 1, \qquad S(\rho) = -\lambda_0 \log \lambda_0 = -1 \cdot \log 1 = 0.$$

□

Chapter 25

Matrix Product Operators and Density Operators

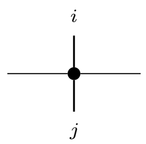
Fix a physical dimension d and a bond dimension D. An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

25.1 MPO tensors

Definition 25.1 (MPO tensor). Lean ✓ Stmt An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \qquad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index i and a bra index j. Diagrammatically,



The black node denotes the tensor M, the horizontal legs are virtual, the upper leg is the ket index i, and the lower leg is the bra index j.

Definition 25.2 (Doubled-index MPS view). Lean ✓ Stmt By combining the pair (i, j) into a single physical index in $\{0, \dots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension d^2 and bond dimension D:

$$M^{\mathrm{MPS}}_{(i,j)} = M^{ij},$$

identifying $\{0, \dots, d-1\} \times \{0, \dots, d-1\}$ with $\{0, \dots, d^2-1\}$ via the product encoding.

Definition 25.3 (Word evaluation). Lean ✓ Stmt For words $u = (i_1, \dots, i_N)$ and $v = (j_1, \dots, j_N)$ in $\{0, \dots, d-1\}^N$, the *word evaluation* is

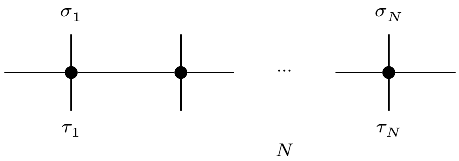
$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \cdots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to $\mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

Definition 25.4 (MPO operator family). Lean ✓ Stmt The operator generated by M on N sites is the matrix $\rho^{(N)}(M)$ indexed by configurations $\sigma, \tau \in \{0, \dots, d-1\}^N$ and given by

$$\rho^{(N)}(M)_{\sigma,\tau} = \mathrm{tr}(M^{\sigma,\tau}).$$

Diagrammatically,



each black node denotes the same local tensor M, and the matrix entry $\rho^{(N)}(M)_{\sigma,\tau}$ is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

Definition 25.5 (Hermitian MPO tensor). Lean ✓ Stmt An MPO tensor is *Hermitian* if, for all $i, j \in \{0, \dots, d-1\}$,

$$M^{ij} = (M^{ji})^\dagger$$

25.2 Transfer maps

Definition 25.6 (MPO transfer map). Lean ✓ Stmt The *transfer map* associated to M is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

Lemma 25.7 (Identification with the doubled-index MPS transfer map). Lean ✓ Stmt *The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view.*

Proof. ✓ Proof After identifying the single physical index of the doubled tensor with a pair (i, j), the single sum defining the MPS transfer map is exactly the double sum defining $\mathcal{E}_M$. □

Theorem 25.8 (Positivity of the MPO transfer map). Lean ✓ Stmt *If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.*

Proof. ✓ Proof By Lemma 25.7, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. □

25.3 Zero correlation length

Definition 25.9 (Zero correlation length for MPO tensors). Lean ✓ Stmt An MPO tensor M has *zero correlation length* when its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

This is the mixed-state analogue of the renormalization fixed-point condition for MPS tensors; see [CPGSV17a, definition of MPDO zero correlation length, lines 736–741] and [CPGSV21, Section II.E.2, lines 937–939].

Theorem 25.10 (MPO ZCL iff doubled-index MPS RFP). Lean ✓ Stmt *An MPO tensor has zero correlation length if and only if its doubled-index MPS view is a renormalization fixed point.*

Proof. ✓ Proof By Lemma 25.7, the transfer map of M agrees with the transfer map of its doubled-index MPS view. Thus $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ holds exactly when the transfer map of that doubled-index MPS tensor is idempotent, which is the RFP condition. □

25.4 MPDOs and LPDOs

Definition 25.11 (MPDO). Lean ✓ Stmt An MPO tensor is an *MPDO* if, for every $N \in \mathbb{N}$,

$$\rho^{(N)}(M) \geq 0$$

This is the global positivity condition of [CPGSV17a, Section 4.1].

Definition 25.12 (LPDO). Lean ✓ Stmt An MPO tensor is an *LPDO* if there exist a Kraus dimension d_K, an inner bond dimension D', a bond-space identification $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ (equivalently, D and $(D')^2$ have the same cardinality), and matrices $A^{(i,k)} \in M_{D'}(\mathbb{C})$ such that, for all physical indices i, j,

$$M^{ij} = \Big(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \Big)_{e,e}$$

where $\otimes$ denotes the Kronecker product, $\overline{(\cdot)}$ denotes entrywise complex conjugation, and $(\cdot)_{e,e}$ denotes reindexing along e back to $\{0, \dots, D-1\}$. See [CPGSV17a, Section 4.3].

Theorem 25.13 (LPDOs are MPDOs). *Lean* *✓ Stmt* *Every LPDO tensor is an MPDO.*

Proof. ✓ Proof By the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, the N-site product decomposes as a sum of Kronecker products after transporting along the bond-space identification e:

$$M^{\sigma_1\tau_1} \cdots M^{\sigma_N\tau_N} = \big(\sum_\kappa P_\kappa \otimes \overline{Q_\kappa}\big)\big|_e,$$

where $P_\kappa = A^{(\sigma_1,\kappa_1)} \cdots A^{(\sigma_N,\kappa_N)}$ and $Q_\kappa = A^{(\tau_1,\kappa_1)} \cdots A^{(\tau_N,\kappa_N)}$. Since the trace is invariant under the reindexing by e, we apply $\mathrm{tr}(X \otimes Y) = \mathrm{tr}(X)\,\mathrm{tr}(Y)$ to obtain

$$\rho^{(N)}(M)_{\sigma,\tau} = \sum_\kappa \mathrm{tr}(P_\kappa)\,\overline{\mathrm{tr}(Q_\kappa)} = \sum_\kappa \psi_\kappa(\sigma)\,\overline{\psi_\kappa(\tau)},$$

where $\psi_\kappa(\sigma) = \mathrm{tr}(P_\kappa)$. Hence $\rho^{(N)}(M) = \sum_\kappa |\psi_\kappa\rangle\langle\psi_\kappa| \geq 0$. □

25.5 Purification RFP

Definition 25.14 (Purifying MPS tensor). Lean ✓ Stmt Given a purifying family $A^{(i,k)} \in M_{D'}(\mathbb{C})$ for $i \in \{0, \dots, d-1\}$ and $k \in \{0, \dots, d_K-1\}$, the *purifying MPS tensor* is

$$A^{\mathrm{pur}}_{(i,k)} = A^{(i,k)},$$

an MPS tensor with physical dimension $d \cdot d_K$ and bond dimension D'.

Definition 25.15 (Purification RFP). Lean ✓ Stmt An MPO tensor M has a *purification RFP* if it admits an LPDO witness whose purifying MPS tensor is a pure-state renormalization fixed point. This is [CPGSV17a, definition of purification RFP, lines 756–759].

Remark 25.16 (PRFP does not imply MPO ZCL). Lean ✓ Stmt Purification RFP is strictly weaker than MPO zero correlation length: there exists an MPO tensor with purification RFP whose transfer map is not idempotent. Equivalently, purification RFP does not imply MPO zero correlation length.

Theorem 25.17 (Purification RFP implies LPDO). *Lean* *✓ Stmt* *A purification-RFP tensor is, in particular, an LPDO.*

Proof. ✓ Proof The PRFP condition provides an LPDO witness by construction. □

25.6 Pure-state recovery inside the MPO formalism

Definition 25.18 (MPDO renormalization fixed point). Lean ✓ Stmt An MPO tensor M is an *MPDO renormalization fixed point* if its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

Theorem 25.19 (MPDO RFP iff MPO ZCL). Lean ✓ Stmt *MPDO renormalization fixed points are exactly MPO tensors with zero correlation length.*

Proof. ✓ Proof Both conditions assert $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$; they are definitionally identical. □

Theorem 25.20 (MPDO RFP iff doubled-index MPS RFP). Lean ✓ Stmt *The MPDO RFP condition is equivalent to the pure-state RFP condition for the doubled-index MPS tensor.*

Proof. ✓ Proof Unfold the MPDO RFP condition to its definitional equivalent, MPO ZCL, then apply Theorem 25.10. □

Definition 25.21 (Diagonal pure-state embedding as an MPO). Lean ✓ Stmt An MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ determines an MPO tensor by placing A^i on the diagonal in the bra and ket indices:

$$M^{ij} = \delta_{ij} A^i.$$

Equivalently, $M^{ii} = A^i$ and $M^{ij} = 0$ for $i \neq j$.

Theorem 25.22 (Transfer map of the diagonal pure-state embedding). Lean ✓ Stmt *The transfer map of the diagonal MPO associated to A agrees with the original MPS transfer map:*

$$\mathcal{E}_{A^{\mathrm{MPO}}} = \mathcal{E}_A.$$

Proof. ✓ Proof In the double sum defining $\mathcal{E}_{A^{\mathrm{MPO}}}$, all off-diagonal terms vanish because $M^{ij} = 0$ for $i \neq j$, while the diagonal terms are exactly $A^i X (A^i)^\dagger$. Thus only the sum over i remains, which is the defining formula for $\mathcal{E}_A$. □

Theorem 25.23 (Pure-state recovery of the RFP condition). Lean ✓ Stmt *For an MPS tensor A, its diagonal MPO embedding is MPDO-RFP if and only if A is an RFP:*

$$A^{\mathrm{MPO}} \text{ is MPDO-RFP} \iff A \text{ is RFP}.$$

Proof. ✓ Proof Both conditions assert idempotence of the corresponding transfer map. Theorem 25.22 identifies those transfer maps, so the two predicates are equivalent. □

Theorem 25.24 (Pure-state recovery of zero correlation length). Lean ✓ Stmt *For an MPS tensor A, the diagonal MPO embedding is MPDO-RFP if and only if A has zero correlation length.*

Proof. ✓ Proof Combine Theorem 25.23 with the characterization of zero correlation length by idempotence of the MPS transfer map (Theorem 19.28). □

25.7 Simple local structure from SAL and ZCL

For the simple MPDO case in Appendix C.2 of [CPGSV17a], the strong-area-law hypothesis is used locally through the equality case of strong subadditivity for a normalized three-site reduced state.

Definition 25.25 (Local η-structure for a three-site reduced state). Lean ✓ Stmt For a three-site density operator ρ_{ABC}, the local η-structure is the quantum Markov decomposition on the middle subsystem B produced by equality in strong subadditivity. Concretely, after a unitary change of basis on B, one has a finite direct-sum decomposition

$$B = \bigoplus_k (b_1^{(k)} \otimes b_2^{(k)})$$

such that

$$\rho_{ABC} = \bigoplus_k \rho_{Ab_1}^{(k)} \otimes \rho_{b_2C}^{(k)}.$$

Theorem 25.26 (SAL implies local η-structure). Lean ✓ Stmt *If a normalized three-site reduced state satisfies the equality case of strong subadditivity — the local entropy form of the simple-MPDO strong area law — then it admits the local η-structure of Definition 25.25.*

Proof. ✓ Proof This is exactly the forward implication of the Hayashi equality characterization: equality in strong subadditivity yields a quantum Markov decomposition on the middle subsystem. □

Definition 25.27 (Injective inverse-map layer for a simple MPO tensor). Lean Lean Lean Lean ✓ Stmt A simple MPO tensor K is called injective if its doubled-index MPS tensor is injective. For such a tensor, the chosen decomposition map of the doubled-index MPS furnishes a concrete inverse tensor K^{-1} together with right- and left-physical realization maps that turn virtual bond insertions back into local physical operators.

Theorem 25.28 (Inverse-map identities for an injective simple MPO tensor). Lean Lean Lean Lean ✓ Stmt *For an injective simple MPO tensor, contracting K^{-1} with K recovers the matrix units on the virtual bond space. Likewise, every right virtual insertion admits a physical realization, this realization is multiplicative, and the analogous left-insertion statement also holds.*

Definition 25.29 (Explicit neighboring η-operator data). Lean Lean ✓ Stmt Fix a Hayashi decomposition witness h_η. An explicit neighboring η-family is a dependent family $(\eta_{k,h})$ indexed by sector pairs, where $\eta_{k,h}$ acts on the neighboring bond space $B_k^R \otimes B_h^L$. Requiring positivity for each pair gives the corresponding positive η-data.

Definition 25.30 (Trace matrices of explicit η-data). Lean Lean Lean Lean ✓ Stmt Every explicit neighboring η-family determines a sector-by-sector trace matrix. We give both its complex-valued form and the real-part version, which is the direct data for the real Perron–Frobenius matrix T used in the rank-one step.

Theorem 25.31 (Positivity of bipartite partial traces). Lean Lean ✓ Stmt *If ρ is positive semidefinite on a bipartite space $A \otimes B$, then both reduced operators $\mathrm{tr}_A\,\rho$ and $\mathrm{tr}_B\,\rho$ are positive semidefinite.*

Proof. ✓ Proof Write each partial trace as the finite sum, over the traced index, of a principal submatrix of ρ. Principal submatrices of a positive semidefinite matrix are positive semidefinite, and finite sums of positive semidefinite matrices remain positive semidefinite. □

Definition 25.32 (Hayashi–Markov extraction of explicit η-operators). Lean ✓ Stmt Every Hayashi decomposition witness h_η canonically produces an explicit neighboring η-family by Kronecker-multiplying the sector-reduced states on the neighboring bond spaces:

$$\eta_{k,h} := \mathrm{tr}_C(\rho^{(k)}_{b_2C}) \otimes \mathrm{tr}_A(\rho^{(h)}_{Ab_1}).$$

Positive semidefiniteness of each $\eta_{k,h}$ follows from the fact that partial traces preserve positivity and that positive semidefiniteness is closed under Kronecker products. This is the sector-reduced extraction layer; the remaining connection to Appendix C.2 is the identification of this family with the inverse-map construction in the original simple-MPDO tensor coordinates.

Theorem 25.33 (Trace matrix of the Hayashi–Markov η-operators). Lean Lean ✓ Stmt *The trace matrix induced by the extracted family is entrywise $T_{k,h} = 1$ in both its complex-valued form and its real-valued form, matching the normalization $T_{k,h} = a_k b_h$ used in $a_k = b_h = 1$.*

Proof. ✓ Proof Each $\rho^{(k)}_{b_2C}$ is a density matrix, hence has unit trace; the partial trace tr_C preserves the trace. The analogous statement holds for $\mathrm{tr}_A \rho^{(h)}_{Ab_1}$. Multiplicativity of the trace under Kronecker products gives $T_{k,h} = 1 \cdot 1 = 1$, and taking real parts gives the real-valued statement. □

Theorem 25.34 (Entrywise nonnegativity of the real trace matrix). Lean ✓ Stmt *Positivity of the neighboring operators gives entrywise nonnegativity of the real trace matrix. Matrix-level positive semidefiniteness of T is a separate structural condition.*

Proof. ✓ Proof The trace of a positive semidefinite neighboring operator is nonnegative in the complex order. Taking real parts gives the stated entrywise nonnegativity. □

Definition 25.35 (Auxiliary matrix predicates for the rank-one step). Lean Lean Lean ✓ Stmt The auxiliary predicates state three matrix-theoretic ingredients used in the rank-one step of Appendix C.2, Lemma C.4: constant traces of all positive powers, existence of a rank-one factorization $T = ab^\top$, and the conditional Perron–Frobenius implication from primitivity together with constant trace powers to such a factorization.

This conditional implication is not a globally provable theorem. As a universal statement over primitive nonnegative matrices it is false; a concrete 3×3 counterexample appears in the archive. The corrected criterion proved here adds positive semidefiniteness and trace normalization, which provide the diagonalizability needed to turn the trace-power condition into a rank-one factorization.

Theorem 25.36 (Trace square of a Hermitian matrix). Lean ✓ Stmt *Let T be a finite Hermitian matrix over a real or complex scalar field, with Hermitian eigenvalues λ_i. Then*

$$\mathrm{tr}(T^2) = \sum_i \lambda_i^2.$$

Proof. ✓ Proof The spectral theorem writes T as a unitary conjugate of the diagonal matrix of its Hermitian eigenvalues. Squaring commutes with this conjugation, and cyclicity of trace reduces the identity to the trace of the squared diagonal matrix. □

Theorem 25.37 (PSD trace-power rank-one criterion). Lean Lean ✓ Stmt *Let T be a finite real square matrix. If T is positive semidefinite, normalized by $\mathrm{tr}(T) = 1$, and has constant traces on all positive powers, then T has a rank-one factorization. Consequently, under the same positive-semidefinite and trace-normalization hypotheses, the conditional rank-one criterion used in Lemma C.4 follows.*

Proof. ✓ Proof The spectral theorem diagonalizes the positive semidefinite matrix with nonnegative real eigenvalues. The trace gives that their sum is 1, while the second trace moment gives that the sum of their squares is also 1. Hence exactly one eigenvalue is equal to 1 and all others vanish, so the diagonal form is rank one. Unitary conjugation preserves the existence of an outer-product factorization. ☐

Theorem 25.38 (SAL and ZCL imply rank-one T (conditional form)). Lean ✓ Stmt *Let T be a finite real square matrix. Assume that T is primitive, that* $\operatorname{tr}(T) = 1$, *and that every positive power of T has the same trace as T. If one further supplies the Perron–Frobenius hypothesis that this particular primitive nonnegative matrix with constant trace powers is rank one, then*

$$T = ab^{\top}$$

for some vectors a and b, with normalization

$$a \cdot b = 1.$$

Proof. ✓ Proof The Perron–Frobenius hypothesis gives the factorization $T = ab^{\top}$. Taking traces and using $\operatorname{tr}(ab^{\top}) = a \cdot b$ converts the normalization $\operatorname{tr}(T) = 1$ into $a \cdot b = 1$. ☐

Theorem 25.39 (PSD-corrected rank-one T criterion). Lean ✓ Stmt *Let T be a finite real square matrix. Assume that T is primitive, positive semidefinite, normalized by* $\operatorname{tr}(T) = 1$, *and has constant traces on all positive powers. Then*

$$T = ab^{\top}$$

for some vectors a and b, with $a \cdot b = 1$.

Proof. ✓ Proof The positive-semidefinite matrix criterion diagonalizes T, uses the first two trace moments to show that exactly one eigenvalue is nonzero, and hence supplies the auxiliary rank-one criterion. The conditional Lemma C.4 statement then gives the factorization and trace normalization. ☐

25.8 Horizontal and Vertical Canonical Forms

Definition 25.40 (Horizontal canonical form for an MPO). Lean Not ready An MPO tensor is in *horizontal canonical form* if its doubled-index MPS tensor generates the same MPV family as a block-diagonal tensor $\bigoplus_k \mu_k A_k$ whose blocks are injective, left-canonical, and have nonzero weights μ_k.

Definition 25.41 (Vertical canonical form for an MPO). Lean Not ready An MPO tensor is in *vertical canonical form* if its diagonal MPS tensor $\{M^{ii}\}_{i=0}^{d-1}$ generates the same MPV family as a block-diagonal tensor obtained from a basis of normal tensors by repeating each block according to a multiplicity and weighting each copy by a positive scalar. This is the flattened form of the decomposition $\bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ from [CPGSV17a, Section 4.4].

Remark 25.42 (Finite witnesses for block-injective canonical form). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let (A_k) be a family of block tensors, and fix a length L. Write A_k^w for the length-L word evaluation of A_k along a word w. The finite-length spanning hypothesis says that the tuple-valued evaluations

$$w \longmapsto (A_k^w)_k$$

span the full product algebra of block matrices. The corresponding finite-length separation property asserts that there exists a length L such that if, for every word w of length L,

$$\sum_k \mathrm{tr}(\Delta_k A_k^w) = 0$$

then these pairings vanish on a spanning set for the product algebra, so the induced linear functional on the product algebra is zero; by nondegeneracy of the product trace pairing, every Δ_k vanishes. In other words, a single finite blocking length already separates the blocks by their word evaluations. Combined with injectivity, left-canonicality, and nonzero block weights, this finite-length separation yields the horizontal canonical-form data of the block-diagonal tensor.

The entrywise scalar family of block-matrix coefficients encodes the same finite-length data. Linear independence of that scalar family implies the product-algebra spanning statement, hence a block-injective canonical-form witness. The same criterion therefore gives horizontal canonical-form data directly. The duplicate-scalar counterexample shows that this finite-length witness is strictly stronger than blockwise injectivity, left-canonicality, and nonzero weights alone: two identical scalar blocks satisfy those hypotheses, but the block-injective canonical-form separation condition fails and no blocking length makes the entrywise family linearly independent.

The block-injectivity proposition in [CPGSV17a, lines 340–345] gives a concrete sufficient criterion for the spanning condition: after a common blocking length each block is block-injective, and after a further finite blocking length one can form linear combinations of word evaluations which equal the identity on one chosen block and vanish on all others. Concatenating an injective prefix with this finite block-separation data produces the product-algebra spanning condition, hence the horizontal canonical-form data.

Theorem 25.43 (Finite block-separation criterion). Lean ✓ Stmt *If a block family admits the finite-length data described in the block-injectivity proposition of [CPGSV17a, lines 340–345], then it is in block-injective canonical form.*

Proof. ✓ Proof This is the direct implication from the finite-length data to the finite-length trace-separation statement. What remains open in the full block-injectivity proposition of [CPGSV17a, lines 340–345] is the derivation of this finite witness from separated canonical-form / BNT data. □

Lemma 25.44 (Blockwise equality from agreement on the MPV family). Lean ✓ Stmt *Let $\bigoplus_k \mu_k A_k$ be a block-diagonal tensor whose blocks are injective, left-canonical, weighted by nonzero scalars, and jointly block-injective in the sense that for some blocking length L the tuple of trace pairings against length-L block words is faithful across all blocks simultaneously. If two physical operators induce the same first-site action on every MPV generated by this tensor, then their first-site insertions agree on each block A_k.*

Proof. ✓ Proof This is Lemma L from [CPGSV17a, Appendix A], using the finite criterion from Theorem 25.43. Expanding the hypothesis over a word of length $L+1$ beginning with the fixed site and folding each block contribution into a trace pairing against the corresponding first-site insertion tensor yields, upon taking the Y-Z difference, a tuple of block matrices whose trace pairings against every length-L block word vanish. The block-injective canonical-form hypothesis then forces each block difference to be zero, and dividing by the nonzero weight μ_k^{L+1} concludes the blockwise equality of the two first-site insertions. □

Theorem 25.45 (Horizontal canonical form implies vertical canonical form). Not ready *Every MPDO in horizontal canonical form is also in vertical canonical form.*

Proof. Starting from the horizontal canonical-form decomposition, one uses the MPDO positivity condition to compare the first-site action of the doubled-index tensor on the diagonal sector. Lemma 25.44 then shows that the diagonal tensor splits along the same normal blocks, with the coefficients rearranged into positive scalar weights. Flattening those positive diagonal coefficients produces the required vertical canonical-form decomposition. □

25.9 Fusion isometries

Following [CPGSV17a, Section 4.5], the next definitions describe the transfer-map content of the fusion-isometry picture. For each blocked size $n \geq 1$, the doubled-index blocked tensor of an MPO has a blocked transfer map E_n acting on bond-space matrices, and the support algebra is modeled by a subspace through which E_n factors as a retract.

Definition 25.46 (Transfer-map fusion-isometry datum). Lean ✓ Stmt A *transfer-map fusion-isometry datum* at blocked size n for an MPO tensor consists of a subspace $\mathcal{A}_n \subseteq M_D(\mathbb{C})$ together with linear maps

$$T_n : M_D(\mathbb{C}) \to \mathcal{A}_n, \qquad S_n : \mathcal{A}_n \to M_D(\mathbb{C})$$

such that

$$T_n \circ S_n = \mathrm{id}_{\mathcal{A}_n}, \qquad S_n \circ T_n = E_n,$$

where E_n denotes the blocked transfer map.

Definition 25.47 (Fusion-isometry RFP predicate). Lean ✓ Stmt An MPO tensor M satisfies the *fusion-isometry formulation* of the renormalization fixed-point condition when for every blocked size $n \geq 1$ there exists transfer-map fusion-isometry data for the blocked transfer map E_n.

Theorem 25.48 (Fusion-isometry data imply the MPDO RFP condition). Lean Lean ✓ Stmt *A one-site fusion-isometry datum implies the MPDO RFP condition. Hence the fusion-isometry formulation, which supplies such data at every positive blocked size, implies the same condition.*

Proof. ✓ Proof Apply the definition at blocked size $n = 1$. If $E_1 = S_1 \circ T_1$ and $T_1 \circ S_1 = \mathrm{id}$, then

$$E_1^2 = S_1 T_1 S_1 T_1 = S_1 (T_1 S_1) T_1 = S_1 T_1 = E_1.$$

Since E_1 is the original transfer map, this is exactly the MPDO RFP condition. □

Theorem 25.49 (One-site fusion retract criterion). Lean ✓ Stmt *A one-site transfer-map fusion-isometry datum exists if and only if the MPO transfer map is idempotent.*

Proof. ✓ Proof The forward implication is the retract calculation $E_1^2 = S_1 T_1 S_1 T_1 = S_1 T_1$. Conversely, if E_1 is idempotent, then it factors through its range, giving the required one-site datum. □

Theorem 25.50 (Transfer-map fusion criterion for MPDO RFP). Lean ✓ Stmt *The fusion-isometry formulation is equivalent to the MPDO RFP condition.*

Proof. ✓ Proof The forward implication is Theorem 25.48. Conversely, if the transfer map E is idempotent, then every positive blocked transfer map equals E and therefore factors through its range. This range-factorization gives the required fusion-isometry data. □

Corollary 25.51 (Pure-state recovery). Lean ✓ Stmt *For the diagonal MPO associated to a pure MPS tensor, the fusion-isometry formulation reduces to the usual pure-state RFP condition.*

Proof. ✓ Proof Combine Theorem 25.50 with the diagonal pure-state recovery theorem for the MPO predicate. □

Theorem 25.52 (All-blocked fusion reduces to one-site fusion). Lean ✓ Stmt *The fusion-isometry formulation for all positive blocked sizes is equivalent to the existence of a one-site transfer-map fusion-isometry datum.*

Proof. ✓ Proof Combine the one-site criterion with the equivalence between the all-blocked fusion formulation and transfer-map idempotence. □

25.10 Algebra structure

Definition 25.53 (Algebra-structure data for an MPDO). Lean ✓ Stmt A family of support $*$-algebras $\{\mathcal{A}_n\}_{n\geq 0}$, one at every blocking size, equipped with blocking maps $m_n : \mathcal{A}_n \otimes \mathcal{A}_n \to \mathcal{A}_{2n}$ and inclusion maps $\iota_n : \mathcal{A}_n \to \mathcal{A}_{n+1}$ such that, after viewing all elements inside the ambient bond-space matrix algebra, m_n is ordinary matrix multiplication and ι_n is the ambient inclusion.

Definition 25.54 (Algebra-structure RFP predicate). Lean ✓ Stmt An MPO tensor M satisfies the *algebra-structure formulation* used here when there exist algebra-structure data whose support algebra at every positive blocking size coincides with the fixed-point algebra of the adjoint blocked transfer map. This is a nontrivial algebraic condition, but it is still weaker than the full coefficient formulation of [CPGSV17a, Theorem IV.13 (ii)]; that formulation also includes the coefficient family $c^{(L)}_{\alpha,\beta,\gamma}$.

Theorem 25.55 (RFP yields a stationary algebra tower under a faithful fixed point). Lean ✓ Stmt *Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer map is idempotent, then there exists a stationary family of support algebras, all equal to the fixed-point algebra of the adjoint transfer map, with multiplication given by matrix multiplication and inclusions given by the identity.*

Proof. ✓ Proof Under the trace-preserving normalization and the faithful fixed point, Wolf's fixed-point-algebra theorem gives a $*$-subalgebra of adjoint fixed points. Idempotence identifies every positive blocked transfer map with the original one, so the same fixed-point algebra works at every blocking size. □

Theorem 25.56 (Fusion yields a stationary algebra tower under a faithful fixed point). Lean ✓ Stmt *Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer-map fusion criterion holds, then the MPO satisfies the algebra-structure formulation of the renormalization fixed-point condition.*

Proof. ✓ Proof Combine Theorem 25.48 with Theorem 25.55. □

Theorem 25.57 (Stabilized blocked adjoint fixed points yield algebra data). Lean ✓ Stmt *Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If, for every positive blocked size n, the adjoint fixed-point space of the blocked transfer map agrees with the adjoint fixed-point space of the original transfer map, then the algebra-structure formulation holds.*

Proof. ✓ Proof Use the adjoint fixed-point algebra of the original transfer map as a constant support-algebra tower. The stabilization hypothesis identifies that same algebra with the adjoint fixed-point space of every positive blocked transfer map, so the stationary family satisfies the definition. □

This does not give the full converse direction of [CPGSV17a, Theorem IV.13]: the compatibility condition above only sees stabilized adjoint fixed-point algebras, not the coefficient/BNT data. In particular, Theorem IV.13(ii) also requires the special diagonal matrices $\chi_{\alpha,\beta,\gamma}$. The coefficient data attached to the algebra tower are: a chosen basis of each support algebra, coordinate maps into a finite scalar family, and the corresponding multiplication / inclusion coefficient arrays.

Definition 25.58 (Blocked coefficients for the algebra tower). Lean ✓ Stmt For each blocked size n, once a basis of $\mathcal{A}_n$ is chosen, every element of $\mathcal{A}_n$ is identified with a finite coefficient family on that basis.

Theorem 25.59 (Reconstruction from blocked coefficients). Lean ✓ Stmt *Reconstructing a coefficient family gives the corresponding basis expansion $\sum_i a_i b_i$ in $\mathcal{A}_n$.*

Proof. ✓ Proof This is exactly the finite-basis reconstruction formula for the chosen basis of $\mathcal{A}_n$. □

Definition 25.60 (Blocked multiplication coefficients). Lean ✓ Stmt The blocking multiplication map m_n determines a coefficient family for the product of any two chosen basis elements of $\mathcal{A}_n$, now viewed in the basis of $\mathcal{A}_{2n}$.

Theorem 25.61 (Blocked multiplication reconstructs from its coefficients). Lean ✓ Stmt *In the ambient bond-space matrix algebra, the product of two chosen basis elements of $\mathcal{A}_n$ is the sum of the basis elements of $\mathcal{A}_{2n}$ weighted by these extracted coefficients.*

Proof. ✓ Proof Apply the reconstruction formula of Theorem 25.59 to the element $m_n(b_i, b_j) \in \mathcal{A}_{2n}$. □

Theorem 25.62 (Compatible adjoint fixed points reconstruct from blocked coefficients). Lean ✓ Stmt *If the algebra tower is compatible with an MPO tensor M, then every adjoint blocked fixed point of the blocked transfer map E_n belongs to $\mathcal{A}_n$ and is recovered exactly from its extracted coefficient family.*

Proof. ✓ Proof Compatibility identifies $\mathcal{A}_n$ with the adjoint fixed-point algebra of E_n, so the fixed point first becomes an element of $\mathcal{A}_n$ and then Theorem 25.59 reconstructs it. □

Theorem 25.63 (Adjoint fixed points descend through the algebra tower). Lean ✓ Stmt *If an MPO tensor satisfies the algebra-structure formulation, then every adjoint fixed point of the blocked transfer map E_n at a positive blocking size n is already an adjoint fixed point of the unblocked transfer map E_1.*

Proof. ✓ Proof Compatibility at size n places X in $\mathcal{A}_n$. The inclusion map ι_n lifts X into $\mathcal{A}_{n+1}$, and since ι_n is just an inclusion it does not change the underlying operator, so we continue to denote the lifted element by X. Thus compatibility at size $n+1$ yields $E_{n+1}^\dagger X = X$. Rewriting $E_{n+1} = E_n \circ E_1$ and applying $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$ together with the hypothesis $E_n^\dagger X = X$ extracts $E_1^\dagger X = X$. □

Lemma 25.64 (Reverse inclusion of adjoint fixed points). Lean ✓ Stmt *Every adjoint fixed point of the unblocked transfer map E_1 is an adjoint fixed point of every blocked transfer map E_n.*

Proof. ✓ Proof Induction on n using $E_{n+1} = E_n \circ E_1$ and $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$; the base case is $E_0 = \mathrm{id}$. This step does not require the algebra-structure data. □

Lemma 25.65 (Blocked powers of adjoint eigenvectors). Lean ✓ Stmt *If $E = E_1$ and $E^\dagger X = \lambda X$, then $E_n^\dagger X = \lambda^n X$ for every n.*

Proof. ✓ Proof The case $n = 0$ is $E_0 = \mathrm{id}$. For the induction step, $E_{n+1} = E_n \circ E_1$, hence

$$E_{n+1}^\dagger X = E_1^\dagger E_n^\dagger X = E_1^\dagger(\lambda^n X) = \lambda^n E_1^\dagger X = \lambda^{n+1} X.$$

□

Theorem 25.66 (Finite-order adjoint eigenvalues for algebra towers). Lean ✓ Stmt *Assume the algebra-structure formulation holds. If $X \neq 0$, $E_1^\dagger X = \lambda X$, and $\lambda^n = 1$ for some $n > 0$, then $\lambda = 1$.*

Proof. ✓ Proof By Lemma 25.65, $E_n^\dagger X = \lambda^n X = X$. The descent theorem gives $E_1^\dagger X = X$. Thus $\lambda X = X$, so $(\lambda - 1)X = 0$. Since $X \neq 0$, one obtains $\lambda = 1$. □

Theorem 25.67 (Adjoint fixed-point equality for algebra towers). Lean ✓ Stmt *If an MPO tensor satisfies the algebra-structure formulation, then for every positive blocking size n the adjoint fixed points of E_n are exactly the adjoint fixed points of E_1.*

Proof. ✓ Proof Combine Theorem 25.63 with Lemma 25.64. □

Theorem 25.68 (Algebra data give the stationary fixed-point tower). Lean ✓ Stmt *Under the trace-preserving normalization and a positive-definite fixed point, any algebra-structure witness forces the stationary adjoint fixed-point tower itself to be compatible with the MPO tensor.*

Proof. ✓ Proof Apply the adjoint fixed-point equality obtained from the algebra-structure predicate to the stationary tower constructed from the faithful fixed point. □

Remark 25.69 (Why algebra data do not imply fusion data). Write $E = E_1$ for the one-site transfer map. The present algebra predicate gives, for every $n > 0$,

$$\mathrm{Fix}((E^n)^\dagger) = \mathrm{Fix}(E^\dagger).$$

Hence a vector X with $E^\dagger X = \lambda X$ and $\lambda^n = 1$ lies in $\mathrm{Fix}(E^\dagger)$, so $(\lambda - 1)X = 0$ and $\lambda = 1$ by Theorem 25.66. It does not imply $E^2 = E$. For instance, if Π_{diag} is the projection onto diagonal matrices and $0 < \varepsilon < 1$, set

$$E(X) = \varepsilon\, X + (1 - \varepsilon)\, \Pi_{\mathrm{diag}}(X),$$

so that

$$E^n(X) = \Pi_{\mathrm{diag}}(X) + \varepsilon^n (X - \Pi_{\mathrm{diag}}(X)), \qquad E^2 - E = (\varepsilon^2 - \varepsilon)(\mathrm{id} - \Pi_{\mathrm{diag}}).$$

The fixed-point algebra is diagonal for every $n > 0$, but $E^2 \neq E$.

The missing source step is the coefficient argument in [CPGSV17a, Appendix C.4]. One needs the same-length product law

$$O_L(M_\alpha) O_L(M_\beta) = \sum_\gamma \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}) O_L(M_\gamma)$$

and the length-one idempotent trace identity

$$m_\gamma = \sum_{\alpha,\beta} \mathrm{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

The proof of [CPGSV17a, Theorem IV.13] uses these identities, positivity, and the simple-matrix lemma [CPGSV17a, lines 1155–1163] to give

$$\{\nu_{\gamma,k}\}_k = \{\mu_{\alpha,k_1}\mu_{\beta,k_2}\chi_{\alpha,\beta,\gamma,k_3}\}_{\alpha,\beta,k_1,k_2,k_3}, \qquad \operatorname{tr}(\nu_\gamma) = m_\gamma.$$

The reconstruction maps in [CPGSV17a, lines 2065–2085] then use

$$\widetilde{R}_2\Big(\bigoplus_\gamma X_\gamma\Big) = \bigoplus_\gamma \frac{\nu_\gamma}{m_\gamma} \otimes X_\gamma, \qquad \widetilde{R}_1\Big(\bigoplus_\gamma X_\gamma\Big) = \bigoplus_\gamma \frac{\mu_\gamma}{m_\gamma} \otimes X_\gamma.$$

Thus $\operatorname{tr}(\nu_\gamma) = m_\gamma$ and the defining equality $m_\gamma = \operatorname{tr}(\mu_\gamma)$ normalize the factors ν_γ/m_γ and μ_γ/m_γ, respectively, in $T = \widetilde{R}_2\widetilde{T}R_1$ and $S = \widetilde{R}_1\widetilde{S}R_2$. The scalar Newton–Girard power-sum identity needed to obtain the trace-power form is already available as Theorem 12.17; what remains is the MPDO construction of the BNT-label witness carrying these two displayed equations.

25.10.1 Diagonal χ-matrices and the trace-power formula

The following statements isolate the form of [CPGSV17a, Theorem IV.13 (ii)]: the structure coefficients of the length-L operator algebra are trace-powers of positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$. They state the coefficient identity and the elementary trace-power identity for diagonal matrices.

Definition 25.70 (Diagonal χ family). Lean ✓ Stmt A family of diagonal complex matrices $\chi_{\alpha,\beta,\gamma}$ indexed by ordered triples drawn from a common index type, each of a specified size. This is the matrix $\chi_{\alpha,\beta,\gamma}$ of [CPGSV17a, Theorem IV.13(ii)].

Theorem 25.71 (Trace of χ^L equals the sum of L-th powers of entries). *Lean* *✓ Stmt* *For every triple (α,β,γ), if $\chi_{\alpha,\beta,\gamma}$ has size $r_{\alpha,\beta,\gamma}$, then for every $L \geq 0$,*

$$\operatorname{tr}(\chi^L_{\alpha,\beta,\gamma}) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi^L_{\alpha,\beta,\gamma,k},$$

where $\chi_{\alpha,\beta,\gamma,k}$ are the diagonal entries.

Proof. ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. □

Definition 25.72 (Trace-power form of a structure-coefficient family). Lean ✓ Stmt A binary compatibility predicate between an abstract structure-coefficient family $c^{(L)}_{\alpha,\beta,\gamma}$ and a diagonal χ family: the pair (c,χ) is said to be in *trace-power form* when

$$c^{(L)}_{\alpha,\beta,\gamma} = \operatorname{tr}(\chi^L_{\alpha,\beta,\gamma}) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi^L_{\alpha,\beta,\gamma,k}$$

for every $L \geq 0$ and every triple (α,β,γ), where $r_{\alpha,\beta,\gamma}$ is the size of $\chi_{\alpha,\beta,\gamma}$. This is the formal analogue of the target identity of [CPGSV17a, Theorem IV.13 (ii)]; the existential version—"there exists χ for which (c,χ) has trace-power form"—is obtained by quantifying over χ.

Definition 25.73 (BNT-label coefficient family). Lean Lean Lean Lean ✓ Stmt The same-length coefficient system $c^{(L)}_{\alpha,\beta,\gamma}$ from Theorem IV.13(ii), indexed by fixed BNT labels α,β,γ. The coefficient is attached to the length-L product $O_L(M_\alpha)O_L(M_\beta)$, not to the blocked-basis multiplication $\mathcal{A}_n \times \mathcal{A}_n \to \mathcal{A}_{2n}$. When a diagonal χ-family has already been constructed, it also canonically determines the coefficient family $c^{(L)}_{\alpha,\beta,\gamma} = \operatorname{tr}(\chi^L_{\alpha,\beta,\gamma}) = \sum_r \chi^L_{\alpha,\beta,\gamma,r}$, as in [CPGSV17a, Appendix C.4].

Definition 25.74 (BNT-label operator family). Lean ✓ Stmt A BNT-label operator family records, for each chain length L, the operators $O_L(M_\alpha)$ indexed by the fixed BNT labels α. The ambient algebra at length L is left abstract here; the later comparison step must relate it to the chosen blocked support algebras.

Definition 25.75 (Same-length BNT product form). Lean ✓ Stmt A BNT-label operator family and a BNT-label coefficient system have same-length product form when, for every positive length L,

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c^{(L)}_{\alpha,\beta,\gamma} O_L(M_\gamma).$$

This is the product identity appearing in Theorem IV.13(ii), stated independently of the blocked-basis multiplication $\mathcal{A}_n \times \mathcal{A}_n \to \mathcal{A}_{2n}$.

Theorem 25.76 (Same-length BNT product expansion). Lean ✓ Stmt *Under same-length BNT product form, the product of two length-L BNT-label operators is the corresponding finite linear combination of length-L BNT-label operators.*

Proof. ✓ Proof This is exactly the defining equality of same-length product form. □

Definition 25.77 (BNT-label trace-scalar family). Lean ✓ Stmt A BNT-label trace-scalar family records the scalars $m_\alpha = \mathrm{tr}(\mu_\alpha)$ indexed by the fixed BNT labels α, as in the idempotent condition of Theorem IV.13(ii).

Definition 25.78 (BNT idempotent coefficient form). Lean ✓ Stmt A BNT-label trace-scalar family and a BNT-label coefficient system have idempotent coefficient form when, for every BNT label γ,

$$m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta.$$

This is the idempotent condition in Theorem IV.13(ii), stated separately from the trace-power formula.

Theorem 25.79 (BNT idempotent coefficient expansion). Lean ✓ Stmt *Under BNT idempotent coefficient form, each m_γ is the corresponding finite linear combination of products $m_\alpha m_\beta$ with the length-one coefficients $c^{(1)}_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof This is exactly the defining equality of BNT idempotent coefficient form. □

Definition 25.80 (Blocked-basis BNT-label assignment). Lean Lean ✓ Stmt The BNT-label assignment consists, for every positive blocked length n, of a source label map σ_n from the chosen basis of $\mathcal{A}_n$ and a target label map τ_n from the chosen basis of $\mathcal{A}_{2n}$ to the fixed BNT labels. It also gives the combined label map $\sigma_n \sqcup \tau_n$ on the disjoint union of these two blocked bases.

Definition 25.81 (BNT comparison with blocked coefficients). Lean Lean Lean ✓ Stmt A comparison between the BNT-label coefficients and the chosen blocked-basis coefficients consists of a BNT-label assignment together with the equality

$$c^{(n)}_{i,j,k} = c^{(n)}_{\sigma_n(i),\sigma_n(j),\tau_n(k)}.$$

This is a blocked-basis comparison: the left hand side is expanded in the chosen basis of $\mathcal{A}_{2n}$, so the statement is not itself the same-length BNT product law. This records the coefficient-comparison statement separately from the construction of the label maps from the MPDO tensor.

Theorem 25.82 (Blocked coefficients are pulled back from BNT labels). Lean ✓ Stmt *Under a BNT comparison with blocked coefficients, each chosen blocked-basis multiplication coefficient is the corresponding BNT-label coefficient.*

Proof. ✓ Proof This is exactly the defining equality of the comparison. □

Definition 25.83 (Positive-length BNT-label trace-power form). Lean ✓ Stmt A BNT-label coefficient system and a diagonal χ-family are compatible when, for every positive length L,

$$c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}).$$

The same matrix $\chi_{\alpha,\beta,\gamma}$ is used for all positive L.

Theorem 25.84 (BNT-label coefficient as a positive-length trace power). Lean ✓ Stmt *Under positive-length BNT-label trace-power form, for every $L > 0$, $c^{(L)}_{\alpha,\beta,\gamma}$ is the trace of $\chi^L_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof Combine the defining positive-length trace-power identity with the trace formula for powers of a diagonal matrix. □

Theorem 25.85 (Canonical BNT coefficients from a χ-family). Lean ✓ Stmt *The coefficient family canonically determined by a diagonal χ-family satisfies the positive-length trace-power condition with respect to the same χ-family.*

Proof. ✓ Proof This is immediate from the definition of the canonical coefficient family. □

Definition 25.86 (Positive BNT-label χ trace-power witness). Lean Lean ✓ Stmt A positive BNT-label χ witness consists of a length-independent diagonal $\chi_{\alpha,\beta,\gamma}$-family, positivity of every diagonal entry, and the positive-length trace-power identity for the BNT-label coefficients. This is the coefficient statement of Theorem IV.13(ii); constructing the witness from an MPDO tensor and comparing it to blocked bases remain separate obligations. If the coefficient family is chosen canonically from the same χ-family, positivity of the diagonal entries alone gives such a witness.

Theorem 25.87 (Positive BNT-label witness gives trace powers). Lean ✓ Stmt *A positive BNT-label χ witness gives, for every $L > 0$, the formula $c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})$.*

Proof. ✓ Proof Apply the trace reformulation of the positive-length trace-power identity from the witness. □

Theorem 25.88 (Same-length BNT product with chi-trace coefficients). Lean ✓ Stmt *If the same-length BNT product identity holds and the BNT-label coefficients come from a positive length-independent χ-witness, then for every $L > 0$,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})O_L(M_\gamma).$$

Proof. ✓ Proof Substitute the positive-length trace-power formula into the same-length BNT product expansion. □

Theorem 25.89 (Same-length BNT product for canonical chi coefficients). Lean ✓ Stmt *If the same-length BNT product identity holds with the coefficient family canonically determined by a diagonal χ-family, then for every $L > 0$,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})O_L(M_\gamma).$$

Proof. ✓ Proof Substitute the canonical trace-power formula for the coefficients into the same-length BNT product expansion. □

Theorem 25.90 (BNT idempotent coefficients as chi traces). Lean ✓ Stmt *If the BNT trace scalars satisfy the idempotent coefficient condition and the BNT-label coefficients come from a positive length-independent χ-witness, then*

$$m_\gamma = \sum_{\alpha,\beta} \operatorname{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

Proof. ✓ Proof Substitute the length-one trace-power formula into the idempotent coefficient identity. □

Theorem 25.91 (BNT idempotent coefficients for canonical chi coefficients). Lean ✓ Stmt *If the BNT trace scalars satisfy the idempotent coefficient condition with the coefficient family canonically determined by a diagonal χ-family, then*

$$m_\gamma = \sum_{\alpha,\beta} \operatorname{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

Proof. ✓ Proof Substitute the length-one canonical trace formula into the idempotent coefficient identity. □

Theorem 25.92 (Blocked coefficients inherit BNT trace powers). Lean Lean Lean ✓ Stmt *If the chosen blocked-basis coefficients are compared with the fixed BNT-label coefficient system, and that BNT-label system has a positive length-independent χ-witness, then each positive-length blocked coefficient is the trace power of the corresponding BNT-label χ-matrix:*

$$c^{(n)}_{i,j,k} = \operatorname{tr}(\chi^n_{\sigma_n(i),\sigma_n(j),\tau_n(k)}).$$

Proof. ✓ Proof Use the comparison equality to replace the blocked-basis coefficient by the corresponding BNT-label coefficient, and then apply the BNT-label trace-power formula. In the special case where the compared BNT-label coefficient family is already the canonical family determined by χ, the same conclusion follows directly from the canonical trace formula. □

Definition 25.93 (Blocked χ family for multiplication coefficients). Lean ✓ Stmt A dependent diagonal matrix family indexed by the blocked multiplication triples (n, i, j, k), where i and j label basis elements of $\mathcal{A}_n$ and k labels a basis element of $\mathcal{A}_{2n}$. This is a blocked-basis analogue of the uniform BNT-label family in [CPGSV17a, Theorem IV.13(ii)]; the remaining uniformity gap is recorded in `docs/paper-gaps/cpgsv17_blocked_chi_uniformity.tex`.

Theorem 25.94 (Trace of a blocked χ power). Lean ✓ Stmt *For each blocked multiplication triple (n, i, j, k) and every exponent L, the trace of the L-th power of its diagonal matrix is the sum of the L-th powers of the corresponding diagonal entries.*

Proof. ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. □

Definition 25.95 (Trace-power form for blocked multiplication coefficients). Lean ✓ Stmt The blocked multiplication coefficients are in trace-power form when, for every positive blocked size n and every multiplication triple (i, j, k), the coefficient $c^{(n)}_{i,j,k}$ equals

$$\sum_r \chi^n_{n,i,j,k,r}.$$

Equivalently, by Theorem 25.94, it is the trace of the n-th power of the corresponding diagonal matrix. The exponent n is the source blocking length of the two factors in $\mathcal{A}_n$; the product is expanded in the basis of $\mathcal{A}_{2n}$.

Theorem 25.96 (Blocked coefficient as a trace power). Lean ✓ Stmt *Under trace-power form for the blocked multiplication coefficients, for every positive blocked size n, the coefficient $c^{(n)}_{i,j,k}$ is the trace of the n-th power of the diagonal matrix attached to (n, i, j, k).*

Proof. ✓ Proof Combine the defining trace-power identity with the trace formula for powers of a diagonal matrix. □

Definition 25.97 (Positive blocked χ trace-power witness). Lean ✓ Stmt A positive blocked χ trace-power witness consists of a blocked χ family, positivity of all its diagonal entries, and the positive-length trace-power identity for the blocked multiplication coefficients. This is the blocked-basis analogue of the positive diagonal matrices in [CPGSV17a, Theorem IV.13(ii)]; the uniform BNT-label gap is recorded in `docs/paper-gaps/cpgsv17_blocked_chi_uniformity.tex`.

Theorem 25.98 (BNT witnesses give blocked χ witnesses). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Suppose that the chosen blocked-basis coefficients are compared with a fixed BNT-label coefficient system, and that this BNT-label system has a positive length-independent χ-witness. Then the chosen blocked bases carry a positive blocked χ trace-power witness. The construction is obtained by pulling back the BNT-label $\chi_{\alpha,\beta,\gamma}$-matrices along the source and target label maps.*

Proof. ✓ Proof Reindex the positive BNT-label χ-family along the blocked-basis label map. Positivity is preserved by reindexing, and the trace-power identity follows by substituting the blocked-basis comparison into the BNT-label trace-power formula. □

Definition 25.99 (BNT-label theorem data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The BNT-label theorem data consist of the fixed BNT-label coefficient system, the same-length BNT operator family and product identity, the trace scalars and idempotent condition, the positive length-independent χ-witness, and the comparison with the chosen blocked bases. They also determine the pulled-back blocked-basis χ-family. These data are the source-side hypotheses from which the blocked-basis consequences below are derived. In the source-side special case where the coefficient family is canonically determined by the same χ-family, theorem data are built from that χ-family, its positivity, and the product, idempotent, and blocked-comparison statements for the resulting canonical coefficients. The same product and idempotent predicates may also be rephrased using the canonical coefficient family determined by the χ-matrices carried by the data, with the corresponding displayed equations.

Theorem 25.100 (BNT theorem data give the same-length product law). Lean ✓ Stmt *BNT-label theorem data give the same-length product equation*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c^{(L)}_{\alpha,\beta,\gamma} O_L(M_\gamma)$$

for every positive length L.

Proof. ✓ Proof This is the same-length product law carried by the theorem data. □

Theorem 25.101 (BNT theorem data give the idempotent scalar law). Lean ✓ Stmt *BNT-label theorem data give the idempotent scalar equation*

$$m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta.$$

Proof. ✓ Proof This is the idempotent coefficient law carried by the theorem data. □

Theorem 25.102 (BNT theorem data give the blocked coefficient comparison). Lean Lean Lean ✓ Stmt *BNT-label theorem data give the comparison between the blocked-basis coefficients and the BNT-label coefficients through the source and target label maps. Since the positive-length coefficients agree with the canonical coefficient family determined by the same χ-matrices, the comparison can also be written with those canonical coefficients.*

Proof. ✓ Proof This is the blocked-basis coefficient comparison carried by the theorem data. □

Theorem 25.103 (BNT theorem data give coefficient trace powers). Lean Lean ✓ Stmt *BNT-label theorem data give the coefficient identity $c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})$ for every positive length L. Equivalently, at every positive length, the coefficient family agrees with the canonical coefficient family determined by the same χ-matrices.*

Proof. ✓ Proof This is the positive-length χ-trace identity belonging to the positive BNT-label χ-witness in the theorem data. □

Theorem 25.104 (BNT theorem data give positive chi entries). Lean ✓ Stmt *BNT-label theorem data give positivity of every diagonal entry of $\chi_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof This is the positivity condition carried by the positive BNT-label χ-witness in the theorem data. □

Theorem 25.105 (BNT theorem data give the chi-trace product law). Lean ✓ Stmt *BNT-label theorem data give the same-length product expansion with coefficients written as traces of powers of the length-independent $\chi_{\alpha,\beta,\gamma}$-matrices.*

Proof. ✓ Proof Apply the chi-trace product law to the product identity and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 25.106 (BNT theorem data give the chi-trace idempotent law). Lean ✓ Stmt *BNT-label theorem data give the idempotent scalar identity with the length-one coefficients written as traces of the corresponding χ-matrices.*

Proof. ✓ Proof Apply the chi-trace idempotent law to the idempotent condition and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 25.107 (BNT theorem data give blocked coefficient trace powers). Lean ✓ Stmt *BNT-label theorem data give the blocked-basis coefficient trace-power formula obtained by pulling the BNT-label $\chi_{\alpha,\beta,\gamma}$-matrices back along the blocked-basis comparison maps.*

Proof. ✓ Proof Apply the blocked-basis coefficient trace-power theorem to the comparison and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 25.108 (BNT theorem data give a positive blocked χ witness). Lean Lean Lean Lean Lean ✓ Stmt *BNT-label theorem data give a positive blocked-basis χ trace-power witness by pulling the length-independent BNT-label χ-family back along the blocked-basis comparison maps.*

Proof. ✓ Proof Apply the BNT-to-blocked witness construction to the comparison and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 25.109 (BNT theorem data give blocked trace-power form). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from BNT-label theorem data satisfies the blocked trace-power predicate.*

Proof. ✓ Proof This is the trace-power part of the positive blocked-basis χ-witness obtained from the theorem data. □

Theorem 25.110 (BNT theorem data give positive blocked chi families). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from BNT-label theorem data has positive diagonal entries.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the theorem data. □

Theorem 25.111 (BNT theorem data give positive pulled-back blocked entries). Lean ✓ Stmt *The pulled-back blocked-basis χ-entries obtained from BNT-label theorem data are positive.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the theorem data. □

Theorem 25.112 (BNT theorem data give pulled-back blocked trace powers). Lean ✓ Stmt *The positive-length blocked-basis coefficients are traces of powers of the pulled-back blocked-basis χ-matrices obtained from BNT-label theorem data.*

Proof. ✓ Proof Apply the trace-power identity of the positive blocked-basis χ-witness obtained from the theorem data. □

Definition 25.113 (Existential BNT-label theorem witness). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt An existential BNT-label theorem witness consists of the finite BNT-label type, the same-length operator spaces, their algebraic structure, and the corresponding BNT-label theorem data, including the pulled-back blocked-basis χ-family. The construction of such a witness from an MPDO tensor is the outstanding step toward Theorem IV.13(ii). The proposition-level form is the nonemptiness of this witness type. In the source-side special case where the coefficients are canonically determined by a χ-family, the witness is obtained from that same χ-family together with the remaining product, idempotent, and blocked-comparison statements. For any such witness, the product and idempotent predicates can be rephrased with the canonical coefficient family determined by its χ-matrices, with the corresponding displayed equations.

Theorem 25.114 (BNT theorem witnesses give source equations). Lean Lean Lean Lean Lean Lean ✓ Stmt *A proposition-level BNT-label theorem witness contains the source-side same-length product law, the idempotent scalar law, the positive-length trace-power law, and positivity of the diagonal $\chi_{\alpha,\beta,\gamma}$-entries. After unpacking the witness, the same-length product and idempotent equations can first be written with the BNT-label coefficients $c^{(L)}_{\alpha,\beta,\gamma}$, and then with the coefficients expressed as* $\mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})$ *and* $\mathrm{tr}(\chi_{\alpha,\beta,\gamma})$*, respectively. At every positive length, the BNT-label coefficient family also agrees with the canonical coefficient family determined by these χ-matrices. Consequently, the product and idempotent predicates may be stated directly using that canonical coefficient family, and the same displayed source equations hold with those canonical coefficients.*

Proof. ✓ Proof Unpack the existential witness and apply the corresponding witness-level equations. □

Theorem 25.115 (BNT theorem witnesses give coefficient trace powers). Lean Lean ✓ Stmt *An existential BNT-label theorem witness gives the formula $c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})$ for every positive length L. Equivalently, at positive lengths, its coefficient family agrees with the canonical coefficient family determined by the same χ-matrices.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 25.116 (BNT theorem witnesses give positive chi entries). Lean ✓ Stmt *An existential BNT-label theorem witness gives positivity of every diagonal entry of $\chi_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof This follows from the positivity condition in the positive BNT-label χ-witness. □

Theorem 25.117 (BNT theorem witnesses give the same-length product law). Lean ✓ Stmt *An existential BNT-label theorem witness gives the product identity $O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c^{(L)}_{\alpha,\beta,\gamma} O_L(M_\gamma)$ for every positive length L.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 25.118 (BNT theorem witnesses give the idempotent scalar law). Lean ✓ Stmt *An existential BNT-label theorem witness gives the idempotent scalar identity $m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta$.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 25.119 (BNT theorem witnesses give the blocked coefficient comparison). Lean Lean Lean Lean Lean ✓ Stmt *An existential BNT-label theorem witness gives the comparison between the blocked-basis coefficients and the BNT-label coefficients through the source and target label maps. The same comparison is also available with the canonical coefficient family determined by the witness's χ-matrices. The proposition-level nonempty form of the witness gives the same comparisons after choosing a witness.*

Proof. ✓ Proof Apply the corresponding theorem-data comparison carried by the witness. □

Theorem 25.120 (BNT theorem witnesses give the chi-trace product law). Lean ✓ Stmt *An existential BNT-label theorem witness gives, for every positive length L, the identity $O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}) O_L(M_\gamma)$.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 25.121 (BNT theorem witnesses give the chi-trace idempotent law). Lean ✓ Stmt *An existential BNT-label theorem witness gives the idempotent scalar identity $m_\gamma = \sum_{\alpha,\beta} \mathrm{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta$.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 25.122 (BNT theorem witnesses give blocked coefficient trace powers). Lean ✓ Stmt *An existential BNT-label theorem witness gives the blocked-basis coefficient trace-power formula obtained by pulling the BNT-label χ-matrices back along the blocked-basis comparison maps.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 25.123 (BNT theorem witnesses give positive blocked χ witnesses). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *An existential BNT-label theorem witness gives a positive blocked-basis χ trace-power witness for the chosen algebra-structure data. The blocked-basis χ-family is the pullback of the BNT-label χ-family along the source and target comparison maps. The proposition-level nonempty form of the witness gives the corresponding existential blocked-basis trace-power family, both in coefficient form and in matrix-trace form.*

Proof. ✓ Proof Apply the corresponding theorem-data consequence carried by the witness. □

Theorem 25.124 (BNT theorem witnesses give blocked trace-power form). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from an existential BNT-label theorem witness satisfies the blocked trace-power predicate.*

Proof. ✓ Proof This is the trace-power part of the positive blocked-basis χ-witness obtained from the existential witness. □

Theorem 25.125 (BNT theorem witnesses give positive blocked chi families). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from an existential BNT-label theorem witness has positive diagonal entries.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the existential witness. □

Theorem 25.126 (BNT theorem witnesses give positive pulled-back blocked entries). Lean ✓ Stmt *The pulled-back blocked-basis χ-entries obtained from an existential BNT-label theorem witness are positive.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the existential witness. □

Theorem 25.127 (BNT theorem witnesses give pulled-back blocked trace powers). Lean ✓ Stmt *The positive-length blocked-basis coefficients are traces of powers of the pulled-back blocked-basis χ-matrices obtained from an existential BNT-label theorem witness.*

Proof. ✓ Proof Apply the trace-power identity of the positive blocked-basis χ-witness obtained from the existential witness. □

Theorem 25.128 (Positive blocked χ witness gives trace powers). Lean ✓ Stmt *A positive blocked χ trace-power witness gives, for every positive blocked size n, the formula $c^{(n)}_{i,j,k} = \mathrm{tr}(\chi^n_{n,i,j,k})$.*

Proof. ✓ Proof Apply the trace reformulation of the trace-power identity from the witness. □

25.11 Commuting form and GSNNCH

Definition 25.129 (Commuting-form data for an MPDO). Lean ✓ Stmt For a fixed chain length $N \geq 2$, this consists of a positive semidefinite two-site operator B such that its translated copies on the periodic chain commute pairwise.

Definition 25.130 (Commuting-form property). Lean ✓ Stmt An MPO tensor has the commuting-form property if for every $N \geq 2$ its N-site operator is a positive scalar multiple of the product of the translated copies of some commuting two-site operator B.

Definition 25.131 (GSNNCH). Lean ✓ Stmt An MPO tensor is a Gibbs state of a nearest-neighbor commuting Hamiltonian if every finite-chain operator admits the equivalent positive-operator presentation $\rho^{(N)} = c\prod_i B_{i,i+1}$ with $c > 0$ and commuting nearest-neighbor factors. The projector-limit convention in [CPGSV17a] identifies this with the exponential Gibbs form.

Theorem 25.132 (GSNNCH iff commuting form). Lean ✓ Stmt *For an MPO tensor, the GSNNCH condition is equivalent to the existence of commuting-form data on every finite chain.*

Proof. ✓ Proof A positive normalization constant converts commuting-form data into a GSNNCH witness, and every GSNNCH witness remembers its underlying commuting-form data. □

Definition 25.133 (GSNNCH with zero correlation length). Lean ✓ Stmt This is the conjunction of the GSNNCH condition with MPO zero correlation length, matching item (iii) of the simple-MPDO equivalence in [CPGSV17a].

Theorem 25.134 (GSNNCH–ZCL iff commuting form with ZCL). Lean ✓ Stmt *The GSNNCH–ZCL branch is equivalent to the conjunction of the commuting-form property with MPO zero correlation length.*

Proof. ✓ Proof Unfold the definition of GSNNCH with zero correlation length and apply Theorem 25.132. □

Definition 25.135 (Translation-invariant two-site bond). Lean ✓ Stmt This records a single positive two-site bond together with its translated copies on every finite periodic chain, requiring those translated copies to commute pairwise.

Definition 25.136 (η-local commuting-form data). Lean ✓ Stmt This defines the eta-local form of Appendix C.2: the translation-invariant positive two-site bond data together with the requirement that the corresponding product realizes the MPO at every finite length.

Definition 25.137 (Finite-chain data from a translation-invariant bond). Lean ✓ Stmt A translation-invariant positive two-site bond determines finite-chain commuting-form data at every length $N \geq 2$.

Theorem 25.138 (Component identities for finite-chain bond data). Lean Lean Lean ✓ Stmt *The finite-chain data associated to a translation-invariant bond retains the original local bond and the supplied lower bound $2 \leq N$, and the factor at site i is the translated copy of that local two-site bond.*

Proof. ✓ Proof These are the component identities for the finite-chain construction. □

Theorem 25.139 (η-local structure carries commuting form). Lean Lean Lean Lean Lean ✓ Stmt *The η-local structure itself determines a commuting-form witness for every chain length $N \geq 2$. In particular, the translated nearest-neighbor bonds commute and the MPDO at length N is a positive scalar multiple of their product, as in [CPGSV17a, Proposition C.6].*

Proof. ✓ Proof At length N, apply the commuting two-site bond and realization identity provided by the η-local structure. □

Theorem 25.140 (η-local structure gives GSNNCH). Lean ✓ Stmt *The same η-local structure gives a GSNNCH witness on every finite periodic chain.*

Proof. ✓ Proof Evaluate the local structure at each chain length $N \geq 2$. □

Theorem 25.141 (η-local structure with ZCL gives GSNNCH–ZCL). Lean Lean ✓ Stmt *Once the η-local structure gives the commuting nearest-neighbor product form, adding zero correlation length gives the GSNNCH–ZCL case of the simple-MPDO equivalence.*

Proof. ✓ Proof Use the commuting-form witness carried by the η-local structure and combine it with the ZCL hypothesis. □

Theorem 25.142 (η-local structure implies commuting form). Lean ✓ Stmt *Once the explicit neighboring operators $\eta_{k,h}$ have been assembled into an η-local structure, the global commuting-form property follows.*

Proof. ✓ Proof This is the commuting-form consequence already carried by the η-local structure. □

Theorem 25.143 (η-local structure implies GSNNCH). Lean ✓ Stmt *Once the explicit neighboring operators $\eta_{k,h}$ have been assembled into the η-local structure, the GSNNCH condition follows.*

Proof. ✓ Proof This is the GSNNCH witness already carried by the η-local structure. □

Remark 25.144 (Entropy-side construction). The remaining entropy-side step is the construction of a single η-local product form from SAL(K). The local $\eta_{k,h}$ are already supplied by Definition 25.32. What remains is the tensor-coordinate assembly

$$B = \sum_{k,h} \mathrm{Id}_{H_L^{(k)}} \otimes \eta_{k,h} \otimes \mathrm{Id}_{H_R^{(h)}}, \qquad B \geq 0,$$

and, for every $N \geq 2$,

$$\sigma^{(N)}(K) = c_N \prod_{n=1}^{N} B_{n,n+1}, \qquad c_N > 0, \qquad [B_{i,i+1}, B_{j,j+1}] = 0.$$

These equations are exactly the η-local product hypothesis; after that, Theorem 25.142 gives the commuting-form conclusion. Zero correlation length enters only in the subsequent RFP and GSNNCH–ZCL conclusions.

25.12 Blocked-RFP construction for simple MPDOs

Definition 25.145 (Local simple-MPDO structure). Lean ✓ Stmt This structure records the local Appendix C.2 ingredients: a normalized three-site reduced state with equality in strong subadditivity, the resulting local η-structure, and the rank-one conclusion for the auxiliary primitive matrix T, supplied here through the conditional rank-one criterion of Theorem 25.38.

Theorem 25.146 (Local simple-MPDO structure from SAL–ZCL hypotheses). Lean Lean ✓ Stmt *The hypotheses of Lemmas C.3–C.4 determine the local simple-MPDO structure. There are two forms. The conditional form assumes the rank-one implication for T. The corrected form assumes instead that $T \geq 0$ as a matrix, with $\mathrm{tr}(T) = 1$ and $\mathrm{tr}(T^m) = 1$ for every $m > 0$, and then applies Theorem 25.39.*

Definition 25.147 (Simple-MPDO blocked-RFP structure). Lean ✓ Stmt For an MPO tensor K, this records the local Appendix C.2 ingredients, the commuting-form property, and zero correlation length. The local component consists of the hypotheses used in the entropy-side argument for commuting form.

Theorem 25.148 (Blocked-RFP structure from SAL–ZCL and commuting form). Lean Lean ✓ Stmt *The SAL–ZCL hypotheses together with a commuting-form witness determine the simple-MPDO blocked-RFP structure. In the PSD-corrected form, the matrix input is $T \geq 0$, $\mathrm{tr}(T) = 1$, and $\mathrm{tr}(T^m) = 1$ for all $m > 0$, rather than an abstract Perron–Frobenius rank-one assumption for T.*

Theorem 25.149 (Blocked-RFP structure from η-local structure). Lean ✓ Stmt *Given the local Appendix C.2 data, ZCL, and an η-local product $\sigma^{(N)}(K) = c_N \prod_n B_{n,n+1}$ with $c_N > 0$, $B_{n,n+1} \geq 0$, and $[B_{i,i+1}, B_{j,j+1}] = 0$, one obtains the simple-MPDO blocked-RFP structure for K.*

Proof. ✓ Proof The η-local product gives the commuting-form property for K; the local Appendix C.2 hypotheses and ZCL fill the other two fields. □

Theorem 25.150 (Blocked-RFP structure from SAL–ZCL and η-local structure). Lean Lean ✓ Stmt *Suppose the local hypotheses of Lemmas C.3–C.4 give ρ_{ABC}, $I(A : C|B)_\rho = 0$, a primitive matrix T with $\mathrm{tr}(T) = 1$ and $\mathrm{tr}(T^m)$ independent of m, together with the conditional rank-one criterion for T. Suppose also that K has zero correlation length and that the assembled η-local data give $\sigma^{(N)}(K) = c_N \prod_{n=1}^{N} B_{n,n+1}$ with $c_N > 0$, $B_{n,n+1} \geq 0$, and $[B_{i,i+1}, B_{j,j+1}] = 0$ for all i, j. Then these data determine the simple-MPDO blocked-RFP structure for K. In the PSD-corrected form, the conditional rank-one criterion is replaced by $T \geq 0$, $\mathrm{tr}(T) = 1$, and $\mathrm{tr}(T^m) = 1$ for all $m > 0$.*

Proof. ✓ Proof First form the local structure from the SAL–ZCL local hypotheses and the conditional rank-one criterion for T. The η-local product formula supplies the commuting-form component, and ZCL supplies the remaining component of the blocked-RFP structure. □

Theorem 25.151 (Zero correlation length gives a blocked fusion-isometry witness). Lean ✓ Stmt *If an MPO tensor K has zero correlation length, then it has a blocked fusion-isometry witness at size 2 in the sense of Definition 25.46.*

Proof. ✓ Proof Zero correlation length is the MPDO RFP condition. Applying the converse direction of Theorem 25.50 to that RFP witness yields fusion-isometry data at every positive blocked size; evaluating it at size 2 produces the blocked witness. □

Theorem 25.152 (Blocked-RFP data give a blocked fusion-isometry witness). Lean ✓ Stmt *Given the blocked-RFP data for an MPO tensor K, one obtains a blocked fusion-isometry witness at size 2.*

Proof. ✓ Proof Apply the zero-correlation-length theorem to the ZCL component of the blocked-RFP data. □

Theorem 25.153 (Simple-MPDO RFP chain from blocked-RFP data). Lean ✓ Stmt *Given the blocked-RFP data for an MPO tensor K, one obtains the GSNNCH-with-ZCL case, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof The commuting-form and zero-correlation-length assumptions give the GSNNCH-with-ZCL case. Zero correlation length also gives the blocked fusion-isometry witness and the transfer-map fusion formulation. □

Theorem 25.154 (Simple-MPDO RFP chain from η-local structure). Lean ✓ Stmt *If the η-local structure has been constructed for an MPO tensor K, then zero correlation length gives the GSNNCH-with-ZCL case, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof The η-local structure supplies the commuting-form property. Combining this with zero correlation length gives the GSNNCH-with-ZCL case; zero correlation length also gives the blocked fusion-isometry witness and the transfer-map fusion formulation. □

Theorem 25.155 (Simple-MPDO RFP chain from SAL–ZCL and η-local structure)**.** Lean Lean ✓ Stmt *Assume the local SAL–ZCL hypotheses of Lemmas C.3–C.4 and the assembled η-local product form, with the conditional rank-one criterion for T, $\sigma^{(N)}(K) = c_N \prod_{n=1}^{N} B_{n,n+1}$ with $c_N > 0$, $B_{n,n+1} \geq 0$, and $[B_{i,i+1}, B_{j,j+1}] = 0$. Then K is GSNNCH with ZCL, admits a blocked fusion-isometry witness at size 2, and satisfies the transfer-map fusion formulation of MPDO RFP. The PSD-corrected form uses $T \geq 0$ together with $\mathrm{tr}(T^m) = 1$ for all $m > 0$ in place of the conditional matrix rank-one hypothesis.*

Proof. ✓ Proof Combine the local SAL–ZCL hypotheses and the η-local product form into the blocked-RFP structure, then apply Theorem 25.153. □

Theorem 25.156 (Simple-MPDO RFP chain from commuting form and ZCL)**.** Lean ✓ Stmt *From the commuting-form property and zero correlation length one simultaneously recovers the GSNNCH-with-ZCL criterion, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof The commuting-form and zero-correlation-length hypotheses give the GSNNCH–ZCL branch via Theorem 25.134. The blocked witness is Theorem 25.151. The full transfer-map fusion formulation follows by applying the converse direction of Theorem 25.50 to zero correlation length, which is the MPDO RFP condition. □

Remark 25.157 (SAL–ZCL implication for the GSNNCH–ZCL criterion). Not ready The implication from SAL and ZCL alone to the conclusion of Appendix C.2 requires a derivation of the commuting-form property from the local simple-MPDO structure of Lemmas C.3–C.4. Once that derivation is available, Theorem 25.156 gives the GSNNCH–ZCL branch and the blocked fusion-isometry witness.

Chapter 26

Source-Faithful PGVWC07 Recordings

This appendix-style chapter records the intermediate construction steps of the single-tensor canonical-form theorem of [PGVWC07, Theorem Th:TIcanonical], following the source proof order. They faithfully transcribe the PGVWC07 single-tensor construction and are kept here only as a reference. None of these recordings is on the main route of the Fundamental Theorem reduction, which instead goes through Theorem 9.35 and the after-blocking statements of Chapter 10.

26.1 Source-faithful PGVWC07 intermediate steps

The theorems below record the intermediate construction steps of the proof of [PGVWC07, Theorem Th:TIcanonical], in the source proof order: single-block unital and dual-diagonal construction; blockwise composition; arbitrary-input form with the explicit zero summand; the length-zero dimension identity; the positive-length form; the positive-length witness; weight normalization; the nonzero-coefficient existence theorem; and the dichotomy that yields Theorem 9.1. These are recordings of the source structure. They are not used by the canonical-form reduction in the proof of the Fundamental Theorem, which goes through Theorem 9.35 and the after-blocking statements of Chapter 10.

Theorem 26.1 (Single irreducible-block PGVWC07 canonical-form theorem)**.** Lean ✓ Stmt *Let A be an irreducible nonzero MPS block with positive bond dimension. Then the Perron–Frobenius eigenvector produces a positive scalar r and a positive definite matrix ρ such that*

$$B^i := r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$$

is unital. In this unital gauge every fixed point of $\mathcal{E}_B$ is a scalar multiple of the identity. Moreover a unitary conjugate $C^i = U^\dagger B^i U$ remains unital and has a diagonal positive definite dual fixed point Λ:

$$\sum_i C^i(C^i)^\dagger = \mathbb{1}, \qquad \sum_i (C^i)^\dagger \Lambda C^i = \Lambda.$$

The construction preserves the finite-ring coefficients up to the spectral-radius weight carried by the rescaling from A to B.

This composes the single-block canonical-form existence steps of [PGVWC07, Theorem Th:TIcanonical]. It is not yet the full theorem, because the recursive finite-ring block decomposition and the final bond-dimension bound have not been combined into this construction.

Proof. ✓ Proof Theorem 6.24 gives the unital gauge. Gauge equivalence preserves irreducibility of the bond-space action, so Theorem 9.17 shows that every fixed point of the unital representative is scalar. Then Theorem 9.31 gives the unitary diagonalization of the dual fixed point. □

Theorem 26.2 (Blockwise PGVWC07 unital and dual-diagonal block decomposition)**.** Lean ✓ Stmt *Let a tensor A be represented, in finite-ring MPV coefficients, by a finite direct sum of nonzero irreducible blocks with unit weights. Then there is a weighted direct sum with the same MPV family such that every block is in the unital orientation,*

$$\sum_i C_k^i (C_k^i)^\dagger = \mathbb{1},$$

every fixed point of the block transfer map is a scalar multiple of the identity, and each block has a diagonal positive definite dual fixed point

$$\sum_i (C_k^i)^\dagger \Lambda_k C_k^i = \Lambda_k.$$

The weights are the positive spectral-radius weights produced by applying the single-block Perron–Frobenius gauge to each summand.

This is the blockwise composition of [PGVWC07, Theorem Th:TIcanonical] after the recursive splitting has already produced the nonzero irreducible summands. It is not yet the full source theorem, because it does not start from an arbitrary representation and does not prove the final total bond-dimension bound.

Proof. ✓ Proof Apply Theorem 26.1 to each nonzero irreducible block. Gauge equivalence and the final unitary conjugation preserve the block MPV coefficients up to the displayed spectral-radius weights, and unitary conjugation carries the scalar fixed-point property of the unital transfer map. □

Theorem 26.3 (Arbitrary-input PGVWC07 unital dual-diagonal form with zero block)**.** Lean ✓ Stmt *Let A be an arbitrary translation-invariant MPS tensor. Then there is a zero-block bond dimension D_0, a finite family of nonzero blocks C_k, and positive real weights μ_k, such that for every chain length $N \geq 1$ and every word $\sigma \in \{0, \dots, d-1\}^N$,*

$$\mathrm{MPV}_A = \mathrm{MPV}_{\oplus_k \mu_k C_k},$$

together with the length-zero dimension identity $D = D_0 + \sum_k D_k$. Each nonzero block is in the unital orientation,

$$\sum_i C_k^i (C_k^i)^\dagger = \mathbb{1},$$

every fixed point of its transfer map is a scalar multiple of the identity, and it has a diagonal positive definite dual fixed point

$$\sum_i (C_k^i)^\dagger \Lambda_k C_k^i = \Lambda_k.$$

Proof. ✓ Proof First apply the zero-block separation theorem to A, retaining the all-zero summands only through their total bond dimension D_0 and extracting the nonzero irreducible summands. Then apply the blockwise PGVWC07 unital and dual-diagonal theorem to this nonzero family. Composing the positive-length equalities and the dimension identities gives the statement. □

Theorem 26.4 (Bond-dimension bound for the zero-block PGVWC07 form). Lean ✓ Stmt *In the decomposition of Theorem 26.3, the zero-block bond dimension and the nonzero block dimensions satisfy*

$$D = D_0 + \sum_k D_k.$$

In particular, the total bond dimension of the nonzero blocks is at most the original bond dimension D.

Proof. ✓ Proof The dimension identity is part of Theorem 26.3, and the inequality follows because $D_0 \geq 0$. □

Theorem 26.5 (Positive-length PGVWC07 unital dual-diagonal form). Lean ✓ Stmt *For any MPS tensor A of bond dimension D, there are nonzero blocks C_k and positive real weights μ_k such that each block is in the unital orientation,*

$$\sum_i C_k^i (C_k^i)^\dagger = \mathbb{1},$$

has only scalar fixed points for its transfer map, and has a diagonal positive-definite fixed point for the dual transfer map,

$$\sum_i (C_k^i)^\dagger \Lambda_k C_k^i = \Lambda_k.$$

Moreover, for every chain length $N > 0$ and word $\sigma \in \{0, \dots, d-1\}^N$,

$$\mathrm{MPV}_A(\sigma) = \mathrm{MPV}_{\bigoplus_k \mu_k C_k}(\sigma),$$

and the nonzero block dimensions satisfy $\sum_k D_k \leq D$.

Proof. ✓ Proof Apply Theorem 26.4. The all-zero summand contributes zero to every positive-length coefficient, so it may be omitted from the displayed MPV identity for $N > 0$. The bound $\sum_k D_k \leq D$ is one of the conclusions of that theorem. □

Definition 26.6 (Positive-length PGVWC07 canonical-form witness). Lean ✓ Stmt A positive-length PGVWC07 canonical-form witness for an MPS tensor A consists of a finite family of nonzero unital blocks, positive real weights, diagonal positive-definite dual fixed points, scalar fixed-point conclusions, positive block dimensions, positive-length MPV equality with A, and the total bond-dimension bound $\sum_k D_k \leq D$.

This definition records the exact-MPV, nonempty-ring part of the PGVWC07 construction. It does not include the source theorem's upper normalization $1 \geq \mu_k$, which depends on the global spectral-radius normalization convention of [PGVWC07, Theorem Th:TIcanonical].

Theorem 26.7 (Weight normalization for a nonempty PGVWC07 witness). Lean ✓ Stmt *Let a positive-length PGVWC07 canonical-form witness have at least one nonzero block. Then its positive real weights may be divided by their largest value so that the new weights ν_k are positive, $|\nu_k| \leq 1$ for all k, and $|\nu_{k_0}| = 1$ for at least one block k_0. If the original weights are $\mu_k = s\nu_k$ with $s > 0$, then for every chain length $N > 0$,*

$$\mathrm{MPV}_A(\sigma) = s^N \,\mathrm{MPV}_{\bigoplus_k \nu_k C_k}(\sigma).$$

This is the scalar normalization corresponding to [PGVWC07, Theorem Th:TIcanonical]; the displayed factor records the global length-dependent rescaling.

Proof. ✓ Proof Choose the largest positive real weight s. Dividing every weight by s gives positive weights bounded above by 1, and a block attaining the maximum has normalized weight 1. The MPV identity follows from Lemma 9.32. $\square$

Theorem 26.8 (Projective form of PGVWC07 weight normalization)**.** Lean ✓ Stmt *Let a positive-length PGVWC07 canonical-form witness have at least one block. After dividing the positive real weights by their largest value, the normalized weights ν_k are positive real numbers, satisfy $|\nu_k| \leq 1$ for all k, and satisfy $|\nu_{k_0}| = 1$ for some block k_0. Moreover the original tensor and the normalized weighted block tensor generate nonzero proportional MPV families: for every length N there is a nonzero scalar c_N such that*

$$\mathrm{MPV}_A(\sigma) = c_N \, \mathrm{MPV}_{\bigoplus_k \nu_k C_k}(\sigma).$$

Proof. ✓ Proof For $N > 0$, take $c_N = s^N$, which is nonzero because $s > 0$. For $N = 0$, use the length-zero MPV coefficient and the positive total dimension of the nonzero blocks to choose the nonzero scalar relating the two traces. $\square$

Lemma 26.9 (Nonzero positive-length coefficient gives a nonempty PGVWC07 witness)**.** Lean ✓ Stmt *Let a positive-length PGVWC07 canonical-form witness for A be given. If some positive-length MPV coefficient of A is nonzero, then the witness has at least one block.*

Proof. ✓ Proof If the block family were empty, the block-diagonal MPV expansion would be an empty sum at every positive length. The positive-length MPV equality in the witness would then force all positive-length coefficients of A to vanish, contradicting the hypothesis. $\square$

Theorem 26.10 (Nonzero projective PGVWC07 normalized form)**.** Lean ✓ Stmt *Suppose that some positive-length MPV coefficient of an MPS tensor A is nonzero. Then A has a nonempty finite family of nonzero PGVWC07 blocks C_k with positive normalized weights ν_k satisfying $|\nu_k| \leq 1$ for all k and $|\nu_{k_0}| = 1$ for some block k_0. Each block is in the unital orientation, has only scalar fixed points for its transfer map, and has a diagonal positive-definite fixed point Λ_k for the dual transfer map. Moreover the original tensor and the normalized weighted block tensor generate nonzero proportional MPV families, and the block dimensions satisfy $\sum_k D_k \leq D$.*

Proof. ✓ Proof First use Theorem 26.13. The nonzero positive-length coefficient makes the witness nonempty by Lemma 26.9. Apply the projective weight-normalization theorem to this nonempty witness. $\square$

Theorem 26.11 (Exact normalized PGVWC07 form after global rescaling)**.** Lean ✓ Stmt *Suppose that some positive-length MPV coefficient of an MPS tensor A is nonzero. Then there is a positive scalar s and a nonempty finite family of PGVWC07 blocks C_k with positive normalized weights ν_k satisfying $|\nu_k| \leq 1$ for all k and $|\nu_{k_0}| = 1$ for some block k_0. Each block is in the unital orientation, has only scalar fixed points for its transfer map, and has a diagonal positive-definite fixed point for the dual transfer map. Each block has positive bond dimension. Moreover the globally rescaled tensor $s^{-1}A$ and the normalized weighted block tensor have the same positive-length MPV coefficients, and the block dimensions satisfy $\sum_k D_k \leq D$.*

Proof. ✓ Proof Start from the positive-length witness and use the nonzero coefficient to make the block family nonempty. Divide the positive weights by their maximum. The exact coefficient identity for A contains the global factor s^N at length N; applying the scalar rescaling s^{-1} to every matrix of A contributes the factor s^{-N}, so the two factors cancel at every positive length. $\square$

Theorem 26.12 (Zero/nonzero dichotomy for exact rescaled PGVWC07 form). Lean ✓ Stmt *For every MPS tensor A, either every positive-length MPV coefficient of A is zero, or the exact rescaled normalized PGVWC07 form of Theorem 26.11 exists.*

Proof. ✓ Proof If some positive-length coefficient is nonzero, apply Theorem 26.11. Otherwise, every positive-length coefficient is zero. □

Theorem 26.13 (Existence of the positive-length PGVWC07 witness). Lean ✓ Stmt *For every MPS tensor A, the conclusions of Theorem 26.5 give a positive-length PGVWC07 canonical-form witness.*

Proof. ✓ Proof Unpack Theorem 26.5 and use its conclusions as the fields of the witness. □

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